MTH632

Topic: ABC

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Heading

Good school is that which most successfully achieves the targets set by the stakeholders by showing high grades and good results in exams.

1. Instructional Leadership
2. Clear and Focused Mission
3. Safe and Orderly Environment
4. Climate of High Expectations

A school is effective not because of the specific nature of what is taught (though it is important) but through the manner in which a positive, supportive, richly and frequently interactive atmosphere is created.
Gammage, 1985.

Source link with pic

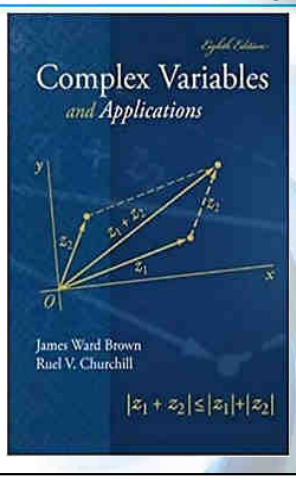
- What is a good school?
- And Effective school?
  What is the difference between 'effective school' and 'good school'?

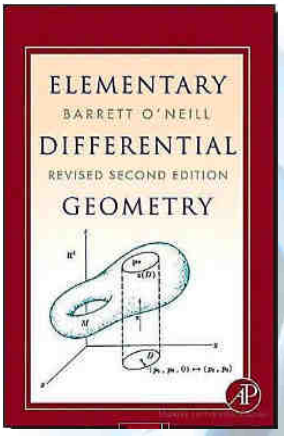
END

And Effective school?
What is the difference between 'effective school' and 'good school'?

Topic No. 01

Complex Analysis & Differential Geometry





Complex Analysis & Differential Geometry

Complex Numbers: An Introduction

Complex Analysis & Differential Geometry

Natural Numbers

$1, 2, 3, 4, 5, 6, 7, 8, 9, \ldots$

Complex Analysis & Differential Geometry

Integers

$0, \pm 1, \pm 2, \pm 3, \pm 4, \pm 5, \pm 6, \pm 7, \pm 8, \pm 9, \ldots$

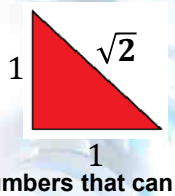
Complex Analysis & Differential Geometry

Rational Numbers $\mathbb{Q}$

$$\mathbb{Q} = \left\{ \frac{a}{b} : a, b \in \mathbb{Z}, b \neq 0 \right\}$$

Complex Analysis & Differential Geometry

Irrational Numbers I



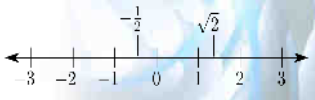
Numbers that can not be written as $\frac{p}{q}$.

The decimal expression is non-terminating and non-periodic

Complex Analysis & Differential Geometry

Real Numbers $\mathbb{R}$

Union of rational numbers and irrational numbers



Complex Analysis & Differential Geometry

Numbers are useful

$6x + 1 = 13$

Do we need more Numbers?

$$x^2 + 2x + 2 = 0$$

We need one more number

$$i = \sqrt{-1}$$

Complex Numbers

$$a + ib: \; a, b \in \mathbb{R}$$

Complex Numbers

"Cardano was the first to introduce complex numbers $a + \sqrt{-1}b$ into algebra, but had misgivings about it"

"To divide 10 in two parts, the product of which is 40".

$$5 + \sqrt{-5}, \qquad 5 - \sqrt{-5}$$

Complex Numbers

Rafael Bombelli authored l'Algebra (1572, and 1579) a set of three books

"At first, the thing seemed to me to be based more on sophism than on truth, but I searched until I found the proof"

END

Complex Numbers

L. Euler (1707-1783) introduced the notation $i = \sqrt{-1}$

Topic No. 02

What are complex numbers?

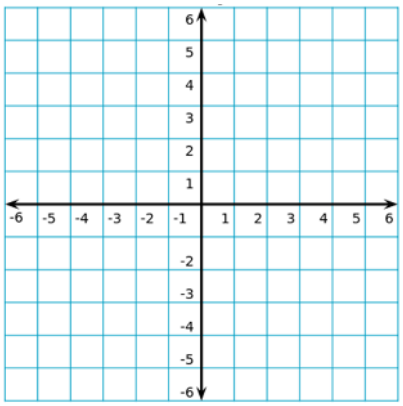
A complex number z can be defined as an ordered pair

$$\mathrm{z} = (\mathrm{x}, \mathrm{y})$$

where x and y are real numbers

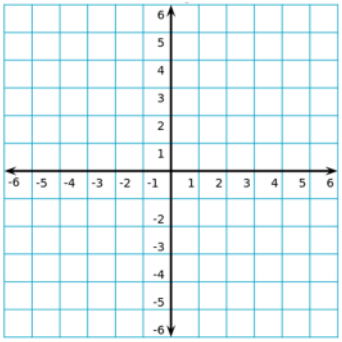
Complex Numbers

Complex numbers can interpreted as points in the complex plane



Complex Numbers

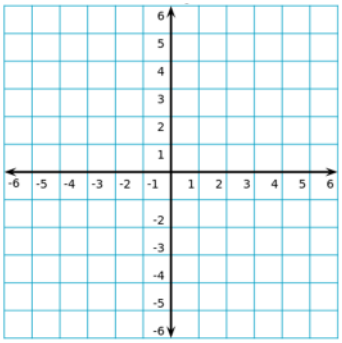
- When real numbers x are displayed as points $(x, 0)$ on the real axis the set of complex numbers includes the real numbers as a subset.
- Complex numbers of the form $(0, y)$ correspond to points on the y-axis and are pure imaginary numbers when $y \neq 0$. The y-axis is then referred to as the imaginary axis.



Complex Numbers

For complex number
$z = (x, y)$
The real numbers x and y are known as the real and imaginary parts of z respectively; and we write

$x = Re\, z \quad y = Im\, z$

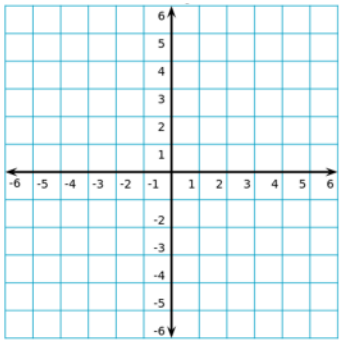


Complex Numbers

Equality of complex numbers

Two complex Numbers z_1 and z_2 are equal whenever they have the same real parts and the same imaginary parts.

Thus the statement $z_1 = z_2$ means that z_1 and z_2 correspond to the same point in the complex, or z, plane.



Complex Numbers

end

The Algebra of Complex Numbers

Topic No. 03

Computations in complex numbers

The Algebra of Complex Numbers

Addition of two complex number

For

$z_1 = (x_1, y_1)$,
$z_2 = (x_2, y_2)$

The addition is defined as

$z_1 + z_2$
$= (x_1, y_1) + (x_2, y_2)$
$= (x_1 + x_2, y_1 + y_2)$

The Algebra of Complex Numbers

Product of two complex number

For $z_1 = (x_1, y_1), z_2 = (x_2, y_2)$, the product is defined as

$z_1 . z_2 = (x_1, y_1).(x_2, y_2) = (x_1 x_2 - y_1 y_2, y_1 x_2 + x_1 y_2)$

The Algebra of Complex Numbers

Product of two complex number

For $z_1 = (x_1, y_1), z_2 = (x_2, y_2)$, the product is defined as

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The following definition of product (denoted by $*$) of complex numbers seems natural and very tempting

$$(x_1, y_1) * (x_2, y_2) = (x_1x_2, y_1y_2).$$

But the above definition has many flaws, including the fact that we can't find inverses with respect to this product.

The Algebra of Complex Numbers

Addition of Complex numbers
and
Addition of Real numbers

The Algebra of Complex Numbers

Product of Complex numbers
and
Product of Real numbers

The Algebra of Complex Numbers

Any complex number $z = (x, y)$ can be written as

$$z = (x, y) = (x, 0) + (0, y)$$

The Algebra of Complex Numbers

Any complex number $z = (x, y)$ can be written as

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Also we have $(0, 1)\ (y, 0) = (0, y)$

The Algebra of Complex Numbers

Any complex number $z = (x, y)$ can be written as

$$z = (x, y) = (x, 0) + (0, y)$$

Also we have $(0, 1)\ (y, 0) = (0, y)$, so

$$z = (x, y) = (x, 0) + (0, 1)(y, 0)\ .$$

The Algebra of Complex Numbers

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The Algebra of Complex Numbers

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The numbers of the form $(x, 0)$ are real numbers so can be simply denoted by x. Since $(0, 1)(0, 1) = (-1, 0)$ so we denote by i the number $(0, 1)$ then

$$z = (x, y) = (x, 0) + (0, 1)(y, 0) = x + iy\ .$$

The Algebra of Complex Numbers

Any complex number $z = (x, y)$ can be written as

$$z = (x, y) = (x, 0) + (0, y)$$

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$$z = (x, y) = (x, 0) + (0, 1)(y, 0) = x + iy\,.$$

The above representation is makes the addition and product of complex numbers simple.

The Algebra of Complex Numbers

Most of the time we use the following representation of complex numbers

$$z = x + iy\,.$$

The representation is due to Gauss.

The Algebra of Complex Numbers

In this representation the addition and multiplication becomes

$$(x_1 + iy_1) + (x_2, +iy_2)$$

The Algebra of Complex Numbers

In this representation the addition and multiplication becomes

$$(x_1 + i\,y_1).(x_2 + iy_2)$$

The Algebra of Complex Numbers

In this representation the addition and multiplication becomes

$$(x_1 + iy_1) + (x_2, +iy_2) = (x_1 + x_2) + i(y_1 + y_2)$$

$$(x_1 + i\,y_1).(x_2 + iy_2) = (x_1x_2 - y_1y_2) + i(y_1\,x_2 + x_1y_2)$$

The Algebra of Complex Numbers

In this representation the addition and multiplication becomes

$$(x_1 + iy_1) + (x_2, +iy_2) = (x_1 + x_2) + i(y_1 + y_2)$$

$$(x_1 + i\,y_1).(x_2 + iy_2) = (x_1x_2 - y_1y_2) + i(y_1\,x_2 + x_1y_2)$$

In this representation the operation of product is natural.

Complex Analysis & Differential Geometry

- Two ways to represent complex numbers.
- Addition of complex numbers
- Product of complex numbers

END

Addition of Complex Numbers

Topic No. 04

Properties of addition of complex numbers

Addition of Complex Numbers

Although the ordered pair representation is the real foundational definition on which complex number system is built.
But due to simplicity we prefer the following representation of complex numbers

$$z = x + iy$$

Addition of Complex Numbers

The addition of complex numbers follow all the properties of real numbers

Addition of Complex Numbers

The commutative Law

$$z_1 + z_2 = z_2 + z_1$$

Addition of Complex Numbers

The associative law

$$(z_1 + z_2) + z_3 = z_1 + (z_2 + z_3)$$

Due to associative laws we can write a sum

$$z_1 + z_2 + z_3$$

without using parenthesis.

Addition of Complex Numbers

Additive Identity

There is unique complex number w such that

$$z + w = w + z = z$$

Such number w is obviously

$$(0,0) \quad \text{or} \quad 0 + i0$$

known as the additive identity and denoted as

$$0 = (0,0) \text{ or } 0 = 0 + i0$$

Addition of Complex Numbers

Additive Inverse

For every complex number

$$z = x + iy$$

there is a unique ξ such that

$$z + \xi = 0$$

Obviously

$$\xi = -x - iy$$

known as additive inverse and denoted as

$$-z = -x - iy$$

Or in order pair notation

$$-z = (-x, -y)$$

Complex Analysis & Differential Geometry

These properties of addition of complex numbers will be used in our algebraic manipulations

END

Multiplication of Complex Numbers

Topic No. 05

Properties of addition of complex numbers

Multiplication of Complex Numbers

We use the following representation of complex number

$$z = x + iy$$

This representation makes it easy to describe properties of multiplication of complex numbers.

Multiplication of Complex Numbers

The product of complex numbers follow all the properties of real numbers

Multiplication of Complex Numbers

The commutative Law

$$z_1 z_2 = z_2 z_1$$

The commutativity implies that

$$iy = yi$$

So we can write complex number in any of the following ways

$$z = x + iy$$

or

$$z = x + yi$$

Multiplication of Complex Numbers

The associative law

$$(z_1 z_2) z_3 = z_1 (z_2 z_3)$$

Due to associative laws we can write a product

$$z_1 z_2 z_3$$

without using parenthesis.

Multiplication of Complex Numbers

Multiplicative Identity

There is complex number w such that

$$zw = wz = z$$

Such number w is obviously unique and is given by

$(1,0)$ or $1 + i0$

known as the multiplicative identity and denoted as

$1 = (1,0)$ or $1 = 1 + i0$

Multiplication of Complex Numbers

Multiplicative Inverse

For every complex number $z = x + iy$ other than $(0,0)$ there is a unique z^{-1} such that

$$zz^{-1} = 1 .$$

In fact

$$z^{-1} = \frac{1}{z} = \frac{1}{x + iy}$$

To write z^{-1} in standard form $x + iy$ we proceed as follows

$$z^{-1} = \frac{1}{x + iy} = \frac{x - iy}{(x + iy)(x - iy)} = \frac{x - iy}{x^2 - i^2 y^2} = \frac{x - iy}{x^2 + y^2}$$

Hence

$$z^{-1} = \frac{x}{x^2 + y^2} - i\frac{y}{x^2 + y^2} .$$

Multiplication of Complex Numbers

Multiplicative Inverse

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Hence

$$z^{-1} = \frac{x}{x^2 + y^2} - i\frac{y}{x^2 + y^2} .$$

Multiplication of Complex Numbers

Division of Complex Numbers

Using

$$z^{-1} = \frac{x}{x^2 + y^2} - i\frac{y}{x^2 + y^2} .$$

We can define division of two complex numbers

$$z_1 = x_1 + iy_1, z_2 = x_2 + iy_2$$

as

$$\frac{z_1}{z_2} = z_1 z_2^{-1}$$

$$\frac{z_1}{z_2} = (x_1 + iy_1)\left(\frac{x_2}{{x_2}^2 + {y_2}^2} - i\frac{y_2}{{x_2}^2 + {y_2}^2}\right)$$

$$= \left(\frac{x_1 x_2 + y_1 y_2}{{x_2}^2 + {y_2}^2}, \frac{y_1 x_2 - x_1 y_2}{{x_2}^2 + {y_2}^2}\right)$$

Multiplication of Complex Numbers

Division of Complex Numbers

If $z_1 = 4 + i$ and $z_2 = 2 - i3$ then $\frac{z_1}{z_2}$ can be calculated using

$$\frac{z_1}{z_2} = \left(\frac{x_1x_2 + y_1y_2}{{x_2}^2 + {y_2}^2}, \frac{y_1x_2 - x_1y_2}{{x_2}^2 + {y_2}^2}\right)$$

we get

$$\frac{z_1}{z_2} = \frac{5}{13} + i\frac{14}{13}$$

Multiplication of Complex Numbers

Distributive Law

The operation of multiplication is distributive over addition, that is,

$z_1(z_2 + z_3) = z_1z_2 + z_1z_3$

Distributive law is a associates addition with multiplication

Multiplication of Complex Numbers

Binomial formula

For any two complex numbers z_1, z_2 we have

$$(z_1 + z_2)^n = \sum_{k=0}^{n} \binom{n}{k} z_1^k z_2^{n-k}$$

for $n = 1, 2, 3, \ldots$ Here

$$\binom{n}{k} = \frac{n!}{k!\,(n-k)!}$$

Complex Analysis & Differential Geometry

These properties of multiplication of complex numbers will be used in our algebraic manipulations

END

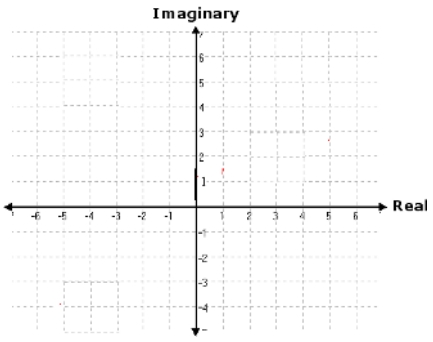
The Geometry of Complex Numbers

Topic No. 06

Algebraic Operations and Geometric Representation of Complex Numbers

The Geometry of Complex Numbers

A complex number $z = (x, y) = x + iy$ can be associated with the directed line segment, or vector, from origin to the point $z = (x, y)$



The Geometry of Complex Numbers

Recall: for

$$z_1 = (x_1, y_1), z_2 = (x_2, y_2)$$

The addition is defined as

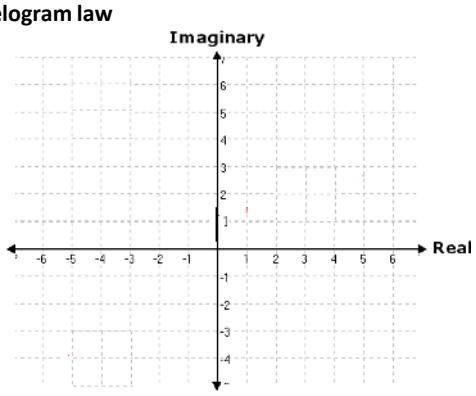
$z_1 + z_2 = (x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2)$

OR

$(x_1 + iy_1) + (x_2, +iy_2) = (x_1 + x_2) + i(y_1 + y_2)$

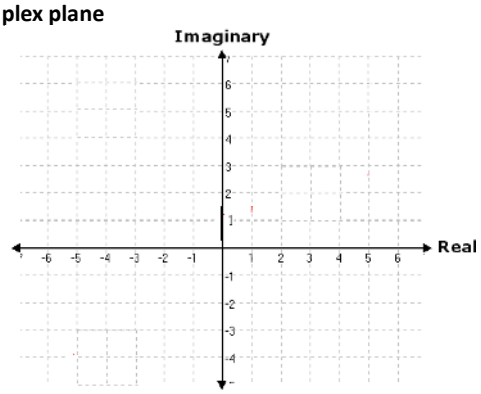
The Geometry of Complex Numbers

Geometrically, in complex plane, $z_1 + z_2$ can be obtained using parallelogram law



The Geometry of Complex Numbers

Similarly we can calculate the difference $z_1 - z_2$ in the complex plane



The Geometry of Complex Numbers

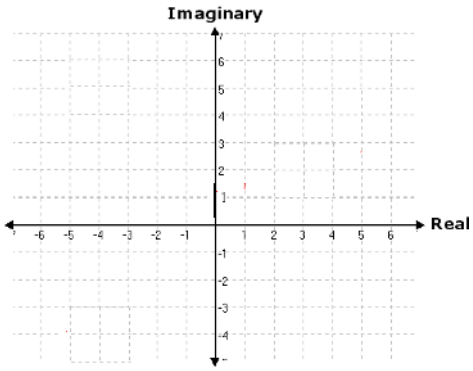
Multiplication of two complex numbers is neither dot product nor vector product

The Geometry of Complex Numbers

The vector interpretation helps us defining modulus or absolute value of a complex number

The Geometry of Complex Numbers

Geometrically the modulus of $z = (x, y) = x + iy$ is distance between the point (x, y) and the origin



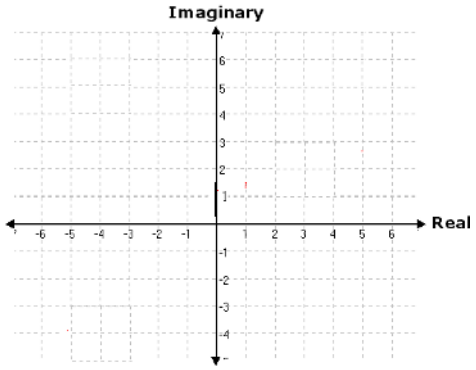
The Geometry of Complex Numbers

The modulus or absolute value of a complex number $z = (x, y) = x + iy$ is defined and denoted as

$$|z| = \sqrt{x^2 + y^2}\,.$$

The Geometry of Complex Numbers

Comparison of two complex numbers



The Geometry of Complex Numbers

- Geometry of algebraic operations
- Modulus of a complex number

END

The Geometry of Complex Numbers

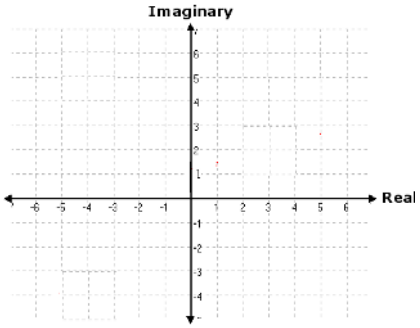
Topic No. 07

Properties of Modulus

The Geometry of Complex Numbers

The modulus or absolute value of a complex number $z = (x, y) = x + iy$ is defined and denoted as

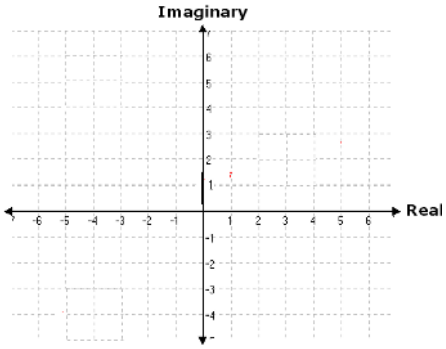
$$|z| = \sqrt{x^2 + y^2}\,.$$



The Geometry of Complex Numbers

Triangle inequality

$$|z_1 + z_2| \le |z_1| + |z_2|$$



The Geometry of Complex Numbers

An important consequence of triangle inequality is

$$|z_1 + z_2| \ge \big||z_1| - |z_2|\big|$$

Proof

The Geometry of Complex Numbers

Using triangle inequality

$$|z_1 + z_2| \le |z_1| + |z_2|$$

and the following inequality

$$|z_1 + z_2| \ge \big||z_1| - |z_2|\big|$$

We get

$$\big||z_1| - |z_2|\big| \le |z_1 + z_2| \le |z_1| + |z_2|$$

The Geometry of Complex Numbers

Example: Consider a complex number z such that $|z| = 1$.

The Geometry of Complex Numbers

Example: Consider a complex number z such that $|z| = 1$.

Using

$$\big||z_1| - |z_2|\big| \le |z_1 + z_2| \le |z_1| + |z_2|$$

we get

$$|z - 2|$$

The Geometry of Complex Numbers

- Properties of Modulus of a complex number

END

Complex Conjugates

Topic No. 08

Complex Conjugates

Complex Conjugates

The complex conjugate or simple the conjugate of a complex number

$$z = x + iy,$$

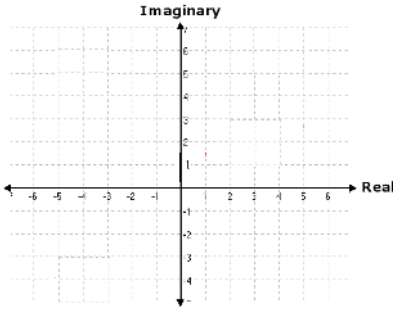
denoted by $\bar{z}$, is given by

$$\bar{z} = (x, -y) = x - iy$$

Complex Conjugates

The complex conjugate or simple the conjugate of a complex number $z = x + iy$, denoted by $\bar{z}$, is given by

$$\bar{z} = (x, -y) = x - iy$$



Complex Conjugates

For two complex number $z_1 = (x_1, y_1) = x_1 + iy_1$ and $z_2 = (x_2, y_2) = x_2 + iy_2$ we have

$$\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}$$

Complex Conjugates

For two complex number $z_1 = (x_1, y_1) = x_1 + iy_1$ and $z_2 = (x_2, y_2) = x_2 + iy_2$ we have

$$\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}$$

Similarly we can show the following

$$\overline{z_1 - z_2} = \overline{z_1} - \overline{z_2}$$

$$\overline{z_1 z_2} = \overline{z_1}\,\overline{z_2}$$

$$\overline{\left(\frac{z_1}{z_2}\right)} = \frac{\overline{z_1}}{\overline{z_2}} \quad if\ z_2 \neq 0$$

Complex Conjugates

Complex conjugate, $\mathrm{Re}(z)$, and $\mathrm{Im}(z)$

Complex Conjugates

Complex conjugate, $\mathrm{Re}\ z$, and $\mathrm{Im}\ z$

For $z = (x, y) = x + iy$ we have

$$Re(z) = \frac{z + \bar{z}}{2}, \qquad Im(z) = \frac{z - \bar{z}}{2i}$$

Similar we have the following

$$Re\,(iz) = -Im\,(z)$$

$$Im(iz) = Re(z)$$

Complex Conjugates

Complex conjugate helps us in proving properties of modulus. First observe that

$$|z|^2 = z\,\bar{z}\,.$$

Complex Conjugates

Complex conjugate helps us in proving properties of modulus. First observe that

$$|z|^2 = z\,\bar{z}\,.$$

For complex numbers z_1 and z_2 we have

$$|z_1 z_2| = |z_1||z_2|$$

Proof:

Complex Conjugates

Complex conjugate helps us in proving properties of modulus.
First observe that

$$|z|^2 = z\,\bar{z}\,.$$

For complex numbers z_1 and z_2 we have

$$|z_1 z_2| = |z_1||z_2|$$

$$\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}$$

Complex Conjugates

Triangle inequality

$$|z_1 + z_2| \le |z_1| + |z_2|$$

Proof:

Complex Conjugates

Example: If a complex number z satisfies $|z| < 2$ then

$$|z^3 + 3z^2 - 2z + 1|$$

has upper bound given by using triangle inequality and properties of modulus in the following way.

The Geometry of Complex Numbers

- Conjugate and its properties
- How conjugate is used to prove different properties of complex numbers

END

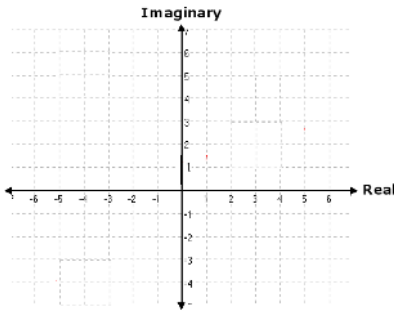
Exponential Form

Topic No. 09

Exponential Form of Complex Numbers

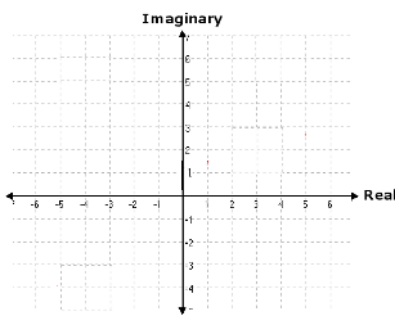
Exponential Form

Recall: A complex number $z = (x, y) = x + iy$ can be associated with the directed line segment, or vector, from origin to the point $z = (x, y)$



Exponential Form

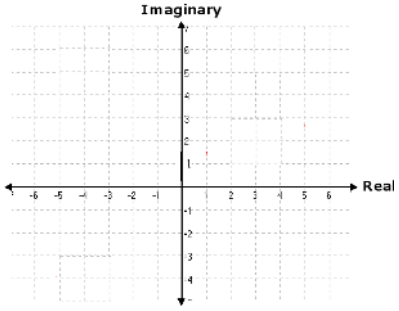
Consider a non-zero complex number $z = x + iy$.
Let r be length of the vector, the length can be obtained by using modulus, i.e. $r = |z|$.



Exponential Form

Let θ be the angle that the vector make with positive x-axis.
Since $x = r\cos\theta$ and $y = r\sin\theta$ so the number z can be written in polar form as

$$z = (r\cos\theta, r\sin\theta) = r\cos\theta + i\sin\theta$$



Exponential Form

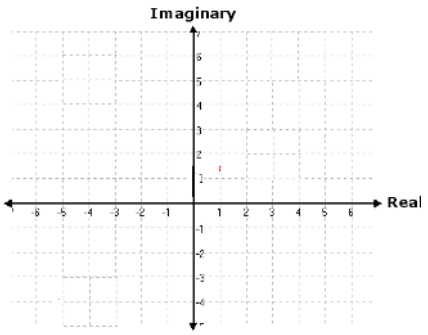
The number $z = x + iy$ can be written in polar form as

$$z = (r\cos\theta, r\sin\theta) = r\cos\theta + i\sin\theta$$

This is known as polar representation of complex numbers and the values r and θ are called polar coordinates of z.

Exponential Form

If $z = 0$ then the coordinate θ is undefined.



Exponential Form

The real number $r = |z|$ is not allowed to be negative since it is length of the radius vector for z.

Exponential Form

The real number θ, represents the angle, measured in radians, that z makes with positive x-axis.

Exponential Form

Example: Let $z = 1 + i$

Exponential Form

For a complex number $z = (x, y)$, the coordinate θ has infinite number of possible values. All these valued differ by integral multiples of 2π.

Exponential Form

So the coordinate θ is any value for which the identities

$$x = r\cos\theta \text{ and } y = r\sin\theta$$

holds. Formally we have the following definition.

Definition (argument of $z = (x, y)$)

If $z \neq 0$,

$$\arg z = \{\theta : z = r(\cos\theta + i\sin\theta)\}$$

If $\theta \in \arg z$ then we say that θ is an argument of z.

Exponential Form

Definition (Principal value of $\arg z$)

Let $z \neq 0$ be a complex number. Then principal value of argument of z, denoted by $\text{Arg } z$, is a unique value θ such that

$$z = r(\cos\theta + i\sin\theta) \text{ and } -\pi < \theta \leq \pi$$

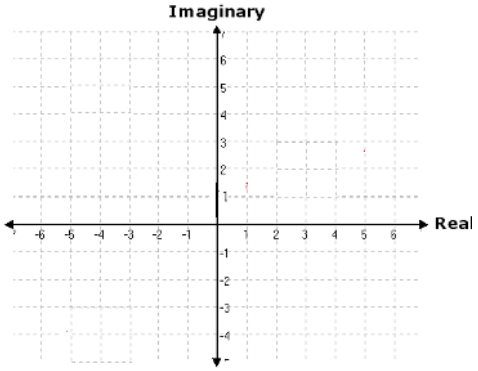
Principal value of argument of z is also referred as "the argument of z"

Exponential Form

$$\arg z = \mathrm{Arg}\, z + 2n\pi, \quad (n = 0, \pm 1, \pm 2, \pm 3, \ldots)$$

Exponential Form

Calculating argument of $z = (x, y)$



Exponential Form

Calculating argument of $z = (x, y)$

If $z = x + iy = r(\cos\theta + i\sin\theta)$, where $x \neq 0$, then

$$\arg z \subset \arctan\left(\frac{y}{x}\right)$$

Exponential Form

Calculating argument of $z = (x, y)$

First step: If $x \neq 0$, then calculate $\arctan\left(\frac{y}{x}\right)$

Second step: Check the quadrant of $z = (x, y)$

Third step: Add/subtract multiples of π according to the quadrant in which z lies

If we want to find the principal valued of argument of z then we add an additional step

Fourth Step: Add/subtract multiples of 2π so that the angle θ satisfies the following

$$-\pi < \theta \leq \pi$$

Exponential Form

Example: Calculate polar representation of $z = -1 - i$

Exponential Form

If $x = 0$ then we can not use the following to find $\arg z$

$$\tan\theta = \left(\frac{y}{x}\right)$$

If $z \neq 0$ and $x = 0$ then we use the following rule

$$Arg\, z = \frac{\pi}{2} \quad if\ Im\, z > 0, and$$

$$Arg\, z = -\frac{\pi}{2} \quad if\ Im\, z < 0.$$

Exponential Form

Example: Find polar representation of $z = (0, 5) = 5i$

The Geometry of Complex Numbers

- Polar representation of complex numbers

END

Exponential Form

Edited by
Mahar Afaq Safdar Muhammadi
MSc Math, Whatsapp 03494873115

Topic No. 10

Exponential Form of Complex Numbers (Part II)

Exponential Form

Recall

A complex number z can be written in polar form as

$$z = (r\cos\theta, r\sin\theta)$$
$$= r\cos\theta + i\sin\theta$$

Exponential Form

Euler formula

The symbol $e^{i\theta}$ or $exp(i\theta)$ is defined by means of Euler formula as

$$e^{i\theta} = \cos\theta + i\sin\theta$$

Exponential Form

Consider a complex number in polar form

$$z = r(\cos\theta + i\sin\theta)$$

The exponential form of z is given by

$$z = r\,e^{i\theta}$$

We will learn more about the exponential functions in our later discussions.

Exponential Form

Example: The polar representation of $z = -1 - i$ is given by

$$z = \sqrt{2}\left(\cos\left(-\frac{3\pi}{4}\right) + i\sin\left(-\frac{3\pi}{4}\right)\right)$$

The exponential form of z becomes

$$z = \sqrt{2}\,exp\left(-\frac{3\pi}{4}\right)$$

Or

$$z = \sqrt{2}\,e^{-i(3\pi/4)}$$

Exponential Form

Example(continued)

$$z = \sqrt{2}\,e^{-i(3\pi/4)}$$

We know that

$$\mathrm{Arg}\,z = \arg z + 2n\pi, \qquad (n = 0, \pm1, \pm2, \pm3, \dots)$$

So there are infinite exponential forms of z, given by

$$z = \sqrt{2}\,exp\left(i\left(-\frac{3\pi}{4} + 2n\pi\right)\right),\ where\ n = 0, \pm1, \pm2, \dots$$

Exponential Form

The case when $r = 1$

The complex numbers for which $r = 1$ are

$$z = e^{i\theta}$$

These complex numbers are located on a circle of radius 1.

Exponential Form

Euler's identity

For $\theta = \pi$, we get

$$z = e^{i\pi} = \cos\pi + i\sin\pi = -1$$

So we have

$$e^{i\pi} + 1 = 0$$

This is known as Euler's identity.

Exponential Form

Euler's identity

$$e^{i\pi} + 1 = 0$$

Exponential Form

The case $r = R$, where R is arbitrary

The equation

$$z = R\, e^{i\theta} \quad 0 \le \theta \le 2\pi$$

Is a parametric representation of the circle $|z| = R$.

Exponential Form

More generally

$$|z - z_0| = R$$

Is a circle of radius R and centre z_0, also written in the following form

$$z = z_0 + R\, e^{i\theta} \quad 0 \le \theta \le 2\pi\,.$$

Here R and z_0 are fixed and θ is the parameter.

Exponential Form

Example: Solve $\left|e^{i\theta} - 1\right| = 2$ geometrically.

Exponential Form

- Calculating exponential form and its consequences

END

Product and Powers in Exponential Form

Topic No. 11

Products and powers in exponential form

Product and Powers in Exponential Form

By using Euler formula

$$e^{i\theta} = cos\ \theta + i\ sin\ \theta$$

We get the following exponential form of complex numbers

$$z = r\, e^{i\theta}$$

Product and Powers in Exponential Form

Using Euler formula and trigonometry we can verify the following

$$e^{i\theta_1}\, e^{i\theta_2} = e^{i(\theta_1 + \theta_2)}$$

Product and Powers in Exponential Form

Product in exponential form

Given

$$z_1 = r_1 e^{i\theta_1}, \qquad z_2 = r_2 e^{i\theta_2}$$

then

$$z_1 z_2 = \left(r_1 e^{i\theta_1}\right)\left(r_2 e^{i\theta_2}\right) = r_1 r_2 e^{i\,(\theta_1+\theta_2)}\,.$$

Product and Powers in Exponential Form

Product in exponential form

Given

$$z_1 = r_1 e^{i\theta_1}, \qquad z_2 = r_2 e^{i\theta_2}$$

then

$$z_1 z_2 = \left(r_1 e^{i\theta_1}\right)\left(r_2 e^{i\theta_2}\right) = r_1 r_2 e^{i\,(\theta_1+\theta_2)}\,.$$

Example:

Product and Powers in Exponential Form

Division in exponential form

If $z = r\,e^{i\theta}$ then

$$z^{-1} =$$

Product and Powers in Exponential Form

Division in exponential form

If $z = r\,e^{i\theta}$ then

$$z^{-1} = \frac{1}{z} = \frac{1}{r\,e^{i\theta}} = \frac{e^{-i\theta}}{r\,e^{i\theta}e^{-i\theta}} = \frac{1}{r}e^{-i\theta}$$

Hence

$$z^{-1} = \frac{1}{r}\left(cos\,(-\theta) + i\,sin\,(-\theta)\right).$$

Product and Powers in Exponential Form

Division in exponential form

If $z = r\,e^{i\theta}$ then

$$z^{-1} = \frac{1}{z} = \frac{1}{r\,e^{i\theta}} = \frac{e^{-i\theta}}{r\,e^{i\theta}e^{-i\theta}} = \frac{1}{r}e^{-i\theta}$$

Hence

$$z^{-1} = \frac{1}{r}\left(cos\,(-\theta) + i\,sin\,(-\theta)\right).$$

We know that

$cos\,(-\theta) = cos\,\theta$, and $sin\,(-\theta) = -sin\,(\theta)$

Product and Powers in Exponential Form

Division in exponential form

If $z = r\,e^{i\theta}$ then

$$z^{-1} = \frac{1}{z} = \frac{1}{r\,e^{i\theta}} = \frac{e^{-i\theta}}{r\,e^{i\theta}e^{-i\theta}} = \frac{1}{r}e^{-i\theta}$$

Hence

$$z^{-1} = \frac{1}{r}\left(cos\,(-\theta) + i\,sin\,(-\theta)\right).$$

We know that

$cos\,(-\theta) = cos\,\theta$, and $sin\,(-\theta) = -sin\,(\theta)$

So

$$z^{-1} = \frac{1}{r}\left(cos\,(\theta) - i\,sin\,(\theta)\right).$$

Product and Powers in Exponential Form

Division in exponential form

If

$$z_1 = r_1 e^{i\theta_1}, \qquad z_2 = r_2 e^{i\theta_2}$$

then

$$\frac{z_1}{z_2} =$$

Product and Powers in Exponential Form

Division in exponential form

If

$$z_1 = r_1 e^{i\theta_1}, \qquad z_2 = r_2 e^{i\theta_2}$$

then

$$\frac{z_1}{z_2} = r_1 r_2 e^{i(\theta_1-\theta_2)}$$

Example

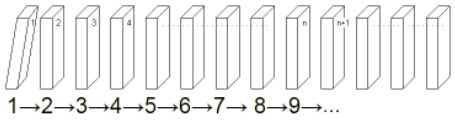
Product and Powers in Exponential Form

Powers in exponential form

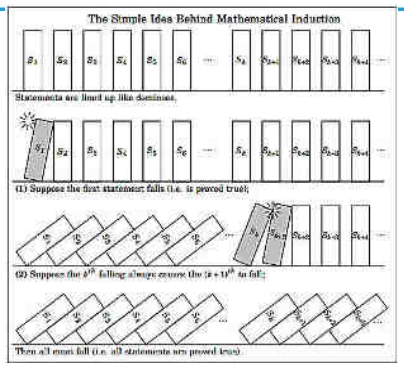
For complex number $z = re^{i\theta}$

$$z^n = r^n e^{in\theta} \ for\ n = 0, \pm 1, \pm 2, \pm 3, \ldots$$

Proof (by mathematical induction)



Product and Powers in Exponential Form



Source: p143 of Book of Proof by Richard Hammack

Product and Powers in Exponential Form

Product in exponential form

For complex number $z = re^{i\theta}$

$$z^n = r^n e^{in\theta} \ for\ n = 0, \pm 1, \pm 2, \pm 3, \ldots$$

Proof (by mathematical induction)

Product and Powers in Exponential Form

Product in exponential form

For complex number $z = re^{i\theta}$

$$z^n = r^n e^{in\theta} \ for\ n = 0, \pm 1, \pm 2, \pm 3, \ldots$$

Proof (by mathematical induction)

Product and Powers in Exponential Form

Product in exponential form

For complex number $z = re^{i\theta}$

$$z^n = r^n e^{in\theta} \ for\ n = 0, \pm 1, \pm 2, \pm 3, \ldots$$

Example:

Product and Powers in Exponential Form

Example: Calculate $\left(\sqrt{3} + i\right)^3$

Product and Powers in Exponential Form

De Moivre's formula

The expression

$$(cos\ \theta + i\ sin\ \theta)^n = cos\ n\theta + i\ sin\ n\theta\ ,$$

Where $n = 0, \pm 1, \pm 2, \ldots$, is known as De Moivre's formula.

Product and Powers in Exponential Form

Example: Show that

$$cos\ 2\theta = cos^2\theta\ - sin^2\theta$$

and

$$sin\ 2\theta = 2\ sin\ \theta\ cos\ \theta$$

Product and Powers in Exponential Form

Exercise: Show that

$$cos\ 5\theta = cos^5\theta\ - cos^3\theta\ sin^2\theta + 5\ cos\ \theta\ sin^4\theta\ .$$

Product and Powers in Exponential Form

We can use De Moivre's formula to prove many trigonometric identities, including Lagrange's trigonometric identity

$$1 + cos\ \theta + cos\ 2\theta + \cdots + cos\ n\theta = \frac{sin\ [(2n+1)\theta/2]}{2\ sin\ (\theta/2)}$$

Product and Powers in Exponential Form

We can use De Moivre's formula to prove many trigonometric identities

$$cos\ n\theta = \sum_{k=0}^{m} \binom{n}{2k} (-1)^k\ cos^{n-2k}\theta\ \ sin^{2k}\ \theta$$

Product and Powers in Exponential Form

END

- Product, division and powers in exponential form of complex numbers
- De Moivre's Formula and its consequences

Roots of Complex Numbers (Part I)

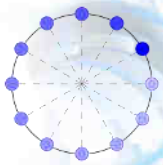
Topic No. 12

Roots of Complex Numbers (PART I)

Source: Azimuth

Roots of Complex Numbers

A number u is said to be an nth root of complex number z if $u^n = z$.



How Many nth Roots

How many nth roots of a complex numbers z exist?

How Many nth Roots

Corollary to the fundamental theorem of algebra

If $p(z)$ is a polynomial of degree n $(n > 0)$ with complex coefficients, then equation $P(z) = 0$ has precisely (counting multiplicities) n solutions.

How Many nth Roots

Corollary to the fundamental theorem of algebra
If $p(z)$ is a polynomial of degree n $(n > 0)$ with complex coefficients, then equation $P(z) = 0$ has precisely (counting multiplicities) n solutions.

Example: $P(z) = z^3 - (4+i)z^2 + (4+i4)z - 4i$

$P(z) = 0$ has following solutions

$P(z) = 0$

$(z-2)^2(z-i) = 0$

$\Rightarrow z_1 = 2,\ z_2 = 2,\ z_3 = i$

How Many nth Roots

- There are precisely n number of nth roots of a complex number z_0

How Many nth Roots

- There are precisely n number of nth roots of a complex number z_0
- We denote the set of all nth roots of a complex number z_0 by $z_0^{1/n}$.

How Many nth Roots

- There are precisely n number of nth roots of a complex number z_0
- We denote the set of all nth roots of a complex number z_0 by $z_0^{1/n}$.
- For positive real number r_0, the symbol $\sqrt[n]{r_0}$ is reserved for one positive root.

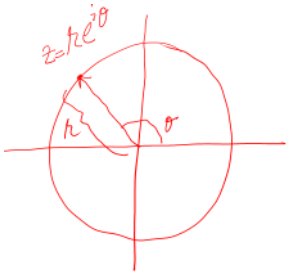
Finding Roots of Complex Numbers

How to find nth roots of a complex number z_0.

Source: momath.org

Finding Roots of Complex Numbers

Equality of complex numbers in exponential form
Consider a complex number $z = r\,e^{i\theta}$



Finding Roots of Complex Numbers

Equality of complex numbers in exponential form
Consider a complex number $z = r\,e^{i\theta}$

Given $z_1 = r_1 e^{i\theta_1}$, $z_2 = r_2 e^{i\theta_2}$

If $z_1 = z_2$

$\Rightarrow r_1 = r_2$

and $\theta_1 = \theta_2 + 2k\pi$, $k \in \mathbb{Z}$.

Finding Roots of Complex Numbers

Calculating powers of complex numbers
Consider a complex number $z = r\,e^{i\theta}$

Finding Roots of Complex Numbers

The nth roots of a complex number

Consider a complex number $z_0 = r_0 e^{i\theta_0}$. Let $z = r\, e^{i\theta}$ be nth root of z_0, that is, we have

$$z^n = z_0$$

Now $z^n = r^n e^{in\theta}$

$z^n = z_0 \Rightarrow r^n e^{in\theta} = r_0 e^{i\theta_0}$

$\Rightarrow r^n = r_0 \qquad n\theta = \theta_0 + 2k\pi, \; k \in \mathbb{Z}$

$\Rightarrow r = \sqrt[n]{r_0}, \qquad \theta = \frac{\theta_0}{n} + \frac{2k\pi}{n}, \; k \in \mathbb{Z}$

So $z = \sqrt[n]{r_0}\, exp\left[i\left(\frac{\theta_0}{n} + \frac{2k\pi}{n}\right)\right], \; k \in \mathbb{Z}$

Finding Roots of Complex Numbers

The nth roots of a complex number

Consider a complex number $z_0 = r_0 e^{i\theta_0}$. The nth root z is given by

$$z = \sqrt[n]{r_0}\, exp\left[i\left(\frac{\theta_0}{n} + \frac{2k\pi}{n}\right)\right], k = 0, \pm 1, \pm 2, \ldots$$

Example: Find square roots of $2i$

$2i = 2\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)$

$z_0 = 2e^{i(\pi/2)}$

$z = \sqrt{2}\, exp\left[i\left(\frac{\pi/2}{2} + \frac{2k\pi}{2}\right)\right], \; k \in \mathbb{Z}$

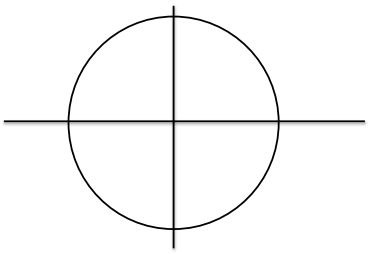
$k = 0: \quad z_1 = 1 + i, \qquad z_2 = -1 - i$

Geometry of Roots

The nth roots of a complex number

Consider a complex number $z_0 = r_0 e^{i\theta_0}$. The nth root z is given by

$$z = \sqrt[n]{r_0}\, exp\left[i\left(\frac{\theta_0}{n} + \frac{2\pi}{n}k\right)\right], k = 0, \pm 1, \pm 2, \ldots$$



Geometry of Roots

The nth roots of a complex number

Consider a complex number $z_0 = r_0 e^{i\theta_0}$. The nth root z is given by

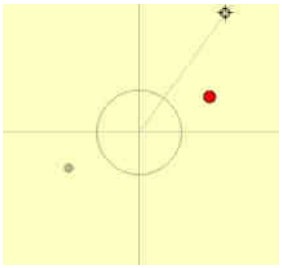
$$z = \sqrt[n]{r_0}\, exp\left[i\left(\frac{\theta_0}{n} + \frac{2\pi}{n}k\right)\right], k = 0, \pm 1, \pm 2, \ldots$$

We denote the distinct nth roots in the following way

$$c_k = \sqrt[n]{r_0}\, exp\left[i\left(\frac{\theta_0}{n} + \frac{2\pi}{n}k\right)\right], \qquad k = 0, 1, 2, \ldots, n-1$$

Geometry of Roots

$$c_k = \sqrt[n]{r_0}\, exp\left[i\left(\frac{\theta_0}{n} + \frac{2\pi}{n}k\right)\right], \qquad k = 0, 1, 2, \ldots, n-1$$



Finding Roots of Complex Numbers

Summary of the method

Consider a complex number z_0.

The following are steps for finding nth roots of z_0

- Write z_0 in exponential form

$$z_0 = r_0 e^{i\theta}$$

Finding Roots of Complex Numbers

Summary of the method

Consider a complex number z_0.

The following are steps for finding nth roots of z_0

- Write z_0 in exponential form

$$z_0 = r_0 e^{i\theta}$$

- Write z_0 in more general form

$$z_0 = r_0 e^{i(\theta_0 + 2k\pi)}, k = 0, \pm 1, \pm 2, \ldots$$

Finding Roots of Complex Numbers

Summary of the method

Consider a complex number z_0.

The following are steps for finding nth roots of z_0

- Write z_0 in exponential form

$$z_0 = r_0 e^{i\theta}$$

- Write z_0 in more general form

$$z_0 = r_0 e^{i(\theta_0 + 2k\pi)}, k = 0, \pm 1, \pm 2, \ldots$$

- The nth roots are

$$z_0^{1/n} = \left[r_0 e^{i(\theta_0 + 2k\pi)}\right]^{1/n}$$

Or

$$c_k = \sqrt[n]{r_0}\, e^{i\left(\frac{\theta_0}{n} + \frac{2\pi}{n}k\right)}, k = 0, 1, \ldots, n-1$$

Roots of Complex Numbers

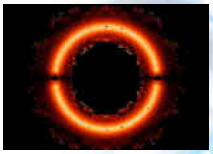
END

- Roots of Complex Numbers
- Geometry of Roots

Roots of Complex Numbers (Part II)

Roots of Complex Numbers (PART II)

Topic No. 13



Source: Azimuth

Finding Roots of Complex Numbers

Summary of the method

Consider a complex number z_0. The following are steps for finding nth roots of z_0

- Write z_0 in exponential form
$$z_0 = r_0 e^{i\theta}$$
- Write z_0 in more general form
$$z_0 = r_0 e^{i(\theta_0 + 2k\pi)}, k = 0, \pm 1, \pm 2, \dots$$
- The nth roots are
$$z_0^{1/n} = \left[r_0 e^{i(\theta_0+2k\pi)}\right]^{1/n} = \sqrt[n]{r_0}\, e^{i\left(\frac{\theta_0}{n}+\frac{2\pi}{n}k\right)}, k = 0, 1, \dots, n-1$$

Roots of Unity

We want to find nth roots of $z_0 = 1$

Step 1 $z_0 = 1 = 1(\cos 0 + i \sin(0))$

$\Rightarrow z_0 = e^{i0}$

Step 2 $z_0 = e^{i(0+2k\pi)}$, $k \in \mathbb{Z}$.

Step 3 $z_0^{1/n} = \sqrt[n]{1}\, e^{i\left(\frac{0}{n}+\frac{2k\pi}{n}\right)}$, $k = 0, 1, \dots, n-1$

$\Rightarrow z_0^{1/n} = e^{i\left(\frac{2k\pi}{n}\right)}$, $k = 0, 1, \dots, n-1$

Roots of Unity

So the nth roots of unity are

$$z_0^{1/n} = e^{i\left(\frac{2k\pi}{n}\right)}, k = 0, 1, 2, \dots, n-1$$

If we denote

$$\omega_n = e^{i\left(\frac{2\pi}{n}\right)}$$

Then the nth roots of unity become

Roots of Unity

So the nth roots of unity are

$$z_0^{1/n} = e^{i\left(\frac{2k\pi}{n}\right)}, k = 0, 1, 2, \dots, n-1$$

If we denote

$$\omega_n = e^{i\left(\frac{2\pi}{n}\right)}$$

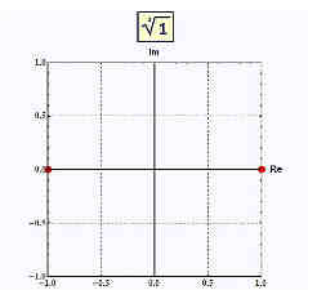
Then the nth roots of unity become

$$1, \omega_n, \omega_n^2, \dots, \omega_n^{n-1}$$

Here ω_n is called the primitive nth root of unity.

Roots of Unity

Example: Find all square roots of unity.



Roots of Unity

Example: Find all cube roots of unity.

$z_0 = 1$

$z_0^{1/3} = e^{i\left(\frac{2k\pi}{3}\right)}$, $k = 0, 1, 2$.

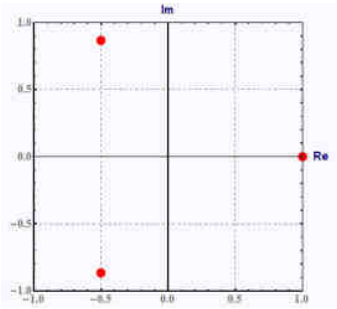
$c_0 = 1$

$c_1 = e^{i\left(\frac{2\pi}{3}\right)} = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$.

$c_2 = e^{i\left(\frac{4\pi}{3}\right)} = -\frac{1}{2} - i\frac{\sqrt{3}}{2}$.

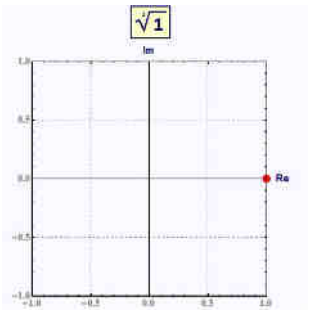
Roots of Unity

Example: Find all cube roots of unity.



Roots of Unity

Geometry of nth roots of unity



Roots of Complex Number Revisited

Relation between roots of a complex number and roots of unity

For a complex number $z_0 = r_0 e^{i\theta_0}$, the nth roots are

$$c_k = \sqrt[n]{r_0}\, exp\left[i\left(\frac{\theta_0}{n} + \frac{2\pi}{n}k\right)\right], \qquad k = 0, 1, 2, \dots, n-1$$

$\Rightarrow c_k = \sqrt[n]{r_0}\, exp\left(\frac{i\theta_0}{n}\right) exp\left(\frac{2\pi i}{n}k\right)$ → ω_n^k Roots of unity

$\Rightarrow c_k = \sqrt[n]{r_0}\, exp\left(\frac{i\theta_0}{n}\right) \omega_n^k$

$k = 0, 1, 2 \dots, n-1$

$\Rightarrow c_k = c_0 \omega_n^k$

Roots of Complex Number Revisited

Relation between roots of a complex number and roots of unity

For a complex number $z_0 = r_0 e^{i\theta_0}$. If ζ is one of the nth roots of z_0 then the other nth roots are

$$\zeta, \zeta\omega_n, \zeta\omega_n^2, \dots, \zeta\omega_n^{n-1}$$

Roots of Complex Number Revisited

Relation between roots of a complex number and roots of unity

Example: Find all values of $(2i)^{1/3}$.

Step1 $z_0 = 2i = 2\left(\cos(\pi/2) + i\sin(\pi/2)\right)$

$\Rightarrow z_0 = 2e^{i(\pi/2)}$

Step2 $z_0 = 2e^{i(\pi/2 + 2k\pi)}$, $k \in \mathbb{Z}$.

Step3 $z_0^{1/3} = \sqrt[3]{2}\, e^{i\left(\frac{\pi/2}{3} + \frac{2k\pi}{3}\right)}$, $k \in \mathbb{Z}$.

$\Rightarrow c_k = \sqrt[3]{2}\, e^{i\left(\pi/6 + \frac{2k\pi}{3}\right)}$, $k = 0, 1$

$\rightarrow c_0 = \sqrt[3]{2}\, e^{i(\pi/6)} = \sqrt[3]{2}\left(\frac{\sqrt{3}}{2} + \frac{i}{2}\right)$

$\Rightarrow c_1 = c_0\omega_3$, $c_2 = c_0\omega_3^2$

Quadratic Formula

Consider equation

$$az^2 + bz + c = 0$$

Then the quadratic formula states the following solutions of the above equation

$$\left\{-\frac{b + (b^2 - 4ac)^{1/2}}{2a}\right\}$$

Quadratic Formula

Example: Find roots of the following equation

$$z^2 + (2+i)z + \frac{3}{4} = 0$$

Here $a = 1$, $b = 2+i$, $c = 3/4$

By quadratic formula

$$\frac{-(2+i) + \left((2+i)^2 - 4(1)(3/4)\right)^{1/2}}{2(1)} = \frac{-(2+i) + (4i)^{1/2}}{2}$$

Now $(4i)^{1/2} = \{\sqrt{2} + i\sqrt{2}, -\sqrt{2} - i\sqrt{2}\}$

$\Rightarrow \frac{-(2+i) + (\sqrt{2} + \sqrt{2}i)}{2}$, $\frac{-(2+i) + (-\sqrt{2} - \sqrt{2}i)}{2}$

Roots of Complex Numbers (Part II)

END

- Roots of Unity
- Relation between roots of complex numbers and roots of unity
- Quadratic formula

Regions in the Complex Plane

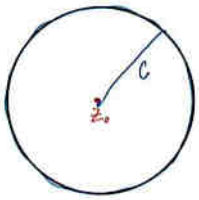
Topic No. 14

Regions in the Complex Plane (Part I)

Circles in the Complex Plane

A circle in plane is a set of points whose distance from a fixed point, say z_0, is constant c.

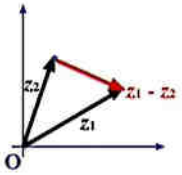
The point z_0 is called center and the constant c is called radius of the circle.



Circles in the Complex Plane

Given two points $z_1 = (x_1, y_1)$ and $z_2 = (x_2, y_2)$ the distance between them is given by

$$|z_1 - z_2|$$



Circles in the Complex Plane

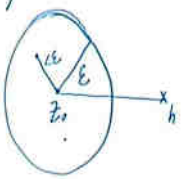
Equation of a circle of radius ε and centre z_0 is given by

$$|z - z_0| = \varepsilon$$

Neighbourhoods

A closed disc with centre z_0 and radius ε is defined by

$$|z - z_0| \leq \varepsilon$$



Neighbourhoods

A open disc with centre z_0 and radium ε is defined by

$$|z - z_0| < \varepsilon$$

Neighbourhoods

ε Neighbourhood of a point z_0 is the following open disc

$$|z - z_0| < \varepsilon$$

When value of ε is irrelevant then we just call it neighbourhood of z_0.

Edited by Mahar Afaq Safdar Muhammadi,

MSc Math, Whatsapp No. 03494873115

Neighbourhoods

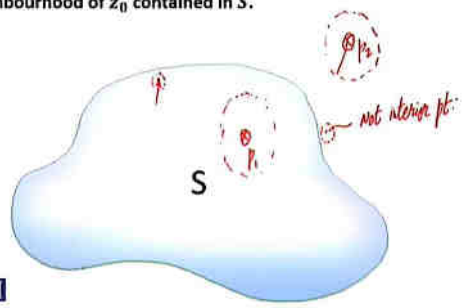
Deleted neighbourhood or punctured open disc

A deleted neighbourhood of z_0 contains all points in an ε neighbourhood of z_0 except the point z_0. Symbolically we have

$$0 < |z - z_0| < \varepsilon$$

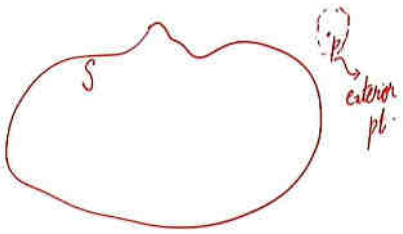
Interior Point

A point z_0 is interior point of a set S whenever there is a neighbourhood of z_0 contained in S.



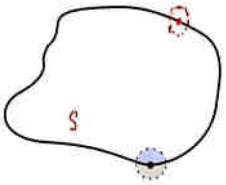
Exterior Point

A point z_0 is exterior point of a set S whenever there is a neighbourhood of z_0 not contained in S.



Boundary Point

A point z_0 is boundary point of a set S if it is neither an interior point nor an exterior point.



Boundary Point

A point z_0 is boundary point of a set S if it is neither an interior point nor an exterior point.

Set of all boundary points of a set S is called boundary of S.

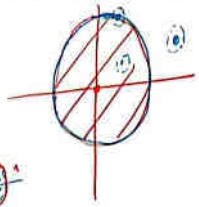
Example: $S: |z| \leq 1$

$|z-0| \leq 1$

Boundary of S: $|z| = 1$

$S_1: |z| < 1$

Boundary = $|z|$.



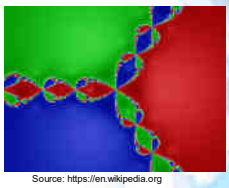
Roots of Complex Numbers (Part II)

- Circles in the complex plane
- Neighbourhoods
- Interior, exterior and boundary points

Regions in the Complex Plane

Regions in the Complex Plane (Part II)

Topic No. 15



Source: https://en.wikipedia.org

Regions in the Complex Plane

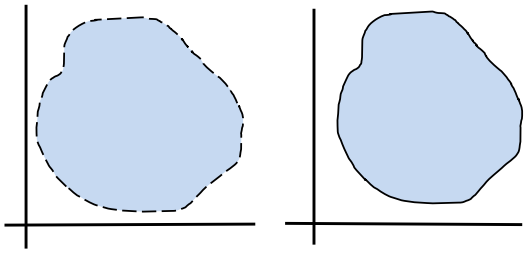
Open set

A set S is called an open set if every point of S is an interior point of S.

Regions in the Complex Plane

Open set

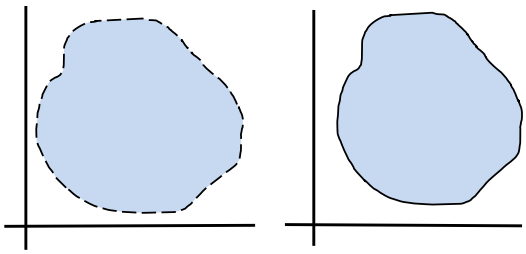
A set S is called an open set if every point of S is an interior point of S.



Regions in the Complex Plane

Closed set

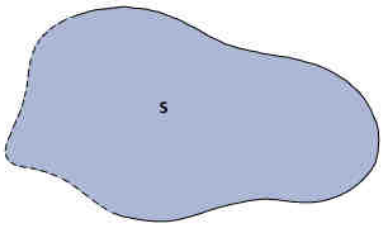
A set S is called closed set if it contains all its boundary points.



Regions in the Complex Plane

Accumulation point

A point z_0 is called an accumulation point of a set S is each deleted neighbourhood of z_0 contains at least one point of S.



Regions in the Complex Plane

Closed sets and Accumulation points

If a set S is closed then it contains all its accumulation point.

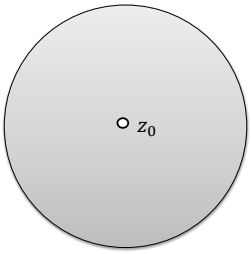
Regions in the Complex Plane

Closure of a set

Closure of a set S is the closed set consisting of all points in S together with boundary points.

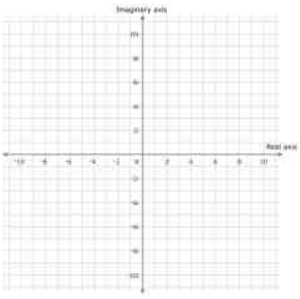
Regions in the Complex Plane

Some sets are neither open nor closed



Regions in the Complex Plane

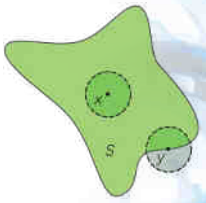
Fact: Some sets are open as well as closed



Regions in the Complex Plane

- **Open sets**
- **Closed sets**
- **Closure of a set**

END

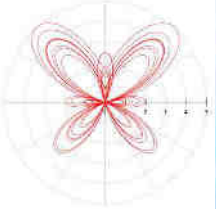


Source: wikipedia

Regions in the Complex Plane

Regions in the Complex Plane (Part III)

Topic No. 16



Source: https://en.wikipedia.org

Curves

What is a curve
Intuitively a curve is a piece of string placed on a flat surface

Curves

Curve
A curve C is the range of a function given by
$z(t) = (x(t), y(t)) = x(t) + iy(t)$, where $t \in [a, b]$
where both $x(t)$ and $y(t)$ are continuous real-valued functions.

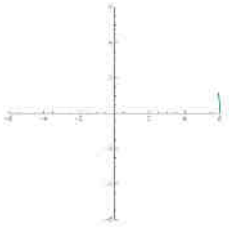
We say that $z(t)$ is parametrization of curve C.

Curves

Example
Consider a curve C_1 defines as
$z_1(t) = 6\cos t + i\,6\sin t, where\ t \in [0, 2\pi]$

Curves

Example
Consider a curve C_1 defines as
$z_1(t) = 6\cos t + i\,6\sin t, where\ t \in [0, 2\pi]$



Contributed by: Milana Dabic

Curves

Example
Consider a curve C_2 defines as
$z_2(t) = 6\cos t - i\,6\sin t, where\ t \in [0, 2\pi]$

Closed Curves

Curve
A curve C is the range of a function given by
$z(t) = (x(t), y(t)) = x(t) + iy(t)$, where $t \in [a, b]$
where both $x(t)$ and $y(t)$ are continuous real-valued functions.

As the parameter increases we move from *initial point* $z(a)$ to the *terminal point* $z(b)$ on the curve.

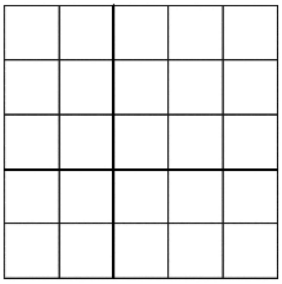
Closed curve
If initial point and terminal points are equal then the curve C is said to be closed curve.

Closed Curves

Example: The curve C_2 defines as
$z(t) = 6\cos t - i\,6\sin t, where\ t \in [0, 2\pi]$
is a closed curve.

Closed Curves

Example: The curve C_2 defines as
$z(t) = 2t + i\,3t, where\ t \in [0, 1]$
is not a closed curve.



Simple Curves

Simple curve

A curve is called simple if it does not cross itself.

Simple Curves

Simple curve

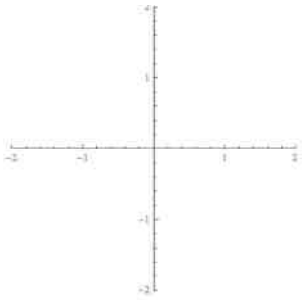
A curve is called simple if it does not crosses itself. In other words the curve

$$C: z(t) = \big(x(t), y(t)\big), \text{for } t \in [a, b]$$

is simple provided $z(t)$ is one-to-one function or

$$z(t_1) = z(t_2) \overset{\Rightarrow}{} t_1 = t_2$$

Examples



Contributed by: Richard Mercer

Orientation of Curves

A simple closed curve C bounds an area A in complex plane.
As the parameter increases we move along the curve.
During this travel if the interior of A is on your left then the curve is said to be positively oriented.

Orientation of Curves

A simple closed curve C bounds an area A in complex plane.
As the parameter increases we move along the curve.
During this travel if the interior of A is on your left then the curve is said to be positively oriented.
The Jordan curve theorem guarantees that a simple closed curve must enclose a region.

Connected Set

A set S is said to be connected set if for every pair of points $z_1, z_2 \in S$ there is a curve C joining them which is entirely contained in S.

Connected Set

A set S is said to be connected set if for every pair of points $z_1, z_2 \in S$ there is a curve C joining them which is entirely contained in S.

Example: The annulus $A = \{z: 1 < |z| < 2\}$ is connected.

Domain

A connected open set is called a domain.

Example:

Check whether the half plane $H = \{z: Re(z) > 0\}$ is a domain or not.

Bounded Set

A set S is said to be bounded set if it can be completely contained in some closed disk, that is, there exist and $R > 0$ such that

$$|z| \leq R \text{ for all } z \in S$$

A set which is not bounded is called unbounded.

Bounded Set

Example: The region

$$\{z: 1 < Re\, z < 2\}$$

is not bounded.

Bounded Set

Example: The region

$$\{z: |z| < 1\}$$

is bounded.

Regions in the Complex Plane

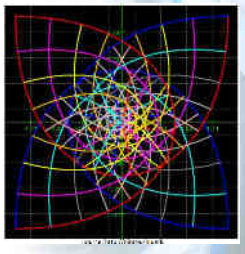
- Curves
- Closed curves
- Simple curves
- Orientation of curves
- Connected sets
- Domain
- Bounded sets

END

Complex Functions

Complex Functions
Part I

Topic No. 17

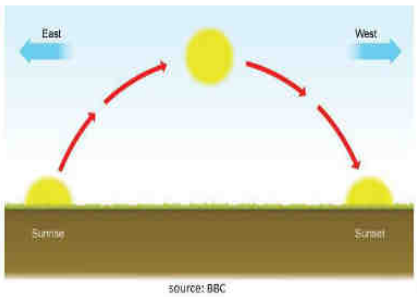


Complex Functions

What is a function?

Complex Functions

What is a function?



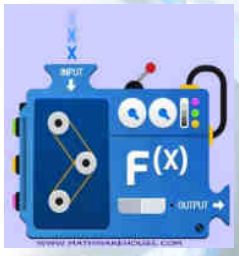
source: BBC

Complex Functions

A temperature function T provides us temperature during different times of a day. In this case we have $A=[0,24]$ and $B=[-20,50]$, and the rule T relates a time with a value of temperature.

Complex Functions

What is a function?



Complex Functions

What is a function?

A function from set A to set B is a rule that assigns each value of A with a unique value of B

Complex Functions

Why complex functions

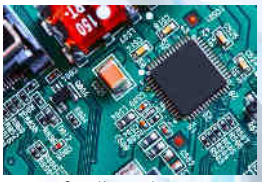
Complex Functions

Why complex functions

Source: MIT OpenCourseware

Complex Functions

Why complex functions



Source: blog.gesrepair.com

Complex Functions

Complex function

A **complex-valued function** f of a complex variable z is a rule that assigns to each value z in a set D one and only one complex value w.

We write $w = f(z)$ and call w the **image of z under f**. Here the set D is the **domain of f** and the set of all images $\{w = f(z): z \in D\}$ is called the **range of f**.

In some contexts functions are referred to as a **mappings** or **transformations**.

Complex Functions

Complex function

A **complex-valued function** f of a complex variable z is a rule that assigns to each value z in a set D one and only one complex value w.

We write $w = f(z)$ and call w the **image of z under f**. Here the set D is the **domain of f** and the set of all images $\{w = f(z): z \in D\}$ is called the **range of f**.

Complex Functions

Complex function

Example: A simple example of a complex valued function $f: \mathbb{C} \to \mathbb{C}$ is given by the rule $w = f(z) = z^2$.

Complex Functions

Complex function

A **complex-valued function** f of a complex variable z is a rule that assigns to each value z in a set D one and only one complex value w.

A function has two parts:

1. Domain
2. Rule

Complex Functions

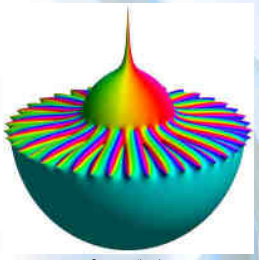
Complex function

Example: A simple example of a complex valued function $f: \mathbb{C} \to \mathbb{C}$ is given by the rule $w = f(z) = z^2$.

Complex Functions

- **Complex Functions**

END

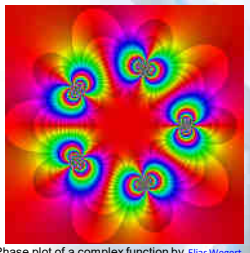


Source: mathworks

Complex Functions

Complex Functions
Part II

Topic No. 18



Phase plot of a complex function by Elias Wegert

Complex Functions

Complex function

A **complex-valued function** f of a complex variable z is a rule that assigns to each value z in a set D one and only one complex value w.

Complex Functions

Real and imaginary parts of a complex function (In Cartesian coordinates)

A complex number z can be expressed by its real and imaginary parts.

Complex Functions

Real and imaginary parts of a complex function (In Cartesian coordinates)

A complex number z can be expressed by its real and imaginary parts.

A complex function $w = f(z)$ can be written as

$$w = f(z) = f(x + iy) = u + iv$$

here u and v depend on x and y.
So we can write

$$w = f(z) = u(x, y) + i\, v(x, y)$$

Complex Functions

Example Express the function

$$f(z) = \bar{z}\ \mathrm{Re}(z) + z^2 + \mathrm{Im}(z)$$

in the form

$$f(z) = u(x, y) + i\, v(x, y).$$

Complex Functions

Example Express the function

$$f(z) = \bar{z}\ \mathrm{Re}(z) + z^2 + \mathrm{Im}(z)$$

in the form

$$f(z) = u(x,y) + i\,v(x,y).$$

Solution

$$f(z) = (x - iy)x + (x + iy)^2 + y,$$

Complex Functions

Example Express the function

$$f(z) = \bar{z}\ \mathrm{Re}(z) + z^2 + \mathrm{Im}(z)$$

in the form

$$f(z) = u(x,y) + i\,v(x,y).$$

Solution

$$f(z) = (x - iy)x + (x + iy)^2 + y,$$
$$f(z) = (x - iy)x + (x^2 - y^2 + i2xy) + y,$$

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$$f(z) = (2x^2 - y^2 + y) + i(xy)$$

So

$$u(x,y) = 2x^2 - y^2 + y \text{ and } v(x,y) = xy.$$

Complex Functions

Two real valued functions $u(x,y)$ and $v(x,y)$ determine a complex -valued function

$$f(x,y) = u(x,y) + iv(x,y)$$

We can use the formulas

$$x = \frac{z + \bar{z}}{2}, \quad \text{and} \quad y = \frac{z - \bar{z}}{2i}$$

to find a formula for f involving variables z and $\bar{z}$.

Complex Functions

Example: Express $f(z) = 4x^2 + i4y^2$ by a formula involving the variables z and $\bar{z}$.

Solution: Calculation reveals that

$$f(z) = 4\left(\frac{z + \bar{z}}{2}\right)^2 + i\,4\left(\frac{z - \bar{z}}{2i}\right)^2$$

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$$= z^2(1 - i) + z\bar{z}\,(2 + 2i) + (1 - i)\,\bar{z}^2.$$

Complex Functions

Sometimes input of a function is a complex number but output is a real number, that is, $v(x,y) = 0$.

The following example shows one such real-valued function.

Example: $f: \mathbb{C} \to \mathbb{R}$, defined by

$$f(x + iy) = \sqrt{x^2 + y^2}$$

Or

$$f(z) = |z|$$

Complex Functions

Real and imaginary parts of a complex function (In polar coordinates)

Using $z = re^{i\theta}$ in the expression of a complex function

$$w = f(z)$$

gives us the polar representation

$$f(z) = f\left(re^{i\theta}\right) = u\,(r,\theta) + iv(r,\theta)$$

where u and v are real functions of the real variables r and θ.

Complex Functions

Real and imaginary parts of a complex function (In polar coordinates)

Example: Express $f(z) = z^2$ in imaginary and real parts using and polar form.

Solution For Polar Form, we use $z = re^{i\theta}$

$$f\left(re^{i\theta}\right) = \left(re^{i\theta}\right)^2 = r^2\cos 2\theta + i\,r^2\sin 2\theta$$
$$= U\,(r,\theta) + iV\,(r,\theta),$$

So that

$U\,(r,\theta) = r^2\cos 2\theta$, and $V\,(r,\theta) = r^2\sin 2\theta$.

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Complex Functions

Example 1: Express $f(z) = z^2 + z + 1$ in imaginary and real parts using both Cartesian and polar form.

Solution: For Cartesian Form, we use $z = x + iy$

$$f(z) = f(x+iy) = (x+iy)^2 + (x+iy) + 1$$

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So that

$u\,(x,y) = x^2 - y^2 + x + 1$, $v\,(x,y) = (2xy + y)$.

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$$= U(r,\theta) + iV(r,\theta),$$

So that

$$U(r,\theta) = r^2\cos 2\theta + r\cos\theta + 1$$

and

$$V(r,\theta) = r^2 \sin 2\theta + r\sin\theta$$

Complex Functions

Example 1: Express $f(z) = z^2 + z + 1$ in imaginary and real parts using both Cartesian and polar form.

Solution:

In Cartesian coordinates

$$f(z) = (x^2 - y^2 + x + 1) + i\,(2xy + y)$$

In polar coordinates

$$f(z) = (r^2\cos 2\theta + r\cos\theta + 1) + i\,(r^2 \sin 2\theta + r\sin\theta),$$

Polynomial Functions

A polynomial of degree n, over complex numbers, is an expression of the form

$$P_n(z) = a_0 + a_1 z + \cdots a_{n-1} z^{n-1} + a_n z^n,$$

where the a_i are complex numbers.

Domain of polynomial functions is the whole complex plane.

Polynomial Functions

Linear polynomial functions

$$f(z) = az + b$$

where a, b are complex numbers, is called linear polynomial function, or just linear transformation.

Polynomial Functions

Example

$$f(z) = 1 + z + z^2$$

is a polynomial function.

Polynomial Functions

Example

$$f(z) = 1 + z + z^2 + z^3 + z^4 + \cdots$$

is NOT a polynomial.

Polynomial Functions

Example

$$f(z) = \frac{1}{z}$$

is NOT a polynomial.

Rational Functions

Rational functions

A rational function is an expression of the following form

$$\frac{P(z)}{Q(z)},$$

where $P(x)$ and $Q(x)$ are polynomials.

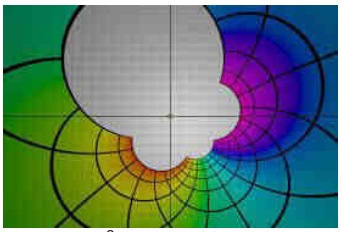
Rational Functions

Linear fractional transformations

A transformation of the form

$$T(z) = \frac{az + b}{cz + d}, ad - bc \neq 0$$

where $a, b, c, d \in \mathbb{C}$.



Source: Semantic Scholar

Multivalued Functions

Example: Consider

$$f(z) = z^{1/2} = \pm\sqrt{r}\exp\left[i\frac{\Theta}{2}\right],$$

where $r = |z|$ and Θ is principal argument of z.

Multivalued Functions

Example: Consider

$$f(z) = z^{1/2} = \pm\sqrt{r}\exp\left[i\frac{\Theta}{2}\right],$$

where $r = |z|$ and Θ is principal argument of z.

Strictly speaking $f(z)$ is not a function but we can construct two single-valued functions from here, give below

$$f_1(z) = z^{1/2} = \sqrt{r}\exp\left[i\frac{\Theta}{2}\right],$$

$$f_2(z) = z^{1/2} = -\sqrt{r}\exp\left[i\frac{\Theta}{2}\right].$$

To explore properties of $f(z)$ we instead consider $f_1(z)$ and $f_2(z)$.

Complex Functions

Complex Functions
Part II

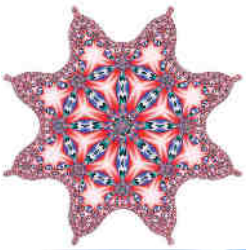
End

Phase plot of a complex function by Elias Wegert

Mappings

Topic No. 19

Mappings



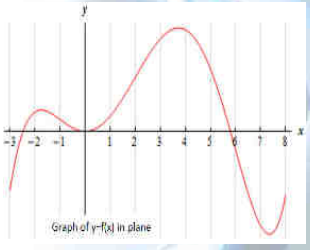
Source: IEEE computer society

Mappings

Properties of a real-valued function

$$y = f(x)$$

of a real variable are often exhibited by the graph of the function

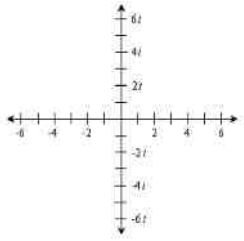


Mappings

For complex functions

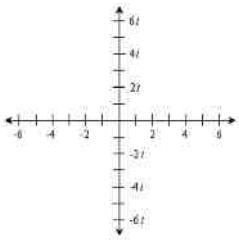
$$w = f(z)$$

no such convenient graphical representation of the function f is available

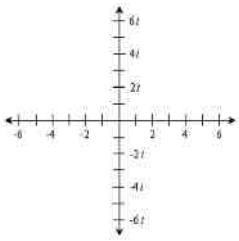


Mappings

To overcome this problem we represent z in one plane, known as z-plane, and its image w is represented in a second plane, known as w-plane.



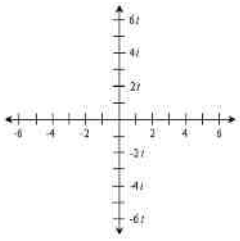
z plane



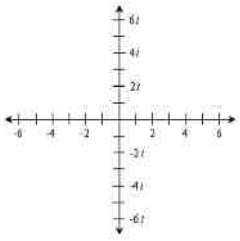
w plane

Mappings

We call a "complex function" a "**complex mapping**" or simply "**mapping**" when it is considered using such correspondence between z-plane and w-plane.



z plane



w plane

Mappings

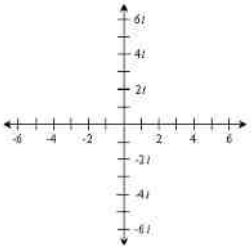
Given a complex function

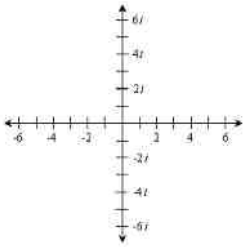
$$f(z) = u(x,y) + iv(x,y)$$

with domain D. The equations

$$u = u(x,y) \quad \text{and} \quad v = v(x,y)$$

describe a transformation (or mapping) from a set D in the z plane onto the range R in the w plane





Mappings

Example: Consider the following complex valued function

$$w = f(z) = z + 1$$

with domain

$$D = \{z: |z| \leq 1\}.$$

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Example: The image of the set $A = \{z: |z| \leq \frac{1}{2}\}$ is under

$$f(z) = z + 1$$

is the set $B = f(A) = \{z: |z - 1| \leq \frac{1}{2}\}$.

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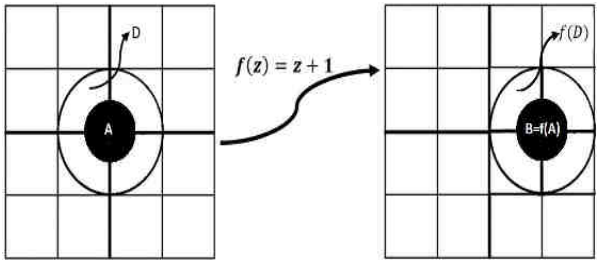

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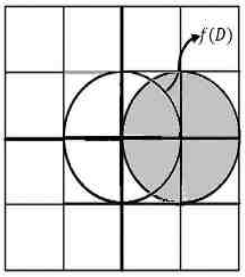
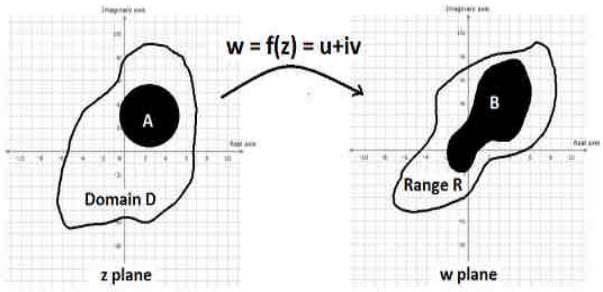
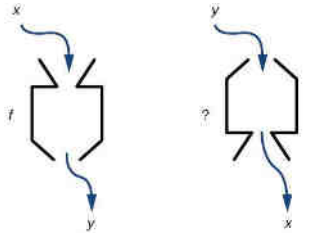


Mappings



Inverse Functions

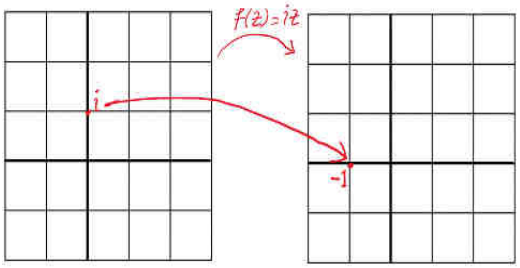
Consider a complex function $w = f(z)$.
The **inverse image** of a point w is the set of all points z in the domain of definition of f that have w as their image.



Source: archive.cnx.org

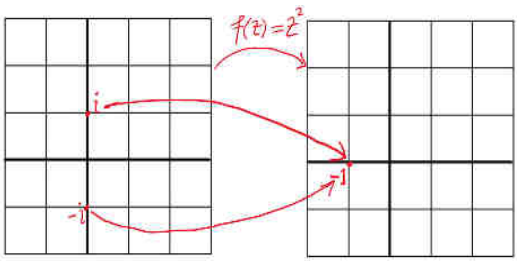
Inverse Functions

Example: If $w = f(z) = iz$, the inverse image of the point -1 is the single point i, because $f(i) = i\,(i) = -1$, and i is the only point that maps to -1.



Inverse Functions

Example: If $w = f(z) = z^2$, the inverse image of the point -1 is the set $\{i, -i\}$.



Inverse Functions

If we have a function f from domain D to range R then its inverse image is a function form R to D only if f is one-to-one.

Inverse Functions

Definition: f is one-to-one if we suppose that $f(z_1) = f(z_2)$ then from this assumption deduce that $z_1 = z_2$.

Inverse Functions

Definition: f is one-to-one if we suppose that $f(z_1) = f(z_2)$ then from this assumption deduce that $z_1 = z_2$.

If f is not one-to-one then the inverse image of a point is not unique.

Inverse Functions

Example: The function $w = f(z) = iz$ is one-to-one

Inverse Functions

Example: The function $w = f(z) = iz$ is one-to-one because

$$f(z_1) = f(z_2)$$

then

$$iz_1 = iz_2$$

dividing both sides by i gives

$$z_1 = z_2$$

Inverse Functions

Example: The function $w = f(z) = z^2$ is not one-to-one

Inverse Functions

Example: The function $w = f(z) = z^2$ is not one-to-one because

$$-i \neq i$$

but

$$f(i) = f(-i) = -1$$

Inverse Functions

If $w = f(z)$ maps the set A one-to-one and onto the set B, then for each w in B there exists exactly one point z in A such that $w = f(z)$.

Inverse Functions

If $w = f(z)$ maps the set A one-to-one and onto the set B, then for each w in B there exists exactly one point z in A such that $w = f(z)$.

Then we can solve the equation $w = f(z)$ by solving for z as a function of w.

That is, the **inverse function** $z = f^{-1}(w)$ can be found, and the following equations hold

$$f^{-1}(f(z)) = z \text{ for all } z \in A,$$

and

$$f(f^{-1}(w)) = w \text{ for all } w \in B$$

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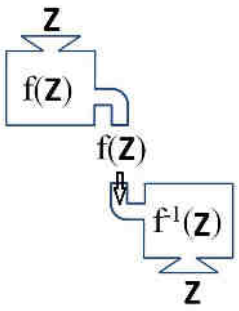

Inverse Functions

Example: The function $w = f(z) = iz$ for any complex number z, find $f^{-1}(w)$.

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Example: The function $w = f(z) = iz$ for any complex number z, find $f^{-1}(w)$.

Solution: Observe that f is one to one and onto the entire complex plane. Now solving for z, we have

$$w = f(z) = iz$$
$$z = \frac{w}{i} = -iw$$

We have

$f^{-1}(w) = -iw$ for all complex numbers w.

Inverse Functions

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We have

$f^{-1}(w) = -iw$ for all complex numbers w.

Also the following equations hold

$$f^{-1}(f(z)) = z \text{ for all } z \in A,$$

and

$$f(f^{-1}(w)) = w \text{ for all } w \in B$$

Behavior of Mappings

Different properties of mappings can be noticed by observing images of carefully chosen subsets of its domain. These subsets are usually described by equations or inequalities. In this situation inverse functions help us in describing images of subsets.

Behavior of Mappings

Example: Show that the function $f(z) = -z$ maps the line $y = -x + 1$ to the line $v = -u - 1$.

In this case subset of domain is $A = \{x + iy : y = -x + 1\}$.

Let $B = f(A)$, the defining conditions for B can be found as follows

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$$\Leftrightarrow -u - iv \in A$$
$$\Leftrightarrow (-u, -v) \in A$$

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Example: Show that the function $f(z) = -z$ maps the line $y = -x + 1$ to the line $v = -u - 1$.
In this case subset of domain is $A = \{x + iy : y = -x + 1\}$.
Let $B = f(A)$, the defining conditions for B can be found as follows

$$w = f(z) = u + iv \in B \Leftrightarrow f^{-1}(w) \in A$$
$$\Leftrightarrow -w \in A$$
$$\Leftrightarrow -u - iv \in A$$
$$\Leftrightarrow (-u, -v) \in A$$
$$\Leftrightarrow -v = u + 1$$

Behavior of Mappings

Example: Show that the function $f(z) = -z$ maps the line $y = -x + 1$ to the line $v = -u - 1$.
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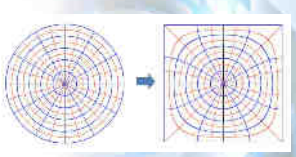
$$\Leftrightarrow -u - iv \in A$$
$$\Leftrightarrow (-u, -v) \in A$$
$$\Leftrightarrow -v = u + 1$$
$$\Leftrightarrow v = -u - 1$$

Mappings

Mappings

End

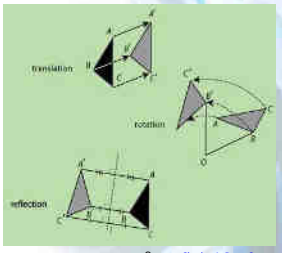


Source: squircular.blogspot.com

Mappings (Linear Transformations)

Mappings
Linear Transformation
Part I

Topic No. 20



Source: Charles A. Dana Center

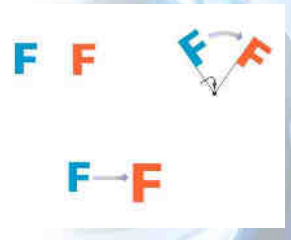
Linear Transformations

Linear transformation

A complex valued function of the form

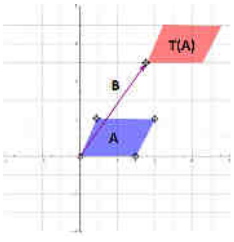
$$L(z) = Az + B$$

where A, B are complex numbers.



Translation

Translation



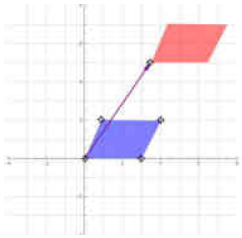
Translation

Translation

Let $B = a + ib$ denote a fixed complex constant then the linear transformation

$$w = T(z) = z + B$$
$$= x + a + i(y + b)$$

is called a **translation**.



Translation

Inverse Translation

Translation is a one-to-one and onto function on the entire complex plane.

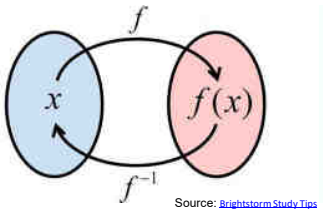
Translation

Inverse Translation

The inverse mapping is given by

$$z = T^{-1}(w) = w - B$$
$$= u - a + i(v - b)$$

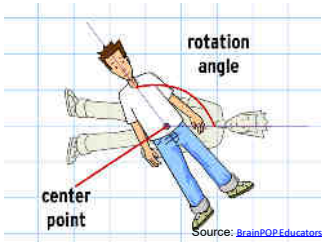
and shows that T is a one-to-one mapping from the z plane onto the w plane.



Source: Brightstorm Study Tips

Rotation

Rotation



Source: BrainPOP Educators

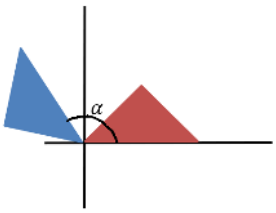
Rotation

Rotation

Let α be a fixed real number. Then for $z = re^{i\theta}$ the transformation

$$w = R(z) = ze^{i\alpha} = re^{i\theta}e^{i\alpha}$$
$$= re^{i(\theta+\alpha)}$$

is called a **rotation.**



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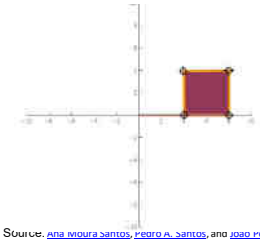


Source: Ana Moura Santos, Pedro A. Santos, and João Pedro Pargana

Rotation

Inverse Rotation

Rotation

Inverse Rotation

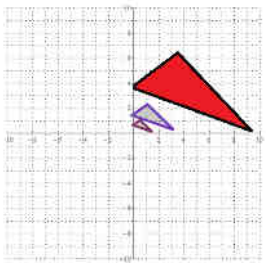
If we use polar coordinates $w = \rho e^{i\varphi}$ in the w plane, then the inverse mapping is given by

$$z = R^{-1}(w) = we^{-i\alpha} = \rho e^{i\varphi}e^{-i\alpha}$$
$$= \rho e^{i(\varphi-\alpha)}$$

This shows that the rotation R is one-to-one mapping of the z plane onto the w plane.

Magnification

Magnification



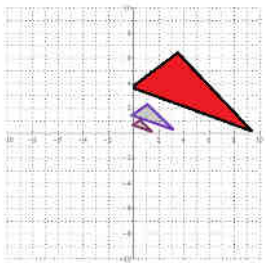
Magnification

Magnification

Let $K > 0$ be a fixed positive real number. Then the transformation

$$w = S(z) = Kz = Kx + iKy$$

is called a **magnification**.



Magnification

Magnification

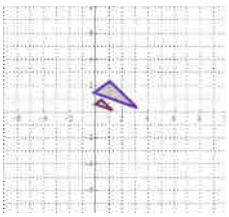
Let $K > 0$ be a fixed positive real number. Then the transformation

$$w = S(z) = Kz = Kx + iKy$$

is called a **magnification**.

It has following cases.

1. If $K < 1$, then it reduces the distance between points by the factor K.
2. If $K > 1$, it has the effect of stretching the distance between points by the factor K.



Linear Transformations

If $A = Ke^{i\alpha}$ and $B = a + ib$, where $K > 0$ is a positive real number then the transformation

$$w = L(z) = Az + B$$

is a one-to-one mapping of the z plane onto the w plane and is called a **linear transformation**.

Linear Transformations

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In terms of composition of a rotation, a magnification, and a translation it can be written as the effect of rotating the plane through an angle given by

$$\alpha = \operatorname{Arg} A,$$

Linear Transformations

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Linear Transformations

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$$\alpha = \operatorname{Arg} A,$$

followed by a magnification by the factor

$$K = |A|,$$

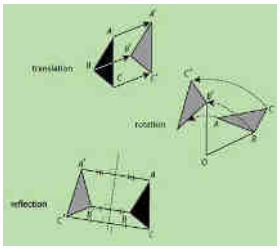
followed by a translation by the vector

$$B = a + ib.$$

Linear Transformations

Hence a linear transformation is composition of

1. Translation
2. Rotation
3. Magnification



Source: Charles A. Dana Center

Linear Transformations

Inverse transformation

L is a one-to-one mapping from the z plane onto the w plane.

Linear Transformations

Inverse transformation

The inverse mapping is given by

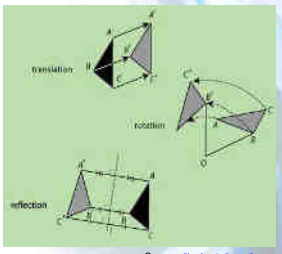
$$z = L^{-1}(w) = \frac{1}{A}w - \frac{B}{A}$$

and shows that L is a one-to-one mapping from the z plane onto the w plane.

Mappings (Linear Transformations)

Mappings
Linear Transformation
Part I

End

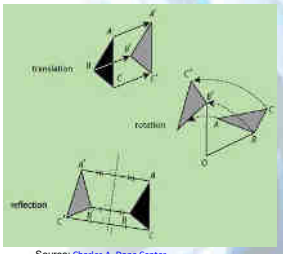


Source: Charles A. Dana Center

Mappings

Mappings
Linear Transformation
Part II

Topic No. 21



Source: Charles A. Dana Center

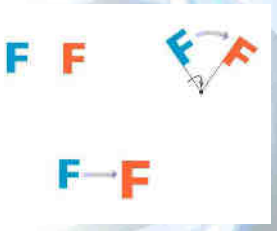
Linear Transformation

Linear transformation

A complexed valued function of the form

$$L(z) = Az + B$$

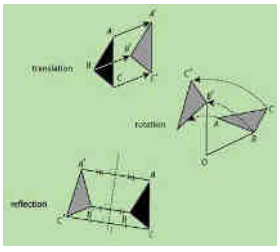
where A, B are complex numbers.



Linear Transformation

Hence a linear transformation is composition of

1. Translation
2. Rotation
3. Magnification



Source: Charles A. Dana Center

Example

Example: The ellipse centered at the origin with a horizontal major axis of four units and vertical minor axis of two units can be represented by the parametric equation

$$s(t) = 2\cos t + i \sin t = (2\cos t, \sin t), \qquad \text{for} \quad 0 \le t \le 2\pi$$

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Suppose that you wanted to rotate the ellipse

1. by an angle of $\frac{\pi}{3}$ radians and
2. shift the center of ellipse 3 units to the right and 1 unit up.

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1. by an angle of $\frac{\pi}{3}$ radians and
2. shift the center of ellipse 3 units to the right and 1 unit up.

We generate a parametric equation $r(t)$ which is given by

$$r(t) = e^{\frac{i\pi}{3}}\, s(t) \; + (3 + i)$$

Example

Example:

$$r(t) = e^{\frac{i\pi}{3}}\, s(t) \; + (3 + i)$$

$$r(t) = (2\cos t + i\sin t)\;\left(\cos\tfrac{\pi}{3} + i\sin\tfrac{\pi}{3}\right) + (3 + i)$$

Example

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$$= \left(2\cos t\cos\tfrac{\pi}{3} - \sin t\sin\tfrac{\pi}{3}\right) + i\left(2\cos t\sin\tfrac{\pi}{3} + \sin t\cos\tfrac{\pi}{3}\right) + (3 + i)$$

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Linear Transformations

Example: Let $= f(z) = (3 + 4i)z - 2 + i$.

1. Find the image of the disk $|z - 1| < 1$.

Linear Transformations

Example: Let $w = f(z) = (3+4i)z - 2 + i$.

1. Find the image of the disk $|z-1| < 1$.

Solution:

Let A be a disk, of the form

$$A = \{ z : |z-1| < 1 \}$$

The inverse transformation is

$$z = \frac{w+2-i}{3+4i}$$

Linear Transformations

Example: Let $= f(z) = (3+4i)z - 2 + i$.

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Solution:

As $w = f(z)$ then the inverse is

$$f^{-1}(w) = z \in A$$

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$$\Leftrightarrow |w-1-5i| < 5|.$$

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Linear Transformations

Example: Let $= f(z) = (3+4i)z - 2 + i$.

2. Find the image of the half plane $\text{Im}(z) > 1$.

Linear Transformations

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Solution:

Write $w = f(z)$ in Cartesian form as

Linear Transformations

Example: Let $= f(z) = (3+4i)z - 2 + i$.

2. Find the image of the half plane $\text{Im}(z) > 1$.

Solution:

Write $w = f(z)$ in Cartesian form as

$$w = u + iv = (3+4i)(x+iy) - 2 + i$$

$$u + iv = 3x + 3iy + 4ix - 4y - 2 + i$$

$$u + iv = 3x - 4y - 2 + i(3y + 4x + 1)$$

Here $u = 3x - 4y - 2$ and $v = 3y + 4x + 1$.

Linear Transformations

Example: Let $= f(z) = (3+4i)z - 2 + i$.

2. Find the image of the half plane $\text{Im}(z) > 1$.

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$u = 3x - 4y - 2$ and $v = 3y + 4x + 1$.

Linear Transformations

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2. Find the image of the half plane $\text{Im}(z) > 1$.

Solution:

$u = 3x - 4y - 2$ and $v = 3y + 4x + 1$.

Which implies

$$y = \frac{v-4x-1}{3}$$

Linear Transformations

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Linear Transformations

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$$\frac{v-4x-1}{3} = y > 1$$

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$$\frac{v-4x-1}{3} = y > 1$$

$$\frac{v-4x-1}{3} > 1$$

$$v - 4x - 1 > 3$$

$$v > 4 + 4x$$

Linear Transformations

Example: Let $= f(z) = (3+4i)z - 2 + i$.

2. Find the image of the half plane $\text{Im}(z) > 1$.

Solution:

Now taking out the value of x from the value of u, we get

$$u = 3x - 4y - 2$$

Linear Transformations

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$$x = \frac{u + 2 + 4y}{3}$$

Linear Transformations

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2. Find the image of the half plane $\text{Im}(z) > 1$.

Solution:

Now taking out the value of x from the value of u, we get

$$u = 3x - 4y - 2$$

$$x = \frac{u + 2 + 4y}{3}$$

Putting the value of y in u

$$x = \frac{u + 2 + 4\left(\frac{v - 4x - 1}{3}\right)}{3}$$

Linear Transformations

Example: Let $= f(z) = (3+4i)z - 2 + i$.

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Putting the value of y in u

$$x = \frac{u + 2 + 4\left(\frac{v - 4x - 1}{3}\right)}{3}$$

$$x = \frac{u + v - 2}{19}$$

Linear Transformations

Example: Let $= f(z) = (3+4i)z - 2 + i$.

2. Find the image of the half plane $\text{Im}(z) > 1$.

Solution:

Putting value of x in $v > 4 + 4x$

$$v > 4 + 4\left(\frac{u + v - 2}{19}\right)$$

Linear Transformations

Example: Let $= f(z) = (3+4i)z - 2 + i$.

2. Find the image of the half plane $\text{Im}(z) > 1$.

Solution:

Putting value of x in $v > 4 + 4x$

$$v > 4 + 4\left(\frac{u + v - 2}{19}\right)$$

$$v > \frac{4u + 68}{3}$$

Hence the image of half plane $\text{Im}(z) > 1$ is $v > \frac{4u+68}{3}$.

Linear Transformations

Example: Let $= f(z) = (3 + 4i)z - 2 + i$.

2. Find the image of the half plane $\mathrm{Im}(z) > 1$.

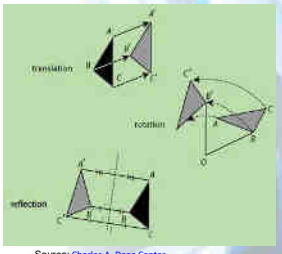
Solution:

Hence the image of half plane $\mathrm{Im}(z) > 1$ is $v > \frac{4u+68}{3}$.

Mappings

Mappings
Linear Transformation
Part II

End



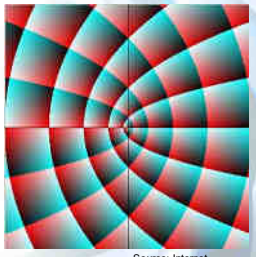
Source: Charles A. Dana Center

The Mappings $w = z^n$ and $w = z^{\frac{1}{n}}$

The Mappings $w = z^n$ and $w = z^{\frac{1}{n}}$

Part I (n=2)

Topic No. 22



Source: Internet

The Mappings $w = z^n$ and $w = z^{\frac{1}{n}}$

A real function of the form

$$f(x) = x^a,$$

where a is a real constant is called power function.

The Mappings $w = z^n$ and $w = z^{\frac{1}{n}}$

Complex power function is a function of the form

$$f(z) = z^\alpha$$

where α is a complex constant.

The Mappings $w = z^2$ and $w = z^{\frac{1}{2}}$

The Mappings
$w = z^2$
and
$w = z^{\frac{1}{2}}$

The Mappings $w = z^2$ and $w = z^{\frac{1}{2}}$

The mapping $f(z) = z^2$

By replacing $z = re^{i\theta}$, we get

$$w = f(z) = z^2 = r^2 e^{i2\theta}$$

The Mappings $w = z^2$ and $w = z^{\frac{1}{2}}$

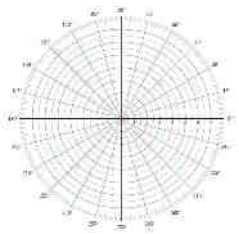
The mapping $f(z) = z^2$

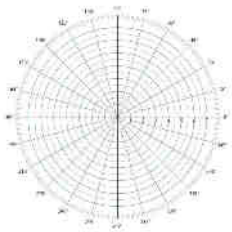
By replacing $z = re^{i\theta}$, we get

$$w = f(z) = z^2 = r^2 e^{i2\theta}.$$

If $w = \rho e^{i\varphi}$ then $f(z)$ is given by following equations

$$\rho = r^2 \quad \text{and} \quad \varphi = 2\theta$$

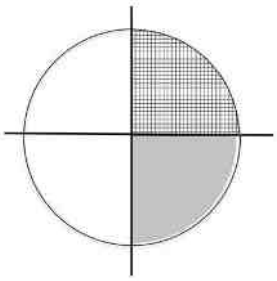




The Mappings $w = z^2$ and $w = z^{\frac{1}{2}}$

Example: If the domain of definition D for $f(z) = z^2$ is restricted to be the region

$$A = \left\{ re^{i\theta} : 0 < r \leq 1 \ and \ \frac{-\pi}{2} < \theta \leq \frac{\pi}{2} \right\}$$



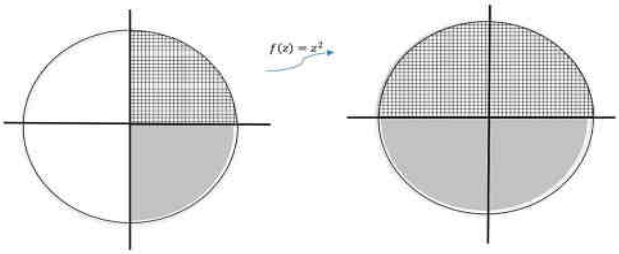
The Mappings $w = z^2$ and $w = z^{\frac{1}{2}}$

If the domain of definition D for $f(z) = z^2$ is restricted to be the region

$$A = \left\{ re^{i\theta} : 0 < r \leq 1 \ and \ \frac{-\pi}{2} < \theta \leq \frac{\pi}{2} \right\}$$

Then the image of A is

$$B = \{ \rho e^{i\varphi} : 0 < \rho \leq 1 \ and \ -\pi < \varphi \leq \pi \}$$



The Mappings $w = z^2$ and $w = z^{\frac{1}{2}}$

If the domain of definition D for $f(z) = z^2$ is restricted to be the region

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Then the image of A is

$$B = \{ \rho e^{i\varphi} : 0 < \rho \leq 1 \ and \ -\pi < \varphi \leq \pi \}$$

The inverse mapping of f, which we denote by g, is then

$$z = g(w) = w^{\frac{1}{2}} = \rho^{\frac{1}{2}} e^{i\frac{\varphi}{2}}$$

where $w \in B$.

The Mappings $w = z^2$ and $w = z^{\frac{1}{2}}$

If the domain of definition D for $f(z) = z^2$ is restricted to be the region

$$A = \left\{ re^{i\theta} : 0 < r \leq 1 \ and \ \frac{-\pi}{2} < \theta \leq \frac{\pi}{2} \right\}$$

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The inverse mapping of f, which we denote by g, is then

$$z = g(w) = w^{\frac{1}{2}} = \rho^{\frac{1}{2}} e^{i\frac{\varphi}{2}}$$

where $w \in B$. That is

$$z = g(w) = w^{\frac{1}{2}} = \rho^{\frac{1}{2}} e^{i\frac{Arg(w)}{2}}, \qquad w \neq 0 .$$

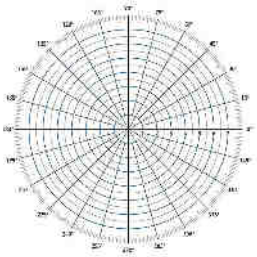
The Mappings $w = z^2$ and $w = z^{\frac{1}{2}}$

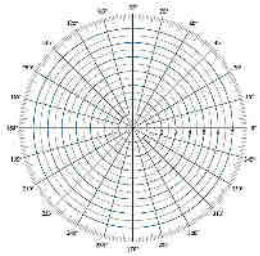
Definition (Principal square root)

The function

$$g(w) = w^{\frac{1}{2}} = \rho^{\frac{1}{2}} e^{i\frac{Arg(w)}{2}}, \qquad w \neq 0$$

is called the principal square root function.

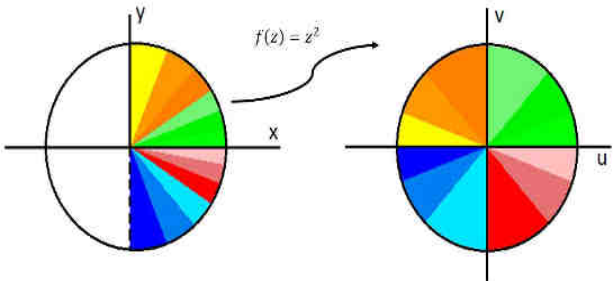




The Mappings $w = z^2$ and $w = z^{\frac{1}{2}}$

It can be checked that f and g satisfy equations

$$g(f(z)) = z \text{ and } f(g(w)) = w$$

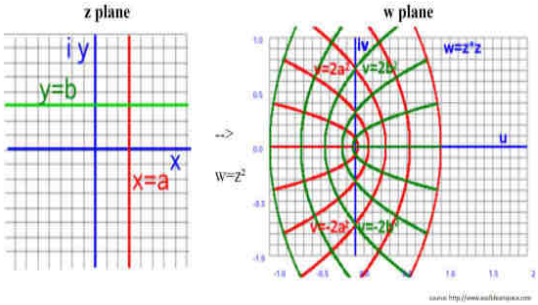


Geometry of the Mapping $w = z^2$

The mapping $f(z) = z^2$

Geometry of the Mapping $w = z^2$

We see images in w-plane of some special lines in the z-plane.

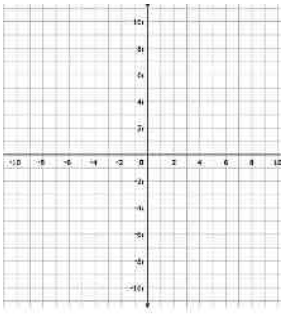


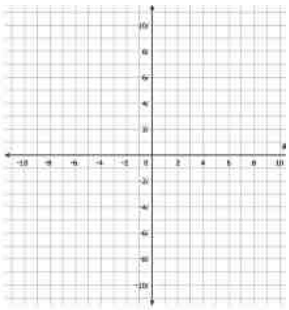
Geometry of the Mapping $w = z^2$

The mapping $w = z^2$ can be thought of as the transformation

$$u = x^2 - y^2 , \quad v = 2xy$$

from the xy plane into uv plane.





Geometry of the Mapping $w = z^2$

Image of a vertical line

The vertical line $x = a$ is mapped onto the set of points given by the equations

$$u = a^2 - y^2 , \quad v = 2ay.$$

If $a \neq 0$, then $y = \frac{v}{2a}$. Putting the value of y in u we get

$$u = a^2 - \frac{v^2}{4a^2}$$

Geometry of the Mapping $w = z^2$

Image of a vertical line

The vertical line $x = a$ is mapped onto the set of points given by the equations

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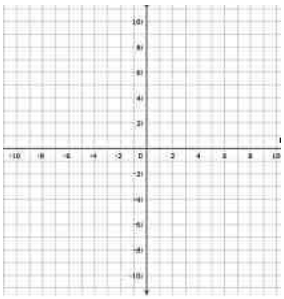
If $a \neq 0$, then $y = \frac{v}{2a}$. Putting the value of y in u we get

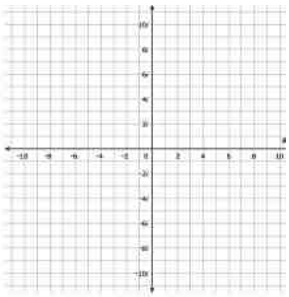
$$u = a^2 - \frac{v^2}{4a^2}$$

which represents a parabola with a vertex at a^2, oriented horizontally, and opening to the left.

Geometry of the Mapping $w = z^2$

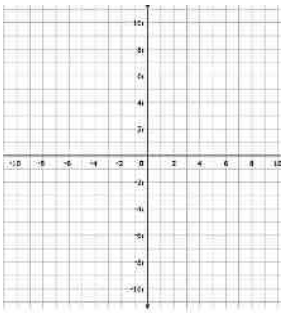
The vertical line $x = a$ is mapped onto $u = a^2 - \frac{v^2}{4a^2}$, which represents a parabola with a vertex at a^2, oriented horizontally, and opening to the left.

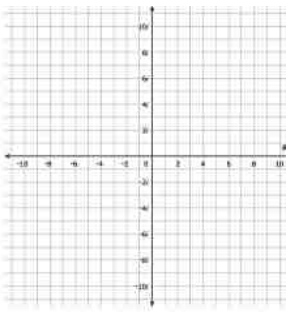




Geometry of the Mapping $w = z^2$

If $a > 0$, the set $\{(u, v): u = a^2 - y^2, v = 2ay\}$ has $v > 0$ whenever $y > 0$, so the part of line $x = a$ lying above the x axis is mapped onto the top half of parabola.





Geometry of the Mapping $w = z^2$

Image of a horizontal line

The horizontal line $y = b$ is mapped onto the parabola given by the equations

$$u = x^2 - b^2 \qquad \text{and} \qquad v = 2xb$$

Geometry of the Mapping $w = z^2$

Image of a horizontal line

The horizontal line $y = b$ is mapped onto the parabola given by the equations

$$u = x^2 - b^2 \qquad \text{and} \qquad v = 2xb$$

If $b \neq 0$, then $x = \frac{v}{2b}$, we get

$$u = -b^2 + \frac{v^2}{4b^2} , \qquad (2)$$

Geometry of the Mapping $w = z^2$

Image of a horizontal line

The horizontal line $y = b$ is mapped onto the parabola given by the equations

$$u = x^2 - b^2 \qquad \text{and} \qquad v = 2xb$$

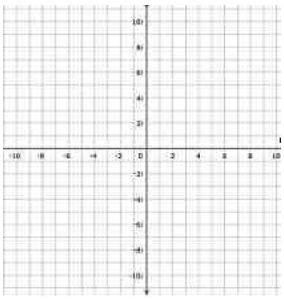
If $b \neq 0$, then $x = \frac{v}{2b}$, we get

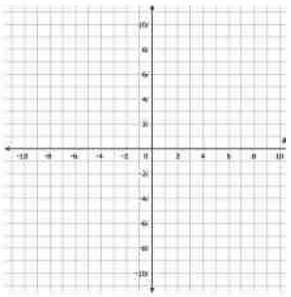
$$u = -b^2 + \frac{v^2}{4b^2}, \qquad (2)$$

which represents a parabola with vertex at $-b^2$, oriented horizontally and opening to the right.

Geometry of the Mapping $w = z^2$

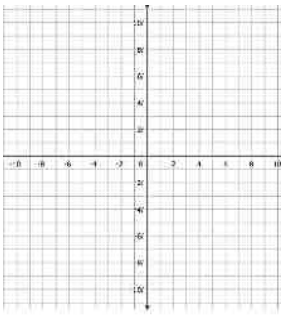
The horizontal line $y = b$ is mapped onto $u = -b^2 + \frac{v^2}{4b^2}$, which represents a parabola with vertex at $-b^2$, oriented horizontally and opening to the right.

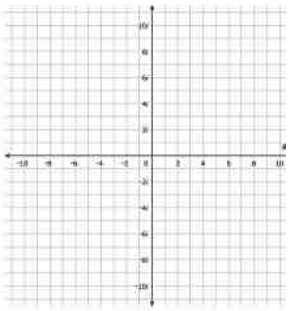




Geometry of the Mapping $w = z^2$

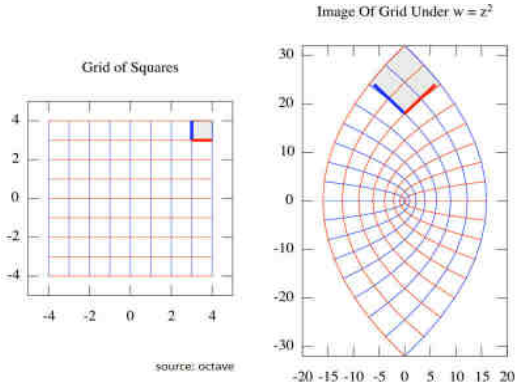
If $b > 0$, the part of the line $y = b$ to the right of y axis is mapped to the top half of the parabola because the set $\{(u, v): u = x^2 - b^2, v = 2bx\}$ has $v > 0$ when $x > 0$.





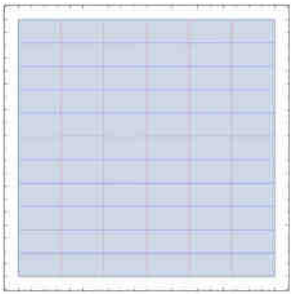
Geometry of the Mapping $w = z^2$

Image of vertical and horizontal lines under mapping z^2



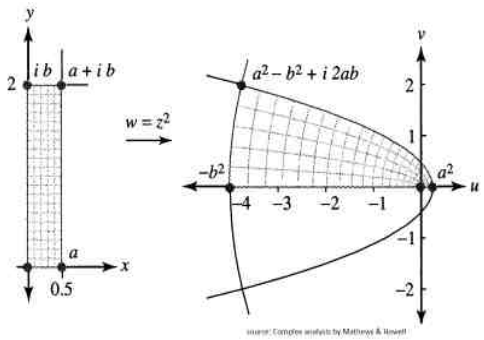
Geometry of the Mapping $w = z^2$

Quadrant 1 is mapped onto Quadrant 1 and Quadrant 2 by $w = z^2$, so the rectangle $0 < x < a, 0 < y < b$ is mapped onto the region bounded by the top halves of the parabolas



Geometry of the Mapping $w = z^2$

The vertices $0, a, a + ib$ and ib of the rectangle are mapped onto the four points $0, a^2,\ a^2 - b^2 + i2ab$ and $-b^2$ respectively .



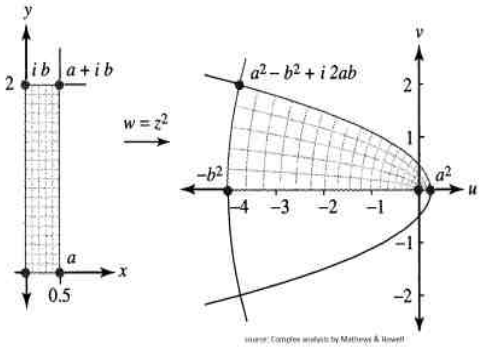
Geometry of the Mapping $w = z^2$

The vertical line $x = 0, y \neq 0$ is mapped to the set $\{(-y^2, 0): y \neq 0\}$ which is the set of negative real numbers.

The vertical line $x = 0, y \neq 0$ is mapped to the set $\{(-y^2, 0): y \neq 0\}$ which is the set of negative real numbers.

Geometry of the Mapping $w = z^2$

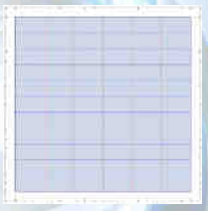
Similarly the horizontal line $y = 0,\ x \neq 0$ is mapped to the set $\{(x^2, 0): x \neq 0\}$ which is the set of positive real numbers.



The Mappings $w = z^n$ and $w = z^{\frac{1}{n}}$

- **The mappings z^2 and $z^{\frac{1}{2}}$**
- **Geometry of the mapping $f(z) = z^2$**

End

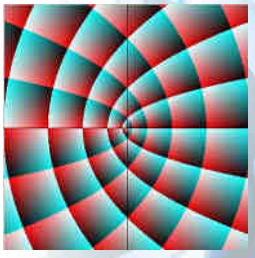


The Mappings $w = z^n$ and $w = z^{\frac{1}{n}}$

The mappings z^n and $z^{\frac{1}{n}}$

Part II

Topic No. 23



Source: Internet

The Mappings $w = z^n$ and $w = z^{\frac{1}{n}}$

The Mappings
$w = z^2$
and
$w = z^{\frac{1}{2}}$

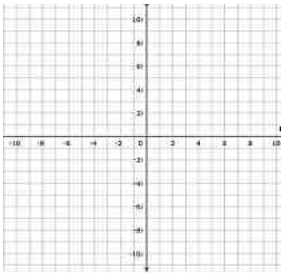
Geometry of the Mapping $w = z^2$

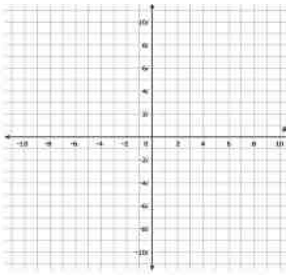
Geometry of the Mapping $w = z^2$

Each branch of hyperbola

$$x^2 - y^2 = c_1 \qquad (c_1 > 0)$$

is mapped in one-to-one manner onto the vertical line $x = c_1$.





Geometry of the Mapping $w = z^2$

From $x^2 - y^2 = c_1$ we get

$$x = \pm\sqrt{c_1 + y^2}$$

Geometry of the Mapping $w = z^2$

From $x^2 - y^2 = c_1$ we get

$$x = \pm\sqrt{c_1 + y^2}$$

For finding v, we put the value of x in $v = 2xy$

$$v = \pm 2y\sqrt{c_1 + y^2}$$

Geometry of the Mapping $w = z^2$

From $x^2 - y^2 = c_1$ we get

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Then the image of right hand side expressed in the form of u and v parametrically as

$$u = c_1, \qquad v = 2y\sqrt{c_1 + y^2} \qquad (-\infty < y < \infty)$$

Geometry of the Mapping $w = z^2$

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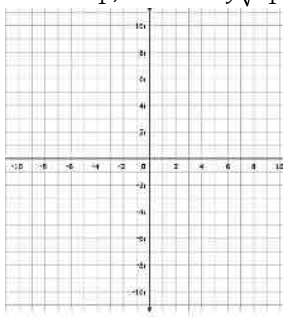
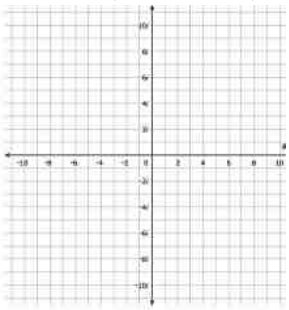
The parametric representation for the image of the left-branch of the hyperbola expressed as

$$u = c_1, \qquad v = -2y\sqrt{c_1 + y^2} \qquad (-\infty < y < \infty).$$

Geometry of the Mapping $w = z^2$

The image of right hand side expressed in the form of u and v parametrically as

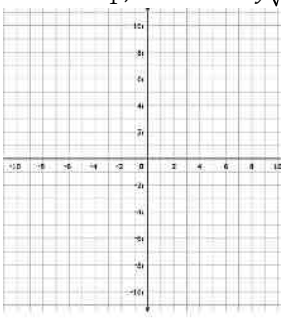
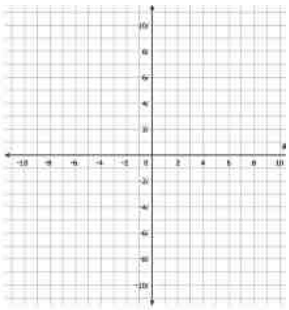
$$u = c_1\,, \qquad v = 2y\sqrt{c_1 + y^2} \qquad (-\infty < y < \infty)$$

Geometry of the Mapping $w = z^2$

The parametric representation for the image of the left-branch of the hyperbola expressed as

$$u = c_1\,, \qquad v = -2y\sqrt{c_1 + y^2} \qquad (-\infty < y < \infty)\,.$$

Geometry of the Mapping $w = z^2$

Each branch of hyperbola

$$2xy = c_2 \qquad (c_2 > 0)$$

is transformed into the line horizontal line $v = c_2$.

Geometry of the Mapping $w = z^2$

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Form $v = 2xy$ we get $v = c_2$.

Geometry of the Mapping $w = z^2$

Each branch of hyperbola

$$2xy = c_2 \qquad (c_2 > 0)$$

is transformed into the line horizontal line $v = c_2$.

Form $v = 2xy$ we get $v = c_2$.

Using

$$y = \frac{c_2}{2x}$$

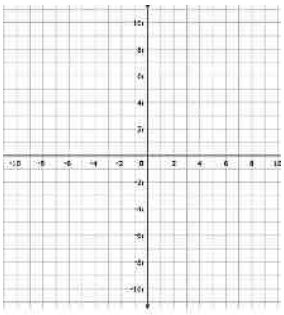
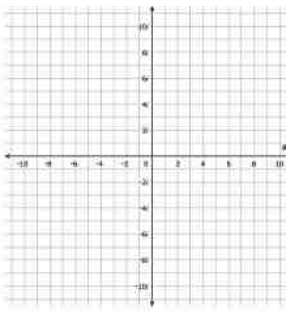
in $u = x^2 - y^2$, we get

$$u = x^2 - \frac{{c_2}^2}{4x^2}, \qquad v = c_2 \qquad 0 < x < \infty$$

Geometry of the Mapping $w = z^2$

The following is a horizontal line

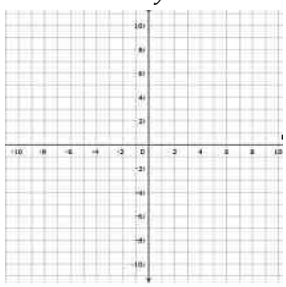
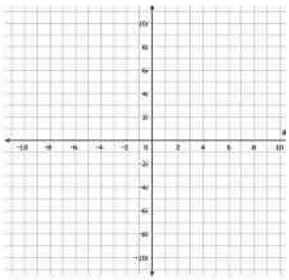
$$u = x^2 - \frac{{c_2}^2}{4x^2}, \qquad v = c_2 \qquad 0 < x < \infty$$

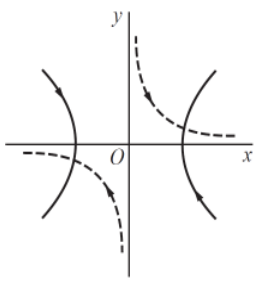
Geometry of the Mapping $w = z^2$

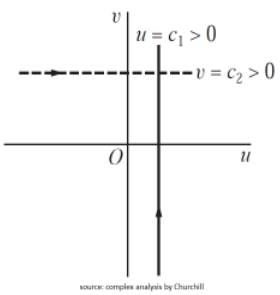
The image of the lower branch has the parametric representation

$$u = \frac{{c_2}^2}{4y^2} - y^2 \qquad v = c_2 \qquad (-\infty < y < 0)$$

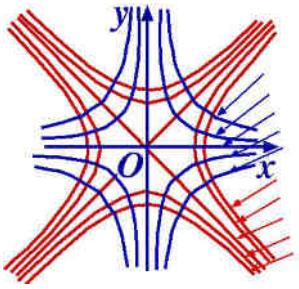
Geometry of the Mapping $w = z^2$



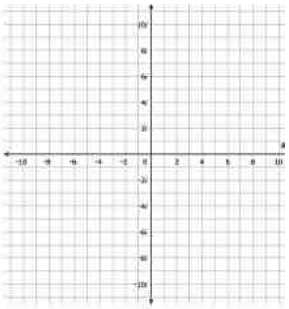


source: complex analysis by Churchill

Geometry of the Mapping $w = z^2$



Source: http://paulscottinfo.ipage.com



Geometry of the Mapping $w = z^{\frac{1}{2}}$

Geometry of the mapping $w = z^{\frac{1}{2}}$

If $z = r\,e^{i\theta}$ then $f(z) = z^{\frac{1}{2}}$ becomes

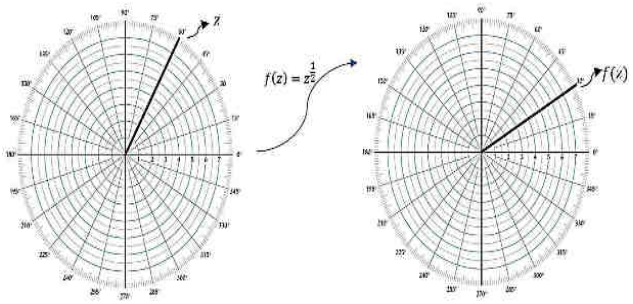
$$w = f(z) = |z|^{\frac{1}{2}} e^{i\frac{Arg(z)}{2}} = r^{\frac{1}{2}} e^{i\frac{\theta}{2}} \quad \text{for } z = re^{i\theta} \neq 0\,.$$

If $w = \rho e^{i\phi}$ then image under f is given by

$$\rho = r^{\frac{1}{2}}\,, \qquad \phi = \frac{\theta}{2}.$$

Geometry of the Mapping $w = z^{\frac{1}{2}}$

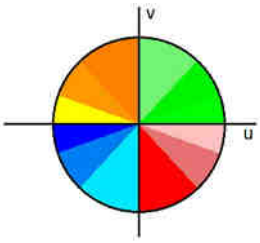
Geometry of the mapping $w = z^{\frac{1}{2}}$



Geometry of the Mapping $w = z^{\frac{1}{2}}$

Example: If the domain of definition D for $f(z) = z^{\frac{1}{2}}$ is restricted to be the region

$$B = \left\{ re^{i\theta} : 0 < r \leq 1 \ \ and \ \ -\pi < \theta \leq \pi \right\}$$



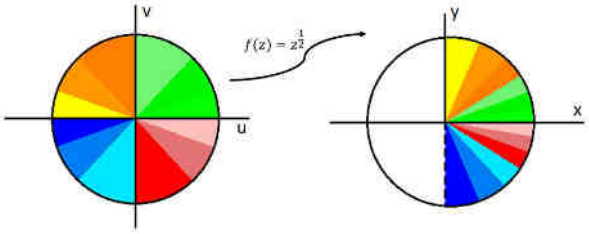
Geometry of the Mapping $w = z^{\frac{1}{2}}$

Example: If the domain of definition D for $f(z) = z^{\frac{1}{2}}$ is restricted to be the region

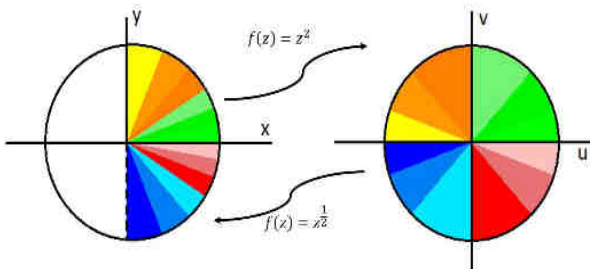
$$B = \left\{ re^{i\theta} : 0 < r \leq 1 \ \ and \ \ -\pi < \theta \leq \pi \right\}$$

Then the image of A under then mapping $f(z) = z^{\frac{1}{2}}$ is

$$A = \left\{ \rho e^{i\phi} : 0 < \rho \leq 1 \ \ and \ \ \frac{-\pi}{2} < \phi \leq \frac{\pi}{2} \right\}.$$

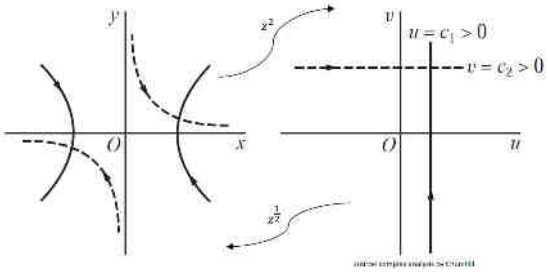


Geometry of the Mapping $w = z^{\frac{1}{2}}$



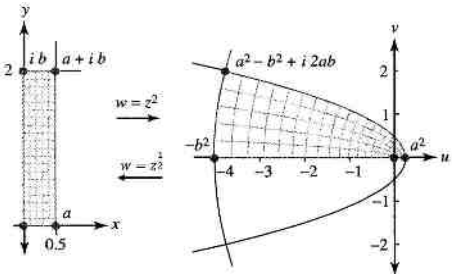
Geometry of the Mapping $w = z^{\frac{1}{2}}$

Example: We have seen that hyperbolas are mapped onto lines under z^2. So the map $f(z) = z^{\frac{1}{2}}$ maps lines on hyperbolas (exercise).



Geometry of the Mapping $w = z^{\frac{1}{2}}$

Example: Similarly, the parabolas are mapped onto lines under $z^{\frac{1}{2}}$ since lines are mapped onto parabolas under z^2(exercise).

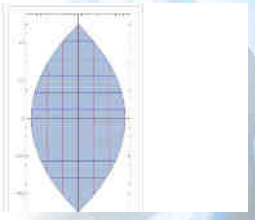


The Mappings $w = z^n$ and $w = z^{\frac{1}{n}}$

- **Geometry of the mapping z^2 and $z^{\frac{1}{2}}$**

Part II

End



The Mappings $w = z^n$ and $w = z^{\frac{1}{n}}$

The mappings z^n and $z^{\frac{1}{n}}$

Topic No. 24

Part III

The Mappings $w = z^n$ and $w = z^{\frac{1}{n}}$

For any positive integer n, the function

$$f(z) = z^n, z = re^{i\theta} \neq 0,$$

can be expressed as

$$w = f(z) = z^n = r^n e^{in\theta}$$

If $w = \rho\, e^{i\phi}$ in the w-plane

The Mappings $w = z^n$ and $w = z^{\frac{1}{n}}$

For any positive integer n, the function

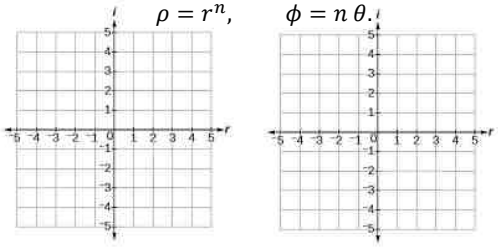
$$f(z) = z^n, z = re^{i\theta} \neq 0,$$

can be expressed as

$$w = f(z) = z^n = r^n e^{in\theta}$$

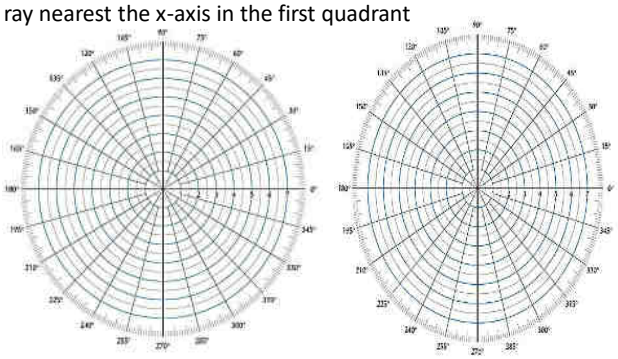
If $w = \rho\, e^{i\phi}$ in the w-plane then the mapping $f(z) = z^n$ can be given by the following equations

$\rho = r^n,$ $\phi = n\,\theta.$



The Mappings $w = z^n$ and $w = z^{\frac{1}{n}}$

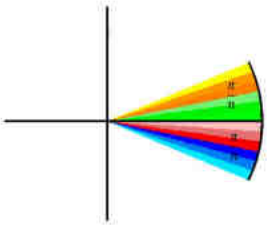
Example: Consider the mapping $f(z) = z^3$. The mapping $w = z^3$ increases the argument of a point by a factor of 3, the ray nearest the x-axis in the first quadrant



The Mappings $w = z^n$ and $w = z^{\frac{1}{n}}$

If we restrict the domain of $f(z) = z^n$ to the region

$$A = \left\{re^{i\theta}: 0 < r \leq 1 \text{ and } -\frac{\pi}{n} \leq \theta \leq \frac{\pi}{n}\right\}$$



The Mappings $w = z^n$ and $w = z^{\frac{1}{n}}$

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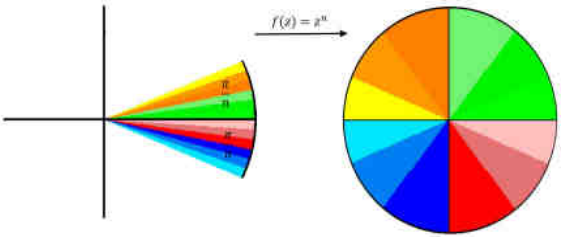
$$A = \left\{ re^{i\theta} : 0 < r \leq 1 \text{ and } -\frac{\pi}{n} \leq \theta \leq \frac{\pi}{n} \right\}$$

Then the image of A under the under the mapping $f(z)$ is given by

$$B = \left\{ \rho\, e^{i\,\phi} : 0 < \rho \leq 1 \text{ and } -\pi < \phi \leq \pi \right\}.$$



The Mappings $w = z^n$ and $w = z^{\frac{1}{n}}$

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$$A = \left\{ re^{i\theta} : 0 < r \leq 1 \text{ and } -\frac{\pi}{n} \leq \theta \leq \frac{\pi}{n} \right\}$$

Then the image of A under the under the mapping $f(z)$ is given by

$$B = \left\{ \rho\, e^{i\,\phi} : 0 < \rho \leq 1 \text{ and } -\pi < \phi \leq \pi \right\}.$$

The inverse mapping of $f(z)$, denoted by $g(z)$, is then

$$g(w) = w^{\frac{1}{n}} = \rho^{\frac{1}{n}}\, e^{i\frac{\phi}{n}}, \qquad w \neq 0$$

where $w \in B$. That is

$$z = g(w) = w^{\frac{1}{n}} = |w|^{\frac{1}{n}}\, e^{i\frac{Arg(w)}{n}}, w \neq 0.$$

The Mappings $w = z^n$ and $w = z^{\frac{1}{n}}$

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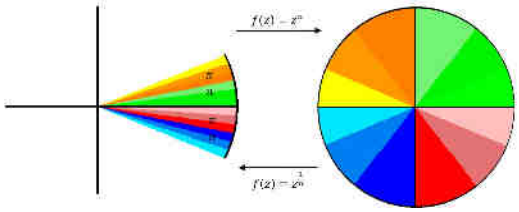

The Mappings $w = z^n$ and $w = z^{\frac{1}{n}}$

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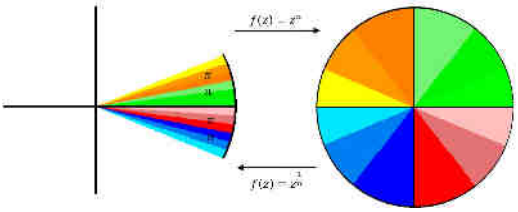


The Mappings $w = z^n$ and $w = z^{\frac{1}{n}}$

Definition (Principal nth root)

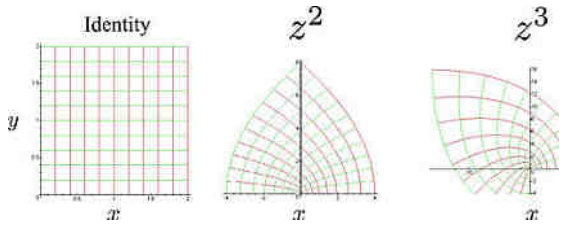
The function

$$g(w) = |w|^{\frac{1}{n}} = \rho^{\frac{1}{n}}\, e^{i\frac{Arg(w)}{n}}, \qquad w \neq 0$$

is called the principal nth root function.

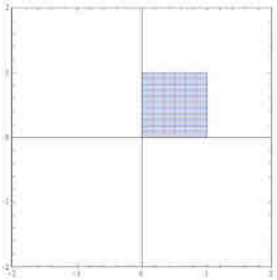
The Mappings $w = z^n$ and $w = z^{\frac{1}{n}}$

The mapping z^3



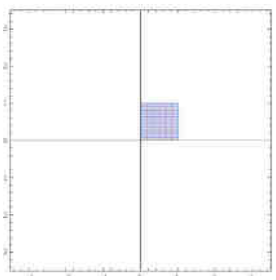
The Mappings $w = z^n$ and $w = z^{\frac{1}{n}}$

The mapping z^3



The Mappings $w = z^n$ and $w = z^{\frac{1}{n}}$

The mapping z^4



The Mappings $w = z^n$ and $w = z^{\frac{1}{n}}$

Example: Find image of the following region

$$A = \left\{ r\, e^{i\theta} : r > 2, and\ \frac{\pi}{4} < \theta < \frac{\pi}{3} \right\}$$

Solution: Using $z = re^{i\theta}$ in $w = f(z) = z^4$, we get

$$w = r^4 e^{i4\theta}$$

The Mappings $w = z^n$ and $w = z^{\frac{1}{n}}$

Example: Find image of the following region

$$A = \left\{ r\, e^{i\theta} : r > 2, and\ \frac{\pi}{4} < \theta < \frac{\pi}{3} \right\}$$

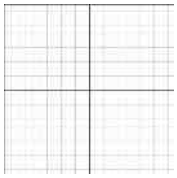
Solution: Using $z = re^{i\theta}$ in $w = f(z) = z^4$, we get

$$w = r^4 e^{i4\theta}$$

If we use polar coordinates $w = \rho\, e^{i\phi}$ in w-plane then

$$\rho = r^4 > 16$$

$$\Rightarrow \rho > 16$$



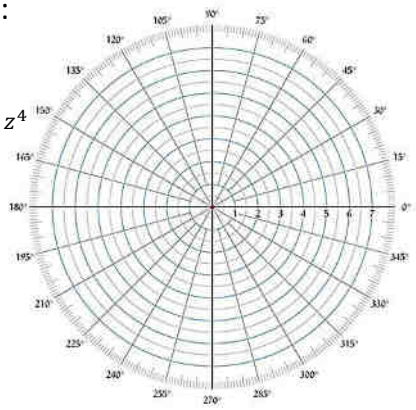
The Mappings $w = z^n$ and $w = z^{\frac{1}{n}}$

Solution(cont.):

The image of

$$\frac{\pi}{4} < \theta < \frac{\pi}{3}$$

by applying $f(z) = z^4$ becomes



The Mappings $w = z^n$ and $w = z^{\frac{1}{n}}$

Solution(cont.):

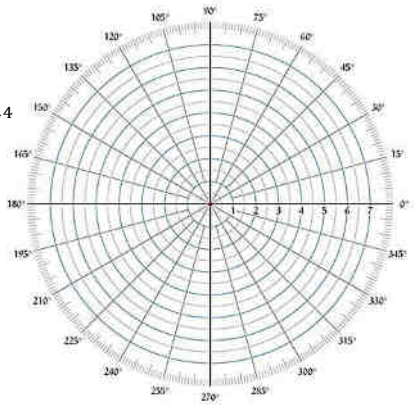
The image of

$$\frac{\pi}{4} < \theta < \frac{\pi}{3}$$

by applying $f(z) = z^4$ becomes

$$\frac{4\pi}{4} < 4\theta < \frac{4\pi}{3}$$

$$\Rightarrow \pi < \phi < \frac{4\pi}{3}$$



The Mappings $w = z^n$ and $w = z^{\frac{1}{n}}$

Example: Find the image of the sector $r > 0, -\pi < \theta < \frac{2\pi}{3}$ under the mapping $z^{\frac{1}{4}}$.

Solution:

Using exponential form $z = re^{i\theta}$, we get

$$w = r^{\frac{1}{4}} e^{i\frac{1}{4}\theta}$$

The Mappings $w = z^n$ and $w = z^{\frac{1}{n}}$

Example: Find the image of the sector $r > 0, -\pi < \theta < \frac{2\pi}{3}$ under the mapping $z^{\frac{1}{4}}$.

Solution:

Using exponential form $z = re^{i\theta}$, we get

$$w = r^{\frac{1}{4}} e^{i\frac{1}{4}\theta}$$

If we use polar coordinates $w = \rho e^{i\phi}$ in w-plane. Then

$$r > 0$$

$$\Rightarrow \quad r^{\frac{1}{4}} > 0$$

The Mappings $w = z^n$ and $w = z^{\frac{1}{n}}$

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Using exponential form $z = re^{i\theta}$, we get

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If we use polar coordinates $w = \rho e^{i\phi}$ in w-plane. Then

$$r > 0$$

$$\Rightarrow \quad r^{\frac{1}{4}} > 0$$

$$\Rightarrow \quad \rho > 0$$

The Mappings $w = z^n$ and $w = z^{\frac{1}{n}}$

Example: Find the image of the sector $r > 0, -\pi < \theta < \frac{2\pi}{3}$ under the mapping $z^{\frac{1}{4}}$.

Solution: And the image of

$$-\pi < \theta < \frac{2\pi}{3}$$

The Mappings $w = z^n$ and $w = z^{\frac{1}{n}}$

Example: Find the image of the sector $r > 0, -\pi < \theta < \frac{2\pi}{3}$ under the mapping $z^{\frac{1}{4}}$.

Solution: And the image of

$$-\pi < \theta < \frac{2\pi}{3}$$

$$\Rightarrow \quad -\frac{\pi}{4} < \frac{\theta}{4} < \frac{2\pi}{12}$$

.

The Mappings $w = z^n$ and $w = z^{\frac{1}{n}}$

Example: Find the image of the sector $r > 0, -\pi < \theta < \frac{2\pi}{3}$ under the mapping $z^{\frac{1}{4}}$.

Solution: And the image of

$$-\pi < \theta < \frac{2\pi}{3}$$

$$\Rightarrow \quad -\frac{\pi}{4} < \frac{\theta}{4} < \frac{2\pi}{12}$$

$$\Rightarrow \quad -\frac{\pi}{4} < \phi < \frac{\pi}{6}$$

The Mappings $w = z^n$ and $w = z^{\frac{1}{n}}$

Example: Find the image of the sector $r > 0, -\pi < \theta < \frac{2\pi}{3}$ under the mapping $z^{\frac{1}{4}}$.

Solution: And the image of

$$-\pi < \theta < \frac{2\pi}{3}$$

$$\Rightarrow \quad -\frac{\pi}{4} < \frac{\theta}{4} < \frac{2\pi}{12}$$

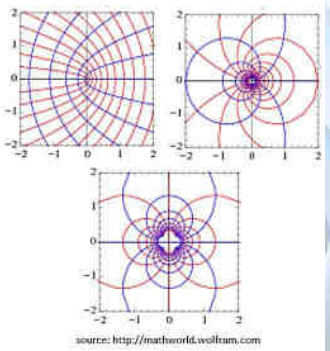
$$\Rightarrow \quad -\frac{\pi}{4} < \phi < \frac{\pi}{6}$$

So the image of mapping $w = z^{\frac{1}{4}}$ is the set

$$\left\{\rho e^{i\phi}:\ \rho > 0 \text{ and } -\frac{\pi}{4} < \phi < \frac{\pi}{6}\right\}.$$

The Mappings $w = z^n$ and $w = z^{\frac{1}{n}}$

The mappings z^n and $z^{\frac{1}{n}}$



End

Limits

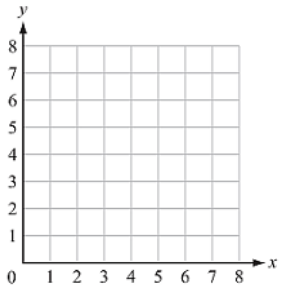
Topic No. 25

Edited by
Mahar Afaq Safdar Muhammadi
MSc Math, Whatsapp 03494873115

Source: https://steemit.com

Real Limits

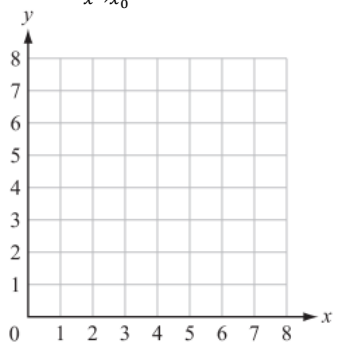
Consider a real valued function $y = f(x)$ of real variable x. We can calculate its graph in $\mathbb{R}^2$.



Real Limits

The limit

$$\lim_{x \to x_0} f(x) = L$$



Real Limits

The limit

$$\lim_{x \to x_0} f(x) = L$$

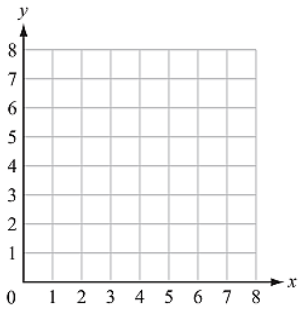
means that values $f(x)$ of the function f can be made **arbitrary close** to the real number L if values of x are chosen **sufficiently close** to, but not equal to, the real number x_0.

Real Limits

On the real line, the points x and x_0 are close if $|x - x_0|$ is a small positive number.

Real Limits

On the real line, the points x and x_0 are close if $|x - x_0|$ is a small positive number.
Similarly, the points $f(x)$ and L are close if $|f(x) - L|$ is a small positive number.

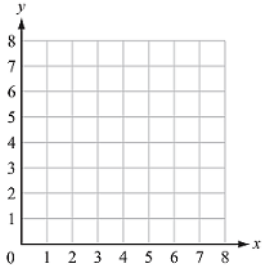


Real Limits

The limit

$$\lim_{x \to x_0} f(x) = L$$

means that values $f(x)$ of the function f can be made **arbitrary close** to the real number L if values of x are chosen **sufficiently close** to, but not equal to, the real number x_0.



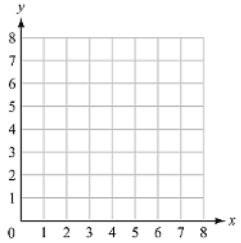
Real Limits

"$f(x)$ can be made arbitrary close to the L"
for any $\varepsilon > 0$, x can be chosen in that way so that

$$|f(x) - L| < \varepsilon$$

"sufficiently close to, but not equal to, x_0"
There exist $\delta > 0$ such that if x is within distance δ of x_0 and $x \neq x_0$, then

$$|f(x) - L| < \varepsilon$$



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The limit

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means that values $f(x)$ of the function f can be made **arbitrary close** to the real number L if values of x are chosen **sufficiently close** to, but not equal to, the real number x_0.

For every $\varepsilon > 0$ there exists a $\delta > 0$ such that
if

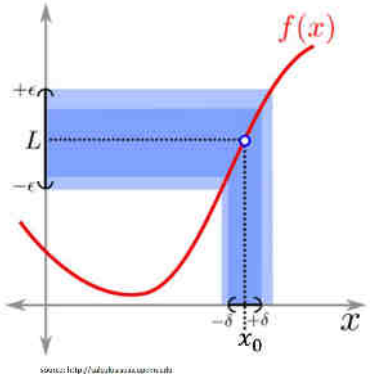
$$0 < |x - x_0| < \delta$$

then

$$|f(x) - L| < \varepsilon$$

Real Limits

For every $\varepsilon > 0$ there exists a $\delta > 0$ such that
if $0 < |x - x_0| < \delta$ then $|f(x) - L| < \varepsilon$



Real Limits

For every $\varepsilon > 0$ there exists a $\delta > 0$ such that
if $0 < |x - x_0| < \delta$ then $|f(x) - L| < \varepsilon$

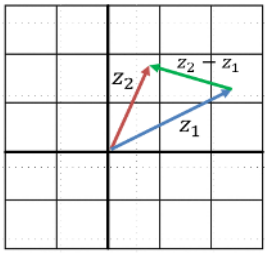
Complex Limits

The complex limit is the same than that of real limit except the notion "close" in the complex plane.

Distance between of z_1 and z_2 written as

$$|z_2 - z_1|$$

and is a real number.



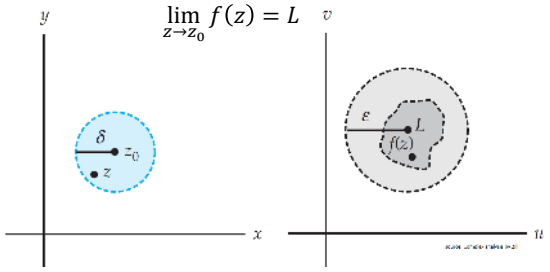
Complex Limits

The limit of f as $z \to z_o$ exits and is equal to L if

"For every $\varepsilon > 0$ there exists a $\delta > 0$ such that $|f(z) - L| < \varepsilon$ whenever $0 < |z - z_0| < \delta$."

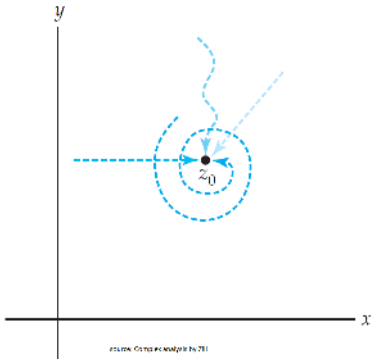
This fact is denoted as

$$\lim_{z \to z_0} f(z) = L$$



Complex Limits

There are many ways to approach z_0 in complex plane. We can choose any path in the neighbourhood of z_0 to reach z_0.

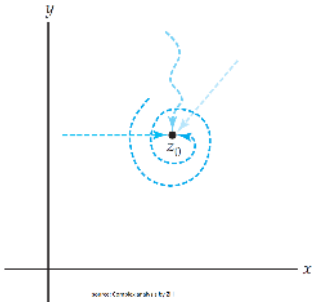


Complex Limits

If

$$\lim_{z \to z_0} f(z) = L$$

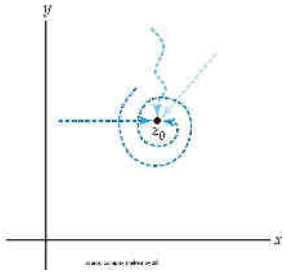
we want that $f(z)$ approaches the same complex number L along every possible curve through z_0.



Complex Limits

Nonexistence of limits

If f approaches two complex numbers $L_1 \neq L_2$ for two different curves or paths through z_0, then $\lim_{z \to z_0} f(z)$ does not exist.



Complex Limits

Nonexistence of limits

Example: Show that the limit

$$\lim_{z \to 0} \frac{z}{\bar{z}}$$

Does not exist.

Complex Limits

Nonexistence of limits

Example: Show that the limit

$$\lim_{z \to 0} \frac{z}{\bar{z}}$$

Does not exist.

Solution: First we calculate the limit when z approaches to 0 along the real axis

$$\lim_{z \to 0} \frac{z}{\bar{z}} = \lim_{x \to 0} \frac{x + i0}{x - i0}.$$

Complex Limits

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Example: Show that the limit

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Does not exist.

Solution: First we calculate the limit when z approaches to 0 along the real axis

$$\lim_{z \to 0} \frac{z}{\bar{z}} = \lim_{x \to 0} \frac{x + i0}{x - i0} = \lim_{x \to 0} 1 = 1 \ .$$

Complex Limits

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If z approaches 0 along the imaginary axis.

$$\lim_{z \to 0} \frac{z}{\bar{z}} = \lim_{y \to 0} \frac{0 + iy}{0 - iy}$$

Complex Limits

Nonexistence of limits

Example: Show that the limit

$$\lim_{z\to 0} \frac{z}{\bar{z}}$$

Does not exist.

Solution: First we calculate the limit when z approaches to 0 along the real axis

$$\lim_{z\to 0} \frac{z}{\bar{z}} = \lim_{x\to 0} \frac{x+i0}{x-i0} = \lim_{x\to 0} 1 = 1 \ .$$

If z approaches 0 along the imaginary axis.

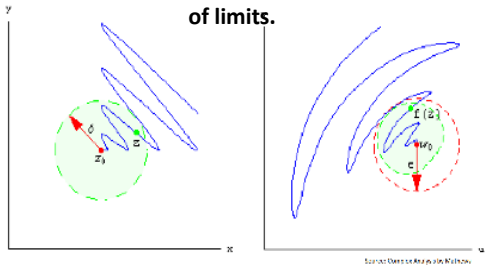
$$\lim_{z\to 0} \frac{z}{\bar{z}} = \lim_{y\to 0} \frac{0+iy}{0-iy} = \lim_{x\to 0} -1 = -1 \ .$$

Complex Limits

Nonexistence of limits

If f approaches two complex numbers $L_1 \neq L_2$ for two different curves or paths through z_0, then $\lim_{z\to z_0} f(z)$ does not exist.

The above criterion can not be used to prove the existence of limits.



Complex Limits

To show existence of limit

$$\lim_{z\to z_0} f(z) = L$$

we need to show the following

"For every $\varepsilon > 0$ there exists a $\delta > 0$ such that $|f(z) - L| < \varepsilon$ whenever $0 < |z - z_0| < \delta$."

Complex Limits

Example: Show that

$$\lim_{z\to z_0} \bar{z} = z_0$$

Solution: We need to show that

"For every $\varepsilon > 0$ there exists a $\delta > 0$ such that $|\bar{z} - \bar{z_0}| < \varepsilon$ whenever $0 < |z - z_0| < \delta$."

Complex Limits

Example: Show that

$$\lim_{z\to z_0} \bar{z} = z_0$$

Solution:

Limit

- **Real Limits**
- **Complex limits**

End

Source: https://www.rangerabbit.com

Limits

Topic No. 26

Source: https://steemit.com

Recall: Complex Limits

The limit of f as $z \to z_o$ exits and is equal to L if

"For every $\varepsilon > 0$ there exists a $\delta > 0$ such that $|f(z) - L| < \varepsilon$ whenever $0 < |z - z_0| < \delta$."

This fact is denoted as

$$\lim_{z\to z_0} f(z) = L$$

Complex Limits

The definition of limit of a complex function does not provide a convenient way to calculate limits.

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Our next aim is to provide a practical method for computing complex limits.

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The definition of limit of a complex function does not provide a convenient way to calculate limits.

Our next aim is to provide a practical method for computing complex limits.

We want to use the connection between complex limits and real limits to simplify computation of complex limits.

Complex Limits

A complex function is of the form

$$f(z) = f(x+iy) = u(x,y) + iv(x,y)$$

Complex Limits

A complex function is of the form

$$f(z) = f(x+iy) = u(x,y) + iv(x,y)$$

We find a relation between the complex limit of $f(z)$ and the real limits of the multivariable real function $u(x,y)$ and $v(x,y)$.

Relation between Real and Complex Limits

Main motivation

Theorem

Let $f(z) = u(x,y) + iv(x,y)$ be a complex function that is defined in some neighborhood of z_0, except perhaps at $z_0 = x_0 + iy_0$. Then

$$\lim_{z \to z_0} f(z) = u_0 + iv_0$$

iff

$$\lim_{(x,y)\to(x_0,y_0)} u(x,y) = u_0 \text{ and } \lim_{(x,y)\to(x_0,y_0)} v(x,y) = v_0 .$$

Relation between Real and Complex Limits

Limit of real-valued functions of two variables

What does the following mean

$$\lim_{(x,y)\to(x_0,y_0)} u(x,y) = u_0$$

Relation between Real and Complex Limits

Limit of real-valued functions of two variables

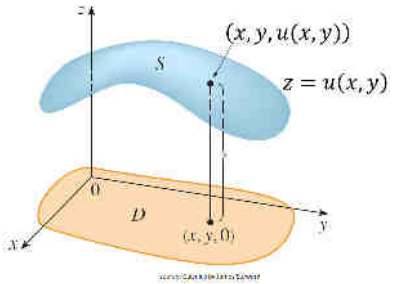
What does the following mean

$$\lim_{(x,y)\to(x_0,y_0)} u(x,y) = u_0$$

The value of $u(x,y)$ can be made to get **as close as we want** to the value u_0 by taking (x,y) to be **sufficiently close** to (x_0,y_0).

Relation between Real and Complex Limits

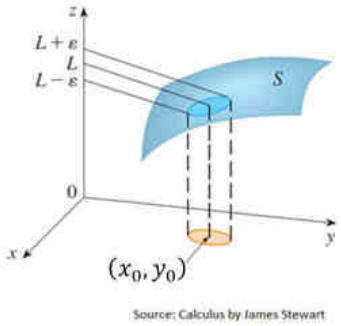
The function $u(x,y)$ can be associated to a surface in 3D.



Relation between Real and Complex Limits

What does the following mean

$$\lim_{(x,y)\to(x_0,y_0)} u(x,y) = u_0$$

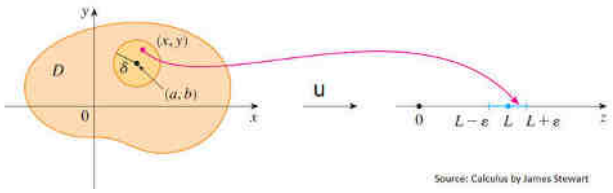


Source: Calculus by James Stewart

Real Limits of Multivariable functions

What does the following mean

$$\lim_{(x,y)\to(x_0,y_0)} u(x,y) = u_0$$



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Real Limits of Multivariable functions

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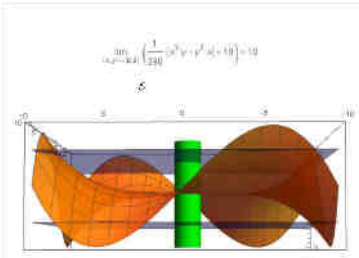


Real Limits of Multivariable functions

What does the following mean

$$\lim_{(x,y)\to(x_0,y_0)} u(x,y) = u_0$$

$|u(x,y) - u_0|$ can be made arbitrarily small by making distance between (x,y) and (x_0,y_0) small.

Real Limits of Multivariable functions

What does the following mean

$$\lim_{(x,y)\to(x_0,y_0)} u(x,y) = u_0$$

$|u(x,y) - u_0|$ can be made arbitrarily small by making distance between (x,y) and (x_0,y_0) small.

Here the point (x,y) is in the xy plane, and the distance between (x,y) and (x_0,y_0) can be calculated using distance formula $\sqrt{(x-x_0)^2+(y-y_0)^2}$.

Real Limits of Multivariable functions

Definition

$$\lim_{(x,y)\to(x_0,y_0)} u(x,y) = u_0$$

means the following

for each $\varepsilon > 0$, there exist $\delta > 0$ such that
$|u(x,y) - u_0| < \varepsilon$ whenever $0 < \sqrt{(x-x_0)^2+(y-y_0)^2} < \delta$.

Real Limits of Multivariable functions

Example: If $u(x,y) = \frac{2x^3}{(x^2+y^2)}$ then show that

$$\lim_{(x,y)\to(0,0)} u(x,y) = 0 \ .$$

Solution:

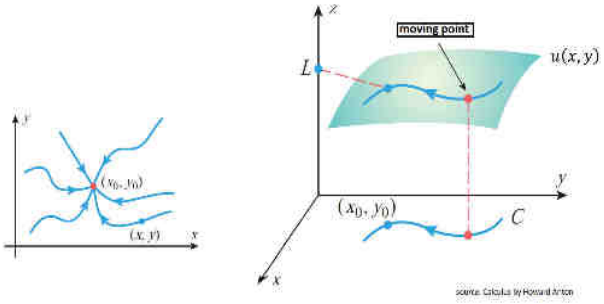
Real Limits of Multivariable functions

Example: If $u(x,y)=\frac{2x^3}{(x^2+y^2)}$ then show that $\lim_{(x,y)\to(0,0)} u(x,y)=0$.

Solution:

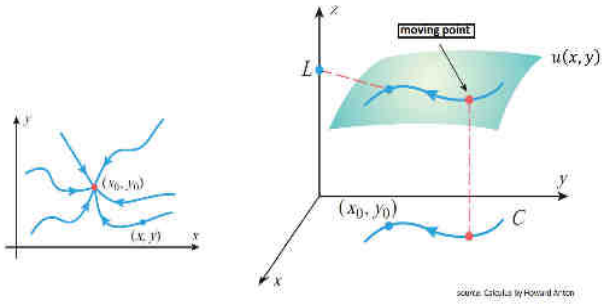
Real Limits of Multivariable functions

The value u_0 of the limit must not depend on how (x,y) approaches (x_0,y_0), so $u(x,y)$ must approach the value u_0 when (x,y) approaches (x_0,y_0) along any curve that ends at the point (x_0,y_0).



Real Limits of Multivariable functions

If we can find two curves L_1 and L_2that end at (x_0,y_0) along which $u(x,y)$ approaches two distinct values u_1 and u_2, then $u(x,y)$ does not have a limit as (x,y) approaches (x_0,y_0).



Real Limits of Multivariable functions

Example: Show that the function $u(x,y)=\frac{xy}{x^2+y^2}$ does not have a limit as (x,y) approaches $(0,0)$.
Solution: Along $y=x$

Real Limits of Multivariable functions

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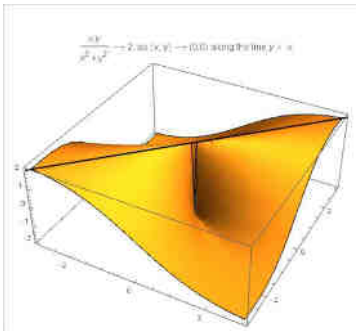

Real Limits of Multivariable functions

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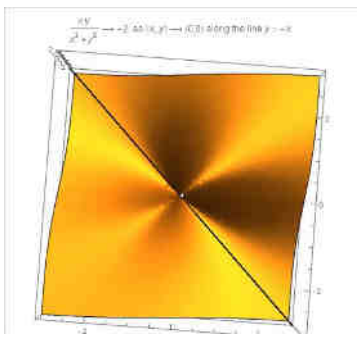

Limit

End

Real Limits of Multivariable functions

Source: https://www.rangerabbit.com

Limits

Limits
Part III

Topic No. 27

Source: https://steemit.com

Complex Limits

A complex function is of the form

$$f(z) = f(x+iy) = u(x,y) + iv(x,y)$$

We find a relation between the complex limit of $f(z)$ and the real limits of the multivariable real function $u(x,y)$ and $v(x,y)$.

Relation between Real & Complex Limits

Theorem

Let $f(z) = u(x,y) + iv(x,y)$ be a complex function that is defined in some neighborhood of z_0, except perhaps at $z_0 = x_0 + iy_0$. Then

$$\lim_{z\to z_0} f(z) = u_0 + iv_0$$

iff

$$\lim_{(x,y)\to(x_0,y_0)} u(x,y) = u_0 \text{ and } \lim_{(x,y)\to(x_0,y_0)} v(x,y) = v_0 .$$

Relation between Real & Complex Limits

Proof: Assume

$$\lim_{z\to z_0} f(z) = u_0 + iv_0$$

To show:

$$\lim_{(x,y)\to(x_0,y_0)} u(x,y) = u_0 \text{ and } \lim_{(x,y)\to(x_0,y_0)} v(x,y) = v_0$$

Relation between Real & Complex Limits

Proof: Assume

$$\lim_{z\to z_0} f(z) = u_0 + iv_0$$

To show:

$$\lim_{(x,y)\to(x_0,y_0)} u(x,y) = u_0 \text{ and } \lim_{(x,y)\to(x_0,y_0)} v(x,y) = v_0$$

Recall

"For every $\varepsilon > 0$ there exists a $\delta > 0$ such that $|f(z) - L| < \varepsilon$ whenever $0 < |z - z_0| < \delta$."

Relation between Real & Complex Limits

Proof: Assume

$$\lim_{z\to z_0} f(z) = u_0 + iv_0$$

To show:

$$\lim_{(x,y)\to(x_0,y_0)} u(x,y) = u_0 \text{ and } \lim_{(x,y)\to(x_0,y_0)} v(x,y) = v_0$$

Recall

$$\lim_{(x,y)\to(x_0,y_0)} u(x,y) = u_0$$

means the following

for each $\varepsilon > 0$, there exist $\delta > 0$ such that

$|u(x,y) - u_0| < \varepsilon$ whenever $0 < \sqrt{(x-x_0)^2 + (y-y_0)^2} < \delta$.

Relation between Real & Complex Limits

Proof: Assume

$$\lim_{z\to z_0} f(z) = u_0 + iv_0$$

To show:

$$\lim_{(x,y)\to(x_0,y_0)} u(x,y) = u_0 \text{ and } \lim_{(x,y)\to(x_0,y_0)} v(x,y) = v_0$$

Relation between Real & Complex Limits

Proof:

Relation between Real & Complex Limits

Proof: Conversely assume

$\lim_{(x,y)\to(x_0,y_0)} u(x,y) = u_0$ and $\lim_{(x,y)\to(x_0,y_0)} v(x,y) = v_0$

To show:

$$\lim_{z\to z_0} f(z) = u_0 + iv_0$$

Relation between Real & Complex Limits

Proof:

Limit

End

Relation between real and complex limits

Source: https://www.rangerabbit.com

Limits

Topic No. 28

Limits Part IV

Source: https://steemit.com

Real Limits of Multivariable functions

Properties of real limits of functions of two variables

The following fact about real limits is easy to prove

$$\lim_{(x,y)\to(x_0,y_0)} 1 = 1 ,$$

Real Limits of Multivariable functions

Properties of real limits of functions of two variables

The following fact about real limits is easy to prove

$$\lim_{(x,y)\to(x_0,y_0)} 1 = 1 ,$$
$$\lim_{(x,y)\to(x_0,y_0)} x = x_0 ,$$

Real Limits of Multivariable functions

Properties of real limits of functions of two variables

The following fact about real limits is easy to prove

$$\lim_{(x,y)\to(x_0,y_0)} 1 = 1 ,$$
$$\lim_{(x,y)\to(x_0,y_0)} x = x_0 ,$$
$$\lim_{(x,y)\to(x_0,y_0)} y = y_0 .$$

Real Limits of Multivariable functions

Properties of real limits of functions of two variables

Theorem: If $\lim_{(x,y)\to(x_0,y_0)} F(x,y) = A$

and

$$\lim_{(x,y)\to(x_0,y_0)} G(x,y) = B$$

then

1. $\lim_{(x,y)\to(x_0,y_0)} cF(x,y) = cA$, where c is a real constant

Real Limits of Multivariable functions

Properties of real limits of functions of two variables

Theorem: If $\lim_{(x,y)\to(x_0,y_0)} F(x,y) = A$

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then

1. $\lim_{(x,y)\to(x_0,y_0)} cF(x,y) = cA$, where c is a real constant
2. $\lim_{(x,y)\to(x_0,y_0)} F(x,y) \pm G(x,y) = A \pm B$

Real Limits of Multivariable functions

Properties of real limits of functions of two variables

Theorem: If $\lim_{(x,y)\to(x_0,y_0)} F(x,y) = A$

and

$$\lim_{(x,y)\to(x_0,y_0)} G(x,y) = B$$

then

1. $\lim_{(x,y)\to(x_0,y_0)} cF(x,y) = cA$, where c is a real constant
2. $\lim_{(x,y)\to(x_0,y_0)} F(x,y) \pm G(x,y) = A \pm B$
3. $\lim_{(x,y)\to(x_0,y_0)} F(x,y).G(x,y) = A.B$

Real Limits of Multivariable functions

Properties of real limits of functions of two variables

Theorem: If $\lim_{(x,y)\to(x_0,y_0)} F(x,y) = A$

and

$$\lim_{(x,y)\to(x_0,y_0)} G(x,y) = B$$

then

1. $\lim_{(x,y)\to(x_0,y_0)} cF(x,y) = cA$, where c is a real constant
2. $\lim_{(x,y)\to(x_0,y_0)} F(x,y) \pm G(x,y) = A \pm B$
3. $\lim_{(x,y)\to(x_0,y_0)} F(x,y).G(x,y) = A.B$
4. $\lim_{(x,y)\to(x_0,y_0)} \frac{F(x,y)}{G(x,y)} = \frac{A}{B}$, $B \neq 0$

Real Limits of Multivariable functions

Example: Calculate the limit $\lim_{(x,y)\to(1,2)} 3xy^2 - y$.

Solution:

Relation between Real & Complex Limits

Theorem

Let $f(z) = u(x,y) + iv(x,y)$ be a complex function that is defined in some neighborhood of z_0, except perhaps at $z_0 = x_0 + iy_0$. Then

$$\lim_{z\to z_0} f(z) = u_0 + iv_0$$

iff

$\lim_{(x,y)\to(x_0,y_0)} u(x,y) = u_0$ and $\lim_{(x,y)\to(x_0,y_0)} v(x,y) = v_0$.

Relation between Real & Complex Limits

Example: Compute the limit

$$\lim_{z\to i} z^2 + z + 1 .$$

Solution:

Properties of Complex Limits

Theorem: Suppose that

$$\lim_{z\to z_0} f(z) = A$$

and

$$\lim_{z\to z_0} g(z) = B$$

then

1. $\lim_{z\to z_0} c\,f(z) = cA$, c a complex constant.

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2. $\lim_{z\to z_0} (f(z) \pm g(z)) = A \pm B.$

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3. $\lim_{z\to z_0} (f(z).g(z)) = A.B$
4. $\lim_{z\to z_0} \frac{f(z)}{g(z)} = \frac{A}{B}$ provided $B \neq 0$.

Properties of Complex Limits

Theorem: Suppose that $\lim_{z\to z_0} f(z) = A$ and $\lim_{z\to z_0} g(z) = B$

then

1. $\lim_{z\to z_0} c\,f(z) = cA$, c a complex constant.

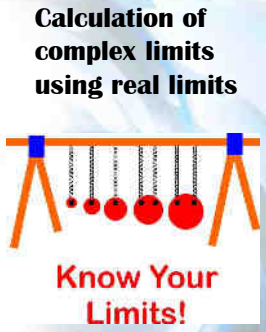
Properties of Complex Limits

Example: Compute the limit

$$\lim_{z\to i} z^2 + z + 1\ .$$

Limit

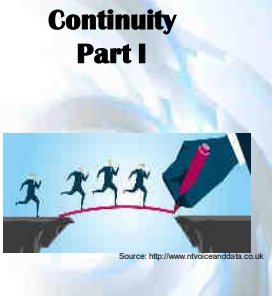
End



Source: https://www.rangerabbit.com

Continuity

Topic No. 29



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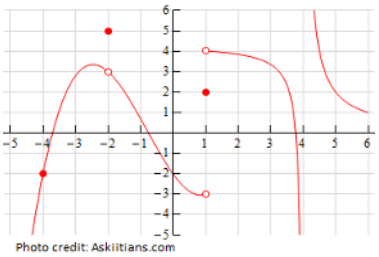
- In mathematics, continuous function also has similar kind of definition.
- If a small change in the input of a function brings a small change in the output of that function, then we say that the function is continuous.

Continuity of Real Functions $f(x)$

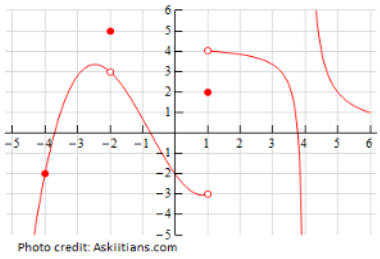
Continuity of real functions $f(x)$



Continuity of Real Functions $f(x)$

A function $f(x)$ is continuous at a point x_0 if

1. $\lim_{x \to x_0} f(x)$ exists



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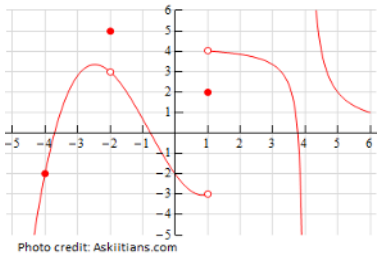

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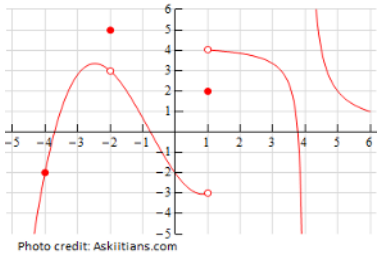

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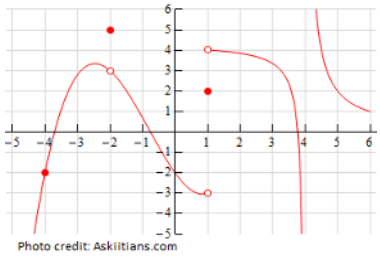

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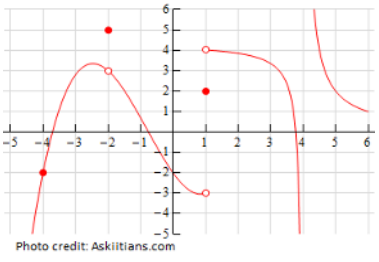
A function is said to be continuous on a set A if it is continuous at each point of A.

Continuity of Real Functions $f(x)$

Note that in order to calculate limit at a point x_0 function $f(x)$ must be defined in some neighbourhood of x_0.



Continuity of Real Functions $f(x)$

Example: Check whether the following function is continuous or not

$$f(x) = \begin{cases} x^2 + 1 & x \le 0 \\ x & x > 0 \end{cases}$$

Solution:

Continuity of Real Functions $f(x)$

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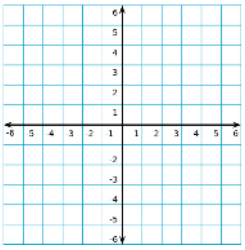

Continuity of $u(x, y)$

Continuity of $u(x, y)$

Let $u(x, y)$ be a real valued function of the two real variables x and y. We say that u is continuous at a point (x_0, y_0) if these three conditions are satisfied

1. $\lim_{(x,y)\to(x_0,y_0)} u(x, y)$ exists

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A function is said to be continuous on a set A if it is continuous at each point of A.

Continuity of $u(x, y)$

Example: Check whether the following function is continuous at $(0,0)$

$$u(x, y) = \begin{cases} \dfrac{x^3}{x^2 + y^2} & (x, y) \neq (0,0) \\ 0 & (x, y) = (0,0) \end{cases}$$

Solution:

Continuity of $u(x, y)$

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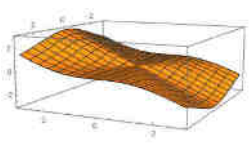

Continuity of Complex Functions

Continuity of complex valued function $f(z)$

Let $f(z)$ be a complex function of the complex variable z that is defined for all values of z in some neighborhood of z_0. We say that f is continuous at z_0 if the following three conditions are satisfied.

1. $\lim_{z \to z_0} f(z)$ exists.

Continuity of Complex Functions

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A function is said to continuous in on a set A if it is continuous at each point of A.

Continuity of Complex Functions

Example: Show that the function $f(z) = \bar{z}$ is continuous.

Solution:

Continuity of Complex Functions

Continuity of complex valued function $f(z)$

Let $f(z)$ be a complex function of the complex variable z that is defined for all values of z in some neighborhood of z_0. We say that f is continuous at z_0 if the following three conditions are satisfied.

1. $\lim_{z \to z_0} f(z)$ exists.
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3. $\lim_{z \to z_0} f(z) = f(z_0)$.

Statements (1) and (2) are consequence of statement (3). Using (ε, δ)- definition of limit we can define continuity in the following way as well.

Continuity of Complex Functions

Recall

"For every $\varepsilon > 0$ there exists a $\delta > 0$ such that $|f(z) - L| < \varepsilon$ whenever $0 < |z - z_0| < \delta$."

Continuity of Complex Functions

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(ε, δ) – definition of continuity of $f(z)$

Let $f(z)$ be a complex function of the complex variable z. We say that f is continuous at z_0 if

for each $\varepsilon > 0$ there exist a $\delta > 0$ such that $|f(z) - f(z_0)| < \varepsilon$ whenever $0 < |z - z_0| < \delta$.

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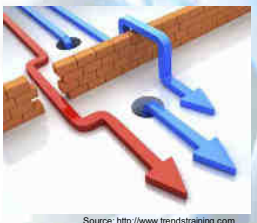
Limit

- **Continuity of real function**
- **Continuity of complex functions**

End



Source: http://www.trendstraining.com

Continuity

Topic No. 30

Continuity Part II

Source: http://www.ntvoiceanddata.co.uk

Continuity of Complex Functions

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1. $\lim_{z \to z_0} f(z)$ exists.
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Continuity of Complex Functions

(ε, δ) – definition of continuity of $f(z)$

Let $f(z)$ be a complex function of the complex variable z. We say that f is continuous at z_0 if

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Continuity of Complex Functions

Theorem: Let $f(z) = u(x, y) + iv(x, y)$ be defined in some neighborhood of z_0. Then f is continuous at $z_0 = x_0 + iy_0$ iff u and v are continuous at (x_0, y_0).

Proof: Assume f is continuous at $z_0 = x_0 + iy_0$

To show: u and v are continuous at (x_0, y_0).

Continuity of Complex Functions

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Continuity of $u(x, y)$

Let $u(x, y)$ be a real valued function of the two real variables x and y. We say that u is continuous at a point (x_0, y_0) if these three conditions are satisfied

1. $\lim_{(x,y) \to (x_0, y_0)} u(x, y)$ exists
2. $u(x_0, y_0)$ exists
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Continuity of Complex Functions

Theorem: Suppose that f and g are continuous at the point z_0. Then the following functions are continuous at z_0

- The sum $f + g$, where $(f + g)(z) = f(z) + g(z)$.

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- The quotient $\frac{f}{g}$, where $\frac{f}{g}(z) = \frac{f(z)}{g(z)}$, provided $g(z_0) \neq 0$

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Proof:

Continuity of Complex Functions

Example: Show that the polynomial function given by

$$w = P(z) = a_0 + a_1 z + \cdots + a_n z^n$$

is continuous at each point z_0 in the complex plane.

Solution:

Continuity of Complex Functions

Example: Show that the polynomial function given by

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Solution:

Continuity of Complex Functions

Example: Find the limit

$$\lim_{z \to i} \frac{z^2 + 2}{z^2 - 2z + 1}.$$

Solution:

Limit

- **Properties of continuous functions**

End

Source: http://www.trendstraining.com

Continuity of Complex Functions

Composition of Functions

Consider two complex valued functions $f(z)$ and $g(z)$ then its composition, denoted by $g \circ f(z)$, is defined as

$$g \circ f(z) = g\big(f(z)\big)$$

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Continuity of Complex Functions

Composition of Functions

Theorem: If f and g are continuous at the point z_0 then $g \circ f$ is continuous at z_0.

Proof:

Continuity of Complex Functions

Consequences of continuity

Theorem: If a function $f(z)$ is continuous and nonzero at a point z_0, then $f(z) \neq 0$ throughout some neighbourhood of that point.

Continuity of Complex Functions

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Theorem: If a function f is continuous throughout a region R that is both closed and bounded, there exists a nonnegative real number M such that

$$|f(z)| \leq M \text{ for all points } z \text{ in } R$$

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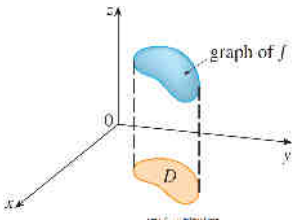

Limit

- **Properties of continuous functions**
- **Consequences of continuity**

End

Source: http://www.trendstraining.com

Continuity

Topic No. 31

Continuity
Part III

Source: http://www.ntvoiceanddata.co.uk

Continuity of Complex Functions

Continuity of complex valued function $f(z)$

Let $f(z)$ be a complex function of the complex variable z that is defined for all values of z in some neighborhood of z_0. We say that f is continuous at z_0 if the following three conditions are satisfied.

1. $\lim_{z \to z_0} f(z)$ exists.
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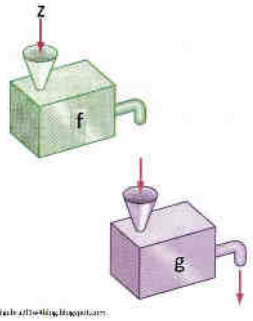

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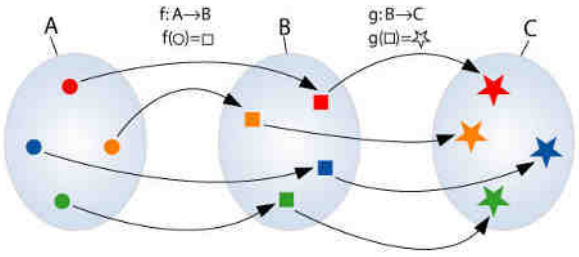

Source: https://m.blog.naver.com

Composition of Functions

Theorem: If f is continuous at a point z_0 and g is continuous at $f(z_0)$ then g $\circ$ f is continuous at z_0.

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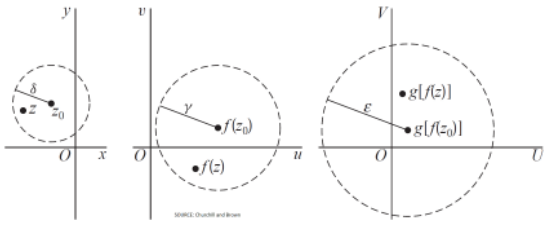
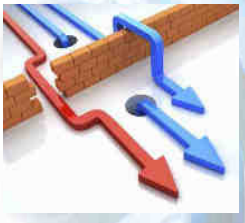


Continuity

End

- **Properties of continuous functions**
- **Consequences of continuity**



Source: http://www.trendstraining.com

Continuity

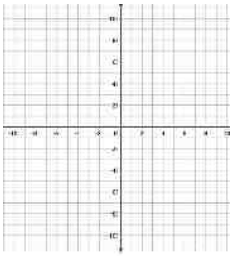
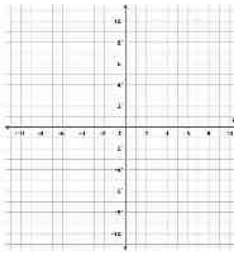
Continuity Part IV

Topic No. 32

Source: http://www.ntvoiceanddata.co.uk

Consequences of Continuity

Theorem: If a function $f(z)$ is continuous and nonzero at a point z_0, then $f(z) \neq 0$ throughout some neighbourhood of that point.

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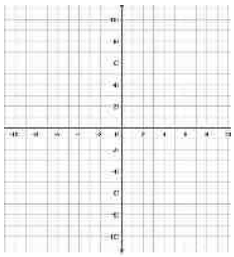
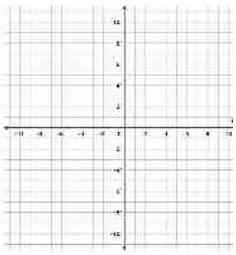
Proof:

Consequences of Continuity

Theorem: If a function f is continuous throughout a region R that is both closed and bounded, there exists a nonnegative real number M such that

$$|f(z)| \leq M \text{ for all points } z \text{ in } R$$

where equality holds for at least one such z.

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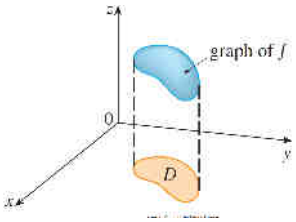
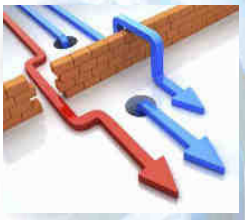


Continuity

End

Consequences of continuity



Source: http://www.trendstraining.com

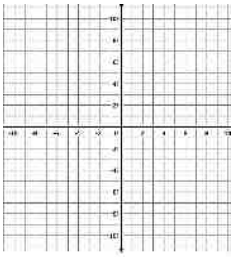
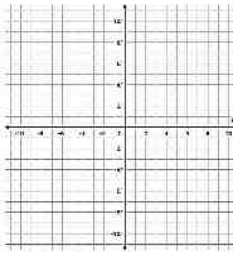
The Reciprocal Transformation

Topic No. 33

The Reciprocal Transformation

The Reciprocal Transformation

The mapping $1/z$ is called the reciprocal function.

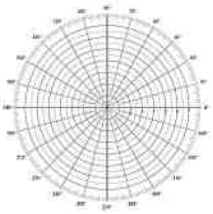
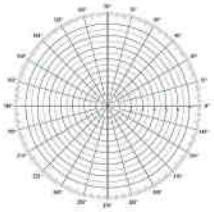



The Reciprocal Transformation

The mapping $1/z$ is called the reciprocal function. In exponential notation it can expressed as

If $z = re^{i\theta}$ then we obtain

$$w = \frac{1}{z} = \frac{1}{re^{i\theta}} = \frac{1}{r}e^{-i\theta}$$

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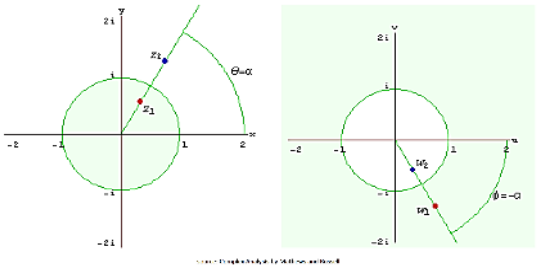

The Reciprocal Transformation

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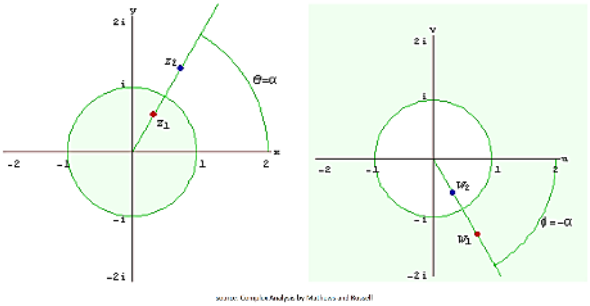
Case1: If z lies on the unit circle.



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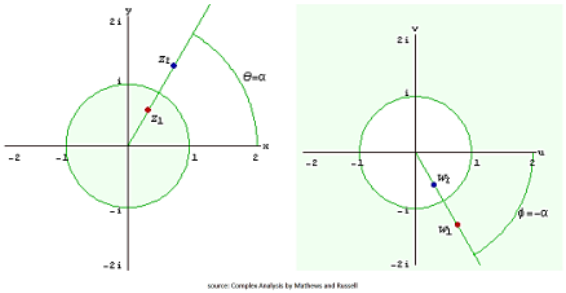
Case2: If z lies outside the unit circle.



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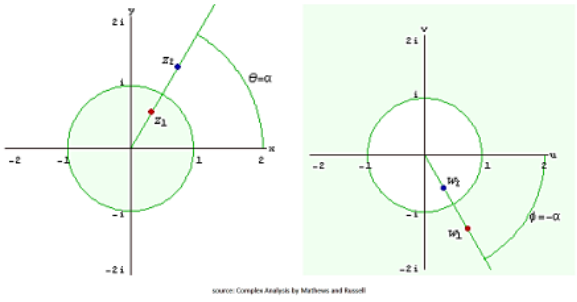

The Reciprocal Transformation

Example: Find the image of the vertical line $x = 1$ under the reciprocal mapping $w = \frac{1}{z}$.

Solution: The vertical line $x = 1$ consists of the set of points

$$z = 1 + iy, \qquad -\infty < y < \infty.$$

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After replacing z with $1 + iy$ in $w = 1/z$ and simplifying, we obtain

$$z = \frac{1}{1+iy} = \frac{1}{1+y^2} - \frac{y}{1+y^2}i.$$

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Extracting u and v

$$u = \frac{1}{1+y^2}, \qquad v = \frac{-y}{1+y^2} \qquad -\infty < y < \infty.$$

The Reciprocal Transformation

Solution(cont.):

$u = \frac{1}{1+y^2}, \qquad v = \frac{-y}{1+y^2} \qquad -\infty < y < \infty.$

From these equations we can write v in terms of u.

$$v = -yu$$
$$y = \frac{-v}{u}$$

The Reciprocal Transformation

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Substituting in $u = \frac{1}{1+y^2}$, we get

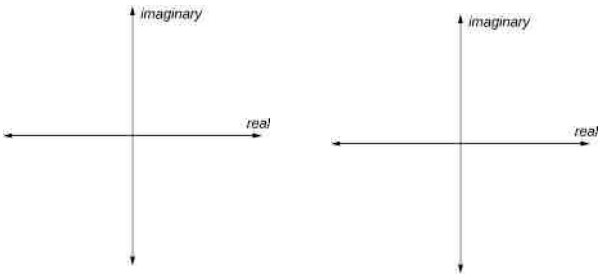
$$u^2 - u + v^2 = 0$$
$$\left(u - \frac{1}{2}\right)^2 + v^2 = \frac{1}{4}, \qquad u \neq 0$$

The Reciprocal Transformation

Solution(cont.): This shows that the image of vertical line maps onto the circle

$$\left|w - \frac{1}{2}\right| = \frac{1}{2}$$

Excluding the point $w = 0$.



The Reciprocal Transformation

Example: Show that the image of the half plane $\left\{z: Re(z) \geq \frac{1}{2}\right\}$ under the mapping $w = \frac{1}{z}$ is the closed disk $\{w: |w-1| \leq 1\}$.

Source: Complex analysis by Mathews

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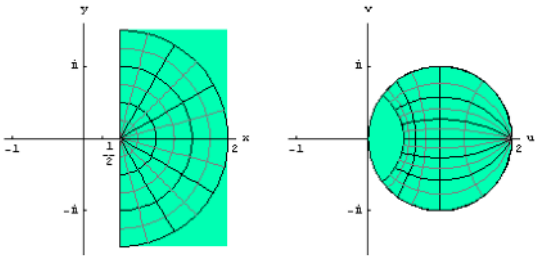

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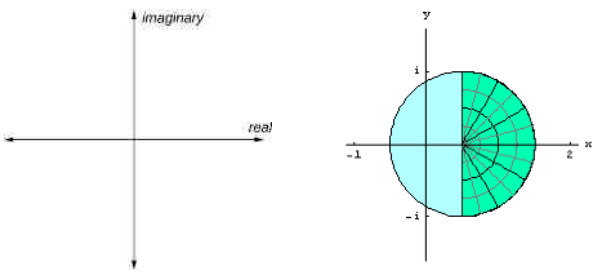
The Reciprocal Transformation

Example: For the transformation $\frac{1}{z}$, find the image of the portion of the half-plane $\mathrm{Re}(z) \geq \frac{1}{2}$ that is inside the closed disk $\left\{z: \left|z - \frac{1}{2}\right| \leq 1\right\}$.

Solution:



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Solution: Note that the given disc is equal to

$$A = \left\{(x,y): x^2 + y^2 - x \leq \frac{3}{4}\right\}$$

The Reciprocal Transformation

Solution (cont.): Because $z = f^{-1}(w) = \frac{1}{w}$, we have

$u + iv = w \in f(A) \Leftrightarrow f^{-1}(w) \in A$

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$\Leftrightarrow \frac{1}{u^2+v^2} - \frac{u}{u^2+v^2} \leq \frac{3}{4}$

$\Leftrightarrow \left(u + \frac{2}{3}\right)^2 + v^2 \leq \left(\frac{4}{3}\right)^2.$

The Reciprocal Transformation

Solution (cont.):

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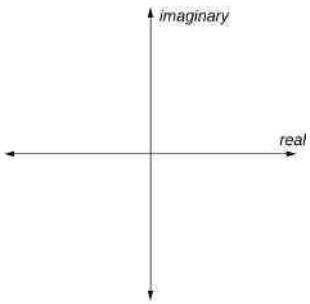

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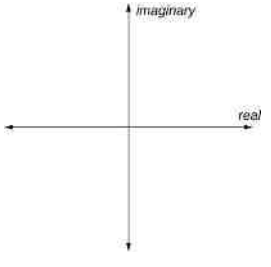

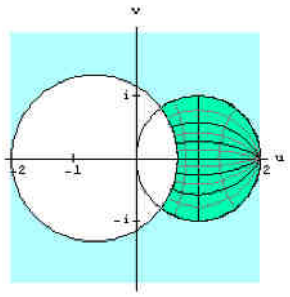

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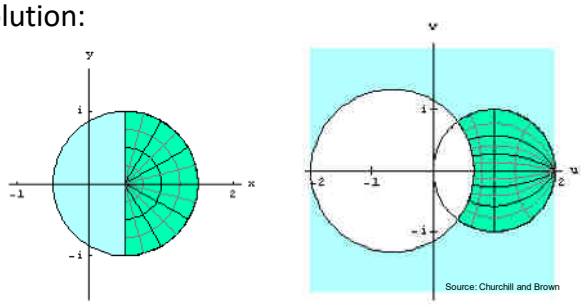


The Reciprocal Transformation

The Reciprocal Transformation

End

Derivatives

Topic No. 34

The Derivative of Complex Functions Part I

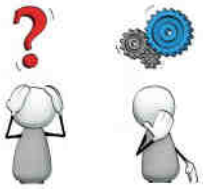
Derivatives

- Does the derivative of a complex function makes sense?

Source: https://www.inc.com

Derivatives

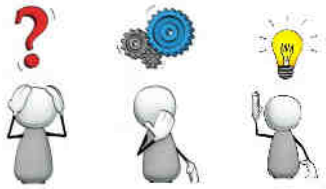
- Does the derivative of a complex function makes sense?
- How derivative is defined.



Source: https://www.inc.com

Derivatives

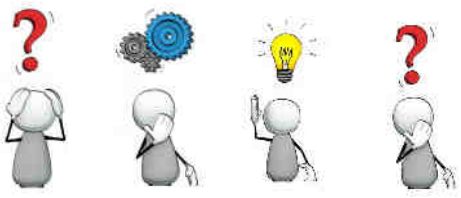
- Does the derivative of a complex function makes sense?
- How derivative is defined.
- What does derivative represent.



Source: https://www.inc.com

Derivatives

- Does the derivative of a complex function makes sense?
- How derivative is defined.
- What does derivative represent.
- Difference between real and complex derivative.



Source: https://www.inc.com

Derivatives of Real Functions

The derivative of f at x_0, written $f'(x_0)$, is

$$f'(x_0) = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}$$

provided the limit exists.

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OR

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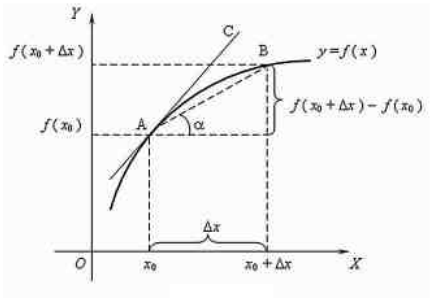

Source: https://www.quora.com

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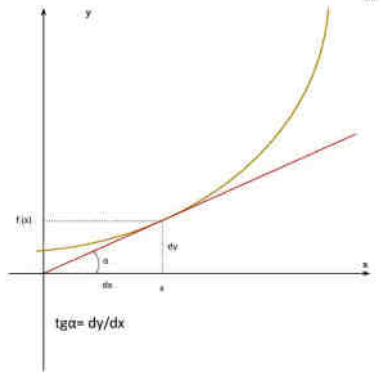

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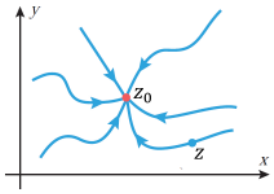

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If we write $\Delta z = z - z_0$ then we express equation (1) in the form

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Derivatives

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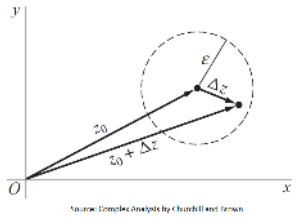

Source: Complex Analysis by Churchill and Brown

Derivatives

Leibniz notation

If we let $w = f(z)$ and $\Delta w = f(z) - f(z_0)$, then we can use the Leibniz notation $\frac{dw}{dz}$ for the derivative

$$f'(z_0) = \frac{dw}{dz} = \lim_{\Delta z \to 0} \frac{\Delta w}{\Delta z}$$

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$$f'(z_0) = \frac{dw}{dz} = \lim_{\Delta z \to 0} \frac{\Delta w}{\Delta z}$$

Hence

$$f'(z_0) = \lim_{\Delta z \to 0} \frac{\Delta w}{\Delta z} = \lim_{\Delta z \to 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}$$

We often drop the subscript on z_0 and write the definition as

$$f'(z) = \lim_{\Delta z \to 0} \frac{\Delta w}{\Delta z} = \lim_{\Delta z \to 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}$$

Derivatives

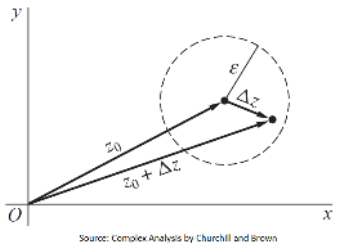
Example: If $f(z) = z^2$ then show that $f'(z) = 2z$.

Solution:

Derivatives

The value of the limit in the following definition must be independent of the manner in which $\Delta z \to 0$.

$$f'(z) = \lim_{\Delta z \to 0} \frac{\Delta w}{\Delta z} = \lim_{\Delta z \to 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}$$



Source: Complex Analysis by Churchill and Brown

Derivatives

If we can find two curves that end at z_0 along which $\frac{\Delta w}{\Delta z}$ approaches distinct values, then $\frac{\Delta w}{\Delta z}$ does not have a limit as $\Delta z \to 0$ and f does not have a derivative at z_0.

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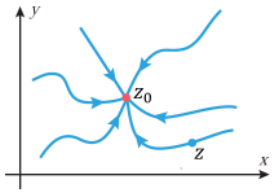
Derivatives

Similar case can be applied to the following definition

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0}$$



Derivatives

Example: Show that the function $f(z) = \mathrm{Re}(z)$ is differentiable nowhere.

Derivatives

Example: Show that the function $f(z) = \mathrm{Re}(z)$ is differentiable nowhere.

Solution: First we approach $z_0 = x_0 + iy_0$ along a line parallel to x −axis. In this case $z = x + iy_0$.

Derivatives

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Solution: First we approach $z_0 = x_0 + iy_0$ along a line parallel to x −axis. In this case $z = x + iy_0$.

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{\Delta z}$$

Derivatives

Example: Show that the function $f(z) = \mathrm{Re}(z)$ is differentiable nowhere.

Solution: First we approach $z_0 = x_0 + iy_0$ along a line parallel to x −axis. In this case $z = x + iy_0$.

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{\Delta z}$$

$$= \lim_{(x+iy_0) \to (x_0+iy_0)} \frac{\mathrm{Re}(x + iy) - \mathrm{Re}(x_0 + iy_0)}{(x + iy_0) - (x_0 + iy_0)}$$

$$= \lim_{(x+iy_0) \to (x_0+iy_0)} \frac{x - x_0}{x - x_0}$$

$$= 1$$

Derivatives

Solution (cont.): Next we approach z_0 along the line parallel to y −axis by forcing z to be of the form $z = x_0 + iy$.

Derivatives

Solution (cont.): Next we approach z_0 along the line parallel to y −axis by forcing z to be of the form $z = x_0 + iy$.

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{\Delta z}$$

Derivatives

Solution (cont.): Next we approach z_0 along the line parallel to y −axis by forcing z to be of the form $z = x_0 + iy$.

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{\Delta z}$$

$$= \lim_{(x_0+iy) \to (x_0+iy_0)} \frac{Re(x_0 + iy) - Re(x_0 + iy_0)}{(x_0 + iy) - (x_0 + iy_0)}$$

$$= \lim_{(x+iy_0) \to (x_0+iy_0)} \frac{x_0 - x_0}{i(y - y_0)}$$

$$= 0$$

Derivatives

Hence the function $f(z) =$Re(z) is differentiable nowhere.

Derivatives

End

- **The Derivative of Complex Functions**
- **Examples**

Derivatives

Topic No. 35

The Derivative of Complex Functions Part II

Derivatives

The derivative of f at z_0, written $f'(z_0)$, is

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

provided the limit exists. The function f must be defined is a neighbourhood of z_0.

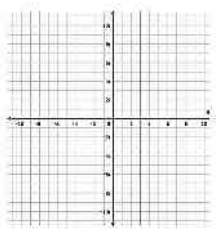
If we write $\Delta z = z - z_0$ then we express equation (1) in the form

$$f'(z_0) = \lim_{\Delta z \to 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}.$$

Derivatives

Example: Consider the real-valued function $f(z) = |z|^2$. Find points where $f(z)$ is differentiable.

Solution:



Derivatives

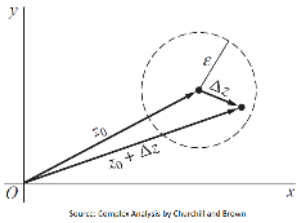
Example: Consider the real-valued function $f(z) = |z|^2$.
Find points where $f(z)$ is differentiable.

Solution: We know that $f'(z) = \lim_{\Delta z \to 0} \frac{f(z+\Delta z) - f(z)}{\Delta z}$

Derivatives

The value of the limit in the following definition must be independent of the manner in which $\Delta z \to 0$.

$$f'(z) = \lim_{\Delta z \to 0} \frac{\Delta w}{\Delta z} = \lim_{\Delta z \to 0} \frac{f(z+\Delta z) - f(z)}{\Delta z}$$



Source: Complex Analysis by Churchill and Brown

Derivatives

Example: Consider the real-valued function $f(z) = |z|^2$.
Find points where $f(z)$ is differentiable.

Solution: Path 1: z moves parallel to x-axis

Derivatives

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Derivatives

Example: Consider the real-valued function $f(z) = |z|^2$.
Find points where $f(z)$ is differentiable.

Solution: Path 2: z moves parallel to y-axis

Derivatives

Example: Consider the real-valued function $f(z) = |z|^2$.
Find points where $f(z)$ is differentiable.

Solution: Path 2: z moves parallel to y-axis

Derivatives

Example: Consider the real-valued function $f(z) = |z|^2$.
Find points where $f(z)$ is differentiable.

Solution:

Derivatives

Example: Consider the real-valued function $f(z) = |z|^2$.
Find points where $f(z)$ is differentiable.

Solution: For $z = 0$

$$f'(z) = \lim_{\Delta z \to 0} \frac{f(z+\Delta z) - f(z)}{\Delta z}$$

Derivatives

Observations

Example: Consider the real-valued function $f(z) = |z|^2$. Find points where $f(z)$ is differentiable.

Derivatives

Theorem: If $f(z)$ is differentiable at z_0 then $f(z)$ is continuous at z_0.

Proof:

Derivatives

Theorem: If $f(z)$ is differentiable at z_0 then $f(z)$ is continuous at z_0.

Proof:

Derivatives

End

- **Examples**
- **Relation Between Continuity and Differentiability**

Derivatives

Topic No. 36

The Derivative of Complex Functions Part III

Derivatives

Recall

The derivative of f at z_0, written $f'(z_0)$, is

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

provided the limit exists. The function f must be defined is a neighbourhood of z_0.

If we write $\Delta z = z - z_0$ then we express equation (1) in the form

$$f'(z_0) = \lim_{\Delta z \to 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}.$$

Differentiation Formulas

The proofs are very similar to differentiation formulas for real-valued functions

Differentiation Formulas

The proofs are very similar to differentiation formulas for real-valued functions

$$\frac{d}{dz} C = 0$$

where C is a constant.

Differentiation Formulas

$$\frac{d}{dz}z^n = n\,z^{n-1}$$

where n is a positive integer.

Example:

Differentiation Formulas

$$\frac{d}{dz}[Cf(z)] = C\frac{d}{dz}f(z)$$

Example:

Differentiation Formulas

$$\frac{d}{dz}[f(z)+g(z)] = \frac{d}{dz}f(z) + \frac{d}{dz}g(z)$$

Example:

Differentiation Formulas

$$\frac{d}{dz}[f(z)g(z)] = f(z)\frac{d}{dz}g(z) + g(z)\frac{d}{dz}f(z)$$

OR

$$\frac{d}{dz}[f(z)g(z)] = f(z)\,g'(z) + g(z)\,f'(z)$$

Differentiation Formulas

$$\frac{d}{dz}[f(z)g(z)] = f(z)\,g'(z) + g(z)\,f'(z)$$

Proof: Let $w = f(z)g(z)$

$$w'(z) = \lim_{\Delta z\to 0}\frac{w(z+\Delta z)-w(z)}{\Delta z}$$

Differentiation Formulas

$$\frac{d}{dz}[f(z)g(z)] = f(z)\,g'(z) + g(z)\,f'(z)$$

Proof: Let $w = f(z)g(z)$

$$w'(z_0) = \lim_{\Delta z\to 0}\frac{w(z+\Delta z)-w(z)}{\Delta z}$$

Now

$$w(z+\Delta z) = f(z+\Delta z)g(z+\Delta z)$$

Differentiation Formulas

$$\frac{d}{dz}[f(z)g(z)] = f(z)\,g'(z) + g(z)\,f'(z)$$

Proof: Let $w = f(z)g(z)$

$$w'(z_0) = \lim_{\Delta z\to 0}\frac{w(z+\Delta z)-w(z)}{\Delta z}$$

Now

$$w(z+\Delta z) = f(z+\Delta z)g(z+\Delta z)$$

$$w(z+\Delta z) - w(z) = f(z+\Delta z)g(z+\Delta z) - f(z)\,g(z)$$

Differentiation Formulas

$$w(z+\Delta z) = f(z+\Delta z)g(z+\Delta z)$$

$$w(z+\Delta z) - w(z) = f(z+\Delta z)g(z+\Delta z) - f(z)\,g(z)$$

Adding and subtracting $f(z)g(z+\Delta z)$, we get

$$w(z+\Delta z) - w(z) = f(z+\Delta z)g(z+\Delta z) - f(z)g(z+\Delta z) + f(z)g(z+\Delta z) - f(z)\,g(z)$$

Differentiation Formulas

$$w(z+\Delta z) = f(z+\Delta z)g(z+\Delta z)$$

$$w(z+\Delta z) - w(z) = f(z+\Delta z)g(z+\Delta z) - f(z)\,g(z)$$

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$$w(z+\Delta z) - w(z) = f(z+\Delta z)g(z+\Delta z) - f(z)g(z+\Delta z) + f(z)g(z+\Delta z) - f(z)\,g(z)$$

$$w(z+\Delta z) - w(z) = g(z+\Delta z)\,[f(z+\Delta z) - f(z)] + f(z)[g(z+\Delta z) - g(z)]$$

Differentiation Formulas

$$w(z+\Delta z) - w(z) = g(z+\Delta z)\,[f(z+\Delta z) - f(z)] + f(z)[g(z+\Delta z) - g(z)]$$

Dividing by Δz

$$\frac{w(z+\Delta z) - w(z)}{\Delta z} = \frac{g(z+\Delta z)\,[f(z+\Delta z) - f(z)]}{\Delta z} + \frac{f(z)[g(z+\Delta z) - g(z)]}{\Delta z}$$

Differentiation Formulas

$$w(z+\Delta z) - w(z) = g(z+\Delta z)\,[f(z+\Delta z) - f(z)] + f(z)[g(z+\Delta z) - g(z)]$$

Dividing by Δz

$$\frac{w(z+\Delta z) - w(z)}{\Delta z} = \frac{g(z+\Delta z)\,[f(z+\Delta z) - f(z)]}{\Delta z} + \frac{f(z)[g(z+\Delta z) - g(z)]}{\Delta z}$$

Taking limit $\Delta z \to 0$

$$\lim_{\Delta z\to 0}\frac{w(z+\Delta z) - w(z)}{\Delta z} = \lim_{\Delta z\to 0} g(z+\Delta z)\,\frac{[f(z+\Delta z) - f(z)]}{\Delta z} + \lim_{\Delta z\to 0} f(z)\,\frac{[g(z+\Delta z) - g(z)]}{\Delta z}$$

Differentiation Formulas

$$\lim_{\Delta z\to 0}\frac{w(z+\Delta z) - w(z)}{\Delta z} = \lim_{\Delta z\to 0} g(z+\Delta z)\,\frac{[f(z+\Delta z) - f(z)]}{\Delta z} + \lim_{\Delta z\to 0} f(z)\,\frac{[g(z+\Delta z) - g(z)]}{\Delta z}$$

Hence proved the following

$$w'(z) = g(z)f'(z) + f(z)g'(z)$$

Differentiation Formulas

Example: Given $f(z) = a_0 + a_1 z + a_2 z^2 + \cdots + a_n z^n$ then find

$$\frac{d}{dx} f(z)$$

Solution:

Differentiation Formulas

Chain Rule

Chain rule is useful when we want to differentiate composition of two functions.

Given two functions $f(z)$ and $g(z)$ composition of f and g is given by

$$f \circ g(z) = f\big(g(z)\big)$$

Differentiation Formulas

Chain Rule

$$\frac{d}{dz}\big[f(g(z))\big] = (f \circ g)'(z) = f'\big(g(z)\big)\,g'(z)$$

Differentiation Formulas

Chain Rule

$$\frac{d}{dz}\big[f(g(z))\big] = (f \circ g)'(z) = f'\big(g(z)\big)\,g'(z)$$

Derivative of outside function

Differentiation Formulas

Chain Rule

$$\frac{d}{dz}\left[f\left(g(z)\right)\right] = (f \circ g)'(z) = f'\left(g(z)\right) g'(z)$$

Derivative of outside function

Derivative of inside function

Differentiation Formulas

Example: Given $f(z) = z^3 + i2z^2 + 4$ then find

$$\frac{d}{dx}[f(z)]^3$$

Solution:

Differentiation Formulas

Example: Given $f(z) = z^3 + i2z^2 + 4$ then find

$$\frac{d}{dx}[f(z)]^3$$

Solution:

Differentiation Formulas

Differentiation formulas

End

The Cauchy-Riemann Equations

Topic No. 37

The Cauchy-Riemann Equations Part I

The Cauchy-Riemann Equations

Source: Wikipedia

Augustin-Louis Cauchy

Source: Wikipedia

Bernhard Riemann

The Cauchy-Riemann Equations

Cauchy-Riemann equations are a pair of equations that first-order partial derivative of the component functions u and v of a function

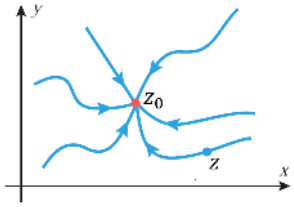
$$f(z) = u(x, y) + iv(x, y)$$

must satisfy at a point (x_0, y_0) when the derivative of f exists there.

The Cauchy-Riemann Equations

Main Idea

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0}$$



The Cauchy-Riemann Equations

Theorem: Suppose that

$$f(z) = f(x+iy) = u(x,y) + iv(x,y)$$

is differentiable at a point $z_0 = x_0 + iy_0$.

The Cauchy-Riemann Equations

Theorem: Suppose that

$$f(z) = f(x+iy) = u(x,y) + iv(x,y)$$

is differentiable at a point $z_0 = x_0 + iy_0$.
Then at $x_0 + iy_0 = (x_0, y_0)$ the first order partial derivatives of u and v exist, and

$$f'(z_0) = u_x(x_0,y_0) + iv_x(x_0,y_0),$$

The Cauchy-Riemann Equations

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$$f'(z_0) = u_x(x_0,y_0) + iv_x(x_0,y_0),$$

and also

$$f'(z_0) = v_y(x_0,y_0) - iu_y(x_0,y_0).$$

The Cauchy-Riemann Equations

Theorem: Suppose that

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Equating the real and imaginary parts of both equations gives **Cauchy-Riemann equations**

The Cauchy-Riemann Equations

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Equating the real and imaginary parts of both equations gives **Cauchy-Riemann equations**

$u_x(x_0,y_0) = v_y(x_0,y_0)$ and $u_y(x_0,y_0) = -v_x(x_0,y_0)$.

The Cauchy-Riemann Equations

Theorem: Suppose that

$$f(z) = f(x+iy) = u(x,y) + iv(x,y)$$

If $f(z)$ differentiable at a point $z_0 = x_0 + iy_0$ then u and v satisfy **Cauchy-Riemann equations**
$u_x(x_0,y_0) = v_y(x_0,y_0)$ and $u_y(x_0,y_0) = -v_x(x_0,y_0)$.

The Cauchy-Riemann Equations

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Observations:

$\Rightarrow$

Diff C R Equations

The Cauchy-Riemann Equations

Theorem: Suppose that

$$f(z) = f(x+iy) = u(x,y) + iv(x,y)$$

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Observations:

$\Rightarrow$

Diff C R Equations

- Contrapositive

The Cauchy-Riemann Equations

Theorem: Suppose that

$$f(z) = f(x+iy) = u(x,y) + iv(x,y)$$

If $f(z)$ differentiable at a point $z_0 = x_0 + iy_0$ then u and v satisfy **Cauchy-Riemann equations**

$u_x(x_0,y_0) = v_y(x_0,y_0)$ and $u_y(x_0,y_0) = -v_x(x_0,y_0)$.

Observations:

$\Rightarrow$

Diff C R Equations

- Contrapositive: If Cauchy-Riemann equations equations are not satisfied at $z_0$then function is not differentiable at z_0.

The Cauchy-Riemann Equations

Theorem: Suppose that

$$f(z) = f(x+iy) = u(x,y) + iv(x,y)$$

If $f(z)$ differentiable at a point $z_0 = x_0 + iy_0$ then u and v satisfy **Cauchy-Riemann equations**

$u_x(x_0,y_0) = v_y(x_0,y_0)$ and $u_y(x_0,y_0) = -v_x(x_0,y_0)$.

Observations:

$\Rightarrow$

Diff C R Equations

- Contrapositive: If Cauchy-Riemann equations equations are not satisfied at $z_0$then function is not differentiable at z_0.
- If Cauchy-Riemann equations are satisfied at a point z_0 then function may or may not be differentiable at z_0.

The Cauchy-Riemann Equations

Theorem: Suppose that

$$f(z) = f(x+iy) = u(x,y) + iv(x,y)$$

is differentiable at a point $z_0 = x_0 + iy_0$.

Then at $x_0 + iy_0 = (x_0,y_0)$ the first order partial derivatives of u and v exist, and

$$f'(z_0) = u_x(x_0,y_0) + iv_x(x_0,y_0),$$

and also

$$f'(z_0) = v_y(x_0,y_0) - iu_y(x_0,y_0).$$

Equating the real and imaginary parts of both equations gives **Cauchy-Riemann equations**

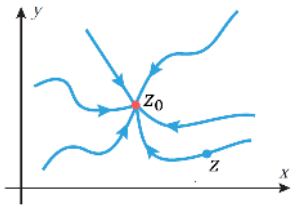
$u_x(x_0,y_0) = v_y(x_0,y_0)$ and $u_y(x_0,y_0) = -v_x(x_0,y_0)$.

Proof:

The Cauchy-Riemann Equations

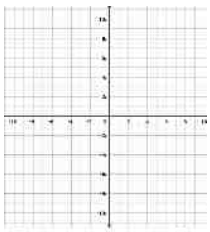
Proof: Given $f(z)$ is differentiable at z_0 so the following limit exist

$$f'(z_0) = \lim_{z\to z_0} \frac{f(z)-f(z_0)}{z-z_0}$$



The Cauchy-Riemann Equations

Proof: Case1: Parallel to x-axis



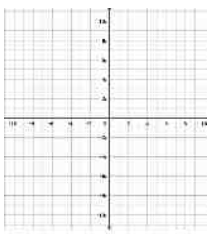
The Cauchy-Riemann Equations

Proof: Case1: Parallel to x-axis

$$f'(z_0) = \lim_{z\to z_0} \frac{f(z)-f(z_0)}{z-z_0}$$

The Cauchy-Riemann Equations

Proof: Case1: Parallel to y-axis



The Cauchy-Riemann Equations

Proof: Case1: Parallel to y-axis

$$f'(z_0) = \lim_{z\to z_0} \frac{f(z)-f(z_0)}{z-z_0}$$

The Cauchy-Riemann Equations

At $x_0 + iy_0 = (x_0, y_0)$ the first order partial derivatives of u and v exist, and

$$f'(z_0) = u_x(x_0, y_0) + iv_x(x_0, y_0),$$

and also

$$f'(z_0) = v_y(x_0, y_0) - iu_y(x_0, y_0).$$

The Cauchy-Riemann Equations

The Cauchy-Riemann Equations

End

The Cauchy-Riemann Equations

Topic No. 38

The Cauchy-Riemann Equations Part II

The Cauchy-Riemann Equations

Recall

Cauchy-Riemann equations are a pair of first-order partial derivative that the components functions u and v of a function

$$f(z) = u(x, y) + iv(x, y)$$

must satisfy at a point (x_0, y_0) when the derivative of f exists there.

The Cauchy-Riemann Equations

Recall

Theorem: Suppose that

$$f(z) = f(x + iy) = u(x, y) + iv(x, y)$$

If $f(z)$ differentiable at a point $z_0 = x_0 + iy_0$ then u and v satisfy **Cauchy-Riemann equations**

$u_x(x_0, y_0) = v_y(x_0, y_0)$ and $u_y(x_0, y_0) = -v_x(x_0, y_0)$.

The Cauchy-Riemann Equations

Example: We know $f(z) = z^2$ is differentiable and $f'(z) = 2z$. Check whether Cauchy-Riemann equations are satisfied by $f(z)$.

Solution:

The Cauchy-Riemann Equations

Example: We Show that $f(z) = \bar{z}$ is nowhere differentiable.

Solution:

The Cauchy-Riemann Equations

Example: We Show that $f(z) = \bar{z}$ is nowhere differentiable.

Solution:

The Cauchy-Riemann Equations

Example: Show that the function

$$f(z)=\begin{cases}\dfrac{(\bar{z})^2}{z} & when\ z\neq 0\\ 0 & when\ z=0\end{cases}$$

is not differentiable at the point $z_0=0$ even though the Cauchy-Riemann equations are satisfied at $(0,0)$.

Solution:

The Cauchy-Riemann Equations

Example: Show that the function

$$f(z)=\begin{cases}\dfrac{(\bar{z})^2}{z} & when\ z\neq 0\\ 0 & when\ z=0\end{cases}$$

is not differentiable at the point $z_0=0$ even though the Cauchy-Riemann equations are satisfied at $(0,0)$.

Solution: We can rewrite $f(z)$ in the following form

$$f(z)=\begin{cases}\dfrac{(\bar{z})^2}{z}=\dfrac{x^3-3xy^2}{x^2+y^2}+i\dfrac{y^3-3x^2y}{x^2+y^2} & when\ z\neq 0\\ 0 & when\ z=0\end{cases}$$

The Cauchy-Riemann Equations

Solution: We can rewrite $f(z)$ in the following form

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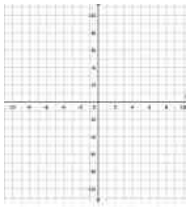

The Cauchy-Riemann Equations

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Now lets check the differentiability of $f(z)$ at $(0,0)$.

$$f'(z_0)=\lim_{z\to z_0}\frac{f(z)-f(z_0)}{z-z_0}$$

The Cauchy-Riemann Equations

Solution: We can rewrite $f(z)$ in the following form

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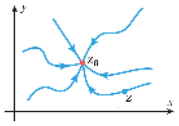

The Cauchy-Riemann Equations

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$$f'(z_0)=\lim_{z\to z_0}\frac{f(z)-f(z_0)}{z-z_0}$$

Path1: Along x-axis.

The Cauchy-Riemann Equations

Solution: We can rewrite $f(z)$ in the following form

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Solution: We can rewrite $f(z)$ in the following form

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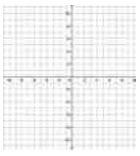
Now lets check the differentiability of $f(z)$ at $(0,0)$.

$$f'(z_0)=\lim_{z\to z_0}\frac{f(z)-f(z_0)}{z-z_0}$$

Path2: Along $y=x$ line.



The Cauchy-Riemann Equations

Solution: We can rewrite $f(z)$ in the following form

$$f(z)=\begin{cases}\dfrac{(\bar{z})^2}{z}=\dfrac{x^3-3xy^2}{x^2+y^2}+i\dfrac{y^3-3x^2y}{x^2+y^2} & when\ z\neq 0\\ 0 & when\ z=0\end{cases}$$

Now lets check the differentiability of $f(z)$ at $(0,0)$.

$$f'(z_0)=\lim_{z\to z_0}\frac{f(z)-f(z_0)}{z-z_0}$$

Path2: Along $y=x$ line.

The Cauchy-Riemann Equations

Solution: We can rewrite $f(z)$ in the following form

$$f(z)=\begin{cases}\dfrac{(\bar{z})^2}{z}=\dfrac{x^3-3xy^2}{x^2+y^2}+i\dfrac{y^3-3x^2y}{x^2+y^2} & when\ z\neq 0\\ 0 & when\ z=0\end{cases}$$

Now lets check the differentiability of $f(z)$ at $(0,0)$.

$$f'(z_0)=\lim_{z\to z_0}\frac{f(z)-f(z_0)}{z-z_0}$$

Path1: Along x-axis.

Path2: Along $y=x$ line.

The Cauchy-Riemann Equations

Cauchy-Riemann equations for differentiability

Theorem: Let $f(z)=f(x+iy)=u(x,y)+iv(x,y)$ be a continuous function that is defined in some neighborhood of the point $z_0=x_0+iy_0$.

The Cauchy-Riemann Equations

Cauchy-Riemann equations for differentiability

Theorem: Let $f(z)=f(x+iy)=u(x,y)+iv(x,y)$ be a continuous function that is defined in some neighborhood of the point $z_0=x_0+iy_0$. If all the partial derivatives u_x, v_y, u_y and v_x are continuous at the point (x_0,y_0)

The Cauchy-Riemann Equations

Cauchy-Riemann equations for differentiability

Theorem: Let $f(z)=f(x+iy)=u(x,y)+iv(x,y)$ be a continuous function that is defined in some neighborhood of the point $z_0=x_0+iy_0$. If all the partial derivatives u_x, v_y, u_y and v_x are continuous at the point (x_0,y_0) and if the Cauchy-Riemann equations hold at (x_0,y_0)

The Cauchy-Riemann Equations

Cauchy-Riemann equations for differentiability

Theorem: Let $f(z)=f(x+iy)=u(x,y)+iv(x,y)$ be a continuous function that is defined in some neighborhood of the point $z_0=x_0+iy_0$. If all the partial derivatives u_x, v_y, u_y and v_x are continuous at the point (x_0,y_0) and if the Cauchy-Riemann equations hold at (x_0,y_0), then f is differentiable at z_0,

The Cauchy-Riemann Equations

Cauchy-Riemann equations for differentiability

Theorem: Let $f(z)=f(x+iy)=u(x,y)+iv(x,y)$ be a continuous function that is defined in some neighborhood of the point $z_0=x_0+iy_0$. If all the partial derivatives u_x, v_y, u_y and v_x are continuous at the point (x_0,y_0) and if the Cauchy-Riemann equations hold at (x_0,y_0), then f is differentiable at z_0, and the derivative $f'(z_0)$ can be computed with either of the following equations

$$f'(z_0)=u_x(x_0,y_0)+iv_x(x_0,y_0),$$

and also

$$f'(z_0)=v_y(x_0,y_0)-iu_y(x_0,y_0).$$

The Cauchy-Riemann Equations

Example: Show that the function

$$f(z)=e^{-y}\cos x+i\,e^{-y}\sin x$$

is differentiable for all z and find its derivative.

Solution:

The Cauchy-Riemann Equations

Example: Show that the function

$$f(z) = e^{-y}\cos x + i\, e^{-y}\sin x$$

is differentiable for all z and find its derivative.

Solution:

The Cauchy-Riemann Equations

The Cauchy-Riemann Equations

End

The Cauchy-Riemann Equations

Topic No. 39

The Cauchy-Riemann Equations Part III

The Cauchy-Riemann Equations

Recall

Theorem: Suppose that

$$f(z) = f(x+iy) = u(x,y) + iv(x,y)$$

If $f(z)$ differentiable at a point $z_0 = x_0 + iy_0$ then u and v satisfy **Cauchy-Riemann equations**

$$u_x(x_0,y_0) = v_y(x_0,y_0) \text{ and } u_y(x_0,y_0) = -v_x(x_0,y_0).$$

The Cauchy-Riemann Equations

Can we find some criteria for checking differentiability ??

Edited by Mahar Afaq Safdar Muhammadi,

MSc Math, Whatsapp No. 03494873115

The Cauchy-Riemann Equations

Cauchy-Riemann Conditions for differentiability

Theorem: Let $f(z) = f(x+iy) = u(x,y) + iv(x,y)$ be defined in some neighborhood of the point $z_0 = x_0 + iy_0$ and suppose that

1. If all the partial derivatives u_x, v_y, u_y and v_x exist everywhere in the neighbourhood.
2. those partial derivatives are continuous at (x_0, y_0) and satisfy Cauchy-Riemann equations

$$u_x = v_y, \qquad u_y = -v_x$$

Then f is differentiable at z_0, and the derivative $f'(z_0)$ can be computed with either of the following equations

$$f'(z_0) = u_x(x_0,y_0) + iv_x(x_0,y_0),$$

and also

$$f'(z_0) = v_y(x_0,y_0) - iu_y(x_0,y_0).$$

The Cauchy-Riemann Equations

Example: Show that the function

$$f(z) = e^{-y}\cos x + i\, e^{-y}\sin x$$

is differentiable for all z and find its derivative.

Solution: Let (x_0, y_0) be an arbitrary point.

defined in some neighborhood of the point $z_0 = x_0 + iy_0$ ✓ and suppose that

$f(z) = f(x+iy) = u(x,y) + iv(x,y)$

$\Rightarrow u(x,y) = e^{-y}\cos x, \quad v(x,y) = e^{-y}\sin x$

$\Rightarrow u(x,y)$ & $v(x,y)$ are defined for all (x,y).

The Cauchy-Riemann Equations

Example: Show that the function

$$f(z) = e^{-y}\cos x + i\, e^{-y}\sin x$$

is differentiable for all z and find its derivative.

Solution:

If all the partial derivatives u_x, v_y, u_y and v_x exist everywhere in the neighbourhood.

$u(x,y) = e^{-y}\cos x$ | $v(x,y) = e^{-y}\sin x$

$u_x(x,y) = -e^{-y}\sin x$ | $v_x(x,y) = e^{-y}\cos x$

$u_y(x,y) = -e^{-y}\cos x$ | $v_y(x,y) = -e^{-y}\sin x$

The Cauchy-Riemann Equations

Example: Show that the function

$$f(z) = e^{-y}\cos x + i\, e^{-y}\sin x$$

is differentiable for all z and find its derivative.

Solution:

those partial derivatives are continuous at (x_0, y_0) and satisfy Cauchy-Riemann equations

$u_x = v_y$, $u_y = -v_x$

$u_x = -e^{-y}\sin x$, $v_x = e^{-y}\cos x$

$u_y = -e^{-y}\cos x$, $v_y = -e^{-y}\sin x$

$u_x = -e^{-y}\sin x = v_y$ ✓, $u_y = -e^{-y}\cos x = -v_x$ ✓

The Cauchy-Riemann Equations

Example: Show that the function

$$f(z) = e^{-y}\cos x + i\, e^{-y}\sin x$$

is differentiable for all z and find its derivative.

Solution:

Then f is differentiable at z_0, and the derivative $f'(z_0)$ can be computed with either of the following equations

$$f'(z_0) = u_x(x_0, y_0) + iv_x(x_0, y_0).$$ ✓

and also

$$f'(z_0) = v_y(x_0, y_0) - iu_y(x_0, y_0).$$ ✓

$u_x = -e^{-y}\sin x$, $v_x = e^{-y}\cos x$

$f'(z_0) = -e^{-y_0}\sin x_0 + i\left(e^{-y_0}\cos x_0\right)$

The Cauchy-Riemann Equations

Topic No. 40

The Cauchy-Riemann Equations in Polar Coordinates

The Cauchy-Riemann Equations

Cauchy-Riemann Conditions for differentiability

Theorem: Let $f(z) = f(x + iy) = u(x, y) + iv(x, y)$ be defined in some neighborhood of the point $z_0 = x_0 + iy_0$ and suppose that

1. If all the partial derivatives u_x, v_y, u_y and v_x exist everywhere in the neighbourhood.
2. those partial derivatives are continuous at (x_0, y_0) and satisfy Cauchy-Riemann equations

$$u_x = v_y, \qquad u_y = -v_x$$

Then f is differentiable at z_0, and the derivative $f'(z_0)$ can be computed with either of the following equations

$$f'(z_0) = u_x(x_0, y_0) + iv_x(x_0, y_0).$$ ✓

and also

$$f'(z_0) = v_y(x_0, y_0) - iu_y(x_0, y_0).$$ ✓

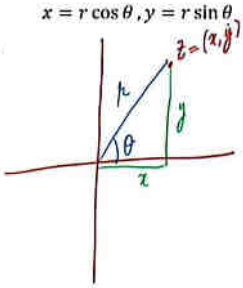
CR Equations in Polar Coordinates

Cauchy-Riemann conditions for differentiability In polar coordinates

CR Equations in Polar Coordinates

We know that

$$x = r\cos\theta, y = r\sin\theta$$



CR Equations in Polar Coordinates

We know that

$$x = r\cos\theta, y = r\sin\theta$$ ✓

Aim

Using these coordinate conversion formulas we want to convert cartesian Cauchy-Riemann equations in polar Cauchy-Riemann equations.

CR Equations in Polar Coordinates

Cauchy-Riemann equations

$$u_x = v_y, \qquad u_y = -v_x$$

Let us write

$$u_x, u_y, v_x, v_y$$

in terms of

$$u_r, u_\theta, v_r, v_\theta.$$

CR Equations in Polar Coordinates

For this purpose we are going to use chain rule for multivariable functions.

CR Equations in Polar Coordinates

Chain Rule

Suppose that $u(x, y)$ is a differentiable function of x and y, where $x = x(r, \theta)$ and $y = y(r, \theta)$ are differentiable functions of r and θ. Then

$$\frac{\partial u}{\partial r} = \frac{\partial u}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial r}$$

and

$$\frac{\partial u}{\partial \theta} = \frac{\partial u}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial \theta}$$

CR Equations in Polar Coordinates

Given $u(x, y)$ and $v(x, y)$ be two component functions of complex valued function

$$f(x + iy) = u(x, y) + iv(x, y)$$

Then

$$\frac{\partial u}{\partial r} = \frac{\partial u}{\partial x}\frac{\partial x}{\partial r} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial r}$$

where $x = r\cos\theta, y = r\sin\theta$

$\frac{\partial x}{\partial r} = \cos\theta, \frac{\partial y}{\partial r} = \sin\theta$

$u_r = u_x \cos\theta + u_y \sin\theta$

CR Equations in Polar Coordinates

So we have $u_r = u_x \cos\theta + u_y \sin\theta$; $v_r = v_x \cos\theta + v_y \sin\theta$

$u_\theta = -u_x r\sin\theta + u_y r\cos\theta$; $v_\theta = -v_x r\sin\theta + v_y r\cos\theta$

$u_\theta = -u_x r\sin\theta + u_y r\cos\theta = -r(u_x \sin\theta - u_y \cos\theta)$

$= -r(v_y \sin\theta + v_x \cos\theta)$

$= -rv_r$

$\Rightarrow \boxed{u_\theta = -rv_r,\ ru_r = v_\theta} \longleftrightarrow u_x = v_y, u_y = -v_x$

CR Equations in Polar Coordinates

Theorem: Let the function

$$f(z) = u(r, \theta) + iv(r, \theta)$$

be defined throughout some neighbourhood of a non-zero point $z_0 = r_0 e^{i\theta_0}$ and suppose that

1. the first order partial derivates of the functions u and v with respect to r and θ exist everywhere in the neighbourhood
2. those partial derivatives are continuous at (r_0, θ_0) and satisfy polar form of Cauchy-Riemann equations

$$ru_r = v_\theta, \quad u_\theta = -rv_r$$

Then $f'(z_0)$ exists and its value is

$$f'(z_0) = e^{-i\theta}(u_r + iv_r)$$

where the right side is to be evaluated at (r_0, θ_0)

CR Equations in Polar Coordinates

Example: Show that if f is the principal cube root function given by

$$f(re^{i\theta}) = f(z) = z^{\frac{1}{3}} = r^{\frac{1}{3}}\cos\frac{\theta}{3} + i\, r^{\frac{1}{3}}\sin\frac{\theta}{3},$$

where the domain is restricted to be

$$\{re^{i\theta} : r > 0 \text{ and } -\pi < \theta < \pi\}$$

then the derivative is given by

$$f'(z) = \frac{1}{3z^{\frac{2}{3}}} = \frac{1}{3}r^{-\frac{2}{3}}\cos\frac{\theta}{3} + i\frac{1}{3}r^{-\frac{2}{3}}\sin\frac{\theta}{3}$$ ✓

for every point in the domain.

CR Equations in Polar Coordinates

Solution:

$$f(re^{i\theta}) = r^{\frac{1}{3}}\cos\frac{\theta}{3} + i\, r^{\frac{1}{3}}\sin\frac{\theta}{3},$$

be defined throughout some neighbourhood of a non-zero point $z_0 = r_0 e^{i\theta_0}$ and suppose that

CR Equations in Polar Coordinates

Solution:

$$f(re^{i\theta}) = r^{\frac{1}{3}}\cos\frac{\theta}{3} + i\, r^{\frac{1}{3}}\sin\frac{\theta}{3},$$

the first order partial derivates of the functions u and v with respect to r and θ exist everywhere in the neighbourhood

$u(r, \theta) = r^{1/3}\cos\frac{\theta}{3}$, $v(r, \theta) = r^{1/3}\sin\frac{\theta}{3}$

$u_r = \frac{1}{3}r^{-2/3}\cos\frac{\theta}{3}$, $v_r = \frac{1}{3}r^{-2/3}\sin\frac{\theta}{3}$

$u_\theta = -\frac{1}{3}r^{1/3}\sin\frac{\theta}{3}$, $v_\theta = \frac{1}{3}r^{1/3}\cos\frac{\theta}{3}$

CR Equations in Polar Coordinates

Solution:

$$f(re^{i\theta}) = r^{\frac{1}{3}}\cos\frac{\theta}{3} + i\, r^{\frac{1}{3}}\sin\frac{\theta}{3},$$

those partial derivatives are continuous at (r_0, θ_0) and satisfy polar form of Cauchy-Riemann equations

$$ru_r = v_\theta, \quad u_\theta = -rv_r.$$

$u_r = \frac{1}{3} r^{-\frac{2}{3}} \cos\frac{\theta}{3}$, $v_\theta = \frac{1}{3} r^{\frac{1}{3}} \cos\frac{\theta}{3} = r\left(\frac{1}{3} r^{-\frac{2}{3}} \cos\frac{\theta}{3}\right)$

$v_r = \frac{1}{3} r^{\frac{1}{3}} \cos\frac{\theta}{3}$

$= \frac{1}{3} r^{\frac{1}{3}} \cos\frac{\theta}{3}$

Similarly $u_\theta = -r v_r$ is satisfied

VU

CR Equations in Polar Coordinates

Solution:

$$f(re^{i\theta}) = r^{\frac{1}{3}}\cos\frac{\theta}{3} + i\, r^{\frac{1}{3}}\sin\frac{\theta}{3},$$

Then $f'(z_0)$ exists and its value is

$$f'(z_0) = e^{-i\theta}(u_r + iv_r)$$

where the right side is to be evaluated at (r_0, θ_0)

$u_r = \frac{1}{3} r^{-\frac{2}{3}} \cos\frac{\theta}{3}$, $v_r = \frac{1}{3} r^{-\frac{2}{3}} \sin\frac{\theta}{3}$

$f'(z_0) = e^{-i\theta_0}\left(\frac{1}{3} r_0^{-\frac{2}{3}} \cos\frac{\theta_0}{3} + i\frac{1}{3} r_0^{-\frac{2}{3}} \sin\frac{\theta_0}{3}\right)$

VU

CR Equations in Polar Coordinates

Solution:

$$f(re^{i\theta}) = r^{\frac{1}{3}}\cos\frac{\theta}{3} + i\, r^{\frac{1}{3}}\sin\frac{\theta}{3},$$

$$f'(z) = e^{-i\theta}\left[\frac{1}{3} r^{-\frac{2}{3}} \cos\frac{\theta}{3} + i\frac{1}{3} r^{-\frac{2}{3}} \sin\frac{\theta}{3}\right]$$

$$= e^{-i\theta}\left[\frac{1}{3} r^{-\frac{2}{3}} e^{\frac{i\theta}{3}}\right]$$

$$= \frac{1}{3} r^{-\frac{2}{3}} e^{-i\frac{2}{3}}$$

$f'(z) = \frac{1}{3z^{\frac{2}{3}}}$

$z^{\frac{2}{3}} = r^{\frac{2}{3}} e^{i\frac{2}{3}\theta}$

$f(z) = z^{\frac{1}{3}}$

VU

Analytic Functions

Analytic Functions
Part I

Topic No. 41

Analytic Functions

Definition (Analytic Function)

A complex function f is **analytic at the point z_0**, provided there is some $\varepsilon > 0$ such that $f'(z)$ exists in ε neighbourhood of z_0.

Analytic Functions

ε Neighbourhood of a point z_0 is the following open disc

$$|z - z_0| < \varepsilon$$

Analytic Functions

Definition (Analytic Function)

A complex function f is **analytic at the point z_0**, provided there is some $\varepsilon > 0$ such that $f'(z)$ exists in ε neighbourhood of z_0.

In other words, f must be differentiable not only at z_0, but also at all points in some ε neighborhood of z_0.

Analytic Functions

Example: The function $\frac{1}{z}$ is analytic at each nonzero point in complex plane.

Analytic Functions

Example: The function $|z|^2$ is only differentiable at $z = 0$.
So it is nowhere analytic function.

Analytic Functions

Analytic Function

Analytic Functions

Analytic Function

Holomorphic Functions

Analytic Functions

Analytic Function

Holomorphic Functions

Regular Functions

Analytic Functions

Definition (Analytic Function)

A complex function f is **analytic at the point z_0**, provided there is some $\varepsilon > 0$ such that $f'(z)$ exists in ε neighbourhood of z_0.

If a complex function f is analytic at a point z_0 then it must be analytic at each point in some neighbourhood of z_0.

Analytic Functions

Definition (Analytic Function)

A complex function f is **analytic at the point z_0**, provided there is some $\varepsilon > 0$ such that $f'(z)$ exists in ε neighbourhood of z_0.

If f is analytic at each point in the region R, then we say that f is analytic on R.

Analytic Functions

Definition (Analytic Function)

A complex function f is **analytic at the point z_0**, provided there is some $\varepsilon > 0$ such that $f'(z)$ exists in ε neighbourhood of z_0.

A function f is analytic in an open set if it has derivative everywhere is that set.

Analytic Functions

Definition (Analytic Function)

A complex function f is **analytic at the point z_0**, provided there is some $\varepsilon > 0$ such that $f'(z)$ exists in ε neighbourhood of z_0.

A complex function f that is analytic in a set S which is not open, it is understood that f is analytic in an open set containing S.

Analytic Functions

We have a special term if f is analytic on the whole complex plane.

Analytic Functions

We have a special term if f is analytic on the whole complex plane.

Definition:

If f is analytic on the whole complex plane, then f is said to be **entire**.

Analytic Functions

We have a special term if f is analytic on the whole complex plane.

Definition:

If f is analytic on the whole complex plane, then f is said to be **entire**.

Example: Since a polynomial is differentiable everywhere on the complex plane. So polynomials are entire functions.

Analytic Functions

Singular Point

If a function f fails to be analytic at a point z_0 but is analytic at some point in every neighbourhood of z_0, then z_0 is called a singular point or singularity of f .

Analytic Functions

Singular Point

If a function f fails to be analytic at a point z_0 but is analytic at some point in every neighbourhood of z_0, then z_0 is called a singular point or singularity of f .

Example: The function $\frac{1}{z}$ has singularity $z = 0$.

Analytic Functions

Analytic Functions
Part II

Topic No. 42

Analytic Functions

RECALL

Definition (Analytic Function)

A complex function f is **analytic at the point z_0**, provided there is some $\varepsilon > 0$ such that $f'(z)$ exists in ε neighbourhood of z_0.

Analytic Functions

RECALL

Definition (Analytic Function)

A complex function f is **analytic at the point z_0**, provided there is some $\varepsilon > 0$ such that $f'(z)$ exists in ε neighbourhood of z_0.

In other words, f must be differentiable not only at z_0, but also at all points in some ε neighborhood of z_0.

Analytic Functions

A necessary condition for a function f to be analytic in a domain D is clearly

continuity of f

Analytic Functions

A necessary condition for a function f to be analytic in a domain D is clearly

Satisfaction of Cauchy-Riemann Equations

Analytic Functions

How to check

Function f is analytic at a point z_0?

Analytic Functions

Differentiation formulas help us in finding analytic functions.

Analytic Functions

Let f and g be two analytic functions in a domain D. Then

- $f + g$ is also analytic in domain D

Analytic Functions

Let f and g be two analytic functions in a domain D. Then

- $f + g$ is also analytic in domain D
- $f\ g$ is also analytic in domain D

Analytic Functions

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- $f + g$ is also analytic in domain D
- $f\ g$ is also analytic in domain D
- $\frac{f}{g}$ is analytic in domain D provided the function in the denominator does not vanish at any point of D

Analytic Functions

Example:

Analytic Functions

Example: At what points the quotient

$$f(z) = \frac{z^4 + 1}{(z^2 + 1)(z - 3)}$$

is analytic?

Solution:

Analytic Functions

Composition of functions

Composite of two analytic functions is an analytic function.

Analytic Functions

Composition of functions

Composite of two analytic functions is an analytic function.

More precisely let

$$f: D \to \mathbb{C}$$

and

$$g: E \to \mathbb{C}$$

be two complex functions such that

$$f(D) \subset E$$

Then $g\big(f(z)\big)$ is an analytic function with derivative

$$\frac{d}{dz} g\big(f(z)\big) = g'\big(f(z)\big)\, f'^{(z)}.$$

Analytic Functions

If a function is given in the following form

$$f(z) = u(x, y) + i\, v(x, y)$$

Then we can apply Cauchy-Riemann criteria for differentiability to check whether f is analytic.

Analytic Functions

RECALL: CR Conditions for differentiability

Theorem: Let $f(z) = f(x + iy) = u(x, y) + iv(x, y)$ be defined in some neighborhood of the point $z_0 = x_0 + iy_0$ and suppose that

1. If all the partial derivatives u_x, v_y, u_y and v_x exist everywhere in the neighbourhood.
2. those partial derivatives are continuous at (x_0, y_0) and satisfy Cauchy-Riemann equations

$$u_x = v_y, \qquad u_y = -v_x$$

Then f is differentiable at z_0, and the derivative $f'(z_0)$ can be computed with either of the following equations

$$f'(z_0) = u_x(x_0, y_0) + iv_x(x_0, y_0),$$

and also

$$f'(z_0) = v_y(x_0, y_0) - iu_y(x_0, y_0).$$

Analytic Functions

Example: Consider function

$$f(z) = \cosh x \cos y + i \sinh x \sin y$$

Find points where f is analytic.

Solution:

Analytic Functions

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Analytic Functions

Example: Consider function

$$f(z) = \cosh x \cos y + i \sinh x \sin y$$

Find points where f is analytic.

Solution:

Then f is differentiable at z_0

Analytic Functions

Other results that can also be used to check whether a function is analytic at a point.

Analytic Functions

CR Conditions for differentiability in Polar Coordinates

Theorem: Let the function

$$f(z) = u(r, \theta) + iv(r, \theta)$$

be defined throughout some neighbourhood of a non-zero point $z_0 = r_0\, e^{i\theta_0}$ and suppose that

1. the first order partial derivates of the functions u and v with respect to r and θ exist everywhere in the neighbourhood
2. those partial derivatives are continuous at (r_0, θ_0) and satisfy polar form of Cauchy-Riemann equations

$$ru_r = v_\theta, \qquad u_\theta = -rv_r$$

Then $f'(z_0)$ exists and its value is

$$f'(z_0) = e^{-i\theta}(u_r + iv_r)$$

where the right side is to be evaluated at (r_0, θ_0)

Analytic Functions

Topic No. 43

Analytic Functions Part III

Analytic Functions

RECALL

Definition (Analytic Function)

A complex function f is **analytic at the point z_0**, provided there is some $\varepsilon > 0$ such that $f'(z)$ exists in ε neighbourhood of z_0.

Analytic Functions

RECALL

Definition (Analytic Function)

A complex function f is **analytic at the point z_0**, provided there is some $\varepsilon > 0$ such that $f'(z)$ exists in ε neighbourhood of z_0.

In other words, f must be differentiable not only at z_0, but also at all points in some ε neighborhood of z_0.

Analytic Functions

Theorem: If $f'(z) = 0$ everywhere in a domain D, then $f(z)$ must be constant throughout D.

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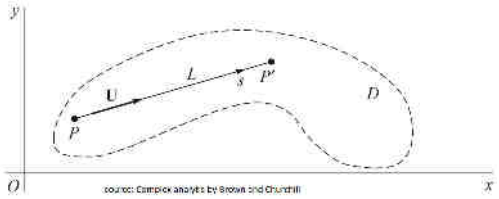

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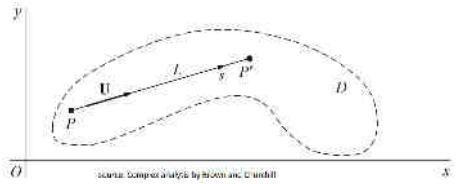

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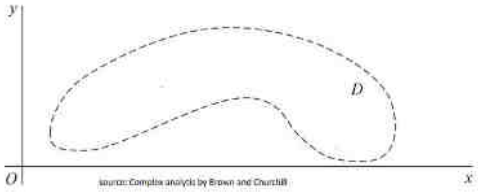

Analytic Functions

Theorem: If $f'(z) = 0$ everywhere in a domain D, then $f(z)$ must be constant throughout D.

Proof:

Analytic Functions

Example: Suppose that a function

$$f(z) = u(x, y) + iv(x, y)$$

and its conjugates

$$\overline{f(z)} = u(x, y) - iv(x, y)$$

are both analytic in a domain D. Show that $f(z)$ is constant throughout D.

Analytic Functions

Solution:

Analytic Functions

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Analytic Functions

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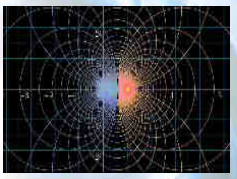
Analytic Functions

Analytic Functions
Part IV

Topic No. 44



Analytic Functions

RECALL

Definition (Analytic Function)

A complex function f is **analytic at the point z_0**, provided there is some $\varepsilon > 0$ such that $f'(z)$ exists in ε neighbourhood of z_0.

In other words, f must be differentiable not only at z_0, but also at all points in some ε neighborhood of z_0.

Analytic Functions

Theorem (L' Hopital's rule)

Assume that f and g are analytic at z_0. If $f(z_0) = 0$, $g(z_0) = 0$ and $g'(z_0) \neq 0$, then

$$\lim_{z \to z_0} \frac{f(z)}{g(z)} = \lim_{z \to z_0} \frac{f'(z)}{g'(z)}.$$

Analytic Functions

Example: Use L' Hopital's rule to compute the limit

$$\lim_{z\to i}\frac{z^7+i}{z^{14}+1}.$$

Solution:

Analytic Functions

Example: Use L' Hopital's rule to compute the limit

$$\lim_{z\to i}\frac{z^7+i}{z^{14}+1}.$$

Solution:

$$f(z)=z^7+i, \qquad f'(z)=7z^6,$$
$$g(z)=z^{14}+1, \qquad g'(z)=14z^{13}$$

Analytic Functions

Example: Use L' Hopital's rule to compute the limit

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$$=-\frac{7}{14i}$$

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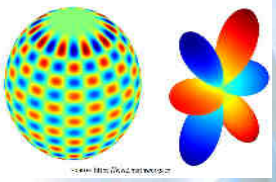
$$=\frac{7i}{14}$$

Harmonic Functions

Harmonic Functions
Part I

Topic No. 45



Harmonic Functions

Definition (Harmonic Functions)

A real-valued function H of two variables x and y is said to be harmonic in a given domain D of the xy plane if

Harmonic Functions

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Harmonic Functions

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A real-valued function H of two variables x and y is said to be harmonic in a given domain D of the xy plane if

- throughout the domain it has continuous partial derivatives of the first and second order
- and satisfies the following partial differential equation

$$H_{xx}(x,y) + H_{yy}(x,y) = 0\,.$$

Or

$$\frac{\partial^2 H}{\partial x^2} + \frac{\partial^2 H}{\partial y^2} = 0$$

Harmonic Functions

Notations

The sum

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2}$$

of the two partial derivatives is denoted by $\nabla^2\phi$ and is called the **Laplacian** of ϕ.

Harmonic Functions

Notations

The sum

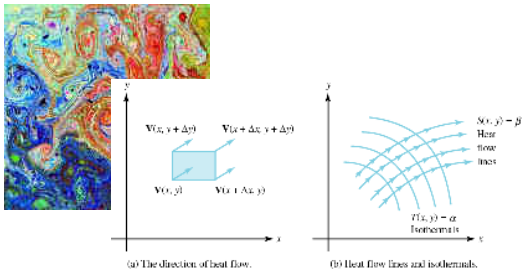
$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2}$$

of the two partial derivatives is denoted by $\nabla^2\phi$ and is called the **Laplacian** of ϕ.

Laplace equation (or sometimes known as **potential equation**) is

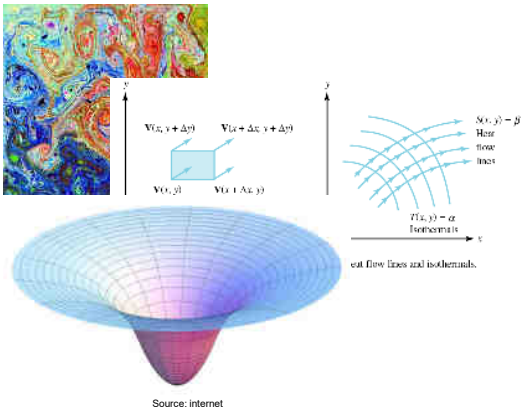
$$\nabla^2\phi = 0\,.$$

Harmonic Functions



Source: internet

Harmonic Functions



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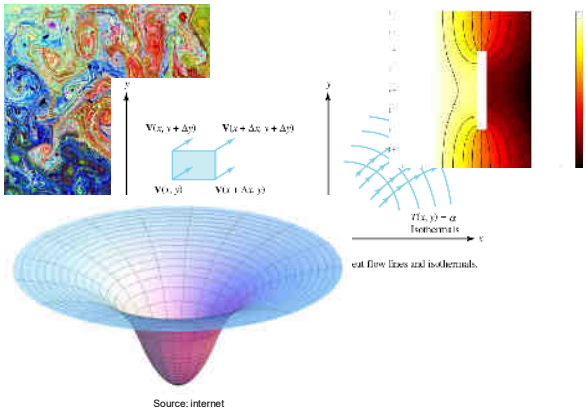
Harmonic Functions



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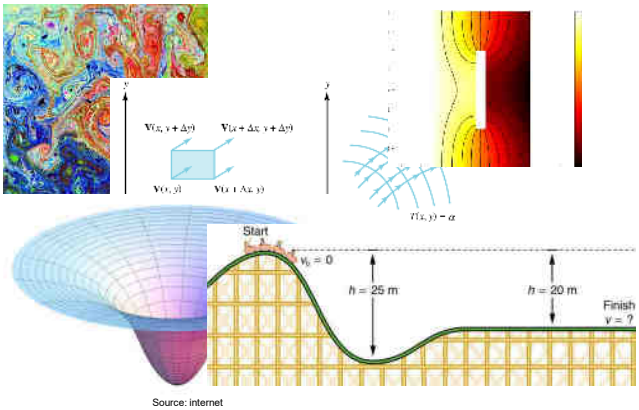
Harmonic Functions



Source: internet

Harmonic Functions

Example: Consider the function

$$T(x,y) = e^{-y}\sin x$$

Show that $T(x,y)$ is harmonic function.

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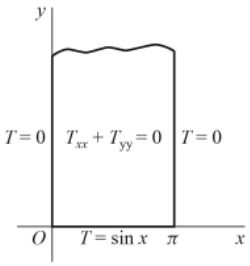

Source: Complex Analysis by Brown and Churchill

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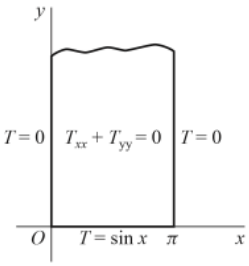

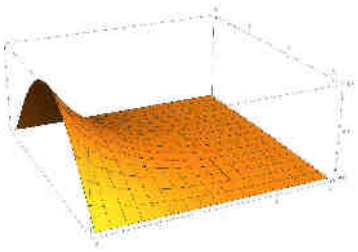

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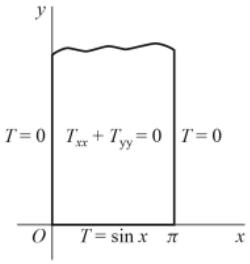

$$T_{xx}(x,y) + T_{yy}(x,y) = 0$$

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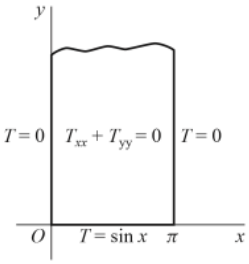

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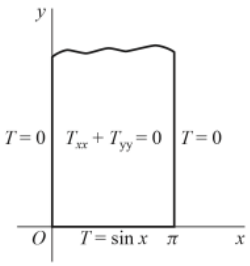

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$$T(x,y) = \sin x\,,$$

Source: Complex Analysis by Brown and Churchill

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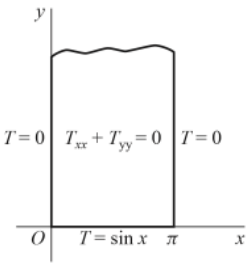

$$T_{xx}(x,y) + T_{yy}(x,y) = 0$$

$$T(0,y) = 0, T(\pi,y) = 0$$

$$T(x,y) = \sin x\,,$$

$$\lim_{y\to\infty} T(x,y) = 0$$

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Harmonic Functions

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$$T(x,y) = e^{-y}\sin x$$

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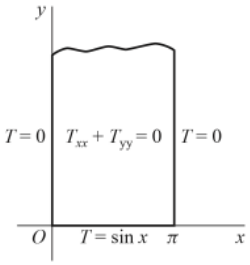
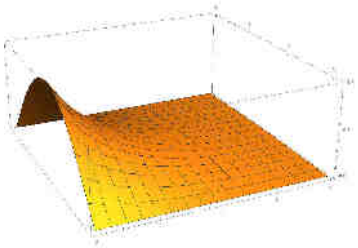
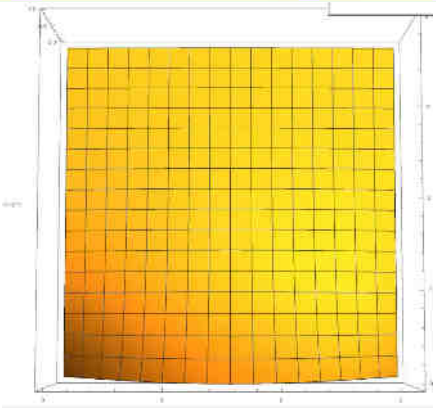




Source: Complex Analysis by Brown and Churchill

Harmonic Functions



Harmonic Functions

What does the following condition mean mathematically

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0 \quad ???$$

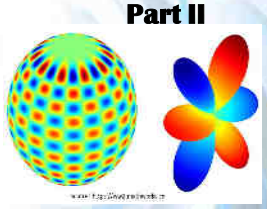
Source: Internet

Harmonic Functions

Harmonic Functions

Topic No. 46

Harmonic Functions
Part II



Harmonic Functions

Definition (Harmonic Functions)

A real-valued function H of two variables x and y is said to be harmonic in a given domain D of the xy plane if

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$$H_{xx}(x,y) + H_{yy}(x,y) = 0\,.$$

Or

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Harmonic Functions

Theorem: If a function $f(z) = u(x,y) + iv(x,y)$ is analytic in a domain D then its component functions u and v are harmonic function in D.

Proof:

Source: internet

Harmonic Functions

Theorem: If a function $f(z) = u(x,y) + iv(x,y)$ is analytic in a domain D then its component functions u and v are harmonic function in D.

Proof:

Harmonic Functions

Theorem: If a function $f(z) = u(x,y) + iv(x,y)$ is analytic in a domain D then its component functions u and v are harmonic function in D.

Example: Consider the function $f(z) = z^2$.

Harmonic Functions

Example: The function

$$f(z) = e^{-y}\sin x + i\,e^{-y}\cos x$$

is entire function.

Harmonic Functions

Theorem: If a function $f(z) = u(x,y) + iv(x,y)$ is analytic in a domain D then its component functions u and v are harmonic function in D.

Harmonic Conjugates

Definition: Let $u(x,y)$ and $v(x,y)$ be two functions such that

1. $u(x,y)$ and $v(x,y)$ are harmonic functions in a domain D
2. the first partial derivatives of u and v satisfy Cauchy-Riemann equations throughout D

then v is said to be harmonic conjugate of u (in D).

Analytic Functions

RECALL: CR Conditions for differentiability

Theorem: Let $f(z) = f(x+iy) = u(x,y) + iv(x,y)$ be defined in some neighborhood of the point $z_0 = x_0 + iy_0$ and suppose that

1. If all the partial derivatives u_x, v_y, u_y and v_x exist everywhere in the neighbourhood.
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and also

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Harmonic Conjugates

Example: Suppose that

$$u(x,y) = x^2 - y^2, v(x,y) = 2xy$$

Harmonic Conjugates

Definition: Let $u(x,y)$ and $v(x,y)$ be two functions such that

1. $u(x,y)$ and $v(x,y)$ are harmonic functions in a domain D

Harmonic Conjugates

Theorem: A function $f(z) = u(x,y) + iv(x,y)$ is analytic in a domain D if and only if v is harmonic conjugate of u.

Proof:

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Harmonic Conjugates

Example: Find harmonic conjugate of

$$u(x,y) = x^3 - 3xy^2 - 5y$$

Harmonic Conjugates

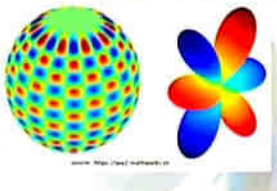
Example: Find harmonic conjugate of

$$u(x,y) = x^3 - 3xy^2 - 5y$$

Harmonic Functions

Topic No. 47

Harmonic Functions
Part III



Harmonic Conjugates

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1. $u(x,y)$ and $v(x,y)$ are harmonic functions in a domain D
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then v is said to be harmonic conjugate of u (in D).

$f(z) = u + iv$ $\quad u_x = v_y, \quad u_y = -v_x.$]

$g(z) = v(x,y) + i\,u(x,y)$

Harmonic Conjugates

Example: Find harmonic conjugate of

$$u(x,y) = x^3 - 3xy^2 - 5y$$

To find $v(x,y)$: u & v are harmonic functions.

Also $u_x = v_y, \quad u_y = -v_x.$

We know: v_y & v_x.

$v_y = u_x = 3x^2 - 3y^2$

Integrate v_y w.r.t. y (partial integration)

$v = \int (3x^2 - 3y^2)\,dy = 3x^2y - \frac{3y^3}{3} + C(x)$

Harmonic Conjugates

Example: Find harmonic conjugate of

$$u(x,y) = x^3 - 3xy^2 - 5y$$

$v = 3yx^2 - y^3 + C(x)$ — (I)

$u_y = -v_x \Rightarrow v_x = -u_y$ — (II)

(I) $\Rightarrow$ $v_x = 6yx + C'(x)$ — (*)

(II) $\Rightarrow$ $v_x = -(-6xy - 5) = 6xy + 5$ — (**)

Comparing (*) & (**), we get

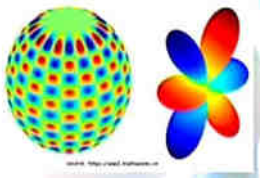
$C'(x) = 5 \Rightarrow C(x) = 5x + d$

$\Rightarrow v = 3yx^2 - y^3 + (5x + d).$

Harmonic Functions

Topic No. 48

Harmonic Functions
Part IV



Harmonic Functions

Definition (Harmonic Functions)

A real-valued function H of two variables x and y is said to be harmonic in a given domain D of the xy plane if

- throughout the domain it has continuous partial derivatives of the first and second order
- and satisfies the following partial differential equation

$$H_{xx}(x,y) + H_{yy}(x,y) = 0.$$

Or

$$\frac{\partial^2 H}{\partial x^2} + \frac{\partial^2 H}{\partial y^2} = 0$$

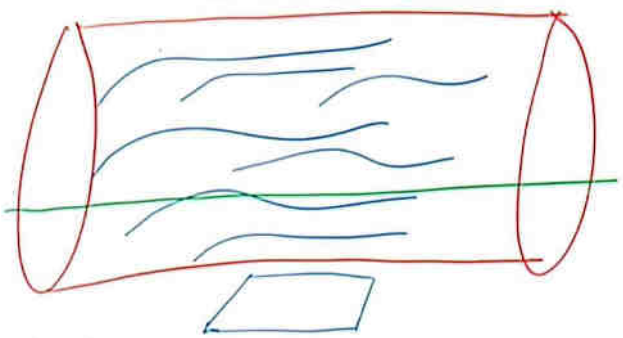
Harmonic Functions

Harmonic functions appear as solutions of many physical problems.

We consider one such problem of fluid flow.

Harmonic Functions

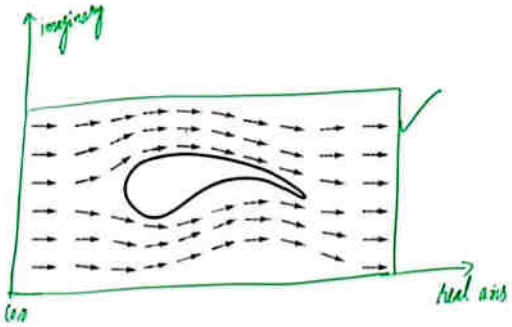
Consider flow of an incompressible and frictionless fluid flow.



VU

Harmonic Functions

Consider flow of an incompressible and frictionless fluid flow.

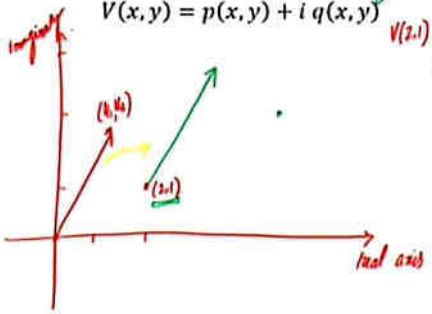


VU

Harmonic Functions

Consider flow of an incompressible and frictionless fluid flow.
The velocity vector at the point (x, y) is

$$V(x, y) = p(x, y) + i\, q(x, y)$$



VU

Harmonic Functions

Consider flow of an incompressible and frictionless fluid flow.
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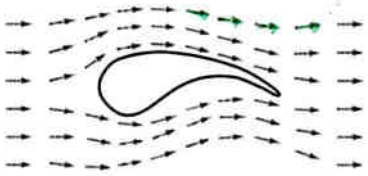


VU

Harmonic Functions

The velocity vector at the point (x, y) is

$$V(x, y) = p(x, y) + i\, q(x, y)$$

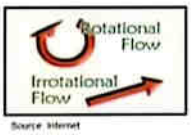
Flow is irrotational so

$$\nabla \times V = 0$$

$$\nabla = \frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}$$

$$\nabla \times V = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ p & q & 0 \end{vmatrix} \Rightarrow \left(\frac{\partial q}{\partial z}\right)\hat{i} + \left(\frac{\partial p}{\partial z}\right)\hat{j} + \left(q_x - p_y\right)\hat{k}$$

$$\Rightarrow q_x - p_y = 0$$



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VU

Harmonic Functions

The velocity vector at the point (x, y) is

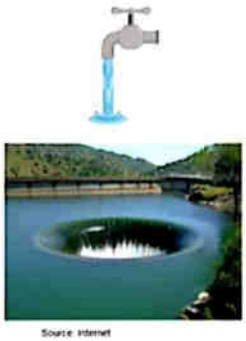
$$V(x, y) = p(x, y) + i\, q(x, y)$$

No source or sink so

$$\nabla . V = 0$$

$$p_x + q_y = 0$$

$$\left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}\right).\left(p\hat{i} + q\hat{j} + 0\hat{k}\right)$$



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VU

Harmonic Functions

The velocity vector at the point (x, y) is

$$V(x, y) = p(x, y) + i\, q(x, y)$$

such that

$$p_x(x, y) = -q_y(x, y) \text{ and } p_y(x, y) = q_x(x, y)$$

$$p_x = -q_y, \quad p_y = q_x$$

VU

Harmonic Functions

The velocity vector at the point (x, y) is

$$V(x, y) = p(x, y) + i\, q(x, y)$$

such that

$$p_x(x, y) = -q_y(x, y) \text{ and } p_y(x, y) = q_x(x, y)$$

Consider

$$f(x, y) = u(x, y) + iv(x, y) = p(x, y) - i\, q(x, y)$$

Such that

$$V(x, y) = \overline{f(x, y)}.$$

VU

Harmonic Functions

Since

$$p_x(x,y) = -q_y(x,y) \text{ and } p_y(x,y) = q_x(x,y)$$

So f satisfy Cauchy-Riemann equations

$$f(x,y) = u(x,y) + iv(x,y) = p(x,y) - i\,q(x,y)$$

$u_x = v_y$ ✓ $u_y = -v_x$ — similarly can be checked.

Here $u = p$, $v = -q$. $\Rightarrow v_y = -q_y$

$u_x = p_x = -q_y = v_y$ ✓

Harmonic Functions

RECALL: CR Conditions for differentiability

Theorem: Let $f(z) = f(x+iy) = u(x,y) + iv(x,y)$ be defined in some neighborhood of the point $z_0 = x_0 + iy_0$ and suppose that

1. If all the partial derivatives u_x, v_y, u_y and v_x exist everywhere in the neighbourhood.
2. those partial derivatives are continuous at (x_0, y_0) and satisfy Cauchy-Riemann equations

$$u_x = v_y, \qquad u_y = -v_x$$

Then f is differentiable at z_0, and the derivative $f'(z_0)$ can be computed with either of the following equations

$$f'(z_0) = u_x(x_0, y_0) + iv_x(x_0, y_0),$$

and also

$$f'(z_0) = v_y(x_0, y_0) - iu_y(x_0, y_0).$$

Harmonic Functions

Since

$$p_x(x,y) = -q_y(x,y) \text{ and } p_y(x,y) = q_x(x,y)$$

So f satisfy Cauchy-Riemann equations

$$f(x,y) = u(x,y) + iv(x,y) = p(x,y) - i\,q(x,y)$$

we assume p & q have continuous partial derivatives.

$\Rightarrow f(x,y)$ is analytic in domain.

Harmonic Functions

Hence

$$f(x,y) = u(x,y) + iv(x,y)$$

is analytic such that

$$V(x,y) = \overline{f(x,y)}$$

Harmonic Functions

Fact from future: An analytic function has analytic antiderivative.

So there exist $F(x,y)$ such that

$$F'(x,y) = f(x,y)$$

Let

$$F(x,y) = \phi(x,y) + i\psi(x,y)$$

and $F(x,y)$ is analytic.

Harmonic Functions

Recall

Theorem: If a function $f(z) = u(x,y) + iv(x,y)$ is analytic in a domain D then its component functions u and v are harmonic function in D.

Harmonic Functions

For

$$F(x,y) = \phi(x,y) + i\psi(x,y)$$

the component $\phi(x,y)$ is harmonic.

Also

$$\phi_y = -\psi_x$$

Hence

$$\begin{aligned} grad\,\phi &= \phi_x + i\,\phi_y \\ &= \phi_x - i\,\psi_x \\ &= \overline{\phi_x + i\,\psi_x} \\ &= \overline{F'(x,y)} \\ &= \overline{f(x,y)} \\ &= V(x,y) \end{aligned}$$

Harmonic Functions

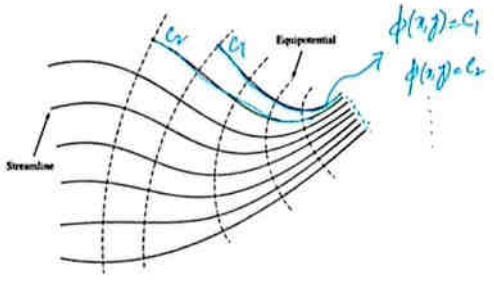
Hence

$$grad\,\phi = V(x,y)$$

The curves given by

$$\{(x,y): \phi(x,y) = constant\}$$

are called equipotentials.



Harmonic Functions

Example: Find streamlines and equipotentials of a fluid flow with velocity vector.

$$V(x,y) = 2x - i2y$$

Solution: Consider

$$V(x,y) = \overline{f(x,y)}$$

$f(x,y) = 2x + i2y = 2(x + iy)$

If $z = x + iy \Rightarrow f(z) = 2z$

$\Rightarrow F(x,y) = \overline{2z}$

$= (x^2 - y^2) + i(2xy)$

Harmonic Functions

Solution (Cont.)

Hence streamlines are:

$2xy = \text{Constant}$

and equipotentials are :

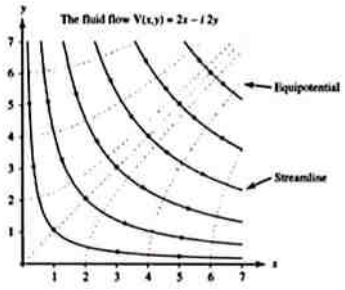
$x^2 - y^2 = \text{Constant}$

Harmonic Functions

Example: Find streamlines and equipotentials of a fluid flow with velocity vector.

$$V(x,y) = 2x - i2y$$

Solution:



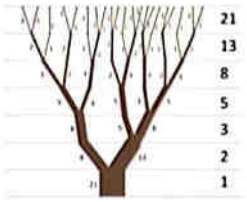
Sequences

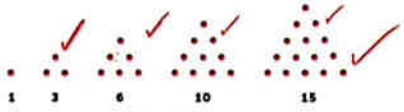
Topic No. 49

Sequences
Part I

Sequences

What is a sequence??



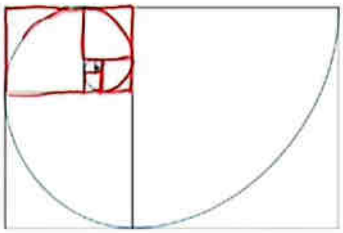


Source: Internet

Sequences

Source: https://insteading.com

Sequences



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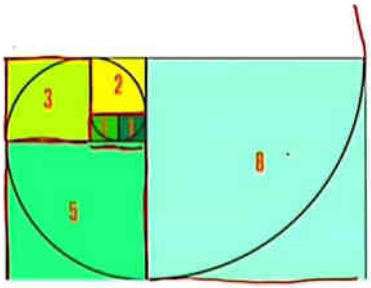
Sequences



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Sequences

So we have the following sequence of numbers

1,1,2,3,5,8,13,21, ...

VU

Sequences

The numbers form a sequence

1, 2, 3, 4, 5, 6, 7, ...

1 3 5 7 9 11 13

$f(n) = 2n-1$ ✓

$n=1$ $f(1) = 2(1)-1 = 2-1 = 1$

$n=3$ $f(3) = 2(3)-1 = 5$

VU

Sequences

The numbers form a sequence

1, 2, 3, 4, 5, 6, 7, ...

2 4 6 8 10 12 14. . . .

$f(n) = 2n$

$n=1$ $f(1) = 2(1) = 2$ ✓

$n=7$ $f(7) = 2(7) = 14.$

$f(100) = 2(100) = 200.$

VU

Sequences

Definition (Sequence)

A sequence is a function whose domain is the positive integers and whose range is a subset of the complex numbers.

$f: \mathbb{Z}^+ \longrightarrow \mathbb{C}$ ✓

$f: \mathbb{N} \longrightarrow \mathbb{C}$ ✓

VU

Sequences

Definition (Sequence)

A sequence is a function whose domain is the positive integers and whose range is a subset of the complex numbers.

Examples:

1. $f_1(n) = n + i2n, \quad n \in \mathbb{N}$

$n=1, \quad f_1(1) = 1 + 2i$

$n=100, \quad f_1(100) = 100 + i200.$

VU

Sequences

Definition (Sequence)

A sequence is a function whose domain is the positive integers and whose range is a subset of the complex numbers.

Examples:

1.

2. $f_2(n) = \left(2 - \frac{1}{n}\right) + i\left(5 - \frac{1}{n}\right), \quad n \in \mathbb{N},$

VU

Sequences

Definition (Sequence)

A sequence is a function whose domain is the positive integers and whose range is a subset of the complex numbers.

Examples:

1.

2.

3. $f_3 = (2+i5)^n, \quad n \in \mathbb{N}$

VU

Sequences

Definition (Sequence)

A sequence is a function whose domain is the positive integers and whose range is a subset of the complex numbers.

For convenience we simply call complex sequences as sequences.

VU

Sequences

Notations

A sequence is usually denoted as

$$\{z_n\}_{n=1}^{\infty} \quad \text{OR} \quad \{z_n\}_1^{\infty}$$

and if the indices are understood then simply as the following

$$\{z_n\}$$

The terms of the sequence $\{z_n\}$ are

$$[z_1, z_2, z_3, \dots]$$

or (roughly speaking) sequence.

Sequences

Sometimes we start a sequence from other integers

$$\{z_n\}_{n=-1}^{\infty}$$

Or

$$\{z_n\}_{n=0}^{\infty}$$

Sequences

Example: The sequence

$$f(n) = \left\{\left(2-\frac{1}{n}\right)+i\left(5+\frac{1}{n}\right)\right\}_{n=1}^{\infty}$$

can also be written in the following ways

$$f(n) = \left\{\left(2-\frac{1}{n+2}\right)+i\left(5+\frac{1}{n+2}\right)\right\}_{n=-1}^{\infty}$$

Sequences

Topic No. 50

Sequences
Part II

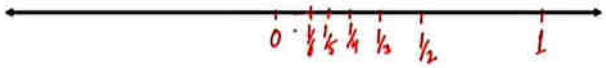
Limit of a Sequence

Limit of a sequence
What is behavior of the sequence $\{z_n\}$ as n gets very large?

Limit of a Sequence

Consider the sequence

$$\left\{\frac{1}{n}\right\}$$



Limit of a Sequence

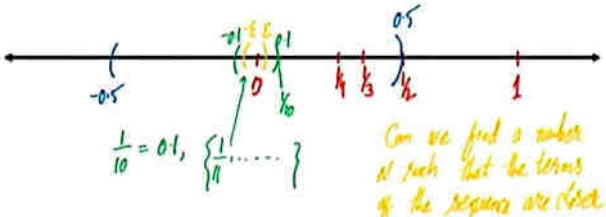
Intuitive definition

A sequence $\{x_n\}$ has p as its limit as n approaches to infinity provided the terms x_n can be made as close as we want to p by making n large enough.

Limit of a Sequence

Consider the sequence

$$\left\{\frac{1}{n}\right\}$$



$\frac{1}{10} = 0.1, \left\{\frac{1}{n}, \dots\right\}$

Can we find a number N such that the terms of the sequence are closer than ε: $\frac{1}{n} - 0 < \varepsilon \Rightarrow \frac{1}{n} < \varepsilon$ or $\frac{1}{\varepsilon} < n$

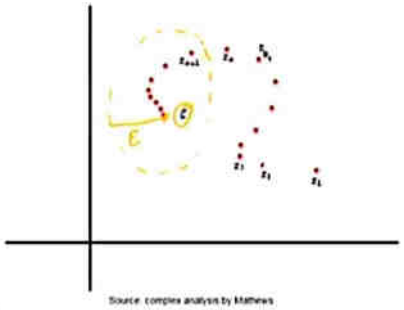
Limit of a Sequence

A sequence of real numbers $\{x_n\}$ converges to p if

for every $\varepsilon > 0$ there exist N_ε such that
$|x_n - p| < \varepsilon$ whenever $n > N_\varepsilon$

Limit of a Sequence

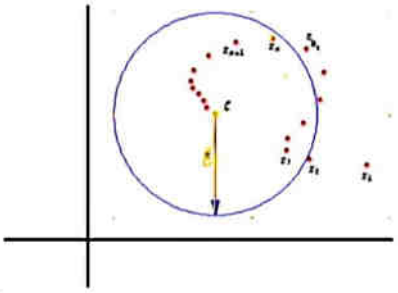
A sequence of complex numbers $\{z_n\}$ converges to ζ



Source: complex analysis by Mathews

Limit of a Sequence

A sequence of complex numbers $\{z_n\}$ converges to ζ



Source: complex analysis by Mathews

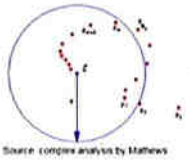
Limit of a Sequence

A sequence of complex numbers $\{z_n\}$ converges to ζ if for each $\varepsilon > 0$ there exist N_ε such that

$|z_n - \zeta| < \varepsilon$ whenever $n > N_\varepsilon$

denoted as

$$\lim_{n\to\infty} z_n = \zeta$$



Source: complex analysis by Mathews

Limit of a Sequence

Can we apply the definition of convergence of complex sequence?

Limit of a Sequence

One solution

Find relation between convergence of real sequences and complex sequences

Limit of a Sequence

Theorem: Let $z_n = x_n + iy_n$ and $\zeta = u + iv$. Then

$$\lim_{n\to\infty} z_n = \zeta$$

if and only if

$$\lim_{n\to\infty} x_n = u \quad \text{and} \quad \lim_{n\to\infty} y_n = v$$

Sequences

Example: Find limit of the following sequence

$$\left\{\left(2 - \frac{1}{n}\right) + i\left(5 + \frac{1}{n}\right)\right\}_{n=1}^{\infty}$$

$z_n = x_n + iy_n$, $x_n = \left(2 - \frac{1}{n}\right)$, $y_n = 5 + \frac{1}{n}$

$\lim_{n\to\infty} \left(2 - \frac{1}{n}\right) = 2$, $\lim_{n\to\infty} \left(5 + \frac{1}{n}\right) = 5$

$\lim_{n\to\infty} z_n = 2 + i5$

Limit of a Sequence

Theorem: Let $z_n = x_n + iy_n$ and $\zeta = u + iv$. Then

$$\lim_{n\to\infty} z_n = \zeta$$

if and only if

$$\lim_{n\to\infty} x_n = u \text{ and } \lim_{n\to\infty} y_n = v$$

Proof: Given $\lim_{n\to\infty} z_n = u+iv$

To show $\lim_{n\to\infty} x_n = u$, $\lim_{n\to\infty} y_n = v$

Given: For $\varepsilon>0$ $\exists$ N_ε such that $|z_n - \zeta| < \varepsilon$, whenever $n > N_\varepsilon$

To show: For $\varepsilon>0$ $\exists$ N_ε such that $|x_n - u| < \varepsilon$ whenever $n > N_\varepsilon$

Similarly for y_n

VU

Limit of a Sequence

Theorem: Let $z_n = x_n + iy_n$ and $\zeta = u + iv$. Then

$$\lim_{n\to\infty} z_n = \zeta$$

if and only if

$$\lim_{n\to\infty} x_n = u \text{ and } \lim_{n\to\infty} y_n = v$$

Proof:

$|x_n - u| = |\mathrm{Re}(z_n - \zeta)|$

$\leq |z_n - \zeta| < \varepsilon$, whenever $n \geq N_\varepsilon$

Similarly for y_n.

VU

Limit of a Sequence

Theorem: Let $z_n = x_n + iy_n$ and $\zeta = u + iv$. Then

$$\lim_{n\to\infty} z_n = \zeta$$

if and only if

$$\lim_{n\to\infty} x_n = u \text{ and } \lim_{n\to\infty} y_n = v$$

Proof: Assume $|x_n - u| < \varepsilon/2$ whenever $n \geq N_\varepsilon'$

$|y_n - v| < \varepsilon/2$ " $n \geq N_\varepsilon''$

$|z_n - \zeta| = |x_n + iy_n - (u+iv)| = |(x_n - u) + i(y_n - v)|$

$\leq |x_n - u| + |i||y_n - v|$

$< \varepsilon/2 + \varepsilon/2 = \varepsilon$, whenever $n \geq \max\{N_\varepsilon', N_\varepsilon''\}$

VU

Sequences

Topic No. 51

Sequences
Part III

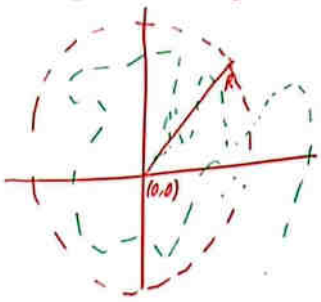
VU

Bounded Sequence

Definition (Bounded sequence)

A complex sequence $\{z_n\}$ is bounded provided that there exist a positive real number R and an integer N such that $|z_n| < R$ for all $n > N$.

$|z_n - 0| < R$



VU

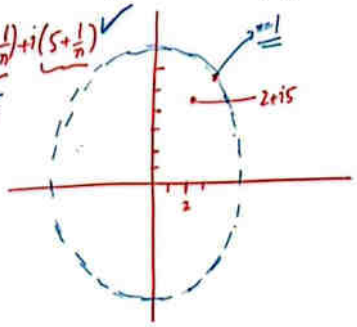
Bounded Sequence

Definition (Bounded sequence)

A complex sequence $\{z_n\}$ is bounded provided that there exist a positive real number R and an integer N such that $|z_n| < R$ for all $n > N$.

Example: $z_n = \left(2 + \frac{1}{n}\right) + i\left(5 + \frac{1}{n}\right)$

$R = |3+i6| = \sqrt{9+36} = \sqrt{45}$



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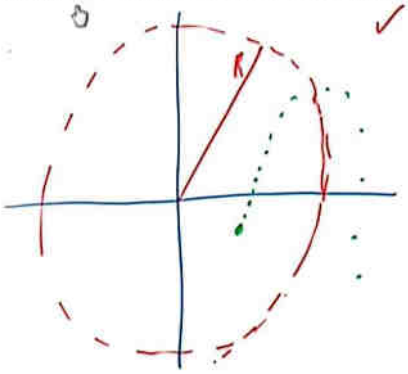
Bounded Sequence

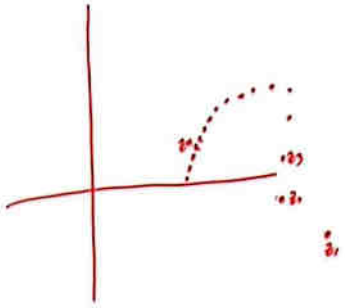
Theorem: If $\{z_n\}$ is a convergent sequence then $\{z_n\}$ is bounded.



VU

Cauchy Sequence

Definition (Cauchy Sequence)



VU

Cauchy Sequence

Definition (Cauchy Sequence)

The sequence $\{z_n\}$ is a Cauchy sequence if for every $\varepsilon > 0$ there is a positive integer N_ε such that

if $n, m > N_\varepsilon$, then $|z_n - z_m| < \varepsilon$.

Cauchy Sequence

Theorem: $\{z_n\}$ is a Cauchy sequence if and only if $\{z_n\}$ converges.

Proof: Given z_n is convergent to z.

For $\varepsilon > 0$ there exist N_ε such that $|z_n - z| < \frac{\varepsilon}{2}$ whenever $n > N_\varepsilon$ } Given

To show:- z_n is a Cauchy sequence.

For $\varepsilon > 0$, $|z_n - z_m| < \varepsilon$ whenever $n, m > N_\varepsilon$

$\exists\ N_\varepsilon$

$$|z_n - z_m| = |z_n - z + z - z_m| = |z_n - z - (z_m - z)|$$
$$\leq |z_n - z| + |z_m - z|$$
$$< \frac{\varepsilon}{2} + \frac{\varepsilon}{2}, \quad \forall\ n, m > N_\varepsilon$$
$$= \varepsilon$$

Cauchy Sequence

Theorem: $\{z_n\}$ is a Cauchy sequence if and only if $\{z_n\}$ converges.

Proof: Given $\Rightarrow$ Assume z_n is Cauchy sequence.

For $\varepsilon > 0$ $\exists\ N_\varepsilon$ such that $|z_n - z_m| < \varepsilon$, whenever $n, m > N_\varepsilon$

To show: $z_n = x_n + iy_n$ is Convergent.

Aim:- Show that $\{x_n\}$ & $\{y_n\}$ are Convergent.

$|x_n - x_m| = |\mathrm{Re}(z_n - z_m)| \leq |z_n - z_m| < \varepsilon$, whenever $n, m > N_\varepsilon$

$\Rightarrow |x_n - x_m| < \varepsilon$, whenever $n, m > N_\varepsilon$

$\Rightarrow \{x_n\}$ is a Cauchy sequence. Similarly $\{y_n\}$ is Cauchy sequence.

Recall

Theorem: Let $z_n = x_n + iy_n$ and $\zeta = u + iv$. Then

$$\lim_{n\to\infty} z_n = \zeta$$

if and only if

$$\lim_{n\to\infty} x_n = u \quad \text{and} \quad \lim_{n\to\infty} y_n = v$$

Cauchy Sequence

Theorem: $\{z_n\}$ is a Cauchy sequence if and only if $\{z_n\}$ converges.

Proof: $z_n = x_n + iy_n$

$\{x_n\}$ is Convergent. $\{y_n\}$ is Convergent

$\Rightarrow z_n$ is Convergent.

Series

Topic No. 52

Series
Part I

Series

Definition (Infinite Series)

The formal expression

$$\sum_{k=1}^{\infty} z_k = z_1 + z_2 + z_3 + \cdots + z_n + \cdots$$

is called infinite series, and $z_1, z_2, \ldots$ are called the terms of the series.

Series

Examples:

$$\sum_{n=1}^{\infty} (n + in^2)$$

In expanded form in becomes

$$(1 + i) + (2 + i4) + (3 + i9) + \cdots$$

Series

Question

$\sum_{k=1}^{\infty} z_k$

is a complex number or not?

Series

To understand

$\sum_{k=1}^{\infty} z_k$

we define sequence of partial sums.

Definition (Sequence of Partial Sums)

For series $\sum_{k=1}^{\infty} z_k$ we define sequence of partial sums as

$S_1 = z_1$

$S_2 = z_1 + z_2$

$S_3 = z_1 + z_2 + z_3$

$\vdots$

$S_n = z_1 + z_2 + \cdots + z_n$

$\vdots$

Series

$S_1 = z_1$

$S_2 = z_1 + z_2$

$S_3 = z_1 + z_2 + z_3$

$\vdots$

$S_n = z_1 + z_2 + \cdots + z_n$

$\vdots$

$\sum_{k=1}^{\infty} z_k$

Series

If there is a complex number S for which

$$S = \lim_{n\to\infty} S_n = \lim_{n\to\infty} \sum_{k=1}^{n} z_k$$

we say that series **converges** and

$$S = \sum_{k=1}^{\infty} z_n$$

S is called the sum of the infinite series.

If a series does not converge, we say that it **diverges**.

Series

Ignoring finitely many terms does not effect the convergence or divergence

$\sum_{k=1}^{\infty} z_k$

$= z_1 + z_2 + \cdots + z_n + \cdots$

Series

If

$\sum_{k=N+1}^{\infty} z_k$

converges then so does the following

$\sum_{k=1}^{\infty} z_k$

And similarly for divergence.

Series

How to check convergence of a series ?

Series

Hint:

We have understanding of series of real numbers.

Series

What is the relation between **series of complex numbers** and **series of real numbers ?**

Series

Theorem: Let $z_n = x_n + iy_n$ and $S = U + iV$. Then

$$S = U + iV = \sum_{k=1}^{\infty} x_n + iy_n$$

if and only if

$$U = \sum_{k=1}^{\infty} x_n \quad \text{and} \quad V = \sum_{k=1}^{\infty} y_n$$

Series

Proof :

Let $u_n = \sum_{k=1}^{n} x_k$ be sequence of partial sums of $\sum_{n=1}^{\infty} x_n$

" $V_n = \sum_{k=1}^{n} y_k$ " " " " " " $\sum_{n=1}^{\infty} y_n$

Then $u_n + iV_n = \sum_{k=1}^{n} (x_k + iy_k)$ is sequence of partial sums of $\sum_{n=1}^{\infty} (x_n + iy_n)$

Theorem: Let $z_n = x_n + iy_n$ and $\zeta = u + iv$. Then

$$\lim_{n\to\infty} z_n = \zeta$$

if and only if

$$\lim_{n\to\infty} x_n = u \text{ and } \lim_{n\to\infty} y_n = v$$

Series

Proof :

$$S = U + iV = \sum_{n=1}^{\infty} (x_n + iy_n)$$

$$\Leftrightarrow \quad S = U + iV = \lim_{n\to\infty} (u_n + iV_n)$$

$$\Leftrightarrow \quad \lim_{n\to\infty} u_n = u \quad \& \quad \lim_{n\to\infty} V_n = V$$

$$\Leftrightarrow \quad u = \sum_{n=1}^{\infty} x_n \quad \& \quad V = \sum_{n=1}^{\infty} y_n$$

Series

Example: Show that the series

$$\sum_{n=1}^{\infty} \frac{1 + i\,n(-1)^n}{n^2}$$

converges.

Solution

$$\sum_{n=1}^{\infty} \left(\frac{1 + i(-1)^n n}{n^2} \right) = \sum \left(\frac{1}{n^2} + i \frac{(-1)^n}{n} \right)$$

$$= \left(\sum_{n=1}^{\infty} \frac{1}{n^2} \right) + i \left(\sum_{n=1}^{\infty} \frac{(-1)^n}{n} \right)$$

$\sum_{n=1}^{\infty} \frac{1}{n^2}$ & $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ Converges so $\sum_{n=1}^{\infty} \frac{1 + i\,n(-1)^n}{n^2}$ (Converges)

Series

Topic No. 53

Series
Part II

Series

Definition (Infinite Series)

The formal expression $\sum_{k=1}^{\infty} z_k$ is called infinite series, and $z_1, z_2, \ldots$ are called the terms of the series.

Series

Sum of series

Given two convergent series

$$\sum_{n=1}^{\infty} z_n \quad \text{and} \quad \sum_{n=1}^{\infty} w_n$$

their sum is defined in the following way

$$\sum_{n=1}^{\infty} (z_n + w_n)$$

Sum of series

Example: Given

$$\sum_{n=1}^{\infty} \left(\frac{1}{n^2} + \frac{i(-1)^n}{n} \right) \quad and \quad \sum_{n=1}^{\infty} \left(\frac{1}{n^2} + \frac{i}{n^3} \right)$$

$$\sum_{n=1}^{\infty} \left(\frac{1}{n^2} + \frac{i(-1)^n}{n} \right) + \sum_{n=1}^{\infty} \left(\frac{1}{n^2} + \frac{i}{n^3} \right)$$

$$= \sum_{n=1}^{\infty} \left(\frac{1}{n^2} + \frac{i(-1)^n}{n} \right) + \left(\frac{1}{n^2} + \frac{i}{n^3} \right)$$

$$= \sum_{n=1}^{\infty} \left(\left(\frac{1}{n^2} + \frac{1}{n^2} \right) + i \left(\frac{(-1)^n}{n} + \frac{1}{n^3} \right) \right)$$

Series

Scalar multiple of series

Given a convergent series

$$\sum_{n=1}^{\infty} z_n \quad \text{and a scalar c}$$

then the scalar multiple of series is defined as

$$\sum_{n=1}^{\infty} (c\, z_n) = c \sum_{n=1}^{\infty} z_n$$

Series

Scalar multiple of series

Example: Given

$$\sum_{n=1}^{\infty} \left(\frac{1}{n^2} + \frac{i(-1)^n}{n} \right) \quad and \; c = 3 + i$$

$$c \sum_{n=1}^{\infty} \left(\frac{1}{n^2} + \frac{i(-1)^n}{n} \right)$$

$$= \sum_{n=1}^{\infty} (3+i) \left(\frac{1}{n^2} + \frac{i(-1)^n}{n} \right)$$

$$= \sum_{n=1}^{\infty} \left(\left(\frac{3}{n^2} - \frac{(-1)^n}{n} \right) + i \left(\frac{1}{n^2} + 3\frac{(-1)^n}{n} \right) \right)$$

Series

Cauchy product of series

Given two convergent series

$$\sum_{n=0}^{\infty} z_n \quad \text{and} \quad \sum_{n=0}^{\infty} w_n$$

their Cauchy product is defined in the following way

$$\sum_{n=0}^{\infty} c_n$$

where

$$c_n = \sum_{k=0}^{n} z_k\, w_{n-k}$$

Series

Cauchy product of series

$$c_n = \sum_{k=0}^{n} z_k\, w_{n-k}$$

$$c_0 = z_0 w_0$$
$$c_1 = z_0 w_1 + z_1 w_0$$
$$c_2 = z_0 w_2 + z_1 w_1 + z_2 w_0$$
$$\vdots$$
$$c_n = z_0 w_n + z_1 w_{n-1} + \cdots + z_n w_0$$

Series

Cauchy product of series

Given

$$\sum_{n=0}^{\infty} z_n = z_0 + z_1 + z_2 + z_3 + \cdots$$

$$\sum_{n=0}^{\infty} w_n = w_0 + w_1 + w_2 + w_3 + \cdots$$

Cauchy product

$$= (z_0 w_0) + (z_0 w_1 + w_0 z_1) + (z_0 w_2 + z_1 w_1 + z_2 w_0) + \cdots$$

Series

Theorem: If the Cauchy product converges, then

$$\sum_{n=0}^{\infty} c_n = \left(\sum_{n=0}^{\infty} z_n\right)\left(\sum_{n=0}^{\infty} w_n\right)$$

If $\sum_{n=0}^{\infty} c_n$ converges then

$$\sum_{n=0}^{\infty} c_n = (z_0 + z_1 + z_2 + z_3 + \cdots)(w_0 + w_1 + w_2 + w_3 + \cdots)$$

$$= (z_0 w_0) + (z_0 w_1 + z_1 w_0) + \cdots$$

VU

Series

Topic No. 54

Series
Part III
(Convergence of series)

VU

Series

Definition
(Infinite Series)

The formal expression

$$\sum_{k=1}^{\infty} z_k$$

is called infinite series, and $z_1, z_2, \ldots$ are called the terms of the series.

VU

Series

Question:
How to check convergence of series

VU

Series

Theorem: If $\sum_{n=1}^{\infty} z_n$ is a convergent series, then

$$\lim_{n\to\infty} z_n = 0.$$

Proof: Given $\sum_{n=1}^{\infty} z_n$ is a convergent series

To show $\lim_{n\to\infty} z_n = 0$

Let

$$\sum_{n=1}^{\infty} z_n = L$$

Let $S_n = z_1 + z_2 + \cdots z_n$, then

$$S_{n-1} = z_1 + z_2 + \cdots z_{n-1}$$

VU

Now

$$S_n = z_1 + z_2 + \cdots z_{n-1} + z_n$$
$$S_{n-1} = z_1 + z_2 + \cdots z_{n-1}$$

Thus

$$\lim_{n\to\infty} z_n = \lim_{n\to\infty} S_n - S_{n-1}$$
$$= L - L = 0$$

Hence

$$\lim_{n\to\infty} z_n = 0.$$

VU

Series

Example: Check whether the following series converges or diverges

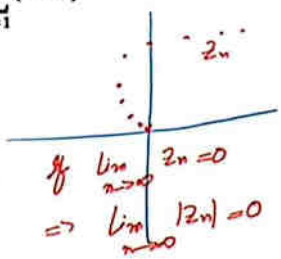
$$\sum_{n=1}^{\infty} z_n = \sum_{n=1}^{\infty} (1+i)^n$$

Want to calculate $\lim_{n\to\infty} z_n = ?$

Here $\lim_{n\to\infty} |z_n| = \lim_{n\to\infty} (\sqrt{2})^n = \infty$

$\Rightarrow \lim_{n\to\infty} z_n \neq 0$

$\sum z_n$ diverges.



If $\lim_{n\to\infty} z_n = 0$

$\Rightarrow \lim_{n\to\infty} |z_n| = 0$

VU

Series

One of the most fundamental test for checking convergence of series is **Comparison Test**

VU

Series

How to compare two complex numbers?

Series

Comparison Test

Let

$$\sum_{n=1}^{\infty} M_n$$

be a convergent series of real non-negative terms. If $\{z_n\}$ is a sequence of complex numbers and

$$|z_n| \leq M_n \text{ for all } n,$$

then

$$\sum_{n=1}^{\infty} z_n = \sum_{n=1}^{\infty} x_n + iy_n$$

converges.

Series

Comparison Test

Example: Show that $\sum_{n=1}^{\infty} \frac{(3+4i)^n}{5^n n^2}$ Converges.

Here $|z_n| = \left|\frac{(3+i4)^n}{5^n n^2}\right| = \frac{(\sqrt{3^2+4^2})^n}{5^n n^2} = \frac{5^n}{5^n n^2} = \frac{1}{n^2}$

we know that $\sum_{n=1}^{\infty} \frac{1}{n^2}$ Converges.

By Comparison Test $\sum \frac{(3+i4)^n}{5^n n^2}$ Converges.

Series

Absolute convergence

The series $\sum_{n=1}^{\infty} z_n$ is said to absolutely convergent provided that the real series of magnitudes

$$\sum_{n=1}^{\infty} |z_n|$$

converges.

Series

Absolute convergence

Theorem: If $\sum_{n}^{\infty} |z_n|$ converges then $\sum_{n=1}^{\infty} z_n$ converges.

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Complex Analysis & Differential Geometry

MTH632

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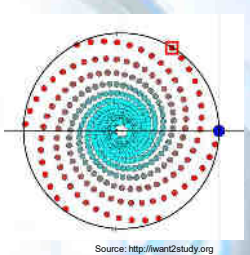
دعاؤں میں یاد رکھیے گا۔

Geometric Series

Topic No. 55

Geometric Series



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Geometric Series

Definition
(Geometric Series)
An infinite series of the form

$$\sum_{n=0}^{\infty} z^n$$

is called geometric series.

Geometric Series

Examples of
Geometric Series

Geometric Series

For what values of z
the
Geometric Series
$$\sum_{n=0}^{\infty} z^n$$
converges?

Geometric Series

Consider the geometric series

$$\sum_{n=0}^{\infty} z^n = 1 + z + z^2 + z^3 + z^4 + \cdots z^n + \cdots$$

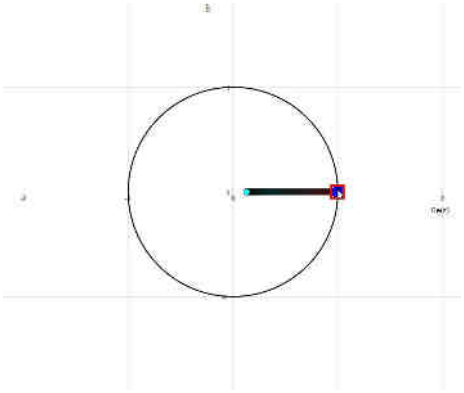
the terms of the series are

$$1, z, z^2, z^3, z^4, \ldots, z^n, \ldots$$

Since the ratio of any two consecutive terms is z so

***z* is called ratio or common ratio**

Geometric Series



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Geometric Series

Theorem: If $|z| < 1$, the series $\sum_{n=0}^{\infty} z^n$ converges to $f(z) = \frac{1}{1-z}$.

Geometric Series

Theorem: If $|z| < 1$, the series $\sum_{n=0}^{\infty} z^n$ converges to $f(z) = \frac{1}{1-z}$. That is, if $|z| < 1$, then

$$\sum_{n=0}^{\infty} z^n = 1 + z + z^2 + z^3 + \cdots z^n + \cdots = \frac{1}{1-z}$$

Geometric Series

Theorem: If $|z| < 1$, the series $\sum_{n=0}^{\infty} z^n$ converges to $f(z) = \frac{1}{1-z}$. That is, if $|z| < 1$, then

$$\sum_{n=0}^{\infty} z^n = 1 + z + z^2 + z^3 + \cdots z^n + \cdots = \frac{1}{1-z}$$

If $|z| \geq 1$ then the series diverges.

Geometric Series

Theorem: If $|z| < 1$, the series $\sum_{n=0}^{\infty} z^n$ converges to $f(z) = \frac{1}{1-z}$. That is, if $|z| < 1$, then

$$\sum_{n=0}^{\infty} z^n = 1 + z + z^2 + z^3 + \cdots z^n + \cdots = \frac{1}{1-z}$$

If $|z| \geq 1$ then the series diverges.

Proof:

Geometric Series

Proof:

Geometric Series

Theorem: If $\sum_{n=1}^{\infty} z_n$ is a convergent series, then $\lim_{n\to\infty} z_n = 0$.

Geometric Series

Proof:

Geometric Series

Following steps of previous proof we can show that

$$\sum_{n=r}^{\infty} z^n = \frac{z^r}{1-z} \quad if\ |z| < 1$$

Geometric Series

In general sum of geometric series with common ratio z is

$$\frac{\text{first term}}{1-z} \quad if\ |z| < 1$$

Geometric Series

In general sum of geometric series with common ratio z is

$$\frac{\text{first term}}{1-z} \quad if\ |z| < 1$$

For series

$$\sum_{n=0}^{\infty} az^n = a + az + az^2 + az^3 + \cdots + az^n + \cdots$$

if $|z| < 1$ then

$$\sum_{n=0}^{\infty} az^n = \frac{a}{1-z}$$

Geometric Series

Corollary: If $|z| > 1$, the series $\sum_{n=1}^{\infty} z^{-n}$ converges to $f(z) = \frac{1}{z-1}$. That is, if $|z| > 1$ then

$$\sum_{n=1}^{\infty} z^{-n} = \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} \ldots + \frac{1}{z^n} + \cdots = \frac{1}{z-1}$$

If $|z| \leq 1$ then the series diverges.

Geometric Series

Example: Evaluate

$$\sum_{n=4}^{\infty} \left(\frac{i}{3}\right)^n$$

Geometric Series

Index shift

Consider a series

$$\sum_{n=4}^{\infty} z_n = z_4 + z_5 + z_6 + z_7 + \cdots + z_n + \cdots$$

Geometric Series

Index shift

Consider a series

$$\sum_{n=4}^{\infty} z_n = z_4 + z_5 + z_6 + z_7 + \cdots + z_n + \cdots$$

The above series can also be written as

$$\sum_{n=0}^{\infty} z_{n+4} = z_4 + z_5 + z_6 + z_7 + \cdots + z_n + \cdots$$

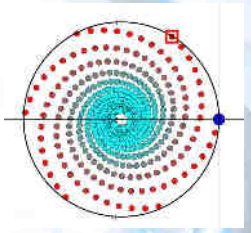
Geometric Series

Example: Evaluate

$$\sum_{n=4}^{\infty} \left(\frac{i}{3}\right)^n$$

Geometric Series

Geometric Series



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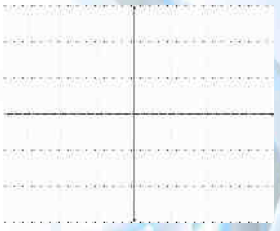
Ratio Test

Topic No. 56

d'Alembert's Ratio Test



Ratio Test

Theorem
(d'Alembert's Ratio test)

If

$$\sum_{n=0}^{\infty} \zeta_n$$

is a complex series

Ratio Test

Theorem
(d'Alembert's Ratio test)

If

$$\sum_{n=0}^{\infty} \zeta_n$$

is a complex series such that

$$\lim_{n\to\infty} \frac{|\zeta_{n+1}|}{|\zeta_n|} = L$$

Ratio Test

Theorem
(d'Alembert's Ratio test)

If

$$\sum_{n=0}^{\infty} \zeta_n$$

is a complex series such that

$$\lim_{n\to\infty} \frac{|\zeta_{n+1}|}{|\zeta_n|} = L$$

then the series is

- absolutely convergent if $L < 1$

Ratio Test

Theorem
(d'Alembert's Ratio test)
If

$$\sum_{n=0}^{\infty} \zeta_n$$

is a complex series such that

$$\lim_{n\to\infty} \frac{|\zeta_{n+1}|}{|\zeta_n|} = L$$

then the series is

- absolutely convergent if $L < 1$
- divergent if $L > 1$.

Ratio Test

Theorem
(d'Alembert's Ratio test)

Proof is similar to the case of real series

Ratio Test

Example: Show that the series

$$\sum_{n=0}^{\infty} \frac{(1-i)^n}{n!}$$

converges.

Ratio Test

Example: Show that the series

$$\sum_{n=0}^{\infty} \frac{(z-i)^n}{2^n}$$

converges for all values of z in the disk $|z-i| < 2$ and diverges if $|z-i| > 2$.

Ratio Test

Example: Show that the series

$$\sum_{n=0}^{\infty} \frac{(z-i)^n}{2^n}$$

converges for all values of z in the disk $|z-i| < 2$ and diverges if $|z-i| > 2$.

Solution:

Ratio Test

Example: Show that the series

$$\sum_{n=0}^{\infty} \frac{(z-i)^n}{2^n}$$

converges for all values of z in the disk $|z-i| < 2$ and diverges if $|z-i| > 2$.

Solution:

Ratio Test

d'Alembert's Ratio Test

Root Test

Root Test Part I

Topic No. 57

Source: google

Root Test

To understand
Root Test
we need to understand
the concept of
limit supremum
for sequences of real
numbers

Root Test

Limit Supremum

Limit Supremum

Limit Supremum

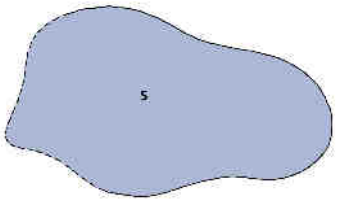
Consider a sequence with terms

$0.5, 1.5, 0.05, 1.05, 0.005, 1.005, 0.0005, 1.0005, \ldots$

Limit Supremum

Accumulation point

A point z_0 is called an accumulation point of a set S is each deleted neighbourhood of z_0 contains at least one point of S.



Limit Supremum

Limit Supremum

Consider a sequence with terms

$A = \{0.5, 1.5, 0.05, 1.05, 0.005, 1.005, 0.0005, 1.0005, \ldots\}$

The points 4 and 5 are limit points of A.

Limit Supremum

Limit Supremum

Consider a sequence with terms

$0.5, 1.5, 0.05, 1.05, 0.005, 1.005, 0.0005, 1.0005, \ldots$

Limit Supremum

Definition (Limit Supremum)

Let $\{t_n\}$ be a sequence of positive real numbers.

Limit Supremum

Definition (Limit Supremum)

Let $\{t_n\}$ be a sequence of positive real numbers. The limit supremum of the sequence, denoted by

$$\lim_{n\to\infty} \sup t_n$$

Limit Supremum

Definition (Limit Supremum)

Let $\{t_n\}$ be a sequence of positive real numbers. The limit supremum of the sequence, denoted by

$$\lim_{n\to\infty} \sup t_n$$

is the smallest real number L having the property that for any $\varepsilon > 0$, there are at most finitely many terms in the sequence that are larger than $L + \varepsilon$

Limit Supremum

Definition (Limit Supremum)

Let $\{t_n\}$ be a sequence of positive real numbers. The limit supremum of the sequence, denoted by

$$\lim_{n\to\infty} \sup t_n$$

is the smallest real number L having the property that for any $\varepsilon > 0$, there are at most finitely many terms in the sequence that are larger than $L + \varepsilon$. If there is no such L then

$$\lim_{n\to\infty} \sup t_n = \infty .$$

Limit Supremum

Example: Consider a sequence

$$\{t_n\} = \{1,2,3,1,2,3,1,2,3,1,2,3, \dots\}$$

Limit Supremum

Example: Consider a sequence

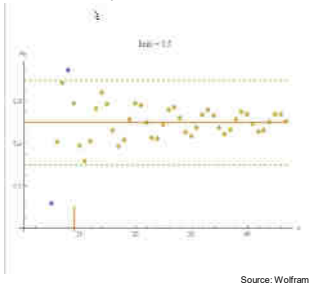
$$\{t_n\} = \{1,2,3,4,5,6, \dots\}$$

Limit Supremum

Relation between limit Supremum and limit of a sequence

If a sequence $\{t_n\}$ is convergent and converging to M that is

$$\lim_{n\to\infty} t_n = M$$



Source: Wolfram

Limit Supremum

Relation between limit Supremum and limit of a sequence

If a sequence $\{t_n\}$ is convergent and converging to M that is

$$\lim_{n\to\infty} t_n = M$$

then

$$\lim_{n\to\infty} \sup t_n = \mathrm{M}$$

Limit Supremum

Root Test
Part I

- **Limit supremum**

Root Test

Root Test
Part II

Topic No. 58

Source: google

Root Test

To understand
Root Test
we need to understand
the concept of
limit supremum
for sequences of real
numbers

Root Test

Definition (Limit Supremum)

Let $\{t_n\}$ be a sequence of positive real numbers. The limit supremum of the sequence, denoted by

$$\limsup_{n\to\infty} t_n$$

is the smallest real number L having the property that for any $\varepsilon > 0$, there are at most finitely many terms in the sequence that are larger than $L + \varepsilon$. If there is no such L then

$$\limsup_{n\to\infty} t_n = \infty .$$

Root Test

Theorem (Root test): Suppose the series $\sum_{n=0}^{\infty} \zeta_n$ has

$$\limsup_{n\to\infty} |\zeta_n|^{\frac{1}{n}} = L$$

Then the series

- absolutely convergent if $L < 1$
- divergent if $L > 1$

Root Test

Example: Check whether the following series converges or diverges

$$\sum_{n=0}^{\infty} \left(\frac{1+i}{2}\right)^n$$

Solution:

Root Test

Theorem (Root test): Suppose the series $\sum_{n=0}^{\infty} \zeta_n$ has

$$\limsup_{n\to\infty} |\zeta_n|^{\frac{1}{n}} = L$$

Then the series

- absolutely convergent if $L < 1$
- divergent if $L > 1$

Proof:

Root Test

Proof: Assume that

$$\limsup_{n\to\infty} |\zeta_n|^{\frac{1}{n}} = L$$

and $L < 1$.

Root Test

Proof: Assume that

$$\limsup_{n\to\infty} |\zeta_n|^{\frac{1}{n}} = L$$

and $L < 1$.

To show: the series $\sum_{n=0}^{\infty} \zeta_n$ converges absolutely.

Root Test

Proof: Assume that

$$\limsup_{n\to\infty} |\zeta_n|^{\frac{1}{n}} = L$$

and $L < 1$.

To show: the series $\sum_{n=0}^{\infty} \zeta_n$ converges absolutely.

We can find r such that

$$L < r < 1$$

Root Test

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To show: the series $\sum_{n=0}^{\infty} \zeta_n$ converges absolutely.
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By definition of limit supremum there are only finitely many terms of the sequence $\left\{ |\zeta_n|^{\frac{1}{n}} \right\}$ greater than r.

Root Test

Proof: Assume that

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To show: the series $\sum_{n=0}^{\infty} \zeta_n$ converges absolutely.
We can find r such that

$$L < r < 1$$

By definition of limit supremum there are only finitely many terms of the sequence $\left\{ |\zeta_n|^{\frac{1}{n}} \right\}$ greater than r. That is there is some integer N such that

$$|\zeta_n|^{\frac{1}{n}} < r \quad \text{for n} \geq N$$

Root Test

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$$\limsup_{n\to\infty} |\zeta_n|^{\frac{1}{n}} = L$$

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$$|\zeta_n|^{\frac{1}{n}} < r \quad \text{for n} \geq N$$

So

$$|\zeta_n| < r^{\text{n}} \quad \text{for n} \geq N$$

Root Test

Theorem: If $|z| < 1$, the series $\sum_{n=0}^{\infty} z^n$ converges to $f(z) = \frac{1}{1-z}$. That is, if $|z| < 1$, then

$$\sum_{n=0}^{\infty} z^n = 1 + z + z^2 + z^3 + \cdots z^n + \cdots = \frac{1}{1-z}$$

If $|z| \geq 1$ then the series diverges.

Root Test

Proof: The geometric series $\sum_{n=0}^{\infty} r^n$ converges since $r < 1$.

Root Test

Proof: The geometric series $\sum_{n=0}^{\infty} r^n$ converges since $r < 1$.
Since

$$|\zeta_n| < r^{\text{n}} \quad \text{for n} \geq N$$

So by comparison test the series

$$\sum_{n=N+1}^{\infty} |\zeta_n|$$

converges and does the following

$$\sum_{n=0}^{\infty} |\zeta_n|$$

Root Test

Proof: Since absolute convergence implies convergence so the following series converges

$$\sum_{n=N+1}^{\infty} \zeta_n$$

Root Test

Proof: Assume that

$$\limsup_{n\to\infty} |\zeta_n|^{\frac{1}{n}} = L$$

and $L > 1$.

Root Test

Proof: Assume that

$$\lim_{n\to\infty} \sup\ |\zeta_n|^{\frac{1}{n}} = L$$

and $L > 1$.
To show: the series $\sum_{n=0}^{\infty} \zeta_n$ diverges.

Root Test

Proof: Assume that

$$\lim_{n\to\infty} \sup\ |\zeta_n|^{\frac{1}{n}} = L$$

and $L > 1$.
To show: the series $\sum_{n=0}^{\infty} \zeta_n$ diverges.
Since $1 < L$ so there exist r such that

$$1 < r < L$$

Root Test

Proof: Assume that

$$\lim_{n\to\infty} \sup\ |\zeta_n|^{\frac{1}{n}} = L$$

and $L > 1$.
To show: the series $\sum_{n=0}^{\infty} \zeta_n$ diverges.
Since $1 < L$ so there exist r such that

$$1 < r < L$$

By definition of limit supremum there exist N such that

$$|\zeta_n|^{\frac{1}{n}} > r \ \ for\ n \geq N$$

Root Test

Proof: Assume that

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Which implies

$$|\zeta_n| > r^n \ \ for\ n \geq N$$

As $r > 1$ so

$$\lim_{n\to\infty} |\zeta_n| \neq 0$$

Root Test

Proof: As $r > 1$ so

$$\lim_{n\to\infty} |\zeta_n| \neq 0$$

Hence

$$\lim_{n\to\infty} \zeta_n \neq 0$$

Root Test

Proof: As $r > 1$ so

$$\lim_{n\to\infty} |\zeta_n| \neq 0$$

Hence

$$\lim_{n\to\infty} \zeta_n \neq 0$$

Which implies

$$\sum_{n=0}^{\infty} \zeta_n$$

diverges.

Root Test

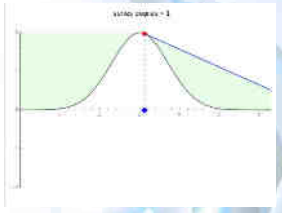
Root Test
Part II

- **Root test**
- **Proof of root test**
- **Examples**

Power Series Functions

Power Series Functions
Part I

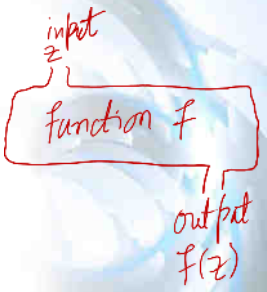
Topic No. 59



Source: Wolfram demonstration

Power Series Functions

What is a function?



Power Series Functions

How can a series be used to define a function?

Source: Google

Power Series Functions

An infinite series an expression of the form

$$\sum_{n=1}^{\infty} a_n$$

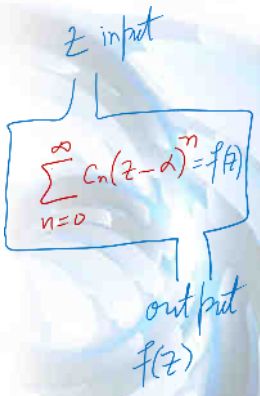
Power Series Functions

Consider an infinite series of the form

$$\sum_{n=0}^{\infty} c_n\,(z-\alpha)^n$$

where c_n and α are fixed complex numbers.

Power Series Functions



Power Series Functions

Warning

The series

$$\sum_{n=0}^{\infty} c_n\,(z-\alpha)^n$$

defines a function only if it converges

Power Series Functions

Power Series Function

is given by

$$f(z) = \sum_{n=0}^{\infty} c_n\,(z-\alpha)^n$$

Power Series Functions

Power Series Function

is given by

$$f(z) = \sum_{n=0}^{\infty} c_n\,(z-\alpha)^n$$

Domain= collection of all points z for which the series converges.

Power Series Functions

Example

$$\alpha = 3, c_n = n!$$

$$f(z) = \sum_{n=0}^{\infty} n!\,(z-3)^n$$

Power Series Functions

Example

$$\alpha = 0\; c_n = 1$$

$$f(z) = \sum_{n=0}^{\infty} (z)^n$$

Power Series Functions

Notation

$$\sum_{n=0}^{\infty} c_n\,(z-\alpha)^n$$
$$= c_0(z-\alpha)^0 + c_1(z-\alpha)^1 + \cdots + c_n(z-\alpha)^n + \cdots$$

Power Series Functions

Notation

$$\sum_{n=0}^{\infty} c_n\,(z-\alpha)^n$$
$$= c_0(z-\alpha)^0 + c_1(z-\alpha)^1 + \cdots + c_n(z-\alpha)^n + \cdots$$

But

$$0^0$$

is undefined.

Power Series Functions

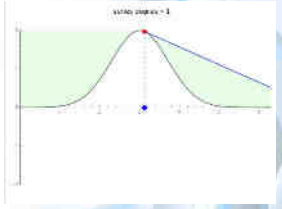
Notation

So we agree on the following convention

$$\sum_{n=0}^{\infty} c_n\,(z-\alpha)^n$$
$$= c_0 + c_1(z-\alpha)^1 + \cdots + c_n(z-\alpha)^n + \cdots$$

Power Series Functions

Power Series Functions
Part I

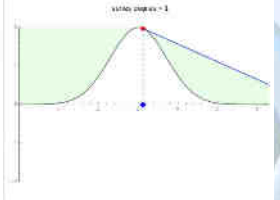


Source: Wolfram demonstration

Power Series Functions

Power Series Functions
Part II

Topic No. 60



Source: Wolfram demonstration

Power Series Functions

Power Series Function

is given by

$$f(z) = \sum_{n=0}^{\infty} c_n (z-\alpha)^n$$

Domain= collection of all points z for which the series converges.

Power Series Functions

Notation

We agree on the following convention

$$\sum_{n=0}^{\infty} c_n (z-\alpha)^n = c_0 + c_1(z-\alpha)^1 + \cdots + c_n(z-\alpha)^n + \cdots$$

Power Series Functions

Theorem: Suppose that $f(z) = \sum_{n=0}^{\infty} c_n (z-\alpha)^n$. Then the set of points z for which the series converges is one of the following:

1. the single point $z = \alpha$

Power Series Functions

Theorem: Suppose that $f(z) = \sum_{n=0}^{\infty} c_n (z-\alpha)^n$. Then the set of points z for which the series converges is one of the following:

1. the single point $z = \alpha$
2. the disk $D_\rho(\alpha) = \{z: |z-\alpha| < \rho\}$, along with part (either none, some or all) of the circle $C_\rho = \{z: |z-\alpha| = \rho\}$

Power Series Functions

Theorem: Suppose that $f(z) = \sum_{n=0}^{\infty} c_n (z-\alpha)^n$. Then the set of points z for which the series converges is one of the following:

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3. the entire complex plane

Power Series Functions

Theorem: Suppose that $f(z) = \sum_{n=0}^{\infty} c_n (z-\alpha)^n$. Then the set of points z for which the series converges is one of the following:

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3. the entire complex plane

Proof:

Power Series Functions

Recall

Theorem (Root test): Suppose the series $\sum_{n=0}^{\infty} \zeta_n$ has

$$\lim_{n\to\infty} \sup |\zeta_n|^{\frac{1}{n}} = L$$

Then the series

- absolutely convergent if $L < 1$
- divergent if $L > 1$

Power Series Functions

Proof: For the series

$$\sum_{n=0}^{\infty} c_n (z-\alpha)^n$$

Power Series Functions

Proof: For the series

$$\sum_{n=0}^{\infty} c_n\,(z-\alpha)^n$$

we need to calculate the following limit

$$\limsup_{n\to\infty} |\zeta_n|^{\frac{1}{n}}\,.$$

were

$$\zeta_n = c_n\,(z-\alpha)^n$$

Power Series Functions

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were

$$\zeta_n = c_n\,(z-\alpha)^n$$

Hence the limit is

$$\limsup_{n\to\infty} |c_n\,(z-\alpha)^n|^{\frac{1}{n}}$$

Power Series Functions

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$$\sum_{n=0}^{\infty} c_n\,(z-\alpha)^n$$

we need to calculate the following limit

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were

$$\zeta_n = c_n\,(z-\alpha)^n$$

Hence the limit is

$$\limsup_{n\to\infty} |c_n\,(z-\alpha)^n|^{\frac{1}{n}}$$

$$= |z-\alpha| \limsup_{n\to\infty} |c_n|^{\frac{1}{n}}$$

Power Series Functions

There are three possibilities to consider for the values of

$$\limsup_{n\to\infty} |c_n|^{\frac{1}{n}}$$

Power Series Functions

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1. The first possibility; if

$$\limsup_{n\to\infty} |c_n|^{\frac{1}{n}} = \infty$$

Power Series Functions

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then the series $\sum_{n=0}^{\infty} c_n\,(z-\alpha)^n$ diverges for all $z \neq \alpha$.

Power Series Functions

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$$|z-\alpha| \limsup_{n\to\infty} |c_n|^{\frac{1}{n}} < 1$$

holds for $z = \alpha$.

Power Series Functions

There are three possibilities to consider for the values of

$$\limsup_{n\to\infty} |c_n|^{\frac{1}{n}}$$

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then the series $\sum_{n=0}^{\infty} c_n\,(z-\alpha)^n$ diverges for all $z \neq \alpha$. But the inequality

$$|z-\alpha| \limsup_{n\to\infty} |c_n|^{\frac{1}{n}} < 1$$

holds for $z = \alpha$.

This gives the first case, that is,

"the single point $z = \alpha$"

Power Series Functions

2. The second possibility; if

$$0 < \limsup_{n\to\infty} |c_n|^{\frac{1}{n}} < \infty$$

Power Series Functions

2. The second possibility; if

$$0 < \limsup_{n\to\infty} |c_n|^{\frac{1}{n}} < \infty$$

According to the root test the series $\sum_{n=0}^{\infty} c_n (z-\alpha)^n$ converges for all z satisfying

$$|z-\alpha| \limsup_{n\to\infty} |c_n|^{\frac{1}{n}} < 1$$

Power Series Functions

2. The second possibility; if

$$0 < \limsup_{n\to\infty} |c_n|^{\frac{1}{n}} < \infty$$

According to the root test the series $\sum_{n=0}^{\infty} c_n (z-\alpha)^n$ converges for all z satisfying

$$|z-\alpha| \limsup_{n\to\infty} |c_n|^{\frac{1}{n}} < 1$$

Since $\boldsymbol{\limsup_{n\to\infty}} |c_n|^{\frac{1}{n}}$ is a finite number so we get the following condition on z

$$|z-\alpha| < \frac{1}{\limsup_{n\to\infty} |c_n|^{\frac{1}{n}}}$$

Power Series Functions

Hence the series $\sum_{n=0}^{\infty} c_n (z-\alpha)^n$ converges for all z satisfying

$$|z-\alpha| < \frac{1}{\limsup_{n\to\infty} |c_n|^{\frac{1}{n}}}$$

Power Series Functions

Hence the series $\sum_{n=0}^{\infty} c_n (z-\alpha)^n$ converges for all z satisfying

$$|z-\alpha| < \frac{1}{\limsup_{n\to\infty} |c_n|^{\frac{1}{n}}}$$

OR

for all $z \in D_\rho(\alpha)$, where

$$D_\rho(\alpha) = \{z: |z-\alpha| < \rho\}$$

and

$$\rho = \frac{1}{\limsup_{n\to\infty} |c_n|^{\frac{1}{n}}}.$$

Power Series Functions

Since the root test does not say anything about the points of C_ρ, where

$$C_\rho = \{z: |z-\alpha| = \rho\}$$

So the series may converge for none, some, or for all points of C_ρ.

Power Series Functions

This gives the second case, that is,

"the disk $D_\rho(\alpha) = \{z: |z-\alpha| < \rho\}$, along with part (either none, some or all) of the circle $C_\rho = \{z: |z-\alpha| = \rho\}$"

Power Series Functions

2. The third possibility; if

$$\limsup_{n\to\infty} |c_n|^{\frac{1}{n}} = 0$$

Power Series Functions

2. The third possibility; if

$$\lim_{n\to\infty} \sup\ |c_n|^{\frac{1}{n}} = 0$$

According to the root test the series $\sum_{n=0}^{\infty} c_n\,(z-\alpha)^n$ converges for all z satisfying

$$|z-\alpha| \lim_{n\to\infty} \sup\ |c_n|^{\frac{1}{n}} < 1$$

Power Series Functions

2. The third possibility; if

$$\lim_{n\to\infty} \sup\ |c_n|^{\frac{1}{n}} = 0$$

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$$|z-\alpha| \lim_{n\to\infty} \sup\ |c_n|^{\frac{1}{n}} < 1$$

Since $\boldsymbol{\lim_{n\to\infty} \sup\ |c_n|^{\frac{1}{n}} = 0}$ so the following is satisfied for all complex numbers

$$|z-\alpha| = 0 < 1$$

Power Series Functions

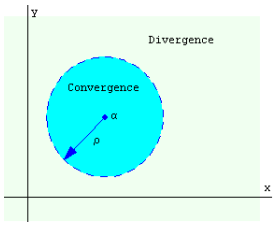
This proves the third case, that is,

"the entire complex plane"

Power Series Functions

Theorem: Suppose that $f(z) = \sum_{n=0}^{\infty} c_n\,(z-\alpha)^n$. Then the set of points z for which the series converges is one of the following:

1. the single point $z = \alpha$
2. the disk $D_\rho(\alpha) = \{z: |z-\alpha| < \rho\}$, along with part (either none, some or all) of the circle $C_\rho = \{z: |z-\alpha| = \rho\}$
3. the entire complex plane



Source: Complex analysis by Mathews

Power Series Functions

Definition: The number ρ such that the series

$$\sum_{n=0}^{\infty} c_n\,(z-\alpha)^n$$

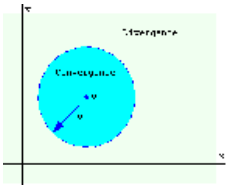
converges for all z satisfying

$$|z-\alpha| < \rho$$

and diverges for all z satisfying

$$|z-\alpha| > \rho$$

is called radius of convergence.



Source: Complex analysis by Mathews

Power Series Functions

If the series

$$\sum_{n=0}^{\infty} c_n\,(z-\alpha)^n$$

converges for

"the single point $z = \alpha$"

the radius of convergence is 0.

Power Series Functions

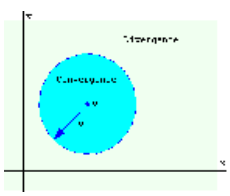
If the series

$$\sum_{n=0}^{\infty} c_n\,(z-\alpha)^n$$

converges for

"the disk $D_\rho(\alpha) = \{z: |z-\alpha| < \rho\}$"

the radius of convergence is ρ.



Source: Complex analysis by Mathews

Power Series Functions

If the series

$$\sum_{n=0}^{\infty} c_n\,(z-\alpha)^n$$

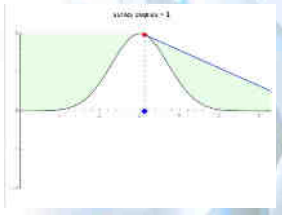
converges for

"the entire complex plane"

the radius of convergence is ∞.

Power Series Functions

Power Series Functions
Part II

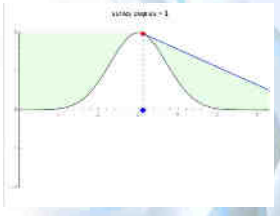


Source: Wolfram demonstration

Power Series Functions

Power Series Functions
Part III

Topic No. 61



Source: Wolfram demonstration

Power Series Functions

Power Series Function
is given by

$$f(z) = \sum_{n=0}^{\infty} c_n (z-\alpha)^n$$

Domain= collection of all points z for which the series converges.

Power Series Functions

Theorem: Suppose that $f(z) = \sum_{n=0}^{\infty} c_n (z-\alpha)^n$. Then the set of points z for which the series converges is one of the following:

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Power Series Functions

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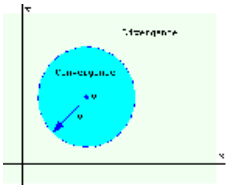

Source: Complex analysis by Mathews

Power Series Functions

Question: How to find radius of convergence?

Source: internet

Power Series Functions

Theorem: For the power series function

$$f(z) = \sum_{n=0}^{\infty} c_n (z-\alpha)^n$$

We can find its radius of convergence, denoted by, ρ using any of the following methods

1. Cauchy's root test

$$\rho = \frac{1}{\lim_{n\to\infty} |c_n|^{\frac{1}{n}}}$$

(provided the limit exist)

Power Series Functions

Theorem: For the power series function

$$f(z) = \sum_{n=0}^{\infty} c_n (z-\alpha)^n$$

We can find its radius of convergence, denoted by, ρ using any of the following methods

2. Cauchy-Hadamard root test

$$\rho = \frac{1}{\lim_{n\to\infty} sup|c_n|^{\frac{1}{n}}}$$

(the limit always exist)

Power Series Functions

Theorem: For the power series function

$$f(z) = \sum_{n=0}^{\infty} c_n\,(z-\alpha)^n$$

We can find its radius of convergence, denoted by, ρ using any of the following methods

2. d'Alembert's ratio test

$$\rho = \frac{1}{\lim\limits_{n\to\infty}\left|\frac{c_{n+1}}{c_n}\right|}$$

(provided the limit exist)

Power Series Functions

Example: Find radius of convergence of the following series

$$\sum_{n=0}^{\infty}\left(\frac{n+2}{3n+1}\right)^n (z-4)^n$$

Power Series Functions

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$$\rho = \frac{1}{\lim\limits_{n\to\infty}|c_n|^{\frac{1}{n}}}$$

Power Series Functions

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Solution: According to Cauchy's root test, the radius of convergence is

$$\rho = \frac{1}{\lim\limits_{n\to\infty}|c_n|^{\frac{1}{n}}}$$

So we calculate the following limit

$$\lim_{n\to\infty}|c_n|^{\frac{1}{n}}$$

where $c_n = \left(\frac{n+2}{3n+1}\right)^n$

Power Series Functions

Solution (cont) :

$$\lim_{n\to\infty}|c_n|^{\frac{1}{n}}$$

Power Series Functions

Solution (cont) :

$$\lim_{n\to\infty}|c_n|^{\frac{1}{n}}$$

$$= \lim_{n\to\infty}\left|\left(\frac{n+2}{3n+1}\right)^n\right|^{\frac{1}{n}}$$

Power Series Functions

Solution (cont) :

$$\lim_{n\to\infty}|c_n|^{\frac{1}{n}}$$

$$= \lim_{n\to\infty}\left|\left(\frac{n+2}{3n+1}\right)^n\right|^{\frac{1}{n}}$$

$$= \lim_{n\to\infty}\frac{1+\frac{2}{n}}{3+\frac{1}{n}}$$

Power Series Functions

Solution (cont) :

$$\lim_{n\to\infty}|c_n|^{\frac{1}{n}}$$

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$$= \frac{1}{3}$$

Power Series Functions

Solution (cont) :

$$\lim_{n\to\infty}|c_n|^{\frac{1}{n}}$$
$$= \lim_{n\to\infty}\left(\frac{n+2}{3n+1}\right)^n$$
$$= \lim_{n\to\infty}\frac{1+\frac{2}{n}}{3+\frac{1}{n}}$$
$$= \frac{1}{3}$$

Hence radius of convergence is

$$\rho = \frac{1}{1/3} = 3$$

Power Series Functions

Example: Find radius of convergence of the following series

$$\sum_{n=1}^{\infty} c_n z^n = 4z + 5^2 z^2 + 4^3 z^3 + 5^4 z^4 + \cdots$$

Solution: In this case

$$\left\{|c_n|^{\frac{1}{n}}\right\} = \{4,5,4,5,4,5,4, \dots\}$$

Power Series Functions

Example: Find radius of convergence of the following series

$$\sum_{n=1}^{\infty} c_n z^n = 4z + 5^2 z^2 + 4^3 z^3 + 5^4 z^4 + \cdots$$

Solution: In this case

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So the limit

$$\lim_{n\to\infty}|c_n|^{\frac{1}{n}}$$

does not exist.

Power Series Functions

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So the limit

$$\lim_{n\to\infty}|c_n|^{\frac{1}{n}}$$

does not exist.

Hence we use Cauchy-Hadamard root test.

Power Series Functions

Solution(cont.) : In this case

$$\left\{|c_n|^{\frac{1}{n}}\right\} = \{4,5,4,5,4,5,4, \dots\}$$

Power Series Functions

Solution(cont.) : In this case

$$\left\{|c_n|^{\frac{1}{n}}\right\} = \{4,5,4,5,4,5,4, \dots\}$$

So the limit

$$\lim_{n\to\infty}\sup|c_n|^{\frac{1}{n}} = 5\,.$$

Power Series Functions

Solution(cont.) : In this case

$$\left\{|c_n|^{\frac{1}{n}}\right\} = \{4,5,4,5,4,5,4, \dots\}$$

So the limit

$$\lim_{n\to\infty}\sup|c_n|^{\frac{1}{n}} = 5\,.$$

Hence radius of convergence is

$$\rho = \frac{1}{5}\,.$$

Power Series Functions

Example: Find radius of convergence of the following series

$$\sum_{n=1}^{\infty} \frac{1}{n!} z^n$$

Power Series Functions

Example: Find radius of convergence of the following series

$$\sum_{n=1}^{\infty} \frac{1}{n!} z^n$$

Solution: According to d'Alembert's ratio test, the radius of convergence is

$$\rho = \frac{1}{\lim_{n\to\infty} \left|\frac{c_{n+1}}{c_n}\right|}$$

where, in this example,

$$c_n = \frac{1}{n!}$$

Power Series Functions

Solution (cont.) :We calculate the following limit

$$\lim_{n\to\infty} \left|\frac{c_{n+1}}{c_n}\right|$$

Power Series Functions

Solution (cont.) :We calculate the following limit

$$\lim_{n\to\infty} \left|\frac{c_{n+1}}{c_n}\right|$$
$$= \lim_{n\to\infty} \left|\frac{1/(n+1)!}{1/n!}\right|$$

Power Series Functions

Solution (cont.) :We calculate the following limit

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$$= \lim_{n\to\infty} \left|\frac{1/(n+1)!}{1/n!}\right|$$
$$= \lim_{n\to\infty} \left|\frac{1}{n+1}\right|$$

Power Series Functions

Solution (cont.) :We calculate the following limit

$$\lim_{n\to\infty} \left|\frac{c_{n+1}}{c_n}\right|$$
$$= \lim_{n\to\infty} \left|\frac{1/(n+1)!}{1/n!}\right|$$
$$= \lim_{n\to\infty} \left|\frac{1}{n+1}\right|$$
$$= 0$$

Power Series Functions

Solution (cont.) :We calculate the following limit

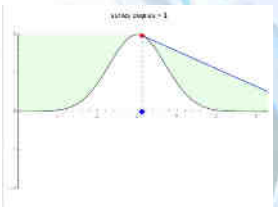
$$\lim_{n\to\infty} \left|\frac{c_{n+1}}{c_n}\right|$$
$$= \lim_{n\to\infty} \left|\frac{1/(n+1)!}{1/n!}\right|$$
$$= \lim_{n\to\infty} \left|\frac{1}{n+1}\right|$$
$$= 0$$

So radius of convergence is

$$\rho = \infty$$

Power Series Functions

Power Series Functions
Part III



Source: Wolfram demonstration

Power Series Functions

Topic No. 62

Power Series Functions
Term by term differentiation
Part I

Power Series Functions

Power Series Function

is given by

$$f(z) = \sum_{n=0}^{\infty} c_n\,(z-\alpha)^n$$

Domain= collection of all points z for which the series converges.

Power Series Functions

Definition: The number ρ such that the series

$$\sum_{n=0}^{\infty} c_n\,(z-\alpha)^n$$

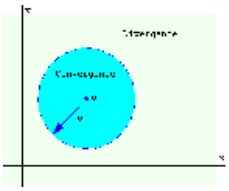
converges for all z satisfying

$$|z-\alpha| < \rho$$

and diverges for all z satisfying

$$|z-\alpha| > \rho$$

is called radius of convergence.



Source: Complex analysis by Mathews

Power Series Functions

Questions

When can we differentiate the power series function?

Source: internet

Power Series Functions

Questions

When can we differentiate the power series function?

What is the derivative, if exists?

Source: internet

Power Series Functions

Notation

For a function $f(z)$ we denote by $f^{(k)}$ the kth derivative of $f(z)$

Power Series Functions

Notation

For a function $f(z)$ we denote by $f^{(k)}$ the kth derivative of $f(z)$

When $k = 0$, $f^{(k)}$ denotes the function itself, that is,

$$f^{(0)}(z) = f(z)$$

Power Series Functions

Theorem: Suppose that the function

$$f(z) = \sum_{n=0}^{\infty} c_n\,(z-\alpha)^n$$

has radius of convergence $\rho > 0$. Then

(i). $f(z)$ is infinitely differentiable for all $z \in D_\rho(\alpha)$. In fact

(ii). for all k

$$f^{(k)}(z) = \sum_{n=k}^{\infty} n(n-1)(n-2)\ldots(n-k+1)\,c_n\,(z-\alpha)^{n-k}$$

and

(iii). $c_k = \frac{f^{(k)}(\alpha)}{k!}$

Power Series Functions

Proof: We first prove the second part (ii), that is,

" for all k

$$f^{(k)}(z) = \sum_{n=k}^{\infty} n(n-1)(n-2)\dots(n-k+1)\, c_n\,(z-\alpha)^{n-k}$$

"

In the **first step** we prove the above statement for $k = 1$. In the **second step** we prove it for any k.

Power Series Functions

Proof (ii): **First step $k = 1$**

We define

$$g(z) = \sum_{n=1}^{\infty} n\, c_n (z-\alpha)^{n-1}$$

Power Series Functions

Proof (ii): **First step $k = 1$**

We define

$$g(z) = \sum_{n=1}^{\infty} n\, c_n (z-\alpha)^{n-1}$$

$$S_j(z) = \sum_{n=0}^{j} c_n (z-\alpha)^n \quad R_j = \sum_{n=j+1}^{\infty} c_n (z-\alpha)^n$$

Here

- $S_j(z)$ is simply the $(j+1)$st partial sum of the series $f(z)$,
- $R_j(z)$ is the sum of the remaining terms of that series.

Power Series Functions

Proof (ii) (cont.):

The radius of convergence of $g(z) = \sum_{n=1}^{\infty} n\, c_n (z-\alpha)^{n-1}$ is ρ.

Power Series Functions

Recall

Theorem: For the power series function

$$f(z) = \sum_{n=0}^{\infty} c_n\,(z-\alpha)^n$$

We can find its radius of convergence, denoted by, ρ using d'Alembert's ratio test as follows

$$\rho = \frac{1}{\lim\limits_{n\to\infty} \left|\frac{c_{n+1}}{c_n}\right|}$$

(provided the limit exist)

Power Series Functions

Proof (ii) (cont.):

The radius of convergence of $g(z) = \sum_{n=1}^{\infty} n\, c_n (z-\alpha)^{n-1}$ is ρ.

Power Series Functions

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Power Series Functions

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Power Series Functions

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Power Series Functions

Proof (ii) (cont.): For fixed $z_0 \in D_\rho(\alpha)$ we show that

$$f'(z_0) = g(z_0)$$

Power Series Functions

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$$f'(z_0) = g(z_0)$$

That is, we want to show that

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0} = g(z_0)$$

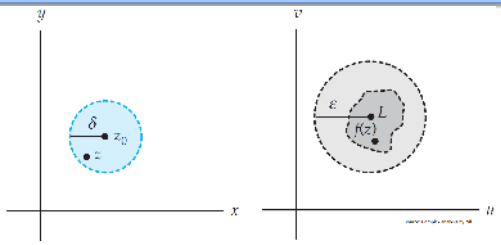
Power Series Functions

Recall the definition of limit

$$\lim_{z \to z_0} f(z) = L$$

is

"For every $\varepsilon > 0$ there exists a $\delta > 0$ such that $|f(z) - L| < \varepsilon$ whenever $0 < |z - z_0| < \delta$."



Power Series Functions

Proof (ii) (cont.): For fixed $z_0 \in D_\rho(\alpha)$ we show that

$$f'(z_0) = g(z_0)$$

That is, we want to show that

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0} = g(z_0)$$

Hence we need to show the following:

For every $\varepsilon > 0$ there exists a $\delta > 0$ such that

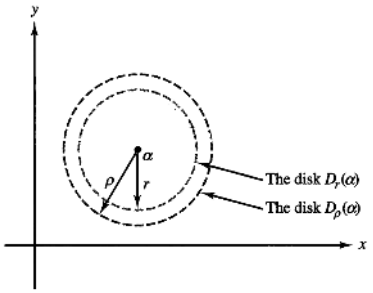
$$\left| \frac{f(z) - f(z_0)}{z - z_0} - g(z_0) \right| < \varepsilon$$

whenever $0 < |z - z_0| < \delta$ and $z \in D_\rho(\alpha)$

Power Series Functions

Proof (ii) (cont.): Let $\varepsilon > 0$ and $z_0 \in D_\rho(\alpha)$.

Choose $r < \rho$ such that $z_0 \in D_r(\alpha)$

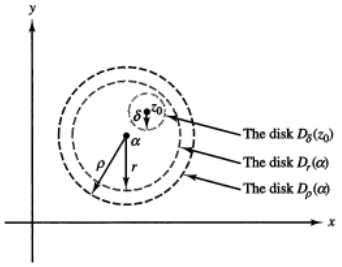


Power Series Functions

Proof (ii) (cont.): Let $\varepsilon > 0$ and $z_0 \in D_\rho(\alpha)$.

Choose $r < \rho$ such that $z_0 \in D_r(\alpha)$. We want to find δ small enough such that

$$D_\delta(\alpha) \subset D_r(\alpha) \subset D_\rho(\alpha)$$



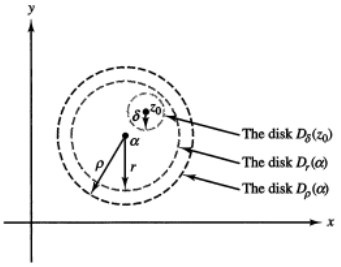
Power Series Functions

Proof (ii) (cont.): Let $\varepsilon > 0$ and $z_0 \in D_\rho(\alpha)$.

Choose $r < \rho$ such that $z_0 \in D_r(\alpha)$. We want to find δ small enough such that

$$D_\delta(\alpha) \subset D_r(\alpha) \subset D_\rho(\alpha)$$

and other conditions that we impose during the proof.



Power Series Functions

Proof (ii) (cont.): Since $f(z) = S_j(z) + R_j(z)$, so

$$\left[\frac{f(z) - f(z_0)}{z - z_0} - g(z_0)\right]$$

Power Series Functions

Proof (ii) (cont.): Since $f(z) = S_j(z) + R_j(z)$, so

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Power Series Functions

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Power Series Functions

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$$\left|\frac{f(z) - f(z_0)}{z - z_0} - g(z_0)\right|$$

Power Series Functions

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Power Series Functions

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Power Series Functions

Proof (ii) (cont.):

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Power Series Functions

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Power Series Functions

Proof (ii) (cont.):

$$\left| \frac{f(z) - f(z_0)}{z - z_0} - g(z_0) \right|$$

Power Series Functions

Proof
To be continued

Power Series Functions

Proof
Term by term differentiation

Power Series Functions

Theorem: Suppose that the function

$$f(z) = \sum_{n=0}^{\infty} c_n (z - \alpha)^n$$

has radius of convergence $\rho > 0$. Then
(i). $f(z)$ is infinitely differentiable for all $z \in D_\rho(\alpha)$. In fact
(ii). for all k

$$f^{(k)}(z) = \sum_{n=k}^{\infty} n(n-1)(n-2)\dots(n-k+1)\, c_n (z - \alpha)^{n-k}$$

and

(iii). $c_k = \frac{f^{(k)}(\alpha)}{k!}$

Power Series Functions

Proof: We first prove the second part (ii), that is,

" for all k

$$\boldsymbol{f^{(k)}(z) = \sum_{n=k}^{\infty} n(n-1)(n-2)\dots(n-k+1)\, c_n (z - \alpha)^{n-k}}$$

"

In the **first step** we prove the above statement for $k = 1$. In the **second step** we prove it for any k.

Power Series Functions

Proof (ii): **First step $k = 1$**

We define

$$g(z) = \sum_{n=1}^{\infty} n\, c_n (z-\alpha)^{n-1}$$

Power Series Functions

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Here

- $S_j(z)$ is simply the $(j+1)$st partial sum of the series $f(z)$,
- $R_j(z)$ is the sum of the remaining terms of that series.

Power Series Functions

Proof (ii) (cont.): For fixed $z_0 \in D_\rho(\alpha)$ we show that

$$f'(z_0) = g(z_0)$$

That is, we want to show that

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0} = g(z_0)$$

Power Series Functions

Proof (ii) (cont.): For fixed $z_0 \in D_\rho(\alpha)$ we show that

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That is, we want to show that

$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0} = g(z_0)$$

Hence we need to show the following:

For every $\varepsilon > 0$ there exists a $\delta > 0$ such that

$\left| \frac{f(z)-f(z_0)}{z-z_0} - g(z_0) \right| < \varepsilon$

whenever $0 < |z - z_0| < \delta$ and $z \in D_\rho(\alpha)$

Power Series Functions

Proof (ii) (cont.):

$$\left| \frac{f(z) - f(z_0)}{z - z_0} - g(z_0) \right|$$

Power Series Functions

Proof (ii) (cont.):

Given

$$g(z) = \sum_{n=1}^{\infty} n\, c_n (z-\alpha)^{n-1} \textbf{ and } S_j(z) = \sum_{n=0}^{j} c_n (z-\alpha)^n$$

Power Series Functions

Proof (ii) (cont.):

Given

$$g(z) = \sum_{n=1}^{\infty} n\, c_n (z-\alpha)^{n-1} \textbf{ and } S_j(z) = \sum_{n=0}^{j} c_n (z-\alpha)^n$$

Since $S_j(z)$ is a polynomial so

$$S_j'(z) = \sum_{n=1}^{j} n\, c_n (z-\alpha)^{n-1}$$

Power Series Functions

Proof (ii) (cont.):

Given

$$g(z) = \sum_{n=1}^{\infty} n\, c_n (z-\alpha)^{n-1} \textbf{ and } S_j(z) = \sum_{n=0}^{j} c_n (z-\alpha)^n$$

Since $S_j(z)$ is a polynomial so

$$S_j'(z) = \sum_{n=1}^{j} n\, c_n (z-\alpha)^{n-1}$$

Hence

$$\lim_{j \to \infty} S_j'(z) = g(z)$$

Power Series Functions

Proof (ii) (cont.):

Given

$$g(z) = \sum_{n=1}^{\infty} n\, c_n (z-\alpha)^{n-1} \textbf{ and } S_j(z) = \sum_{n=0}^{j} c_n (z-\alpha)^n$$

Since $S_j(z)$ is a polynomial so

$$S_j'(z) = \sum_{n=1}^{j} n\, c_n (z-\alpha)^{n-1}$$

Hence

$$\lim_{j\to\infty} S_j'(z) = g(z)$$

which implies

$$\lim_{j\to\infty} S_j'(z_0) = g(z_0)$$

Power Series Functions

Proof (ii) (cont.):

Using definition of limit for the following

$$\lim_{j\to\infty} S_j'(z_0) = g(z_0)$$

Power Series Functions

Proof (ii) (cont.):

Using definition of limit for the following

$$\lim_{j\to\infty} S_j'(z_0) = g(z_0)$$

we get

$\exists\, N_2$ such that $\left|S_j(z_0) - g(z_0)\right| < \frac{\varepsilon}{3}$ for all $j \geq N_2$

Power Series Functions

Proof (ii) (cont.):

Since $S_j(z)$ is a polynomial so it is differentiable.

Power Series Functions

Proof (ii) (cont.):

Since $S_j(z)$ is a polynomial so it is differentiable.

Hence for $\frac{\varepsilon}{3} > 0$ there exist $\delta > 0$ such that

Power Series Functions

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Since $S_j(z)$ is a polynomial so it is differentiable.

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Power Series Functions

Proof (ii) (cont.):

$\exists\, N_2$ such that $\left|S_j(z_0) - g(z_0)\right| < \frac{\varepsilon}{3}$ for all $j \geq N_2$

Power Series Functions

Proof (ii) (cont.):

$$\left|\frac{f(z)-f(z_0)}{z-z_0}-g(z_0)\right|$$

Power Series Functions

Proof (ii) (cont.):

$$\left|\frac{f(z)-f(z_0)}{z-z_0}-g(z_0)\right|$$

Power Series Functions

Proof: So the the following holds for $k=1$

" for all k

$f^{(k)}(z)=\sum_{n=k}^{\infty} n(n-1)(n-2)\dots(n-k+1)\,c_n\,(z-\alpha)^{n-k}$ "

In the **second step** we prove it for any k.

$$f^{(1)}(z)=\sum_{n=1}^{\infty} n\,c_n\,(z-\alpha)^{n-1}$$

$$g(z)=\sum_{n=2}^{\infty} n(n-1)\,c_n\,(z-\alpha)^{n-2}$$

Power Series Functions

Theorem: Suppose that the function

$$f(z)=\sum_{n=0}^{\infty} c_n\,(z-\alpha)^n$$

has radius of convergence $\rho>0$. Then

(ii). for all k

$$f^{(k)}(z)=\sum_{n=k}^{\infty} n(n-1)(n-2)\dots(n-k+1)\,c_n\,(z-\alpha)^{n-k}$$

and

(iii). $c_k=\frac{f^{(k)}(\alpha)}{k!}$

Power Series Functions

Theorem: Suppose that the function

$$f(z)=\sum_{n=0}^{\infty} c_n\,(z-\alpha)^n$$

has radius of convergence $\rho>0$. Then

(i). $f(z)$ is infinitely differentiable for all $z\in D_\rho(\alpha)$. In fact

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and

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Power Series Functions

Power Series Functions
Term by term differentiation
Part I

Power Series Functions

Topic No. 63

Power Series Functions
Term by term differentiation
Part II

Power Series Functions

Power Series Function

is given by

$$f(z)=\sum_{n=0}^{\infty} c_n\,(z-\alpha)^n$$

Domain= collection of all points z for which the series converges.

Power Series Functions

Notation

For a function $f(z)$ we denote by $f^{(k)}$ the kth derivative of $f(z)$
When $k = 0$, $f^{(k)}$ denotes the function itself, that is,

$$f^{(0)}(z) = f(z)$$

Power Series Functions

Theorem: Suppose that the function

$$f(z) = \sum_{n=0}^{\infty} c_n\,(z-\alpha)^n$$

has radius of convergence $\rho > 0$. Then
(i). $f(z)$ is infinitely differentiable for all $z \in D_\rho(\alpha)$. In fact
(ii). for all k

$$f^{(k)}(z) = \sum_{n=k}^{\infty} n(n-1)(n-2)\dots(n-k+1)\,c_n\,(z-\alpha)^{n-k}$$

and
(iii). $c_k = \frac{f^{(k)}(\alpha)}{k!}$

Power Series Functions

Example: Show that

$$\sum_{n=0}^{\infty}(n+1)z^n = \frac{1}{(1-z)^2}$$

For all $z \in D_1(0)$.

Solution:

Power Series Functions

Recall

Theorem: If $|z| < 1$, the series $\sum_{n=0}^{\infty} z^n$ converges to $f(z) = \frac{1}{1-z}$. That is, if $|z| < 1$, then

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If $|z| \geq 1$ then the series diverges.

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Power Series Functions

Example: The Bessel function of order zero is defined as

$$J_0(z) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(n!)^2}\left(\frac{z}{2}\right)^{2n}.$$

Find the derivative $J_0(z)$.

Power Series Functions

Solution: First we need to find radius of convergence of $J_0(z)$.

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Power Series Functions

Solution: Hence the radius of convergence of

$$J_0(z) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(n!)^2}\left(\frac{z}{2}\right)^{2n}$$

is ∞. So we can perform term by term differentiation for any $z \in \mathbb{C}$. Hence

$$J_0(z) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(n!)^2}\left(\frac{z}{2}\right)^{2n} = 1 - \frac{z^2}{2^2} + \frac{z^4}{2^2\,4^2} - \frac{z^6}{2^2 4^2 6^2} + \cdots$$

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$$J_0'(z) = \sum_{n=1}^{\infty} \frac{(-1)^n}{n!\,(n)!}\left(\frac{2n}{2}\right)\left(\frac{z}{2}\right)^{2n-1} = -\frac{z}{2} + \frac{1}{1!\,2!}\left(\frac{z}{2}\right)^3 - \frac{1}{2!\,3!}\left(\frac{z}{2}\right)^5 + \cdots$$

Changing the index, we get

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Also $j_0{}'(z)$ has ∞ as radius convergence.

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The Bessel function of order one, denoted by $J_1(z)$ can be found by solving the following differential equation

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Power Series Functions

Power Series Functions
Term by term differentiation
Part II

Power Series Functions

Topic No. 64

Power Series Functions
Term by term differentiation
Part II

Power Series Functions

Power Series Function
is given by

$$f(z) = \sum_{n=0}^{\infty} c_n\,(z-\alpha)^n$$

Domain= collection of all points z for which the series converges.

Power Series Functions

Notation

For a function $f(z)$ we denote by $f^{(k)}$ the kth derivative of $f(z)$
When $k = 0$, $f^{(k)}$ denotes the function itself, that is,

$$f^{(0)}(z) = f(z)$$

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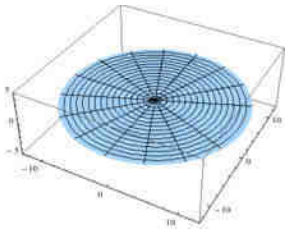

Source: Wikipedia

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Power Series Functions

Power Series Functions
Term by term differentiation
Part II

Elementary Functions

Elementary functions

Topic No. 65

e^x, $\cos x$, $\sin x$
$\log x$, $\tan x$
z^c, $\tanh z$
$\cosh z$, $\sinh z$ . . .

Elementary Functions

Aim

We want to define "complex version" of elementary functions such as

e^x, $\sin x$, $\cos x$,
$\log x$, $\tan x$, *etc*

Elementary Functions

Criteria

1. $\mathbb{R} \subset \mathbb{C}$. So the restriction of these functions to $\mathbb{R}$ must give the corresponding real-valued function on $\mathbb{R}$

Elementary Functions

Criteria

2. The properties of these new functions must correspond to their real counterparts.

$\cos^2 x + \sin^2 x = 1$
$\frac{d}{dx}\cos x = -\sin x$
$\frac{d}{dx} e^x = e^x$
$e^{x_1 + x_2} = e^{x_1} \cdot e^{x_2}$ etc.

Elementary Functions

Strategy

Use expansion of real-valued functions as power series.

Elementary Functions

Strategy

In particular if

$$f(x) = \sum_{n}^{\infty} a_n (x-a)^n$$

where a, a_n are real numbers and x is real variable.

Elementary Functions

Strategy

In particular if

$$f(x) = \sum_{n}^{\infty} a_n (x-a)^n$$

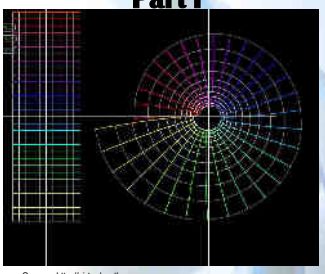
where a, a_n are real numbers and x is real variable. We can extend the domain to $\mathbb{C}$ by defining

$$f(z) = \sum_{n}^{\infty} c_n (z-\alpha)^n$$

Here c_n, α are complex numbers and z is complex variable.

Complex Exponential Function

The Complex exponential function Part I



Source: http://virtualmathmuseum.org

Complex Exponential Function

We know that real exponential function

$$e^x$$

has power series expansion

$$e^x = \sum_{n=0}^{\infty} \frac{1}{n!} x^n$$

Complex Exponential Function

Thus we define complex exponential function e^z (also denoted by $\exp(z)$) as

$$e^z = \exp(z) = \sum_{n=0}^{\infty} \frac{1}{n!} z^n$$

Complex Exponential Function

Our next aim is to prove different properties of complex exponential functions.

Complex Exponential Function

Two key properties that is associated to the associated real exponential functions are:

1. $\frac{d}{dx} e^x = e^x$

Complex Exponential Function

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We want to see if these properties hold for complex exponential functions or not.

Complex Exponential Function

Also we give a proof of Euler formula

$e^{i\theta} = \cos\theta + i\sin\theta$

Complex Exponential Function

Theorem: The function e^z is an entire function satisfying the following conditions:

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Proof:

Complex Exponential Function

Proof ($\frac{d}{dz}e^z = e^z$) : We know that

$$e^z = \exp(z) = \sum_{n=0}^{\infty} \frac{1}{n!} z^n$$

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First we find its radius of convergence.

Complex Exponential Function

Proof ($\frac{d}{dz}e^z = e^z$) : We know that

$$e^z = \exp(z) = \sum_{n=0}^{\infty} \frac{1}{n!} z^n$$

First we find its radius of convergence. In this case

$$c_n = \frac{1}{n!}$$

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$$\lim_{n\to\infty} \left| \frac{c_{n+1}}{c_n} \right| = \lim_{n\to\infty} \left| \frac{\frac{1}{(n+1)!}}{\frac{1}{n!}} \right|$$

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Hence

$$\lim_{n\to\infty} \left| \frac{c_{n+1}}{c_n} \right| = 0$$

Complex Exponential Function

Proof ($\frac{d}{dz}e^z = e^z$) : Hence

$$e^z = \exp(z) = \sum_{n=0}^{\infty} \frac{1}{n!} z^n$$

has infinite radius of convergence.

Complex Exponential Function

Theorem: Suppose that the function

$$f(z) = \sum_{n=0}^{\infty} c_n (z-\alpha)^n$$

has radius of convergence $\rho > 0$. Then

(i). $f(z)$ is infinitely differentiable for all $z \in D_\rho(\alpha)$. In fact

(ii). for all k

$$f^{(k)}(z) = \sum_{n=k}^{\infty} n(n-1)(n-2)\dots(n-k+1)\, c_n (z-\alpha)^{n-k}$$

and

(iii). $c_k = \frac{f^{(k)}(\alpha)}{k!}$

Complex Exponential Function

Proof ($\frac{d}{dz}e^z = e^z$) : Hence

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$$\frac{d}{dz}e^z = \sum_{n=1}^{\infty} \frac{n}{n!} z^{n-1}$$

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$$\frac{d}{dz}e^z = \sum_{n=1}^{\infty} \frac{n}{n!} z^{n-1} = \sum_{n=1}^{\infty} \frac{1}{(n-1)!} z^{n-1}$$

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Complex Exponential Function

Proof ($e^{z_1+z_2} = e^{z_1} + e^{z_2}$) : We define

$g(z) = e^z\, e^{\zeta - z}$, where ζ is any complex number

Complex Exponential Function

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$$g'(z) = e^z\, e^{\zeta - z} + e^z\left(-e^{\zeta - z}\right) = 0.$$

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$$e^{\zeta} = g(z) = e^z\, e^{\zeta - z}$$

we get

$$e^{z_1+z_2} = e^{z_1}\, e^{z_2}$$

Complex Exponential Function

Theorem (Euler Formula): If θ is a real number, then

$$e^{i\theta} = \cos\theta + i\sin\theta$$

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$$= \sum_{n=0}^{\infty}\left[\left[\frac{1}{(2n)!}(i\theta)^{2n}\right] + \left[\frac{1}{(2n+1)!}(i\theta)^{2n+1}\right]\right]$$

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$$= \sum_{n=0}^{\infty}\left[\frac{(-1)^n}{(2n)!}(\theta)^{2n}\right] + i\sum_{n=0}^{\infty}\left[\frac{(-1)^n}{(2n+1)!}(i\theta)^{2n+1}\right]$$

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$$= \cos\theta + i\sin\theta$$

Complex Exponential Function

Using Euler formula and other properties we can now prove DeMoivre's formula

$$(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$$

Complex Exponential Function

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$$(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$$

$$(\cos\theta + i\sin\theta)^n = \left(e^{i\theta}\right)^n = e^{in\theta} = \cos n\theta + i\sin n\theta$$

Complex Exponential Function

If $z = x + iy$ then

$$e^z = e^{x+iy}$$

Complex Exponential Function

If $z = x + iy$ then

$$e^z = e^{x+iy}$$

$$= e^x\, e^{iy}$$

Complex Exponential Function

If $z = x + iy$ then

$$e^z = e^{x+iy}$$
$$= e^x\, e^{iy}$$
$$= e^x(\cos x + i \sin x)$$

Complex Exponential Function

If $z = x + iy$ then

$$e^z = e^{x+iy}$$
$$= e^x\, e^{iy}$$
$$= e^x(\cos x + i \sin x)$$

Hence

$$e^z = e^x(\cos x + i \sin x)\,.$$

Using this expression of e^z we can easily discuss continuity and differentiability of exponential function.

Complex Exponential Function

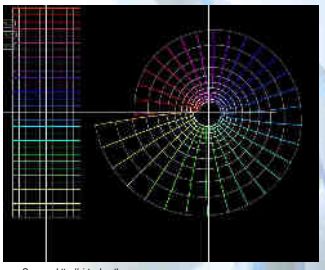
Elementary functions

Complex exponential function
Part I

Complex Exponential Function

Complex exponential function
Part II

Topic No. 66



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Complex Exponential Function

Properties of exponential function

Complex Exponential Function

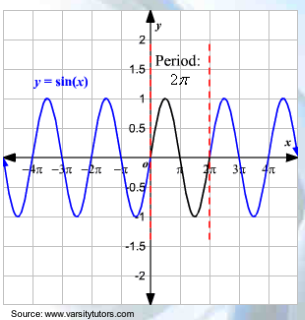
$$\exp(z) = e^x(\cos y + i \sin y)$$

OR

$$e^z = e^x(\cos y + i \sin y)$$

Complex Exponential Function

$\sin x$ is a periodic function



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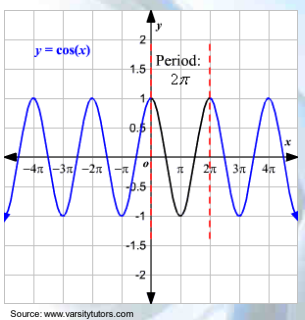
Complex Exponential Function

$\cos x$ is a periodic function



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Complex Exponential Function

For $z = x + iy$, we have

$$e^z = \exp(z) = e^x(\cos y + i \sin y)$$

Complex Exponential Function

For $z = x + iy$, we have

$$\begin{aligned} e^z = \exp(z) &= e^x(\cos y + i \sin y) \\ &= e^x(\cos(2n\pi + y) + i \sin(2n\pi + y)) \end{aligned}$$

Complex Exponential Function

For $z = x + iy$, we have

$$\begin{aligned} e^z = \exp(z) &= e^x(\cos y + i \sin y) \\ &= e^x(\cos(2n\pi + y) + i \sin(2n\pi + y)) \\ &= e^x e^{i\,(2n\pi + y)} \end{aligned}$$

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Complex Exponential Function

Hence

$e^{z+i2n\pi} = e^z$

for all z, provided n is an integer.

Complex Exponential Function

$e^z = 1$

if and only if

$z = i\,2n\,\pi$

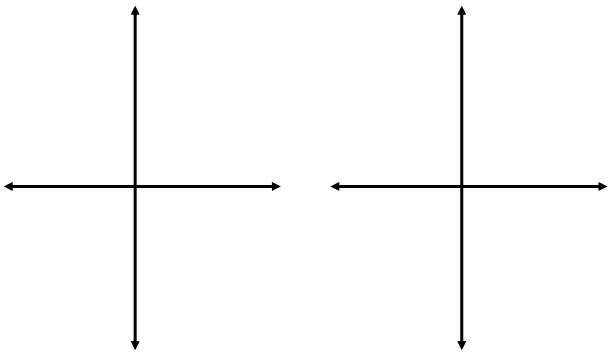
for some integer n

Complex Exponential Function

$e^{z_1} = e^{z_2}$
if and only if
$z_2 = z_1 + i\,2n\pi$
for some integer n

Complex Exponential Function

$e^{z_1} = e^{z_2}$ if and only if $z_2 = z_1 + i\,2n\pi$ for some integer n



Complex Exponential Function

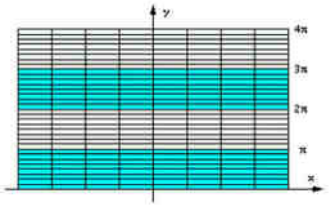
Example: For any integer n, the points

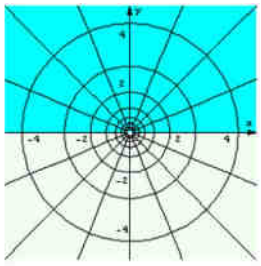
$$z_n = \frac{3}{2} + i\left(\frac{\pi}{4} + 2n\pi\right)$$

In the z-plane. These points are mapped in the w-plane to the one single point.

Complex Exponential Function

$\frac{3}{2} + i\left(\frac{\pi}{4} + 2n\pi\right)$ in z-plane are mapped to $3.1 + i\,3.1$ in w-plane





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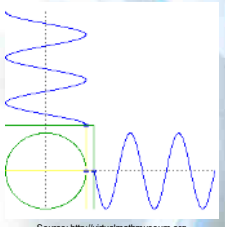
Complex Exponential Function

Complex Exponential function Part II



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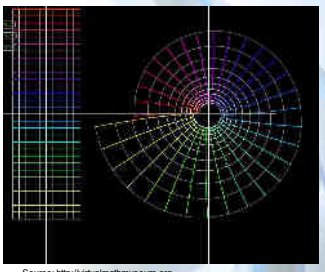
Complex Exponential Function

Complex exponential function Part III

Topic No. 67



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Complex Exponential Function

Range of exponential function

Complex Exponential Function

Is the exponential function
$\exp(z) = e^x(\cos y + i \sin y)$

- One-to-one?
- Onto?

Complex Exponential Function

Onto?

$\exp(z) = e^x(\cos y + i \sin y)$

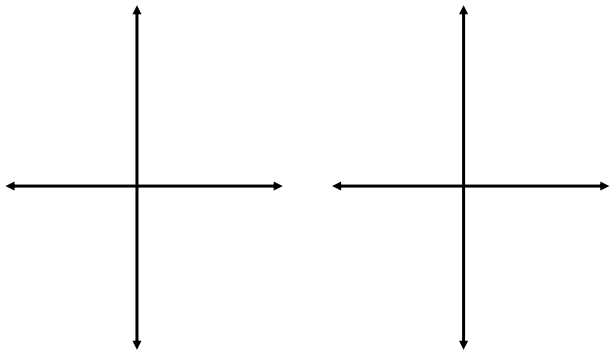
Complex Exponential Function

One-to-one?

$\exp(z) = e^x(\cos y + i \sin y)$

Complex Exponential Function

Exponential function $w = e^z$



Complex Exponential Function

Let $z = x + iy$, and

$$w = e^z = e^{x+iy}$$

Complex Exponential Function

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The equation

$$w = e^z$$

becomes

$$e^x e^{iy} = \rho e^{i\,\phi}$$

Complex Exponential Function

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$e^{z_1} = e^{z_2}$ if and only if $z_2 = z_1 + i\,2n\pi$ for some integer n

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$e^{z_1} = e^{z_2}$ if and only if $z_2 = z_1 + i\,2n\pi$ for some integer n

$$\rho = e^x, \qquad iy = i\phi + i2n\pi$$

Complex Exponential Function

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$$e^x e^{iy} = \rho e^{i\,\phi}$$

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$$\rho = e^x, \qquad iy = i\phi + i2n\pi$$
$$x = \ln \rho\,, \qquad y = \phi + 2n\pi$$

Complex Exponential Function

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The equation

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$$e^x e^{iy} = \rho e^{i\phi}$$

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$$\rho = e^x, \qquad iy = i\phi + i2n\pi$$
$$x = \ln\rho, \qquad y = \phi + 2n\pi$$
$$x = \ln|w|, \qquad y = \phi + 2n\pi$$

Complex Exponential Function

$$x = \ln|w| \quad \text{and} \quad y = \phi + 2n\pi$$

Then $z = x + iy$, becomes

$$z = \ln|w| + i(\phi + 2n\pi)$$

Complex Exponential Function

$$x = \ln|w| \quad \text{and} \quad y = \phi + 2n\pi$$

Then $z = x + iy$, becomes

$$z = \ln|w| + i(\phi + 2n\pi)$$
$$z = \ln|w| + i(Arg(w) + 2n\pi)$$

Complex Exponential Function

$$x = \ln|w| \quad \text{and} \quad y = \phi + 2n\pi$$

Then $z = x + iy$, becomes

$$z = \ln|w| + i(\phi + 2n\pi)$$
$$z = \ln|w| + i(Arg(w) + 2n\pi)$$

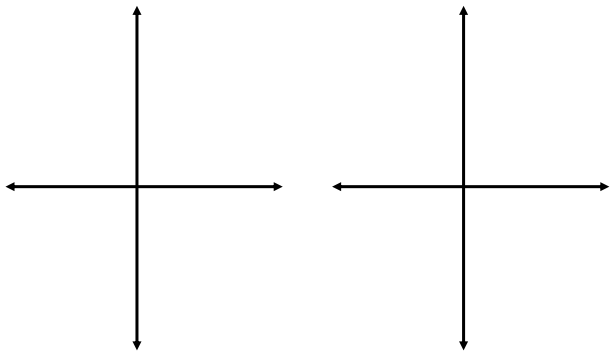
Hence

$$\exp[\ln|w| + i(Arg(w) + 2n\pi)] = w$$

So the complex exponential map e^z is not one-to-one map.

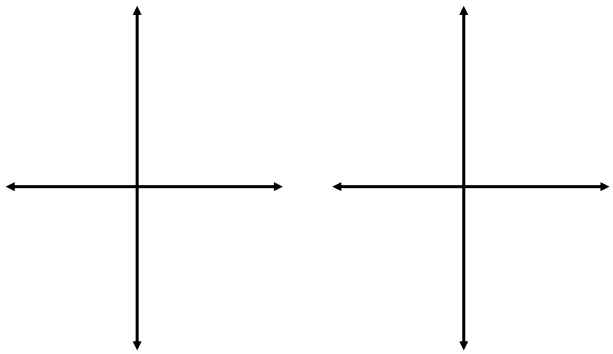
Complex Exponential Function

Range of exponential functions



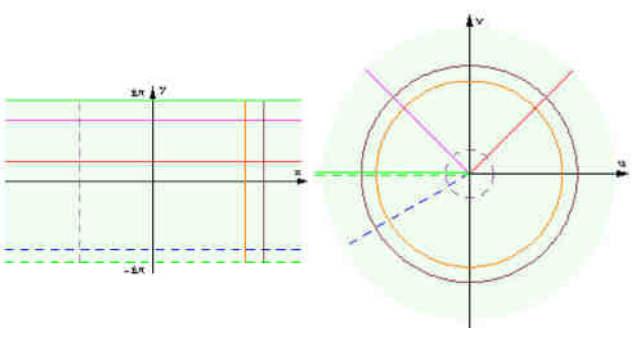
Complex Exponential Function

Range of exponential functions



Complex Exponential Function

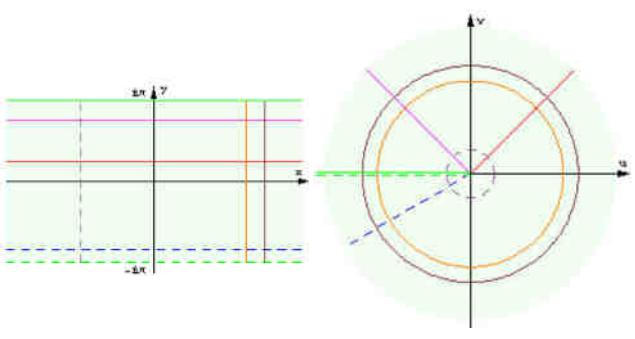
Range of exponential functions



Source: Complex variables and applications by Brown and Churchill

Complex Exponential Function

Fundamental period strip



Source: Complex variables and applications by Brown and Churchill

Complex Exponential Function

Fundamental period strip

If in the solution of $w = e^z = e^{x+iy}$, given by

$$\exp[\ln|w| + i(Arg(w) + 2n\pi)] = w$$

we only consider principal argument

$$-\pi < Arg(w) \leq \pi$$

then the map is one-to-one.

Fundamental period strip
If in the solution of $w = e^z = e^{x+iy}$, given by
$\exp[\ln|w| + i(Arg(w) + 2n\pi)] = w$
we only consider principal argument
$-\pi < Arg(w) \leq \pi$
then the map is one-to-one.

Complex Exponential Function

Example: Find all values of z for which the following equation hold

$$e^z = -4$$

Complex Exponential Function

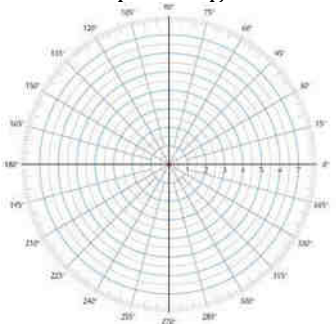
Example: Find image under the complex exponential function of all complex numbers in A

$$A = \left\{x + iy: 1 \leq x \leq 2, -\frac{\pi}{4} \leq y \leq \frac{\pi}{4}\right\}$$

Complex Exponential Function

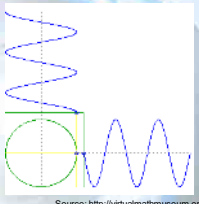
Example: Find image under the complex exponential function of all complex numbers in A

$$A = \left\{x + iy: 1 \leq x \leq 2, -\frac{\pi}{4} \leq y \leq \frac{\pi}{4}\right\}$$



Complex Exponential Function

Complex exponential function Part III



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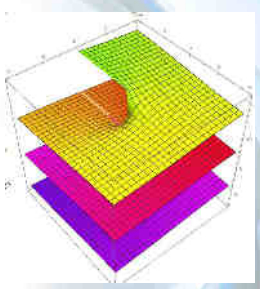
The Complex Logarithm

The complex logarithm Part I

Topic No. 68



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The Complex Logarithm

How to defined logarithm function?

Source: google.com

The Complex Logarithm

Inverse of exponential function?

Source: google.com

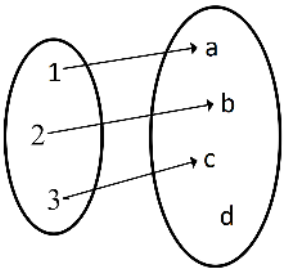
The Complex Logarithm

We have a Problem

Complex exponential function is not one-to-one function

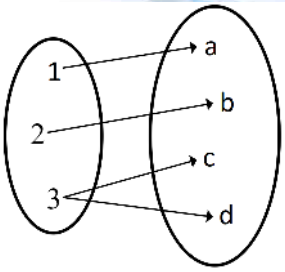
The Complex Logarithm

What is a function $f: A \to B$?



The Complex Logarithm

Multivalued function



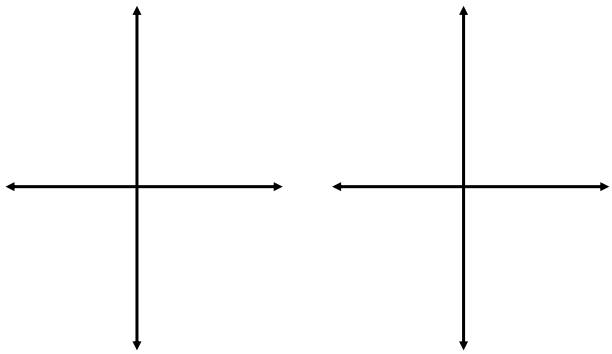
The Complex Logarithm

Multivalued complex logarithm

For $z \neq 0$, we define the multivalued function log as the inverse of the exponential function; that is, $\log(z) = w$ iff $\exp(w) = z$

The Complex Logarithm

Multivalued logarithm is inverse of Exponential function



The Complex Logarithm

We know from our previous discussion:
For $z = x + iy$, and

$$w = e^z = e^{x+iy}$$

we have

$$\exp[\ln|w| + i(Arg(w) + 2n\pi)] = w$$

The Complex Logarithm

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For $z = x + iy$, and

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Hence

$$\log(w) = \ln|w| + i(Arg(w) + 2n\pi)$$

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$$\log(w) = \ln|w| + i(Arg(w) + 2n\pi)$$

if we denote the domain elements by z then

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For $z = x + iy$, and

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if we denote the domain elements by z then

$$\log(z) = \ln|z| + i(Arg(z) + 2n\pi)$$

$$\log(z) = \ln|z| + i\arg(z)$$

The Complex Logarithm

Principal value of the logarithm

For $z \neq 0$, we define Log, the principal value of the logarithm, by

$$\mathrm{Log}(z) = \ln|z| + iArg(z)$$

The Complex Logarithm

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Domain: Set of all nonzero complex numbers.

The Complex Logarithm

Range of $\mathrm{Log}(z) = \ln|z| + iArg(z)$

The Complex Logarithm

Range of $\mathrm{Log}(z) = \ln|z| + iArg(z)$

As

$$-\pi < Arg\,(z) \leq \pi$$

The Complex Logarithm

Range of $\mathrm{Log}(z) = \ln|z| + iArg(z)$

As

$$-\pi < Arg\,(z) \leq \pi$$

So range of w $= Log(z)$ is

$$\{w: -\pi < Im\,(w) \leq \pi\}$$

The Complex Logarithm

$\mathrm{Log}(z)$ is a single value function from nonzero complex numbers to the fundamental period strip.

Complex Exponential Function

Fundamental period strip

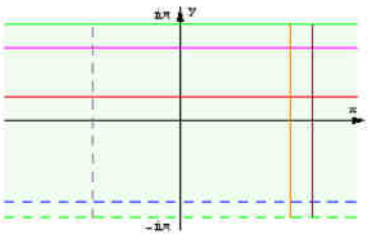
If in the solution of $w = e^z = e^{x+iy}$, given by

$$\exp[\ln|w| + i(Arg(w) + 2n\pi)] = w$$

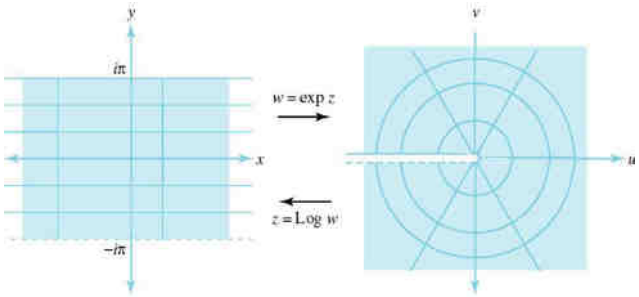
we only consider principal argument

$$-\pi < Arg(w) \leq \pi$$

then the map is one-to-one.



Complex Exponential Function



The Complex Logarithm

The map exp: $FPS \to \mathbb{C}\backslash\{0\}$ is inverse of Log: $\mathbb{C}\backslash\{0\} \to FPS$
That is

$$\exp(\mathrm{Log}(z)) = z$$

and

$$\mathrm{Log}(\exp(z)) = z$$

The Complex Logarithm

$$\exp(\mathrm{Log}(z)) = z$$

The Complex Logarithm

$$\exp(\mathrm{Log}(z)) = z$$

We know that

$$\mathrm{Log}(z) = \ln|z| + iArg(z)$$

The Complex Logarithm

$$\exp(\mathrm{Log}(z)) = z$$

We know that

$$\mathrm{Log}(z) = \ln|z| + iArg(z)$$

Also

$$\exp[\ln|w| + i(Arg(w) + 2n\pi)] = w$$

The Complex Logarithm

$$\exp(\mathrm{Log}(z)) = z$$

We know that

$$\mathrm{Log}(z) = \ln|z| + iArg(z)$$

Also

$$\exp[\ln|w| + i(Arg(w) + 2n\pi)] = w$$

So $\exp(\mathrm{Log}(z)) = \exp[\ln|z| + iArg(z)] = z$

The Complex Logarithm

Similarly we can show the following:

$$\mathrm{Log}(\exp(z)) = z$$

The Complex Logarithm

The map exp: $FPS \to \mathbb{C}\backslash\{0\}$ is inverse of Log: $\mathbb{C}\backslash\{0\} \to FPS$
That is

$$\exp(\mathrm{Log}(z)) = z$$

and

$$\mathrm{Log}(\exp(z)) = z$$

The Complex Logarithm

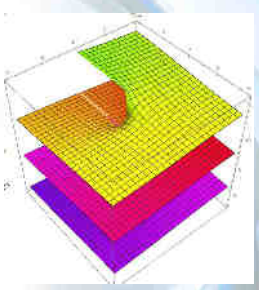
Example: Find the values of $\log(2+i2)$ and $\mathrm{Log}(2+i2)$.

Solution: we know that

$$\log(z) = \ln|z| + i(Arg(z) + 2n\pi)$$
$$\mathrm{Log}(z) = \ln|z| + iArg(z)$$

The Complex Logarithm

The complex logarithm Part I



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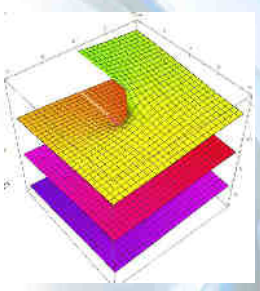
The Complex Logarithm

Topic No. 69

The complex logarithm Part II



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The Complex Logarithm

Multivalued logarithm

For $z \neq 0$, we define the multivalued function log as the inverse of the exponential function; that is,

$\log(z) = w$ iff $\exp(w) = z$

OR

$$\log(z) = \ln|z| + i(Arg(z) + 2n\pi)$$

The Complex Logarithm

Principal value of the logarithm

For $z \neq 0$, we define Log, the principal value of the logarithm, by

$$\mathrm{Log}(z) = \ln|z| + iArg(z)$$

Domain: Set of all nonzero complex numbers.

The Complex Logarithm

Note that

$\log(z) = \ln|z| + i(Arg(z) + 2n\pi)$ is a multivalued function.

But

$$\mathrm{Log}(z) = \ln|z| + iArg(z)$$

is a single valued function

The Complex Logarithm

Properties of $\mathrm{Log}(z)$

- Continuity
- Differentiability

The Complex Logarithm

$Log(z)$ becomes $\ln(x)$ when restricted to real numbers?

The Complex Logarithm

$Log(z)$ becomes $\ln(x)$ when restricted to real numbers?
Answer: Yes

The Complex Logarithm

$Log(z)$ becomes $\ln(x)$ when restricted to real numbers?
Answer: Yes

For $z = x + iy$

$\text{Log}(z) = \ln|z| + iArg(z)$

The Complex Logarithm

$Log(z)$ becomes $\ln(x)$ when restricted to real numbers?
Answer: Yes

For $z = x + iy$

$\text{Log}(z) = \ln|z| + iArg(z)$

Hence for $z = x + i0$ we get

$\text{Log}(z) = \ln|x| + iArg(x)$

The Complex Logarithm

$Log(z)$ becomes $\ln(x)$ when restricted to real numbers?
Answer: Yes

For $z = x + iy$, where $x > 0$

$\text{Log}(z) = \ln|z| + iArg(z)$

Hence for $z = x + i0$ we get

$\text{Log}(z) = \ln|x| + iArg(x)$

$\text{Log}(x) = \ln|x| + 0$

Hence

$\text{Log}(x) = \ln|x|$

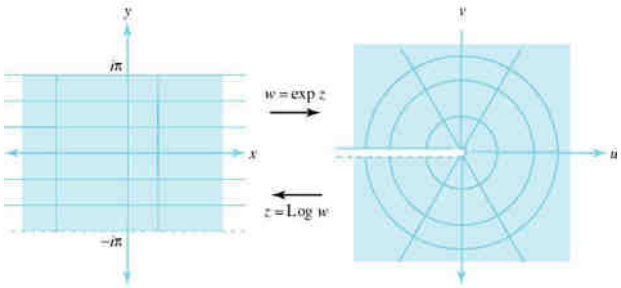
The Complex Logarithm

Continuity of $\text{Log}(z)$

The Complex Logarithm

$\text{Log}(z) = \ln|z| + iArg(z)$

The image of $Log(z)$ is one sheet of complex numbers.



The Complex Logarithm

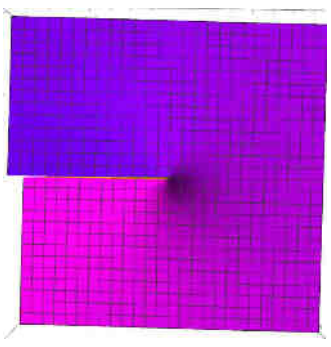
The graph of $\log(z)$ looks like many sheets, one for each n

$\log(z) = \ln|z| + i(Arg(z) + 2n\pi)$

The Complex Logarithm

The graph of looks like many sheets, one for each n

$\log(z) = \ln|z| + i(Arg(z) + 2n\pi)$



The Complex Logarithm

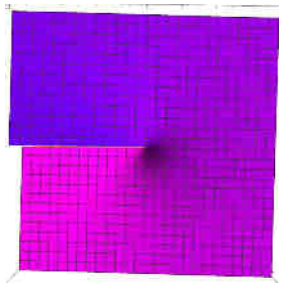
If $w = f(z)$ is a multivalued function.

The Complex Logarithm

If $w = f(z)$ is a multivalued function. A branch of f is any single-valued function f_0 that is continuous is some domain (except perhaps at the boundary).

The Complex Logarithm

If $w = f(z)$ is a multivalued function. A branch of f is any single-valued function f_0 that is continuous is some domain (except perhaps at the boundary). At each point z in the domain it assigns one of the values of $f(z)$.



The Complex Logarithm

The multivalued function

$$\log(z) = \ln|z| + i(Arg(z) + 2n\pi)$$

has branches infinite branches, one for each integer value of n.

The Complex Logarithm

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$$\log(z) = \ln|z| + i(Arg(z) + 2n\pi)$$

has branches infinite branches, one for each integer value of n.
Let

$$\dots, f_{-2}, f_{-1}, f_0, f_1, f_2, \dots$$

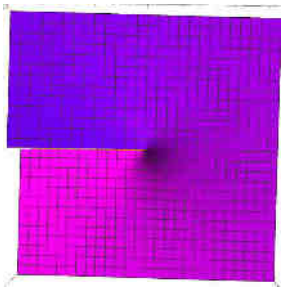
be the branches of $\log(z)$.

The Complex Logarithm

The negative real axis is the branch cut for the functions

$$\dots, f_{-2}, f_{-1}, f_0, f_1, f_2, \dots$$

Each point on the branch cut is a point of discontinuity for all branches of $\log(z)$.



The Complex Logarithm

The principal value of logarithm

$$\mathrm{Log}(z) = \ln|z| + iArg(z)$$

corresponds to just one branch, that is, f_0.

The Complex Logarithm

The principal value of logarithm

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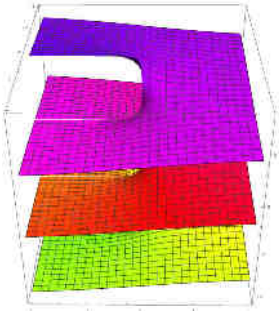
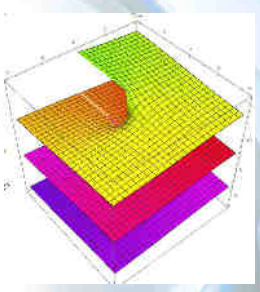
corresponds to just one branch, that is, f_0.
So we can say that the function $Log(z)$ is discontinuous at all the points of negative real axis.



The Complex Logarithm

The complex logarithm Part II



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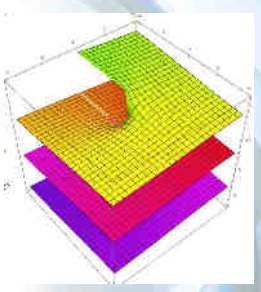
The Complex Logarithm

The complex logarithm Part III

Topic No. 70



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The Complex Logarithm

Multivalued algorithm

For $z \neq 0$, we define the multivalued function log as the inverse of the exponential function; that is,

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OR

$\log(z) = \ln|z| + i(Arg(z) + 2n\pi)$

The Complex Logarithm

Principal value of the logarithm

For $z \neq 0$, we define Log, the principal value of the logarithm, by

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Domain: Set of all nonzero complex numbers.

The Complex Logarithm

Properties of $\mathrm{Log}(z)$

- Continuity
- Differentiability

The Complex Logarithm

The multivalued function

$$\log(z) = \ln|z| + i(Arg(z) + 2n\pi)$$

has branches infinite branches, one for each integer value of n. Let

$$\ldots, f_{-2}, f_{-1}, f_0, f_1, f_2, \ldots$$

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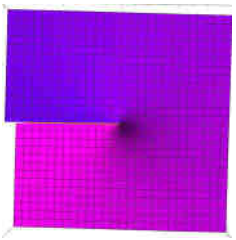

The Complex Logarithm

The negative real axis is the branch cut for the functions

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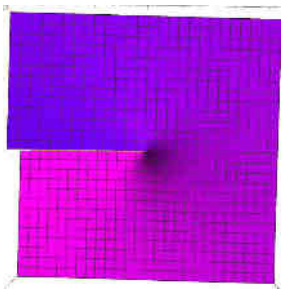
Each point on the branch cut is a point of discontinuity for all branches of $\log(z)$.



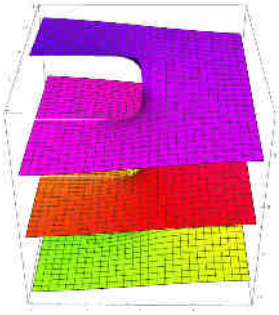
The Complex Logarithm

The principal value of logarithm

$$\mathrm{Log}(z) = \ln|z| + iArg(z)$$

Corresponds to just one branch, that is, f_0.

So we can say that the function $Log(z)$ is discontinuous at all the points of negative real axis.



The Complex Logarithm

Differentiability of $\mathrm{Log}(z)$

The Complex Logarithm

For $z = re^{i\theta}$
$\mathrm{Log}(z) = \ln|z| + iArg(z)$
becomes

The Complex Logarithm

For $z = re^{i\theta}$
$\mathrm{Log}(z) = \ln|z| + iArg(z)$
becomes
$\mathrm{Log}(re^{i\theta}) = \ln r + i\theta$

The Complex Logarithm

For $z = re^{i\theta}$
$\mathrm{Log}(z) = \ln|z| + iArg(z)$
becomes
$\mathrm{Log}(re^{i\theta}) = \ln r + i\theta$
Hence
$\mathrm{Log}(z) = U(r,\theta) + iV(r,\theta)$
where
$U(r,\theta) = \ln r\,, V(r,\theta) = \theta$

The Complex Logarithm

Theorem: Let the function

$$f(z) = u(r,\theta) + iv(r,\theta)$$

be defined throughout some neighbourhood of a non-zero point $z_0 = r_0\, e^{i\theta_0}$ and suppose that

1. the first order partial derivates of the functions u and v with respect to r and θ exist everywhere in the neighbourhood
2. those partial derivatives are continuous at (r_0, θ_0) and satisfy polar form of Cauchy-Riemann equations

$$ru_r = v_\theta, \qquad u_\theta = -rv_r$$

Then $f'(z_0)$ exists and its value is

$$f'(z_0) = e^{-i\theta}(u_r + iv_r)$$

where the right side is to be evaluated at (r_0, θ_0)

The Complex Logarithm

Consider

$$\mathrm{Log}(z) = U(r,\theta) + iV(r,\theta) = \ln r + i\theta$$

where

$$U(r,\theta) = \ln r\,, V(r,\theta) = \theta$$

in domain

$$-\pi < Arg(z) < \pi$$

The Complex Logarithm

Consider

$$\mathrm{Log}(z) = U(r,\theta) + iV(r,\theta) = \ln r + i\theta$$

where

$$U(r,\theta) = \ln r\,, V(r,\theta) = \theta$$

in domain

$$-\pi < Arg(z) < \pi$$

$$U_r(r,\theta) = \frac{1}{r}, \qquad U_\theta(r,\theta) = 0$$

The Complex Logarithm

Consider

$$\mathrm{Log}(z) = U(r,\theta) + iV(r,\theta) = \ln r + i\theta$$

where

$$U(r,\theta) = \ln r\,, V(r,\theta) = \theta$$

in domain

$$-\pi < Arg(z) < \pi$$

$$U_r(r,\theta) = \frac{1}{r}, \qquad U_\theta(r,\theta) = 0$$

and

$$V_r(r,\theta) = 0\,, \qquad V_\theta(r,\theta) = 1$$

The Complex Logarithm

Consider

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in domain

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$$U_r(r,\theta) = \frac{1}{r}, \qquad U_\theta(r,\theta) = 0$$

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Also

$$r\,U_r = V_\theta \quad and \quad -rV_r = U_\theta$$

The Complex Logarithm

Theorem: Let the function

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be defined throughout some neighbourhood of a non-zero point $z_0 = r_0\, e^{i\theta_0}$ and suppose that

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Then $f'(z_0)$ exists and its value is

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where the right side is to be evaluated at (r_0, θ_0)

The Complex Logarithm

Hence

$$\frac{d}{dz}Log\,(z) = e^{-i\theta}(\,U_r + i\,V_r)$$

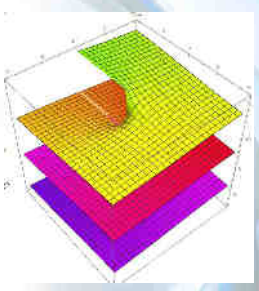
$$\frac{d}{dz}Log\,(z) = e^{-i\theta}\left(\frac{1}{r} + i\,0\right)$$

$$\frac{d}{dz}Log\,(z) = \frac{1}{re^{i\theta}} = \frac{1}{z}$$

Provided $r > 0$ and $-\pi < \theta < \pi$.

The Complex Logarithm

The complex logarithm Part III



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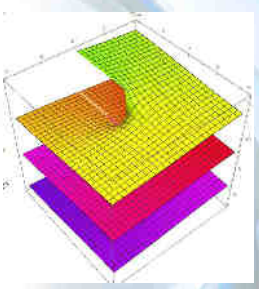
The Complex Logarithm

Topic No. 71

The complex logarithm Part III



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The Complex Logarithm

Multivalued algorithm

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OR

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The Complex Logarithm

Principal value of the logarithm

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Domain: Set of all nonzero complex numbers.

The Complex Logarithm

So far we discussed

Continuity &
Differentiability
Of Log (z)

The Complex Logarithm

Further Properties of
$\text{Log}(z)$ and $\log(z)$

The Complex Logarithm

Theorem: Let z_1 and z_2 be nonzero complex numbers. The multivalued function log obeys the familiar properties of logarithms.

1. $\log(z_1 z_2) = \log(z_1) + \log(z_2)$
2. $\log\left(\frac{z_1}{z_2}\right) = \log(z_1) - \log(z_2)$

The Complex Logarithm

Theorem: Let z_1 and z_2 be nonzero complex numbers. The multivalued function log obeys the familiar properties of logarithms.

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2. $\log\left(\frac{z_1}{z_2}\right) = \log(z_1) - \log(z_2)$

The Complex Logarithm

Proof $(\log(z_1 z_2) = \log(z_1) + \log(z_2))$:

We know that

$$\log(z) = \ln|z| + i(Arg(z) + 2n\pi)$$

Or

$$\log(z) = \ln|z| + i(\arg(z))$$

The Complex Logarithm

Proof $\left(\log\left(\frac{z_1}{z_2}\right) = \log(z_1) - \log(z_2)\right)$:

We know that

$$\log(z) = \ln|z| + i(\arg(z))$$

The Complex Logarithm

Does the following property hold for $Log(z)$?

$$\text{Log}(z_1 z_2) = \text{Log}(z_1) + \text{Log}(z_2)$$

The Complex Logarithm

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$$\text{Log}(z_1 z_2) = \text{Log}(z_1) + \text{Log}(z_2)$$

The Complex Logarithm

Does the following property hold for $Log(z)$?

$$\text{Log}(z_1 z_2) = \text{Log}(z_1) + \text{Log}(z_2)$$

Answer: No

The Complex Logarithm

Example: Let $z_1 = -1$and $z_2 = -2$. Check whether the following holds

$$\text{Log}(z_1 z_2) = \text{Log}(z_1) + \text{Log}(z_2)\ .$$

Solution: We know that

$$\text{Log}(z) = \ln|z| + iArg(z)$$

The Complex Logarithm

Theorem: The identity

$$\text{Log}(z_1 z_2) = \text{Log}(z_1) + \text{Log}(z_2)$$

holds true iff

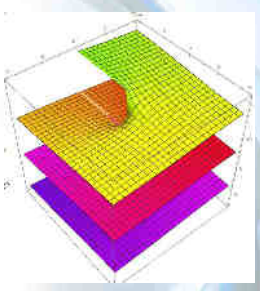
$$-\pi < Arg(z_1) + Arg(z_2) \leq \pi$$

Proof: we know that

$$\text{Log}(z) = \ln|z| + i(\text{Arg}(z))$$

The Complex Logarithm

The complex logarithm Part IV



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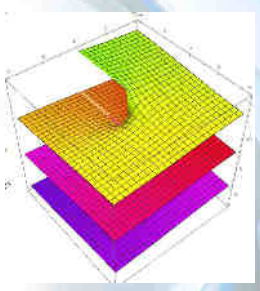
The Complex Logarithm

The complex logarithm Part V

Topic No. 72



Source: http://virtualmathmuseum.org

The Complex Logarithm

Branches of The complex Logarithm

The Complex Logarithm

Multivalued logarithm

For $z \neq 0$, we define the multivalued function log as the inverse of the exponential function; that is,

$\log(z) = w$ iff $\exp(w) = z$

OR

$$\log(z) = \ln|z| + i(Arg(z) + 2n\pi)$$

The Complex Logarithm

Principal value of the logarithm

For $z \neq 0$, we define Log, the principal value of the logarithm, by

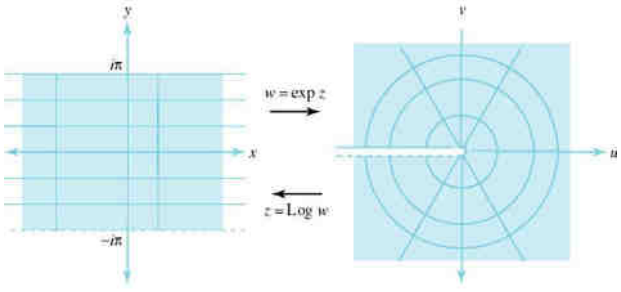
$$\text{Log}(z) = \ln|z| + iArg(z)$$

Domain: Set of all nonzero complex numbers.

The Complex Logarithm

$$\text{Log}(z) = \ln|z| + iArg(z)$$

The image of $\text{Log}(z)$ is one complex sheet/strip of non-zero complex numbers.



The Complex Logarithm

The multivalued function

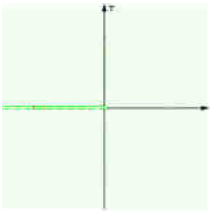
$$\log(z) = \ln|z| + i(Arg(z) + 2n\pi)$$

has infinite branches, one for each integer value of n. Let

$$\dots, f_{-2}, f_{-1}, f_0, f_1, f_2, \dots$$

be the branches of $\log(z)$.

The negative real axis is the branch cut.



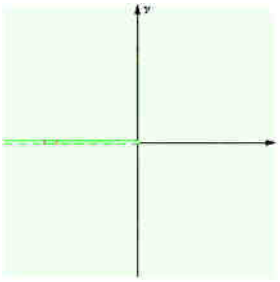
The Complex Logarithm

The principal value of logarithm

$$\mathrm{Log}(z) = \ln|z| + iArg(z)$$

corresponds to just one branch, that is, f_0.

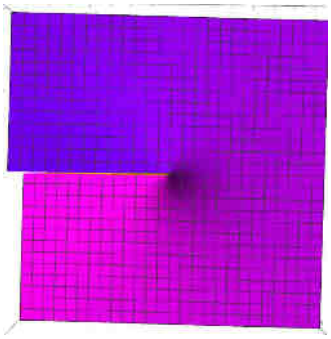
So we can say that the function $\mathrm{Log}(z)$ is discontinuous at all the points of negative real axis or branch cut.



The Complex Logarithm

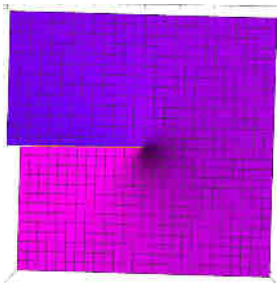
The graph of looks like many sheets, one for each n

$$\log(z) = \ln|z| + i(Arg(z) + 2n\pi)$$



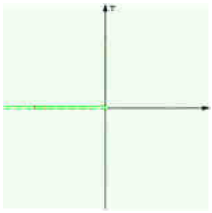
The Complex Logarithm

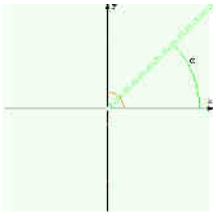
If $w = f(z)$ is a multivalued function. A branch of f is any single-valued function f_0 that is continuous is some domain (except perhaps at the boundary). At each point z in the domain it assigns one of the valued of $f(z)$.



The Complex Logarithm

We can choose branch cut to be any preassigned array $\{r\, e^{i\alpha}: r > 0\}$ (where α is a fixed real number)



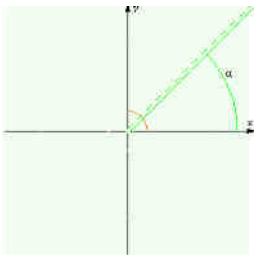


The Complex Logarithm

We can choose branch cut to be any preassigned array

$$\{r\, e^{i\alpha}: r > 0\}$$

So that the branches of the multivalued logarithm function are continuous and differentiable except at the points of this array.



The Complex Logarithm

We define, for $z = r\, e^{i\theta}$

$$\log_\alpha z = \ln r + i\,\theta$$

$$where\ \ \theta \in \arg z\ \ such\ that\ \alpha < \theta \leq \alpha + 2\pi$$

to be a single-valued branch of the complex logarithm. The branch cut in this case is the following ray

$$\{r\, e^{i\alpha}: r > 0\}$$

We define, for $z = r\, e^{i\theta}$

$\log_\alpha z = \ln r + i\,\theta$

$where\ \ \theta \in \arg z\ \ such\ that\ \alpha < \theta \leq \alpha + 2\pi$

to be a single-valued branch of the complex logarithm. The branch cut in this case is the following ray

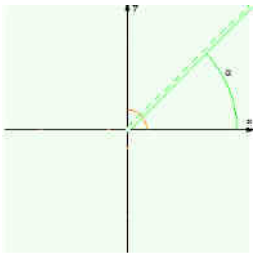
$\{r\, e^{i\alpha}: r > 0\}$

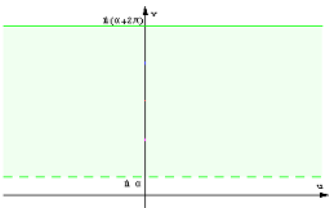
The Complex Logarithm

The mapping $\log_\alpha z$ is

one-to-one mapping from $|z| > 0$ to the horizontal strip

$$\omega = \{\omega: \alpha < \mathrm{Im}\,(\omega) \leq \alpha + 2\pi\}$$

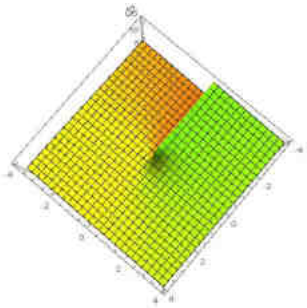




The Complex Logarithm

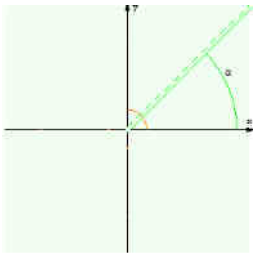
So different branches are separated by the following branch cut

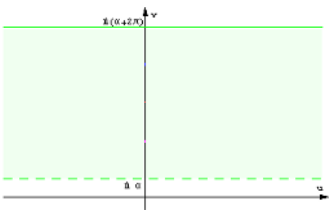
$$\{r\, e^{i\alpha} : r > 0\}$$



The Complex Logarithm

The mapping $\log_\alpha z$ maps different subsets of the z-plane in the following way





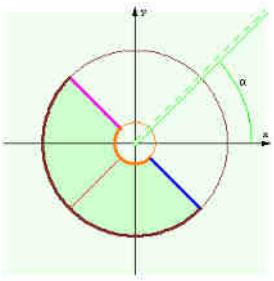
The Complex Logarithm

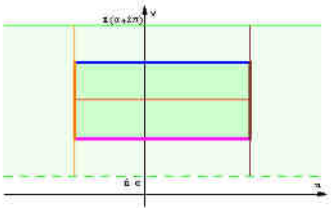
The region

$$\{re^{i\theta} : a < r < b, \qquad c < \theta < d\}$$

is mapped to

$$\{u + iv : \ln a < u < \ln b\,, c < v < d\}$$





The Complex Logarithm

Since the mapping $\log_\alpha z$ is a single valued function so we can talk about its derivative.

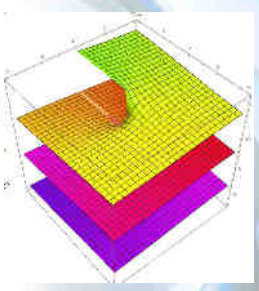
The Complex Logarithm

Show that

$$\frac{d}{dz}\log_\alpha z = \frac{1}{z}, for\ z = re^{i\theta}, \qquad \alpha < \theta < \alpha + 2\pi$$

The Complex Logarithm

The complex logarithm Part V



Source: http://virtualmathmuseum.org

Complex Exponents

Complex exponents Part I

Topic No. 73

Complex Exponents

We can define real exponents like

$$2^2, \pi^{\frac{2}{3}}, \sqrt{87}$$

Complex Exponents

What about complex components

i^i , $\sqrt{1+i}$, 2^{1+i}, etc

Complex Exponents

We use our knowledge about the complex exponential function and complex logarithm to define

complex exponents

Complex Exponents

We know that

$$\exp[\log(z)] = z, for\ z \neq 0$$

Complex Exponents

We know that

$$\exp[\log(z)] = z, for\ z \neq 0$$

Also

$$\log(z^n) = n\log(z)$$

where n is any natural number.

Complex Exponents

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$$\exp[\log(z)] = z, for\ z \neq 0$$

Also

$$\log(z^n) = n\log(z)$$

where n is any natural number.

Hence

$$\exp[\log(z^n)] = \exp[\mathrm{n}\log(z)] = z^n, for\ z \neq 0$$

Complex Exponents

Definition (Complex exponent)

Let $z \neq 0$ and c be a complex number. We define z^c as

$$z^c = \exp[\, c\log(z)]$$

Complex Exponents

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Example: Evaluate $4^{\frac{1}{2}}$.

Complex Exponents

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Example: Evaluate $4^{\frac{1}{2}}$.

Complex Exponents

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Let $z \neq 0$ and c be a complex number. We define z^c as

$$z^c = \exp[\, c \log(z)]$$

Since $\log(z)$ is multivalued so in general z^c is multivalued.

Complex Exponents

Definition (Complex exponent)

Let $z \neq 0$ and c be a complex number. We define z^c as

$$z^c = \exp[\, c \log(z)]$$

Since $\log(z)$ is multivalued so in general z^c is multivalued.

So we can define **the principal branch of the multivalued function z^c** as

$$z^c = \exp[\, c \operatorname{Log}(z)]$$

Complex Exponents

Definition (Complex exponent)

Let $z \neq 0$ and c be a complex number. We define z^c as

$$z^c = \exp[\, c \log(z)]$$

Since $\log(z)$ is multivalued so in general z^c is multivalued.

So we can define **the principal branch of the multivalued function z^c** as

$$z^c = \exp[\, c \operatorname{Log}(z)]$$

sometimes denoted as

$$\text{P.V.}\, z^c = \exp[\, c \operatorname{Log}(z)]\ .$$

Complex Exponents

Example: Evaluate $\text{P.V.}\, i^i$

Complex Exponents

Complex exponents
Part I

Complex Exponents

Complex exponents
Part II

Topic No. 74

Complex Exponents

Complex exponent

Let c be a complex number. We define z^c as

$$z^c = \exp[\, c \log(z)]$$

Complex Exponents

Complex exponent

Let c be a complex number. We define z^c as

$$z^c = \exp[\, c \log(z)]$$

Its principal value is defined as

$$\text{P.V.}\, z^c = \exp[\, c \operatorname{Log}(z)]$$

Complex Exponents

z^c

- When $c \in \mathbb{Q}$
- When c is not rational

Complex Exponents

z^c

Where $c \in \mathbb{Q}$

Complex Exponents

Let $z = re^{i\theta}$

$$z^{\frac{j}{k}} = \exp\left[\frac{j}{k}\log(z)\right]$$

Complex Exponents

Let $z = re^{i\theta}$

$$z^{\frac{j}{k}} = \exp\left[\frac{j}{k}\log(z)\right]$$
$$= \exp\left[\frac{j}{k}\log\left(re^{i\theta}\right)\right]$$

Complex Exponents

Let $z = re^{i\theta}$

$$z^{\frac{j}{k}} = \exp\left[\frac{j}{k}\log(z)\right]$$
$$= \exp\left[\frac{j}{k}\log\left(re^{i\theta}\right)\right]$$
$$= \exp\left[\frac{j}{k}\left(\ln r + i(\theta + 2n\pi)\right)\right]$$

Complex Exponents

Let $z = re^{i\theta}$

$$z^{\frac{j}{k}} = \exp\left[\frac{j}{k}\log(z)\right]$$
$$= \exp\left[\frac{j}{k}\log\left(re^{i\theta}\right)\right]$$
$$= \exp\left[\frac{j}{k}\left(\ln r + i(\theta + 2n\pi)\right)\right]$$
$$= \exp\left[\ln r^{\frac{j}{k}} + i\frac{(\theta + 2n\pi)j}{k}\right]$$

Complex Exponents

Let $z = re^{i\theta}$

$$z^{\frac{j}{k}} = \exp\left[\frac{j}{k}\log(z)\right]$$
$$= \exp\left[\frac{j}{k}\log\left(re^{i\theta}\right)\right]$$
$$= \exp\left[\frac{j}{k}\left(\ln r + i(\theta + 2n\pi)\right)\right]$$
$$= \exp\left[\ln r^{\frac{j}{k}} + i\frac{(\theta + 2n\pi)j}{k}\right]$$
$$= \exp\left[\ln r^{\frac{j}{k}}\right] \exp\left[i\frac{(\theta + 2n\pi)j}{k}\right]$$

Complex Exponents

Let $z = re^{i\theta}$

$$z^{\frac{j}{k}} = \exp\left[\frac{j}{k}\log(z)\right]$$
$$= \exp\left[\frac{j}{k}\log\left(re^{i\theta}\right)\right]$$
$$= \exp\left[\frac{j}{k}\left(\ln r + i(\theta + 2n\pi)\right)\right]$$
$$= \exp\left[\ln r^{\frac{j}{k}} + i\frac{(\theta + 2n\pi)j}{k}\right]$$
$$= \exp\left[\ln r^{\frac{j}{k}}\right] \exp\left[i\frac{(\theta + 2n\pi)j}{k}\right]$$
$$= r^{\frac{j}{k}}\left(\cos\frac{(\theta + 2n\pi)j}{k} + i\sin\frac{(\theta + 2n\pi)j}{k}\right)$$

Complex Exponents

Hence

$$z^{\frac{j}{k}} = r^{\frac{j}{k}}\left(\cos\frac{(\theta+2n\pi)j}{k} + i\sin\frac{(\theta+2n\pi)j}{k}\right)$$

Complex Exponents

Hence

$$z^{\frac{j}{k}} = r^{\frac{j}{k}}\left(\cos\frac{(\theta+2n\pi)j}{k} + i\sin\frac{(\theta+2n\pi)j}{k}\right)$$

$$z^{\frac{j}{k}} = r^{\frac{j}{k}}\left(\cos\left(\frac{\theta}{k}+\frac{2n\pi}{k}\right)j + i\sin\left(\frac{\theta}{k}+\frac{2n\pi}{k}\right)j\right)$$

There are k distinct values of $z^{\frac{j}{k}}$ corresponding to the following values of

$$n = 0,1,2,\dots,k-1$$

Complex Exponents

Particular cases

Case: If $j = mk$

$$z^{\frac{j}{k}} = r^{\frac{j}{k}}\left(\cos\frac{(\theta+2n\pi)j}{k} + i\sin\frac{(\theta+2n\pi)j}{k}\right)$$

Complex Exponents

Particular cases

Case: If $j = mk$

$$z^{\frac{j}{k}} = r^{\frac{j}{k}}\left(\cos\frac{(\theta+2n\pi)j}{k} + i\sin\frac{(\theta+2n\pi)j}{k}\right)$$

Becomes

$$z^{m} = r^{m}(\cos(\theta+2n\pi)m + i\sin(\theta+2n\pi)m)$$

Complex Exponents

Particular cases

Case: If $j = mk$

$$z^{\frac{j}{k}} = r^{\frac{j}{k}}\left(\cos\frac{(\theta+2n\pi)j}{k} + i\sin\frac{(\theta+2n\pi)j}{k}\right)$$

Becomes

$$z^{m} = r^{m}(\cos(\theta+2n\pi)m + i\sin(\theta+2n\pi)m)$$

Which implies

$$z^{m} = r^{m}(\cos(m\theta) + i\sin(m\theta))$$

This is a single valued function which is mth power of z.

Complex Exponents

Particular cases

Case: If $j = 1$

$$z^{\frac{j}{k}} = r^{\frac{j}{k}}\left(\cos\frac{(\theta+2n\pi)j}{k} + i\sin\frac{(\theta+2n\pi)j}{k}\right)$$

Complex Exponents

Particular cases

Case: If $j = 1$

$$z^{\frac{j}{k}} = r^{\frac{j}{k}}\left(\cos\frac{(\theta+2n\pi)j}{k} + i\sin\frac{(\theta+2n\pi)j}{k}\right)$$

Becomes

$$z^{\frac{1}{k}} = r^{\frac{1}{k}}\left(\cos\frac{(\theta+2n\pi)}{k} + i\sin\frac{(\theta+2n\pi)}{k}\right)$$

Complex Exponents

Particular cases

Case: If $j = 1$

$$z^{\frac{j}{k}} = r^{\frac{j}{k}}\left(\cos\frac{(\theta+2n\pi)j}{k} + i\sin\frac{(\theta+2n\pi)j}{k}\right)$$

Becomes

$$z^{\frac{1}{k}} = r^{\frac{1}{k}}\left(\cos\frac{(\theta+2n\pi)}{k} + i\sin\frac{(\theta+2n\pi)}{k}\right)$$

There are k distinct values of $z^{\frac{j}{k}}$ corresponding to the following values of

$$n = 0,1,2,\dots,k-1$$

So $z^{\frac{1}{k}}$ represents multivalued kth root function.

Complex Exponents

z^c

Where c is not rational

Complex Exponents

In this case there are infinitely many values of z^c (if $z \neq 0$)

Complex Exponents

$$2^{\frac{1}{9}+\frac{i}{50}} = \exp\left[\left(\frac{1}{9} + \frac{i}{50}\right)\log(2)\right]$$

Complex Exponents

$$2^{\frac{1}{9}+\frac{i}{50}} = \exp\left[\left(\frac{1}{9} + \frac{i}{50}\right)\log(2)\right]$$
$$= \exp\left[\left(\frac{1}{9} + \frac{i}{50}\right)(\ln 2 + i\,2n\pi)\right]$$

Complex Exponents

$$2^{\frac{1}{9}+\frac{i}{50}} = \exp\left[\left(\frac{1}{9} + \frac{i}{50}\right)\log(2)\right]$$
$$= \exp\left[\left(\frac{1}{9} + \frac{i}{50}\right)(\ln 2 + i\,2n\pi)\right]$$
$$= \exp\left[\frac{\ln 2}{9} - \frac{n\pi}{25} + i\left(\frac{\ln 2}{50} + \frac{2n\pi}{9}\right)\right]$$

Complex Exponents

$$2^{\frac{1}{9}+\frac{i}{50}} = \exp\left[\left(\frac{1}{9} + \frac{i}{50}\right)\log(2)\right]$$
$$= \exp\left[\left(\frac{1}{9} + \frac{i}{50}\right)(\ln 2 + i\,2n\pi)\right]$$
$$= \exp\left[\frac{\ln 2}{9} - \frac{n\pi}{25} + i\left(\frac{\ln 2}{50} + \frac{2n\pi}{9}\right)\right]$$
$$= \exp\left[\frac{\ln 2}{9}\right]\exp\left[-\frac{n\pi}{25}\right]\exp\left[i\left(\frac{\ln 2}{50} + \frac{2n\pi}{9}\right)\right]$$

Complex Exponents

$$2^{\frac{1}{9}+\frac{i}{50}} = \exp\left[\left(\frac{1}{9} + \frac{i}{50}\right)\log(2)\right]$$
$$= \exp\left[\left(\frac{1}{9} + \frac{i}{50}\right)(\ln 2 + i\,2n\pi)\right]$$
$$= \exp\left[\frac{\ln 2}{9} - \frac{n\pi}{25} + i\left(\frac{\ln 2}{50} + \frac{2n\pi}{9}\right)\right]$$
$$= \exp\left[\frac{\ln 2}{9}\right]\exp\left[-\frac{n\pi}{25}\right]\exp\left[i\left(\frac{\ln 2}{50} + \frac{2n\pi}{9}\right)\right]$$
$$= 2^{\frac{1}{9}}e^{-\frac{n\pi}{25}}\left[\cos\left(\frac{\ln 2}{50} + \frac{2n\pi}{9}\right) + i\sin\left(\frac{\ln 2}{50} + \frac{2n\pi}{9}\right)\right]$$

Complex Exponents

Hence

$$2^{\frac{1}{9}+\frac{i}{50}} = 2^{\frac{1}{9}}e^{-\frac{n\pi}{25}}\left[\cos\left(\frac{\ln 2}{50} + \frac{2n\pi}{9}\right) + i\sin\left(\frac{\ln 2}{50} + \frac{2n\pi}{9}\right)\right]$$

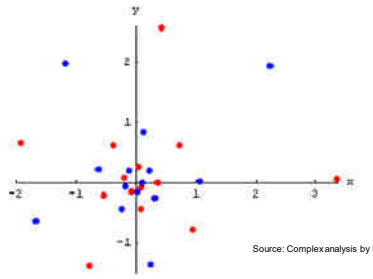
There are distinct values corresponding to the following values of n

$$n = \cdots -3, -2, -1, 0, 1, 2, \ldots$$

Complex Exponents

$$2^{\frac{1}{9}+\frac{i}{50}} = 2^{\frac{1}{9}} e^{-\frac{n\pi}{25}} \left[\cos\left(\frac{\ln 2}{50} + \frac{2n\pi}{9}\right) + i \sin\left(\frac{\ln 2}{50} + \frac{2n\pi}{9}\right)\right]$$

Out put of above for some n is shown below



Source: Complex analysis by Mathews and Russell

Complex Exponents

Rules for exponents

$$\frac{1}{z^c} = z^{-c}$$

$$z^c z^d = z^{c+d}$$

$$\frac{z^c}{z^d} = z^{c-d}$$

Complex Exponents

$$\frac{1}{z^c} = z^{-c}$$

Complex Exponents

$$\frac{1}{z^c} = z^{-c}$$

Holds due to the following

$$\frac{1}{z^c} = \frac{1}{\exp[\, c \log(z)]}$$

$$\frac{1}{z^c} = \exp[-c \log(z)] = z^{-c}$$

Complex Exponents

Similarly the following can be shown

- $z^c z^d = z^{c+d}$

Complex Exponents

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Complex Exponents

Similarly the following can be shown

- $z^c z^d = z^{c+d}$
- $\frac{z^c}{z^d} = z^{c-d}$
- $(z^c)^n = z^{nc}$, only if n is an integer.

Complex Exponents

The following does not hold if n is not an integer

$$(z^c)^n = z^{nc}$$

Complex Exponents

Example: Evaluate and compare the values of $(i^2)^i$ and i^{2i}

Solution:

Complex Exponents

The following rule does not hold in general

$$(z_1 z_2)^c = z_1^c \, z_2^c$$

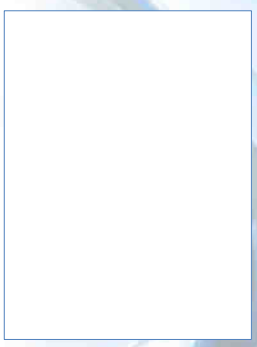
Complex Exponents

Example: $z_1 = 1 + i, z_2 = 1 - i, z_3 = -1 - i$

Solution:

Complex Exponents

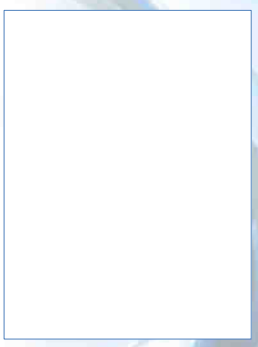
Complex exponents
Part II



Complex Exponents

Topic No. 75

Complex exponents
Part III



Complex Exponents

Complex exponent

Let c be a complex number.

We define z^c as

$$z^c = \exp[\, c \log(z)]$$

Complex Exponents

Complex exponent

Let c be a complex number.

We define z^c as

$$z^c = \exp[\, c \log(z)]$$

Its principal value is defined as

$$\text{P. V. } z^c = \exp[\, c \, \text{Log}(z)]$$

Complex Exponents

Derivative of z^c

Complex Exponents

z^c is multivalued

we consider derivative of the principal branch of z^c

Complex Exponents

$$z^c = \exp[c\ \mathrm{Log} z]$$

Complex Exponents

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Chain rule implies

$$\frac{d}{dz} z^c = \frac{c}{z} \exp[c\ \mathrm{Log} z]$$

Complex Exponents

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Chain rule implies

$$\frac{d}{dz} z^c = \frac{c}{z} \exp[c\ \mathrm{Log} z]$$

Hence, we get

$$\boldsymbol{\frac{d}{dz} z^c = c\, z^{c-1}}$$

Complex Exponents

So far we defined exponential function

$$e^z$$

(which has base e)

Complex Exponents

Using definition of complex exponents we can define exponential function with any base b

Complex Exponents

Exponential function with base $b \neq 0$ is defined as

$$b^z = \exp[z \log(b)]$$

Complex Exponents

Properties of b^z can be explored by using definition

$$b^z = \exp[z \log(b)]$$

and properties of the exponential function e^z

Complex Exponents

Derivative of b^z

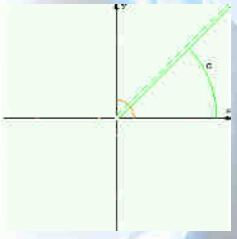
Complex Exponents

The function b^z is a multivalued function
But we can calculate derivative of a branch

Complex Exponents

We choose the branch cut in such a way that b does not lie on the branch cut

$b^z = \exp[z \log_\alpha(b)]$



Complex Exponents

$$\frac{d}{dz} b^z$$
$$= \frac{d}{dz} \exp[z \log_\alpha(b)]$$

Complex Exponents

$$\frac{d}{dz} b^z$$
$$= \frac{d}{dz} \exp[z \log_\alpha(b)]$$
$$= \exp[z \log_\alpha(b)] \frac{d}{dz} [z \log_\alpha(b)]$$

Complex Exponents

$$\frac{d}{dz} b^z$$
$$= \frac{d}{dz} \exp[z \log_\alpha(b)]$$
$$= \exp[z \log_\alpha(b)] \frac{d}{dz} [z \log_\alpha(b)]$$
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Complex Exponents

$$\frac{d}{dz} b^z$$
$$= \frac{d}{dz} \exp[z \log_\alpha(b)]$$
$$= \exp[z \log_\alpha(b)] \frac{d}{dz} [z \log_\alpha(b)]$$
$$= \exp[z \log_\alpha(b)] \log_\alpha(b)$$
$$= b^z \log_\alpha(b)$$

Complex Exponents

Hence

$$\frac{d}{dz} b^z = b^z \log_\alpha(b)$$

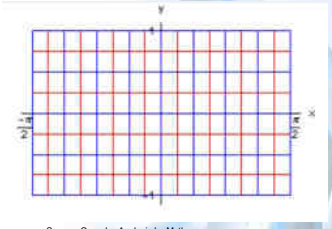
Complex Exponents

Complex exponents
Part III

Trigonometric Functions

Trigonometric functions
Part I

Topic No. 76



Source: Complex Analysis by Mathews

Trigonometric Functions

How to define Trigonometric functions

HOW?

Trigonometric Functions

We know from real analysis

$$\sin(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$$

$$\cos(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$$

Trigonometric Functions

We define

$$\sin(z) = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n+1}}{(2n+1)!}$$

Trigonometric Functions

We define

$$\sin(z) = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n+1}}{(2n+1)!}$$

$$\cos(z) = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n}}{(2n)!}$$

Trigonometric Functions

The other trigonometric functions are

$$\tan z = \frac{\sin z}{\cos z}, \cot z = \frac{\cos z}{\sin z}$$

$$\sec z = \frac{1}{\cos z}, \csc z = \frac{1}{\sin z}$$

Trigonometric Functions

What happens when z is restricted to real numbers ?

Trigonometric Functions

Properties of trigonometric functions

Trigonometric Functions

$$\sin(-z) = -\sin(z)$$

$$\sin(z) = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n+1}}{(2n+1)!}$$

Trigonometric Functions

$$\cos(-z) = \cos(z)$$

$$\cos(z) = \sum_{n=0}^{\infty} \frac{(-1)^n z^{2n}}{(2n)!}$$

Trigonometric Functions

$$\cos^2 z + \sin^2 z = 1$$

Trigonometric Functions

We can define trigonometric functions in another way using the following fact

$$e^{iz} = \cos z + i \sin z$$

Trigonometric Functions

Theorem: If z is a complex number, then

$$e^{iz} = \cos z + i \sin z$$

Trigonometric Functions

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Proof: We know that $e^{iz} = \sum_{n=0}^{\infty} \frac{1}{n!} (iz)^n$

Trigonometric Functions

Theorem: If z is a complex number, then

$$e^{iz} = \cos z + i \sin z$$

Proof: We know that $e^{iz} = \sum_{n=0}^{\infty} \frac{1}{n!} (iz)^n$

$$= \sum_{n=0}^{\infty} \left[\left[\frac{1}{(2n)!} (iz)^{2n} \right] + \left[\frac{1}{(2n+1)!} (iz)^{2n+1} \right] \right]$$

Trigonometric Functions

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$$= \sum_{n=0}^{\infty} \left[\left[\frac{(i)^{2n}}{(2n)!} (z)^{2n} \right] + \left[\frac{i\,(i^{2n})}{(2n+1)!} (z)^{2n+1} \right] \right]$$

Trigonometric Functions

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$$= \sum_{n=0}^{\infty} \left[\frac{(-1)^n}{(2n)!} (z)^{2n} \right] + i \sum_{n=0}^{\infty} \left[\frac{(-1)^2}{(2n+1)!} (z)^{2n+1} \right]$$

Trigonometric Functions

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$$= \cos z + i \sin z$$

Trigonometric Functions

$$e^{iz} = \cos z + i \sin z$$

Trigonometric Functions

$$e^{iz} = \cos z + i \sin z$$

and

$$e^{-iz} = \cos z - i \sin z$$

Trigonometric Functions

$$e^{iz} = \cos z + i \sin z$$

and

$$e^{-iz} = \cos z - i \sin z$$

implies

$$\sin z = \frac{1}{2i}\left(e^{iz} - e^{-iz}\right)$$

Trigonometric Functions

On the same lines

$$e^{iz} = \cos z + i \sin z$$

and

$$e^{-iz} = \cos z - i \sin z$$

implies

$$\cos z = \frac{1}{2}\left(e^{iz} + e^{-iz}\right)$$

Trigonometric Functions

Similarly the following trigonometric functions can be defined

$$\tan z\,,\ \cot z\,,\ \csc z\,,\ \sec z$$

Trigonometric Functions

Consequences

Trigonometric Functions

Theorem: For $z = x + iy$, we have
$$\sin z = \sin x \cosh y + i \cos x \sinh y$$

Trigonometric Functions

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$$\sin z = \sin x \cosh y + i \cos x \sinh y$$

Proof:
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Trigonometric Functions

Theorem: For $z = x + iy$, we have
$$\sin z = \sin x \cosh y + i \cos x \sinh y$$

Proof:
$$\sin z = \frac{1}{2i}\left(e^{iz} - e^{-iz}\right)$$
$$= \frac{1}{2i}\left(e^{i(x+iy)} - e^{-i(x+iy)}\right)$$

Trigonometric Functions

Theorem: For $z = x + iy$, we have
$$\sin z = \sin x \cosh y + i \cos x \sinh y$$

Proof:
$$\sin z = \frac{1}{2i}\left(e^{iz} - e^{-iz}\right)$$
$$= \frac{1}{2i}\left(e^{i(x+iy)} - e^{-i(x+iy)}\right)$$
$$= \frac{1}{2i}\left(e^{-y+ix} - e^{y-ix}\right)$$

Trigonometric Functions

Theorem: For $z = x + iy$, we have
$$\sin z = \sin x \cosh y + i \cos x \sinh y$$

Proof:
$$\sin z = \frac{1}{2i}\left(e^{iz} - e^{-iz}\right)$$
$$= \frac{1}{2i}\left(e^{i(x+iy)} - e^{-i(x+iy)}\right)$$
$$= \frac{1}{2i}\left(e^{-y+ix} - e^{y-ix}\right)$$
$$= \frac{1}{2i}\left(e^{-y}(\cos x + i \sin x) - e^{y}(\cos x - i \sin x)\right)$$

Trigonometric Functions

Theorem: For $z = x + iy$, we have
$$\sin z = \sin x \cosh y + i \cos x \sinh y$$

Proof:
$$\sin z = \frac{1}{2i}\left(e^{iz} - e^{-iz}\right)$$
$$= \frac{1}{2i}\left(e^{i(x+iy)} - e^{-i(x+iy)}\right)$$
$$= \frac{1}{2i}\left(e^{-y+ix} - e^{y-ix}\right)$$
$$= \frac{1}{2i}\left(e^{-y}(\cos x + i \sin x) - e^{y}(\cos x - i \sin x)\right)$$
$$= \sin x \frac{e^{y} + e^{-y}}{2} + i \cos x \frac{e^{y} - e^{-y}}{2}$$

Trigonometric Functions

Theorem: For $z = x + iy$, we have
$$\sin z = \sin x \cosh y + i \cos x \sinh y$$

Proof:
$$\sin z = \frac{1}{2i}\left(e^{iz} - e^{-iz}\right)$$
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$$= \frac{1}{2i}\left(e^{-y}(\cos x + i \sin x) - e^{y}(\cos x - i \sin x)\right)$$
$$= \sin x \frac{e^{y} + e^{-y}}{2} + i \cos x \frac{e^{y} - e^{-y}}{2}$$
$$= \sin x \cosh y + i \cos x \sinh y$$

Trigonometric Functions

On the same lines we can show the following

Theorem: For $z = x + iy$, we have

$$\cos z = \cos x \cosh y - i \sin x \sinh y$$

Trigonometric Functions

Theorem: For all complex numbers $z = x + iy$, we have

$$\sin(z + 2\pi) = \sin z,$$
$$\cos(z + 2\pi) = \cos z,$$
$$\sin(z + \pi) = -\sin z,$$
$$\cos(z + \pi) = -\cos z,$$
$$\tan(z + \pi) = \tan z,$$
$$\cot(z + \pi) = \cot z.$$

Trigonometric Functions

Proof $(\sin(z + 2\pi) = \sin z)$

Trigonometric Functions

Proof $(\sin(z + 2\pi) = \sin z)$

$$\sin(z + 2\pi)$$
$$= \sin(x + iy + 2\pi)$$

Trigonometric Functions

Proof $(\sin(z + 2\pi) = \sin z)$

$$\sin(z + 2\pi)$$
$$= \sin(x + iy + 2\pi)$$
$$= \sin((x + 2\pi) + iy)$$

Recall : $\sin z = \sin x \cosh y + i \cos x \sinh y$

Trigonometric Functions

Proof $(\sin(z + 2\pi) = \sin z)$

$$\sin(z + 2\pi)$$
$$= \sin(x + iy + 2\pi)$$
$$= \sin((x + 2\pi) + iy)$$
$$= \sin(x + 2\pi)\cosh y + i\cos(x + 2\pi)\sinh y$$

Recall : $\sin z = \sin x \cosh y + i \cos x \sinh y$

Trigonometric Functions

Proof $(\sin(z + 2\pi) = \sin z)$

$$\sin(z + 2\pi)$$
$$= \sin(x + iy + 2\pi)$$
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Trigonometric Functions

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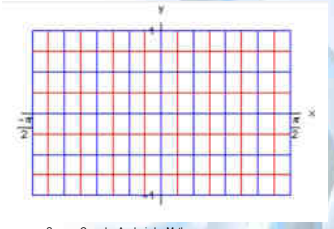
$$\cos(z + 2\pi) = \cos z,$$
$$\sin(z + \pi) = -\sin z,$$
$$\cos(z + \pi) = -\cos z,$$
$$\tan(z + \pi) = \tan z,$$
$$\cot(z + \pi) = \cot z.$$

Proof: Similar to first part.

Trigonometric Functions

Trigonometric functions
Part I

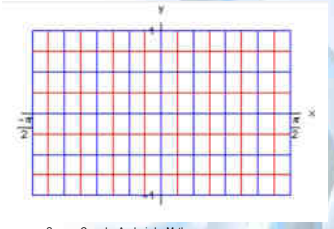


Source: Complex Analysis by Mathews

Trigonometric Functions

Trigonometric functions
Part II

Topic No. 77



Source: Complex Analysis by Mathews

Trigonometric Functions

Further Properties of trigonometric functions

Trigonometric Functions

If z_1 and z_2 are two complex numbers, then

$$\sin(z_1 + z_2) = \sin(z_1)\cos(z_2) + \cos z_1 \sin z_2,$$

Trigonometric Functions

Theorem: If z_1 and z_2 are two complex numbers, then

$$\cos(z_1 + z_2) = \cos(z_1)\cos(z_2) - \sin z_1 \sin z_2$$

and

$$\sin(z_1 + z_2) = \sin(z_1)\cos(z_2) + \cos z_1 \sin z_2,$$

Trigonometric Functions

Theorem: If z_1 and z_2 are two complex numbers, then

$$\cos(z_1 + z_2) = \cos(z_1)\cos(z_2) - \sin z_1 \sin z_2$$

and

$$\sin(z_1 + z_2) = \sin(z_1)\cos(z_2) + \cos z_1 \sin z_2,$$

Proof: Observe that

$$\cos z_1 \cos z_2 = \frac{1}{4}\left(e^{i(z_1+z_2)} + e^{i(z_1-z_2)} + e^{i(z_2-z_1)} + e^{-i(z_1+z_2)}\right)$$

Trigonometric Functions

Theorem: If z_1 and z_2 are two complex numbers, then

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and

$$-\sin z_1 \sin z_2 = \frac{1}{4}\left(e^{i(z_1+z_2)} - e^{i(z_1-z_2)} - e^{i(z_2-z_1)} + e^{-i(z_1+z_2)}\right)$$

Trigonometric Functions

Theorem: If z_1 and z_2 are two complex numbers, then

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and

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Proof: Observe that

$$\cos z_1 \cos z_2 = \frac{1}{4}\left(e^{i(z_1+z_2)} + e^{i(z_1-z_2)} + e^{i(z_2-z_1)} + e^{-i(z_1+z_2)}\right)$$

and

$$-\sin z_1 \sin z_2 = \frac{1}{4}\left(e^{i(z_1+z_2)} - e^{i(z_1-z_2)} - e^{i(z_2-z_1)} + e^{-i(z_1+z_2)}\right)$$

Hence, we get

$$\cos(z_1 + z_2) = \cos(z_1)\cos(z_2) - \sin z_1 \sin z_2$$

Trigonometric Functions

Theorem: If z_1 and z_2 are two complex numbers, then

$$\cos(z_1 + z_2) = \cos(z_1)\cos(z_2) - \sin z_1 \sin z_2$$

and

$$\sin(z_1 + z_2) = \sin(z_1)\cos(z_2) + \cos z_1 \sin z_2,$$

Proof: Similarly we can prove the following

$$\sin(z_1 + z_2) = \sin(z_1)\cos(z_2) + \cos z_1 \sin z_2$$

Trigonometric Functions

If z_1 and z_2 are two complex numbers, then

$$\sin(z_1 + z_2) = \sin(z_1)\cos(z_2) + \cos z_1 \sin z_2,$$

and

$$\cos(z_1 + z_2) = \cos(z_1)\cos(z_2) - \sin z_1 \sin z_2$$

Trigonometric Functions

As a consequence we get

$$\sin(2z) = 2 \sin z \cos z,$$

Trigonometric Functions

As a consequence we get

$$\sin(2z) = 2 \sin z \cos z,$$
$$\cos(2z) = (\cos z)^2 - (\sin z)^2,$$

Trigonometric Functions

$$\sin\left(\frac{\pi}{2} + z\right) = \sin\left(\frac{\pi}{2} - z\right) = \cos z.$$

Trigonometric Functions

Zeros of
$\sin z$ and $\cos z$

Trigonometric Functions

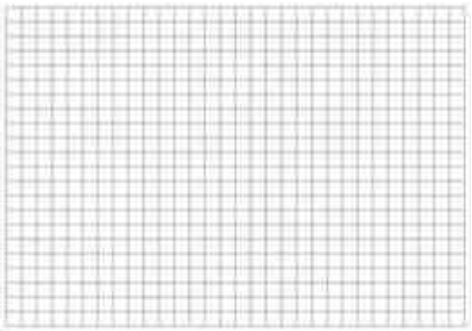
$$\sin z = 0$$
if and only if
$$z = n\pi,\ n \in \mathbb{Z}$$

Trigonometric Functions

Fact: $\sin z = 0$ if and only if $z = n\pi, n \in \mathbb{Z}$.

Trigonometric Functions

Fact: $\sin z = 0$ if and only if $z = n\pi$

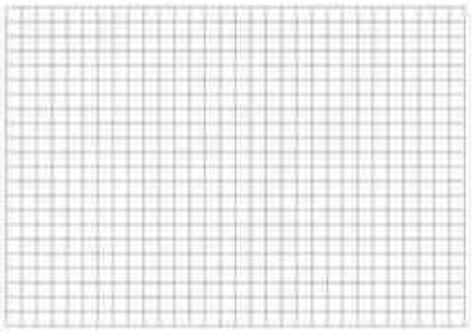


Trigonometric Functions

$\cos z = 0$
if and only if
$z = \left(n + \frac{1}{2}\right)\pi, n \in \mathbb{Z}$

Trigonometric Functions

Fact: $\cos z = 0$ if and only if $z = \left(n + \frac{1}{2}\right)\pi, n \in \mathbb{Z}$



Trigonometric Functions

There is one big difference between real and complex trigonometric functions

Trigonometric Functions

For real variable x
$|\sin x| \leq 1$
But for complex variable z
$\sin z$ is unbounded

Trigonometric Functions

Similarly for cosine function
For real variable x
$|\cos x| \leq 1$
But for complex variable z
$\cos z$ is unbounded

Trigonometric Functions

Fact: For complex variable z, $\sin z$ is unbounded.

$|\sin z|^2 = |\sin(x + iy)|^2$

Trigonometric Functions

Fact: For complex variable z, $\sin z$ is unbounded.

$$|\sin z|^2 = |\sin(x + iy)|^2$$
$$= |\sin x \cosh y + i \cos x \sinh y|^2$$

Trigonometric Functions

Fact: For complex variable z, $\sin z$ is unbounded.

$$|\sin z|^2 = |\sin(x+iy)|^2$$
$$= |\sin x \cosh y + i \cos x \sinh y|^2$$
$$= \sin^2 x \cosh^2 y + \cos^2 x \sinh^2 y$$

Trigonometric Functions

Fact: For complex variable z, $\sin z$ is unbounded.

$$|\sin z|^2 = |\sin(x+iy)|^2$$
$$= |\sin x \cosh y + i \cos x \sinh y|^2$$
$$= \sin^2 x \cosh^2 y + \cos^2 x \sinh^2 y$$
$$= \sin^2 x(\cosh^2 y - \sinh^2 y) + \sinh^2 y(\cos^2 x + \sin^2 x)$$

Trigonometric Functions

Fact: For complex variable z, $\sin z$ is unbounded.

$$|\sin z|^2 = |\sin(x+iy)|^2$$
$$= |\sin x \cosh y + i \cos x \sinh y|^2$$
$$= \sin^2 x \cosh^2 y + \cos^2 x \sinh^2 y$$
$$= \sin^2 x(\cosh^2 y - \sinh^2 y) + \sinh^2 y(\cos^2 x + \sin^2 x)$$

Using $\cosh^2 y - \sinh^2 y = 1$ and $\cos^2 x + \sin^2 x = 1$, we get

$$|\sin z|^2 = \sin^2 x + \sinh^2 y$$

Trigonometric Functions

$$|\sin z|^2 = \sin^2 x + \sinh^2 y$$

$$\lim_{y\to\infty} |\sin(x_0 + iy)|^2$$

Trigonometric Functions

$$|\sin z|^2 = \sin^2 x + \sinh^2 y$$

$$\lim_{y\to\infty} |\sin(x_0 + iy)|^2$$

$$= \sin^2 x_0 + \lim_{y\to\infty} \sinh^2 y = \infty$$

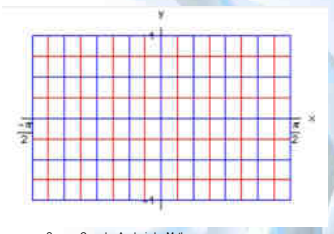
Trigonometric Functions

Fact: For complex variable z, $\cos z$ is unbounded.
Similarly we can prove the following:

$$|\cos z|^2 = \cos^2 x + \sinh^2 y$$

Trigonometric Functions

Trigonometric functions
Part II

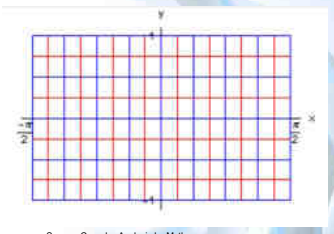


Source: Complex Analysis by Mathews

Trigonometric Functions

Topic No. 78

Trigonometric functions
Part III



Source: Complex Analysis by Mathews

The mapping $w = sin\ z$

Mapping
$w = \sin z$

The mapping $w = sin\ z$

To understand the mapping $w = \sin z$, we study images of line segments from z-plane into w-plane

The mapping $w = sin\ z$

To understand the mapping $w = \sin z$, we study images of line segments from z-plane into w-plane. We consider line segments parallel to y-axis with x-coordinates given by

$$x = -\frac{\pi}{2} + \frac{k\pi}{12}, k = 0,1, \dots, 12$$

and y vary continuously in the interval $-3 \le y \le 3$.

Z-plane

?

W-plane

Source: Complex Analysis by mathews

The mapping $w = sin\ z$

The line segments are parallel to y-axis so the x-coordinate is constant on each line segment. So line segments are given by the equation $x = a$, where

$$-\frac{\pi}{2} \le a \le \frac{\pi}{2}$$

and y vary between $-3 \le y \le 3$.

The line segments are parallel to y-axis so the x-coordinate is constant on each line segment. So line segments are given by the equation x = a, where

$$-\frac{\pi}{2} \le a \le \frac{\pi}{2}$$

and y vary between $-3 \le y \le 3$.

Source: Complex Analysis by mathews

The mapping $w = sin\ z$

Let $z = x + iy$ be coordinates in the z-plane; let

$$w = \sin z = u + iv$$

be the coordinates in the w-plane.

The mapping $w = sin\ z$

Let $z = x + iy$ be coordinates in the z-plane; let

$$w = \sin z = u + iv$$

be the coordinates in the w-plane.

We know that

$$w = \sin z = \sin(x + iy) = \sin x \cosh y + i \cos x \sinh y$$

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Hence

$$u = \sin x \cosh y, \qquad v = \cos x \sinh y$$

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Hence

$$u = \sin x \cosh y, \qquad v = \cos x \sinh y$$

We first consider the line segments $x = a$, where

$$-\frac{\pi}{2} < a < \frac{\pi}{2}, a \neq 0$$

and y vary between $-3 \le y \le 3$.

The mapping $w = sin\ z$

Let $z = x + iy$ be coordinates in the z-plane; let

$$w = \sin z = u + iv$$

be the coordinates in the w-plane.

We know that

$$w = \sin z = \sin(x + iy) = \sin x \cosh y + i \cos x \sinh y$$

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$$u = \sin x \cosh y\,, \qquad v = \cos x \sinh y$$

We first consider the line segments $x = a$, where

$$-\frac{\pi}{2} < a < \frac{\pi}{2}\,, a \neq 0$$

and y vary between $-3 \leq y \leq 3$.

So

$$u = \sin a \cosh y\,, \qquad v = \cos a \sinh y$$

The mapping $w = sin\ z$

So

$$u = \sin a \cosh y\,, \qquad v = \cos a \sinh y$$

Since $a \neq 0, \frac{\pi}{2}, -\frac{\pi}{2}$ so $\sin a \neq 0$ and $\cos a \neq 0$. Hence we get

$$\left(\frac{u}{\sin a}\right)^2 - \left(\frac{v}{\cos a}\right)^2 = 1\,,$$

which is equation of hyperbola with vertices $(\pm \sin a\,, 0)$.

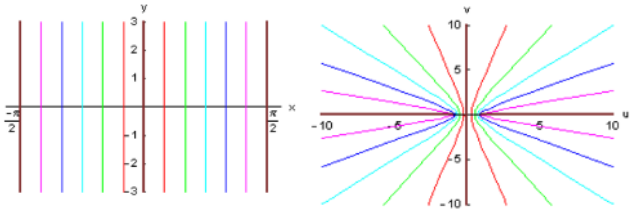
The mapping $w = sin\ z$

So

$$u = \sin a \cosh y\,, \qquad v = \cos a \sinh y$$

Since $a \neq 0, \frac{\pi}{2}, -\frac{\pi}{2}$ so $\sin a \neq 0$ and $\cos a \neq 0$.

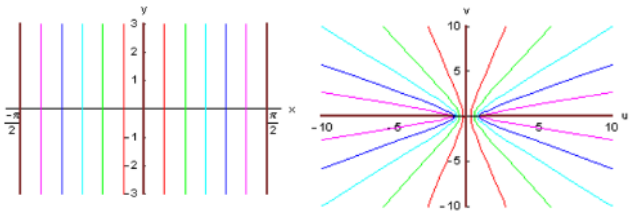
The mapping $w = sin\ z$



The line segment $x = a$, where $0 < a < \frac{\pi}{2}$ maps to branch of hyperbola on the right of v-axis.

Source: Complex Analysis by mathews

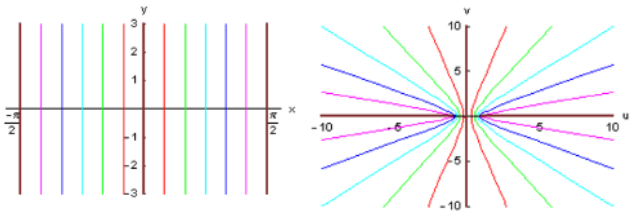
The mapping $w = sin\ z$



The line segment $x = a$, where $0 < a < \frac{\pi}{2}$ maps to branch of hyperbola on the right of v-axis. Similarly $x = -a$, where $0 < a < \frac{\pi}{2}$ maps to branch of hyperbola on the left of v-axis.

Source: Complex Analysis by mathews

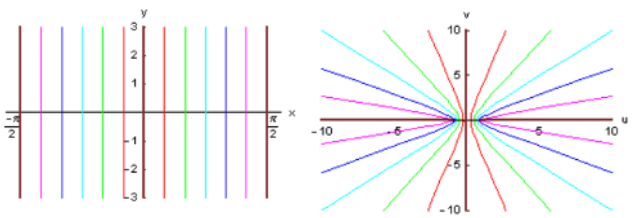
The mapping $w = sin\ z$



The line segment $x = a$, where $0 < a < \frac{\pi}{2}$ maps to branch of hyperbola on the right of v-axis. Similarly $x = -a$, where $0 < a < \frac{\pi}{2}$ maps to branch of hyperbola on the left of v-axis. So pair of lines $x = \pm a$, where $0 < a < \frac{\pi}{2}$, mapped to one hyperbola given by $\left(\frac{u}{\sin a}\right)^2 - \left(\frac{v}{\cos a}\right)^2 = 1$.

Source: Complex Analysis by mathews

The mapping $w = sin\ z$



Now consider $a = 0$, that is, put $x = 0$ in the following

$$u = \sin x \cosh y\,, \qquad v = \cos x \sinh y$$

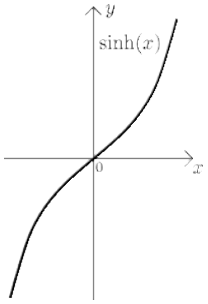
We get

$$u = 0, v = \sinh y\,, \qquad -3 \leq y \leq 3$$

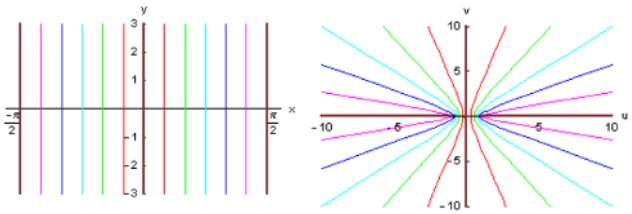
Source: Complex Analysis by mathews

The mapping $w = sin\ z$

Graph of $\sinh x$:



The mapping $w = sin\ z$



Now consider $a = 0$, that is, put $x = 0$ in the following

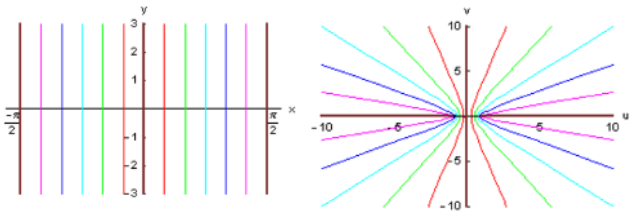
$$u = \sin x \cosh y\,, \qquad v = \cos x \sinh y$$

We get

$$u = 0, v = \sinh y\,, \qquad -3 \le y \le 3$$

Source: Complex Analysis by mathews

The mapping $w = sin\ z$



Now consider $a = \pm\frac{\pi}{2}$, that is, put $x = \pm\frac{\pi}{2}$ in the following

$$u = \sin x \cosh y\,, \qquad v = \cos x \sinh y$$

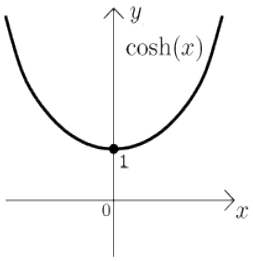
We get

$$u = \pm\cosh y\,, v = 0, \qquad -3 \le y \le 3$$

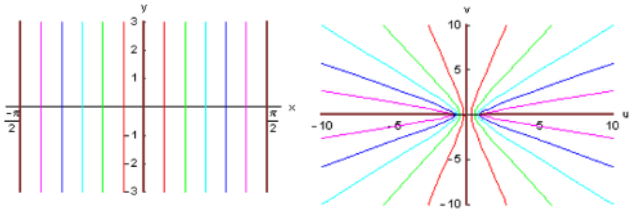
Source: Complex Analysis by mathews

The mapping $w = sin\ z$

Graph of $\cosh x$:



The mapping $w = sin\ z$



Now consider $a = \pm\frac{\pi}{2}$, that is, put $x = \pm\frac{\pi}{2}$ in the following

$$u = \sin x \cosh y\,, \qquad v = \cos x \sinh y$$

We get

$$u = \pm\cosh y\,, v = 0, \qquad -3 \le y \le 3$$

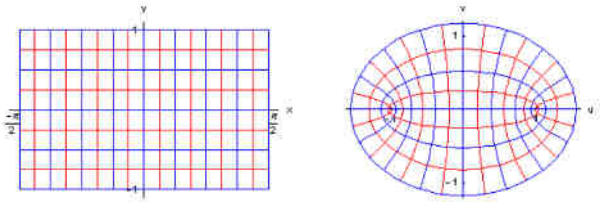
Source: Complex Analysis by mathews

The mapping $w = sin\ z$

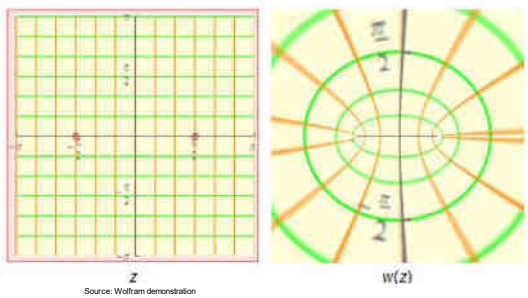
On the same lines we can show that image of horizontal lines

$$y = b, \qquad -\frac{\pi}{2} \le x \le \frac{\pi}{2}\,.$$

are ellipses.

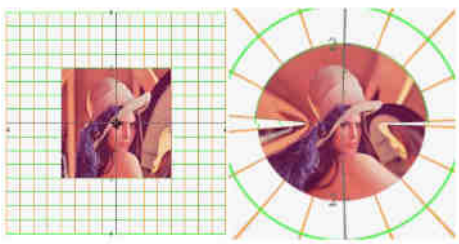


The mapping $w = sin\ z$



Source: Wolfram demonstration

The mapping $w = sin\ z$



Source: Wolfram demonstration

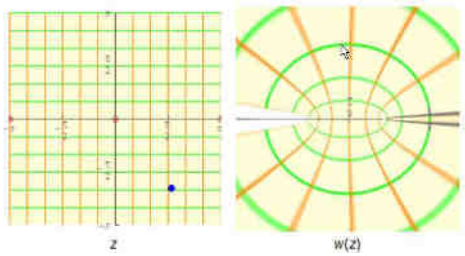
The mapping $w = cos\ z$

The mapping

$w = \cos z$

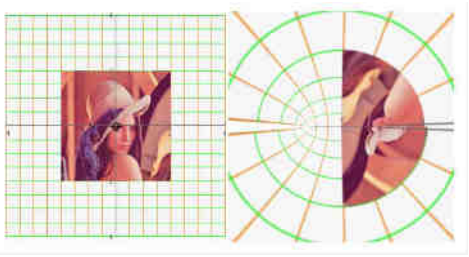
Can be dealt in a similar way.

The mapping $w = cos\ z$



Source: Wolfram demonstration

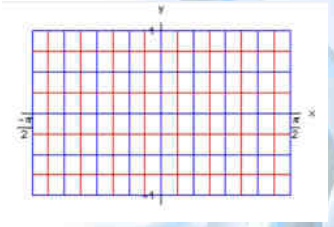
The mapping $w = cos\ z$



Source: Wolfram demonstration

Trigonometric Functions

Trigonometric functions
Part III

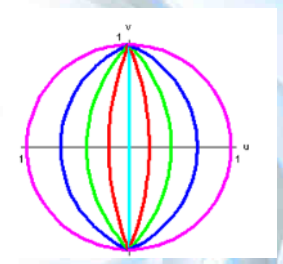


Source: Complex Analysis by Mathews

Trigonometric Functions

Trigonometric functions
Part IV

Topic No. 79



Source: Complex Analysis by Mathews

The mapping $w = tan\ z$

Mapping
$w = \tan z$

The mapping $w = tan\ z$

To understand the mapping $w = \tan z$, we study images of line segments from z-plane into w-plane.

The mapping $w = tan\ z$

To understand the mapping $w = \tan z$, we study images of line segments from z-plane into w-plane. We consider line segments parallel to y-axis with x-coordinates given by

$$x = -\frac{\pi}{4} + \frac{k\pi}{16}, k = 0,1, \dots, 8$$

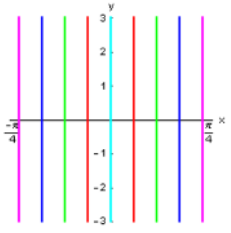
and y vary continuously in the interval $-3 \leq y \leq 3$.

The mapping $w = tan\ z$

The line segments are parallel to y-axis so the x-coordinate is constant on each line segment. So line segments are given by the equation $x = a$, where

$$-\frac{\pi}{4} \leq a \leq \frac{\pi}{4}$$

and y vary between $-3 \leq y \leq 3$.



Source: Complex Analysis by mathews

The mapping $w = tan\ z$

Let $z = x + iy$ be coordinates in the z-plane; let

$$w = \tan z = u + iv$$

be the coordinates in the w-plane.

The mapping $w = tan\ z$

We know that

$$\tan z = \tan(x + iy) = \frac{\sin(x + iy)}{\cos(x + iy)}$$

The mapping $w = tan\ z$

We know that

$$\tan z = \tan(x + iy) = \frac{\sin(x + iy)}{\cos(x + iy)}$$
$$= \frac{\sin x \cosh y + i \cos x \sinh y}{\cos x \cosh y - i \sin x \sinh y}$$

The mapping $w = tan\ z$

We know that

$$\tan z = \tan(x + iy) = \frac{\sin(x + iy)}{\cos(x + iy)}$$
$$= \frac{\sin x \cosh y + i \cos x \sinh y}{\cos x \cosh y - i \sin x \sinh y}$$

Simplifying, we get

$$\tan z = \frac{\cos x \sin x + i \cosh y \sinh y}{\cos^2 x \cosh^2 y + \sin^2 x \sinh^2 y}$$

The mapping $w = tan\ z$

We know that

$$\tan z = \tan(x + iy) = \frac{\sin(x + iy)}{\cos(x + iy)}$$
$$= \frac{\sin x \cosh y + i \cos x \sinh y}{\cos x \cosh y - i \sin x \sinh y}$$

Simplifying, we get

$$\tan z = \frac{\cos x \sin x + i \cosh y \sinh y}{\cos^2 x \cosh^2 y + \sin^2 x \sinh^2 y}$$

Hence

$$u = \frac{\cos x \sin x}{\cos^2 x \cosh^2 y + \sin^2 x \sinh^2 y},$$
$$v = \frac{\cosh y \sinh y}{\cos^2 x \cosh^2 y + \sin^2 x \sinh^2 y}$$

The mapping $w = tan\ z$

So

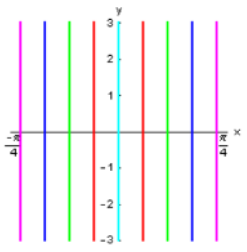
$$u = \frac{\cos x \sin x}{\cos^2 x \cosh^2 y + \sin^2 x \sinh^2 y},$$
$$v = \frac{\cosh y \sinh y}{\cos^2 x \cosh^2 y + \sin^2 x \sinh^2 y}$$

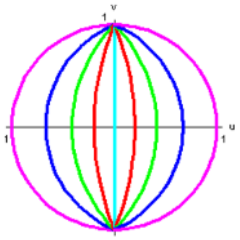
The image of the equation $x = a$, where

$$-\frac{\pi}{4} \le a \le \frac{\pi}{4}$$

and y vary between $-3 \le y \le 3$ is a circular arc.

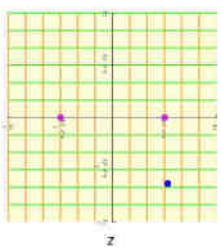
The mapping $w = tan\ z$





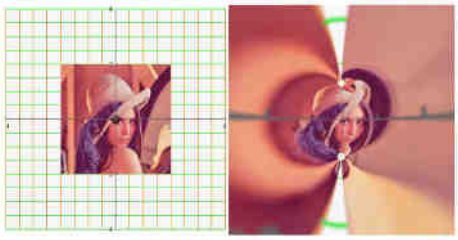
Source: Complex Analysis by mathews

The mapping $w = tan\ z$



Source: Wolfram demonstration

The mapping $w = tan\ z$



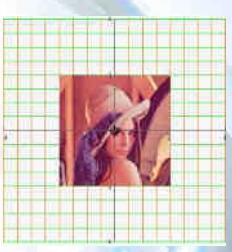
Source: Wolfram demonstration

The mapping $w = cos\ z$

Other mappings of trigonometric functions can be understood in the similar way.

Trigonometric Functions

Trigonometric functions
Part IV



Source: Wolfram demonstration

Trigonometric Functions

Topic No. 80

Trigonometric functions
Part V

Derivatives of Trigonometric Functions

Derivatives of trigonometric functions

Derivatives of Trigonometric Functions

Theorem: $\sin(z)$ and $\cos(z)$ are entire functions with

$$\frac{d}{dz}\sin(z) = \cos(z) \quad \text{and} \quad \frac{d}{dz}\cos(z) = -\sin(z)$$

Proof:

Derivatives of Trigonometric Functions

Proof: $(\frac{d}{dz}\sin(z) = \cos(z))$

$$\frac{d}{\mathrm{dz}}\sin(z) = \frac{d}{\mathrm{dz}}\sum_{n=0}^{\infty}\frac{(-1)^n z^{2n+1}}{(2n+1)!}$$

Derivatives of Trigonometric Functions

Proof: $(\frac{d}{dz}\sin(z) = \cos(z))$

$$\frac{d}{\mathrm{dz}}\sin(z) = \frac{d}{\mathrm{dz}}\sum_{n=0}^{\infty}\frac{(-1)^n z^{2n+1}}{(2n+1)!}$$

$$= \sum_{n=0}^{\infty}\left(\frac{d}{\mathrm{dz}}\frac{(-1)^n}{(2n+1)!}z^{2n+1}\right)$$

Derivatives of Trigonometric Functions

Proof: $(\frac{d}{dz}\sin(z) = \cos(z))$

$$\frac{d}{\mathrm{dz}}\sin(z) = \frac{d}{\mathrm{dz}}\sum_{n=0}^{\infty}\frac{(-1)^n z^{2n+1}}{(2n+1)!}$$

$$= \sum_{n=0}^{\infty}\left(\frac{d}{\mathrm{dz}}\frac{(-1)^n}{(2n+1)!}z^{2n+1}\right)$$

$$= \sum_{n=0}^{\infty}\left(\frac{(-1)^n(2n+1)}{(2n+1)!}z^{2n}\right)$$

Derivatives of Trigonometric Functions

Proof: $(\frac{d}{dz}\sin(z) = \cos(z))$

$$\frac{d}{\mathrm{dz}}\sin(z) = \frac{d}{\mathrm{dz}}\sum_{n=0}^{\infty}\frac{(-1)^n z^{2n+1}}{(2n+1)!}$$

$$= \sum_{n=0}^{\infty}\left(\frac{d}{\mathrm{dz}}\frac{(-1)^n}{(2n+1)!}z^{2n+1}\right)$$

$$= \sum_{n=0}^{\infty}\left(\frac{(-1)^n(2n+1)}{(2n+1)!}z^{2n}\right)$$

$$= \sum_{n=0}^{\infty}\left(\frac{(-1)^n}{(2n)!}z^{2n}\right) = \cos(z)$$

Derivatives of Trigonometric Functions

Theorem: $\sin(z)$ and $\cos(z)$ are entire functions with

$$\frac{d}{dz}\sin(z) = \cos(z) \quad \text{and} \quad \frac{d}{dz}\cos(z) = -\sin(z)$$

Proof: Similar proof can be done for

$$\frac{d}{dz}\cos(z) = \sin(z)\ .$$

Derivatives of Trigonometric Functions

Using the following derivatives

$$\frac{d}{dz}\sin(z) = \cos(z) \quad \text{and} \quad \frac{d}{dz}\cos(z) = -\sin(z)$$

We can prove the following:

$$\frac{d}{\mathrm{dz}}\tan z = (\sec z)^2,$$

$$\frac{d}{\mathrm{dz}}\cot z = -(\csc z)^2,$$

$$\frac{d}{\mathrm{dz}}\sec z = \sec z \tan z\,,$$

$$\frac{d}{\mathrm{dz}}\csc z = -\cot z \csc z\,.$$

Derivatives of Trigonometric Functions

Proof of the fact

$$\frac{d}{\mathrm{dz}}\tan z = (\sec z)^2$$

is given by.

$$\frac{d}{\mathrm{dz}}\tan z = \frac{d}{\mathrm{dz}}\frac{\sin z}{\cos z}$$

Derivatives of Trigonometric Functions

Proof of the fact

$$\frac{d}{\mathrm{dz}}\tan z = (\sec z)^2$$

is given by.

$$\frac{d}{\mathrm{dz}}\tan z = \frac{d}{\mathrm{dz}}\frac{\sin z}{\cos z}$$

$$= \frac{\cos z\frac{d}{\mathrm{dz}}\sin z - \sin z\frac{d}{\mathrm{dz}}\cos z}{(\cos z)^2}$$

Derivatives of Trigonometric Functions

Proof of the fact

$$\frac{d}{\mathrm{dz}}\tan z = (\sec z)^2$$

is given by.

$$\frac{d}{\mathrm{dz}}\tan z = \frac{d}{\mathrm{dz}}\frac{\sin z}{\cos z}$$

$$= \frac{\cos z\frac{d}{\mathrm{dz}}\sin z - \sin z\frac{d}{\mathrm{dz}}\cos z}{(\cos z)^2}$$

$$= \frac{(\cos z)^2 + (\sin z)^2}{(\cos z)^2}$$

Derivatives of Trigonometric Functions

Proof of the fact

$$\frac{d}{\mathrm{dz}}\tan z = (\sec z)^2$$

is given by.

$$\frac{d}{\mathrm{dz}}\tan z = \frac{d}{\mathrm{dz}}\frac{\sin z}{\cos z}$$

$$= \frac{\cos z\frac{d}{\mathrm{dz}}\sin z - \sin z\frac{d}{\mathrm{dz}}\cos z}{(\cos z)^2}$$

$$= \frac{(\cos z)^2 + (\sin z)^2}{(\cos z)^2}$$

$$= \frac{1}{(\cos z)^2}$$

Derivatives of Trigonometric Functions

Proof of the fact

$$\frac{d}{dz}\tan z = (\sec z)^2$$

is given by.

$$\begin{aligned}\frac{d}{dz}\tan z &= \frac{d}{dz}\frac{\sin z}{\cos z}\\ &= \frac{\cos z\frac{d}{dz}\sin z - \sin z\frac{d}{dz}\cos z}{(\cos z)^2}\\ &= \frac{(\cos z)^2+(\sin z)^2}{(\cos z)^2}\\ &= \frac{1}{(\cos z)^2}\\ &= (\sec z)^2\end{aligned}$$

Derivatives of Trigonometric Functions

The following can be proved in a similar way:

$$\frac{d}{dz}\cot z = -(\csc z)^2,$$
$$\frac{d}{dz}\sec z = \sec z\tan z\,,$$
$$\frac{d}{dz}\csc z = -\cot z\csc z\,.$$

Trigonometric Functions

Trigonometric functions
Part V

Hyperbolic Functions

Topic No. 81

Hyperbolic functions

Hyperbolic Functions

How to defined?
$\sinh z$
$\cosh z$
for complex variable z

Hyperbolic Functions

We know for real variable x

$$\sinh x = \frac{e^x - e^{-x}}{2}$$
$$\cosh x = \frac{e^x + e^{-x}}{2}$$

Hyperbolic Functions

So we define

$$\sinh z = \frac{e^z - e^{-z}}{2}$$
$$\cosh z = \frac{e^z + e^{-z}}{2}$$

for complex variable z

Hyperbolic Functions

Other complex hyperbolic functions can be defined using definition of $\sinh z$ and $\cosh z$

Hyperbolic Functions

$$\tanh z = \frac{\sinh z}{\cosh z}$$

Hyperbolic Functions

$$\coth z = \frac{\cosh z}{\sinh z}$$

Hyperbolic Functions

$$\operatorname{sech} z = \frac{1}{\cosh z}$$

Hyperbolic Functions

$$\operatorname{csch} z = \frac{1}{\sinh z}$$

Hyperbolic Functions

Derivatives of hyperbolic functions

Hyperbolic Functions

The derivatives of hyperbolic functions are:

$\frac{d}{dz}\cosh z = \sinh z$, and $\frac{d}{dz}\sinh z = \cosh z$,

$\frac{d}{dz}\tanh z = \operatorname{sech}^2 z$, and $\frac{d}{dz}\coth z = -\operatorname{csch}^2 z$,

$\frac{d}{dz}\operatorname{sech} z = -\operatorname{sech} z \tanh z$, and $\frac{d}{dz}\operatorname{csch} z = -\operatorname{csch} z \coth z$.

Hyperbolic Functions

The derivatives of hyperbolic functions are:

$\frac{d}{dz}\cosh z = \sinh z$, and $\frac{d}{dz}\sinh z = \cosh z$,

$\frac{d}{dz}\tanh z = \operatorname{sech}^2 z$, and $\frac{d}{dz}\coth z = -\operatorname{csch}^2 z$,

$\frac{d}{dz}\operatorname{sech} z = -\operatorname{sech} z \tanh z$, and $\frac{d}{dz}\operatorname{csch} z = -\operatorname{csch} z \coth z$.

Proof:

Hyperbolic Functions

Two important identities

$$\cosh z = \cosh(x + iy) = \cos y \cosh x + i \sinh x \sin y$$
$$\sinh z = \sinh(x + iy) = \sinh x \cos y + i \cosh x \sin y$$

Proof:

Hyperbolic Functions

Two important identities

$$\cosh z = \cosh(x+iy) = \cos y \cosh x + i \sinh x \sin y$$
$$\sinh z = \sinh(x+iy) = \sinh x \cos y + i \cosh x \sin y$$

Proof: Similarly the following can be proved:

$$\sinh z = \sinh(x+iy) = \sinh x \cos y + i \cosh x \sin y$$

Hyperbolic Functions

The following can be proved using definitions.

$$\cosh(iz) = \cos(z)$$
$$\sinh(iz) = i \sin(z)$$
$$\sin(iz) = i \sinh(z)$$
$$\cos(iz) = \cosh(z)$$

Hyperbolic Functions

One of the important identities is:

$$\cosh^2 z - \sinh^2 z = 1 .$$

Proof:

$$\cosh(iz) = \cos(z) \Rightarrow \cosh(z) = \cos(iz)$$

$$\sinh(iz) = i \sin(z) \Rightarrow \sinh(z) = -i \sin(iz)$$

Hyperbolic Functions

Another important identity is following:

$$\cosh(z_1 \pm z_2) = \cosh(z_1)\cosh(z_2) \pm \sinh(z_1)\sinh(z_2)$$

Proof: We know that $\cosh(z) = \cos(iz)$

Hyperbolic Functions

The following important identity can be proved similarly:

$$\sinh(z_1 \pm z_2) = \sinh(z_1)\cosh(z_2) \pm \cosh(z_1)\sinh(z_2)$$

Proof: Similarly using $\sin(iz) = i \sinh(z)$ we prove above.

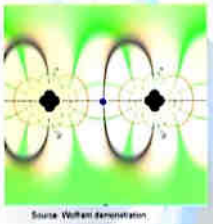
Hyperbolic Functions

Hyperbolic functions

Inverse Trigonometric Functions

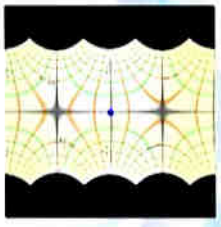
Topic No. 82

Inverse Trigonometric Functions



Inverse Trigonometric Functions

$arcsin(z)$



Inverse Trigonometric Functions

The periodic nature of $\sin(z)$ makes it a multiple-to-one function

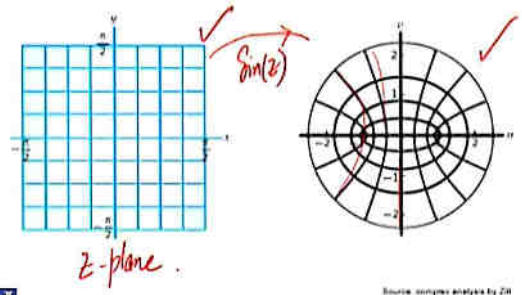
Inverse Trigonometric Functions

Riemann surface of $\sin(z)$

Inverse Trigonometric Functions

Step1: The function $\sin(z)$ is one-to-one on

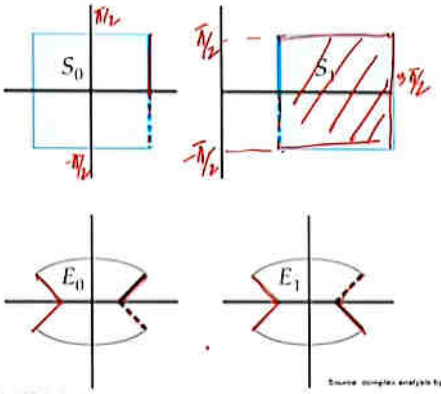
$$\Lambda = \left\{(x, y): -\frac{\pi}{2} \le x \le \frac{\pi}{2}, -\frac{\pi}{2} \le y \le \frac{\pi}{2}\right\}$$



z-plane.

Inverse Trigonometric Functions

Step 2:

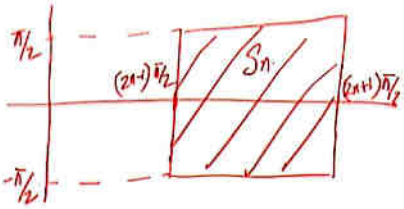


Inverse Trigonometric Functions

Step 3: In general for an integer n an elliptical image of the following square

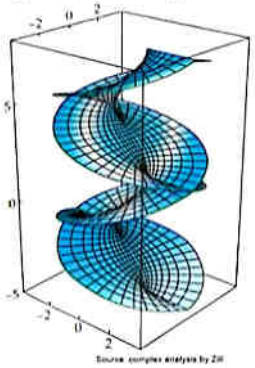
$$S_n = \left\{(x, y): (2n-1)\frac{\pi}{2} \le x \le (2n+1)\frac{\pi}{2}, \frac{\pi}{2} \le y \le \frac{\pi}{2}\right\}$$

is constructed. Each region E_n is cut open, as E_0 and E_1 were, and E_n is glued to E_{n+1} along their boundaries in a manner analogous to that used for E_0 and E_1.



Inverse Trigonometric Functions

Gluing all pieces we get the following Riemann surface.



Inverse Trigonometric Functions

Gluing all pieces we get the following Riemann surface.

Inverse Trigonometric Functions

Hence the inverse of $\sin(z)$, denoted by $arcsin(z)$, is a multivalued function

Inverse Trigonometric Functions

$$arcsin(z) = -i\log\left(iz + (1 - z^2)^{1/2}\right)$$

Inverse Trigonometric Functions

Show that

$$arcsin(z) = -i\log\left(iz + (1 - z^2)^{1/2}\right)$$

Proof:

$w = arcsin(z) \Rightarrow z = \sin(w)$

$\Rightarrow z = \dfrac{e^{iw} - e^{-iw}}{2i}$, Find w in terms of z

$\Rightarrow 2iz = e^{iw} - \dfrac{1}{e^{iw}} \Rightarrow 2iz\,e^{iw} = (e^{iw})^2 - 1$

$\Rightarrow (e^{iw})^2 - 2iz\,e^{iw} - 1 = 0$ → Quadratic in e^{iw}

$\Rightarrow e^{iw} = \dfrac{2iz \pm \sqrt{4i^2z^2 + 4}}{2}$

Inverse Trigonometric Functions

The function is a multivalued function

$$arcsin(z) = -i\log\left(iz + (1 - z^2)^{1/2}\right)$$

Inverse Trigonometric Functions

When principal values of $\log(z)$ and $z^{\frac{1}{2}}$ are used we denote it as

$$Arcssin(z) = -iLog\left(iz + (1 - z^2)^{1/2}\right)$$

Inverse Trigonometric Functions

The function

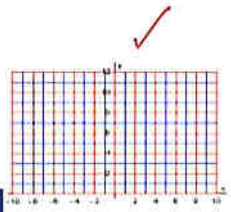
$$Arcssin(z) = -iLog\left(iz + (1 - z^2)^{1/2}\right)$$

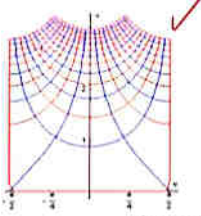
maps upper half plane

$$\{z: Im\,(z) > 0\}$$

onto a portion of upper half plane that lies in the vertical strip

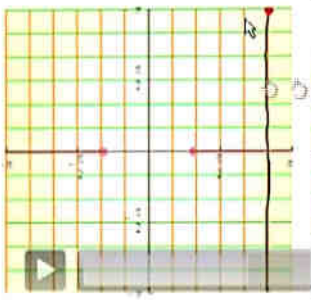
$$\left\{w: -\frac{\pi}{2} < Re(w) < \frac{\pi}{2}\right\}$$

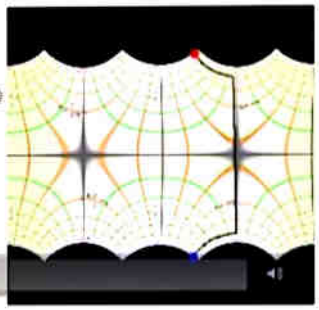




Inverse Trigonometric Functions

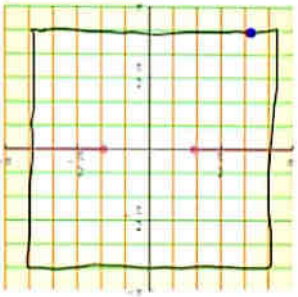
$$Arcssin(z) = -iLog\left(iz + (1 - z^2)^{1/2}\right)$$

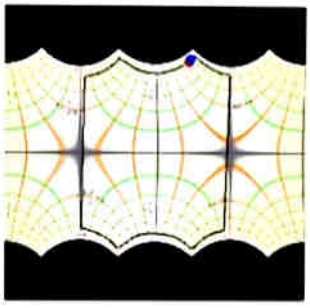




Inverse Trigonometric Functions

$$Arcssin(z) = -iLog\left(iz + (1 - z^2)^{1/2}\right)$$





Inverse Trigonometric Functions

Fact:

$$\arccos z = -i\log\left(z + i(1 - z^2)^{1/2}\right)$$

Proof:

$$\arccos z = \frac{\pi}{2} - \arcsin z = \frac{\pi}{2} + i\log\left(iz + (1 - z^2)^{1/2}\right)$$

$$= \frac{\pi}{2} - i\log\left(-iz + (1 - z^2)^{1/2}\right)$$

$\log(ab) = \log(a) + \log(b)$

$$= -i\frac{\pi i}{2} - i\log\left(-iz + (1 - z^2)^{1/2}\right)$$

$$= -i\log(i) - i\log\left(-iz + (1 - z^2)^{1/2}\right)$$

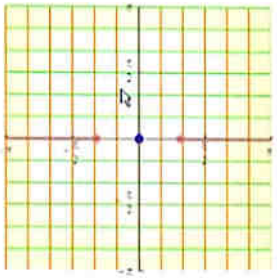
$$= -i\left(\log(i) + \log\left(-iz + (1 - z^2)^{1/2}\right)\right)$$

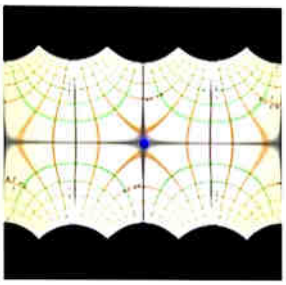
$$= -\log\left(i(-i)z + i(1 - z^2)^{1/2}\right) = -\log\left(z + i(1 - z^2)^{1/2}\right)$$

Inverse Trigonometric Functions

Similarly we can sketch principal branch of the following function

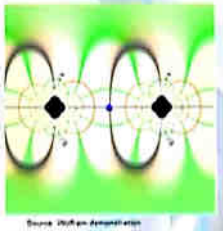
$$\text{Arccos}\, z = -iLog\left(z + i(1 - z^2)^{1/2}\right)$$





Inverse Trigonometric Functions

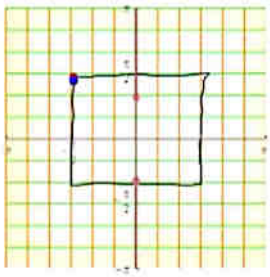
$arctan(z)$

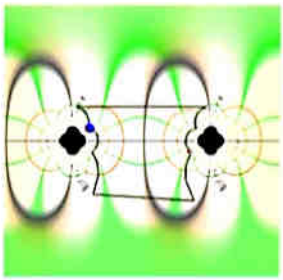


Inverse Trigonometric Functions

we can sketch principal branch of the following function

$$\text{Arctan}\, z = \frac{i}{2}\text{Log}\left(\frac{i+z}{i-z}\right)$$

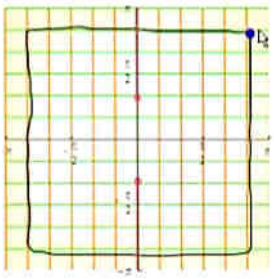


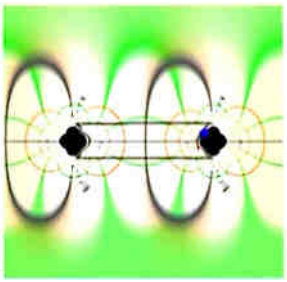


Inverse Trigonometric Functions

we can sketch principal branch of the following function

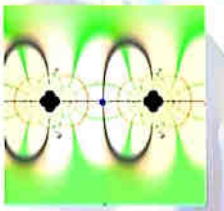
$$\text{Arctan}\, z = \frac{i}{2}\text{Log}\left(\frac{i+z}{i-z}\right)$$





Inverse Trigonometric Functions

Inverse Trigonometric Functions



Inverse Trigonometric Functions

Topic No. 83

Derivatives of inverse trigonometric functions

Inverse Trigonometric Functions

$arcsin(z)$
is a multivalued function

Inverse Trigonometric Functions

But we can talk about the derivative of the principal branch of $\arcsin(z)$

Inverse Trigonometric Functions

$$\frac{d}{\mathrm{dz}}\arcsin z = \frac{1}{\sqrt{1-z^2}}$$

Inverse Trigonometric Functions

Fact:

$$\frac{d}{\mathrm{dz}}\arcsin z = \frac{1}{\sqrt{1-z^2}}$$

Proof:

$$\frac{d}{dz}z = \frac{d}{\mathrm{dz}}\sin w$$

Inverse Trigonometric Functions

Fact:

$$\frac{d}{\mathrm{dz}}\arcsin z = \frac{1}{\sqrt{1-z^2}}$$

Proof:

$$\frac{d}{dz}z = \frac{d}{\mathrm{dz}}\sin w$$
$$1 = \frac{d}{\mathrm{dw}}\sin w\frac{\mathrm{dw}}{\mathrm{dz}}$$

Inverse Trigonometric Functions

Fact:

$$\frac{d}{\mathrm{dz}}\arcsin z = \frac{1}{\sqrt{1-z^2}}$$

Proof:

$$\frac{d}{dz}z = \frac{d}{\mathrm{dz}}\sin w$$
$$1 = \frac{d}{\mathrm{dw}}\sin w\frac{\mathrm{dw}}{\mathrm{dz}}$$
$$1 = \cos w\,\frac{\mathrm{dw}}{\mathrm{dz}}$$

Inverse Trigonometric Functions

Fact:

$$\frac{d}{\mathrm{dz}}\arcsin z = \frac{1}{\sqrt{1-z^2}}$$

Proof:

$$\frac{d}{dz}z = \frac{d}{\mathrm{dz}}\sin w$$
$$1 = \frac{d}{\mathrm{dw}}\sin w\frac{\mathrm{dw}}{\mathrm{dz}}$$
$$1 = \cos w\,\frac{\mathrm{dw}}{\mathrm{dz}}$$
$$\frac{\mathrm{dw}}{\mathrm{dz}} = \frac{1}{\cos w}$$

Inverse Trigonometric Functions

Fact:

$$\frac{d}{\mathrm{dz}}\arcsin z = \frac{1}{\sqrt{1-z^2}}$$

Proof:

$$\frac{d}{dz}z = \frac{d}{\mathrm{dz}}\sin w$$
$$1 = \frac{d}{\mathrm{dw}}\sin w\frac{\mathrm{dw}}{\mathrm{dz}}$$
$$1 = \cos w\,\frac{\mathrm{dw}}{\mathrm{dz}}$$
$$\frac{\mathrm{dw}}{\mathrm{dz}} = \frac{1}{\cos w}$$
$$\frac{dw}{dz} = \frac{1}{\sqrt{1-\sin w^2}} = \frac{1}{\sqrt{1-z^2}}$$

Inverse Trigonometric Functions

$$\frac{d}{\mathrm{dz}}\arccos z = \frac{-1}{\sqrt{1-z^2}}$$

Inverse Trigonometric Functions

$$\frac{d}{\mathrm{dz}}\arctan z = \frac{1}{1+z^2}$$

Inverse Trigonometric Functions

Derivatives of inverse trigonometric functions

Inverse hyperbolic Functions

Topic No. 84

Inverse hyperbolic Functions

Inverse hyperbolic Functions

$arcsinh(z)$
$\operatorname{arccosh}(z)$
$\operatorname{arctanh}(z)$

Inverse hyperbolic Functions

$$\operatorname{arcsinh} z = \log\left[z + (z^2+1)^{1/2}\right]$$

Inverse hyperbolic Functions

$$\operatorname{arccosh} z = \log\left[z + (z^2-1)^{1/2}\right]$$

Inverse hyperbolic Functions

$$\operatorname{arctanh} z = \frac{1}{2}\log\left(\frac{1+z}{1-z}\right)$$

Inverse hyperbolic Functions

Fact

$$\operatorname{arcsinh} z = \log\left[z + (z^2+1)^{1/2}\right]$$

Proof:

Inverse hyperbolic Functions

Similarly we can prove the following

$$\operatorname{arccosh} z = \log\left[z + (z^2-1)^{1/2}\right]$$

Inverse hyperbolic Functions

Similarly we can prove the following

$$\operatorname{arctanh} z = \frac{1}{2}\log\left(\frac{1+z}{1-z}\right)$$

Inverse hyperbolic Functions

Derivatives of inverse hyperbolic functions

Inverse hyperbolic Functions

Derivatives of inverse hyperbolic functions are:

$$\frac{d}{\mathrm{dz}}\operatorname{arcsinh} z = \frac{1}{\sqrt{1+z^2}},$$
$$\frac{d}{\mathrm{dz}}\operatorname{arccosh} z = \frac{1}{\sqrt{z^2-1}},$$
$$\frac{d}{\mathrm{dz}}\operatorname{arctanh} z = \frac{1}{1-z^2}.$$

Inverse hyperbolic Functions

$$\frac{d}{\mathrm{dz}}\operatorname{arcsinh} z = \frac{1}{\sqrt{1+z^2}},$$

Proof:

Inverse hyperbolic Functions

Inverse hyperbolic Functions

Complex Integration

Topic No. 85

Complex Integration

$$\int$$

Complex Integration

Integration

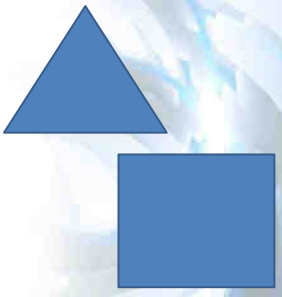
Source: https://educationaltechnology.net

Complex Integration

Real integrals

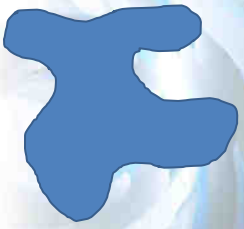
Complex Integration

Area Problem



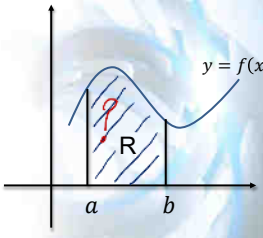
Complex Integration

Area Problem



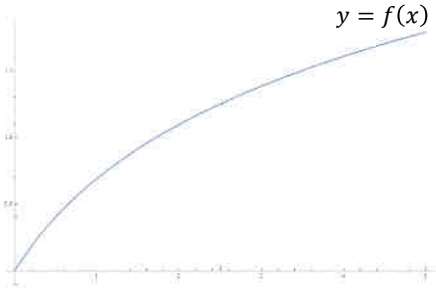
Complex Integration

Area Problem

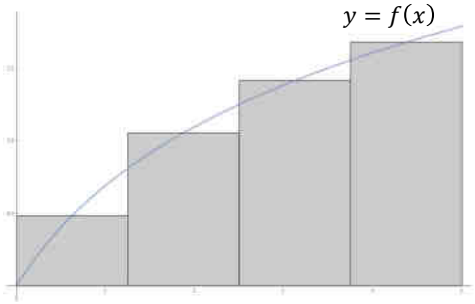


Complex Integration

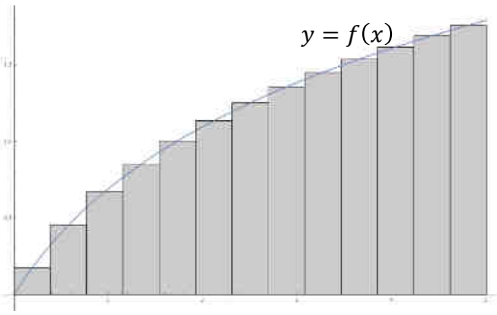
Area Problem: Given a function $y = f(x)$ that is continuous and non-negative on the interval $[a, b]$. Find the area between the graph of $f(x)$ and the interval $[a, b]$ on x-axis.



Complex Integration

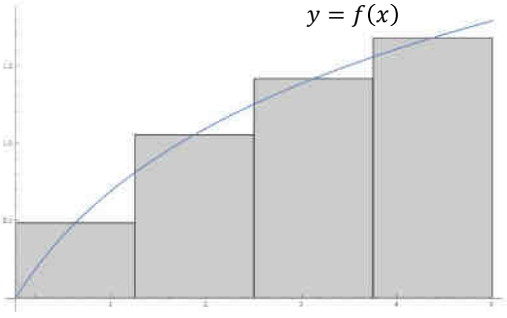


Complex Integration



Complex Integration

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{k=1}^{n} f(x_k^*)\,\Delta x_k$$



Complex Integration

Area Problem: Given a function $y = f(x)$ that is continuous and non-negative on the interval $[a, b]$. Find the area between the graph of $f(x)$ and the interval $[a, b]$ on x-axis.

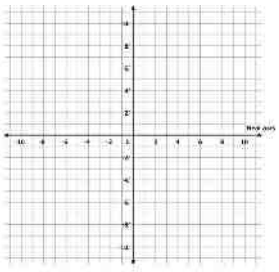
Answer :

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{k=1}^{n} f(x_k^*)\,\Delta x_k$$

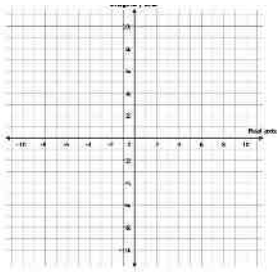
Complex Integration

A planar curve is a two dimensional analogue of the an interval.

Real integrals



Complex integrals

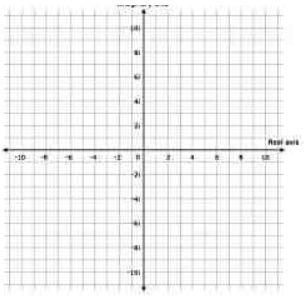


Complex Integration

Aim Define and Evaluate integrals of the form

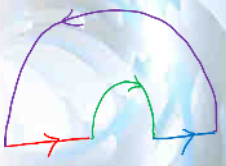
$$\int_C f(z)\,dz\ ,$$

where $f(z)$ is a complex valued function, and C is a contour in the plane.



Complex Integration

Contours



Complex Integration

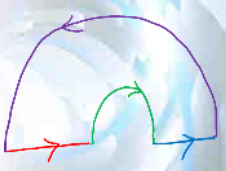
Complex Integration

$$\int$$

Contours

Topic No. 86

Contours

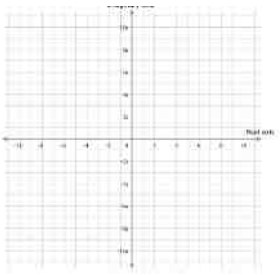


Contours

Aim Define and Evaluate integrals of the form

$$\int_C f(z)\,dz\ ,$$

where $f(z)$ is a complex valued function, and C is a contour in the plane.



Contours

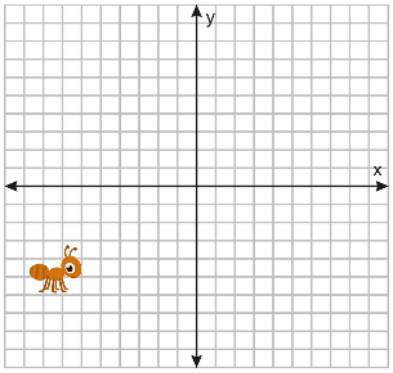
Contours

Contours

What is a **curve** (in plane)?

Contours

Consider motion of an ant in plane.



Contours

A curve in the plane is defined as

$$C: z(t) = x(t) + iy(t), \quad \text{for} \quad a \le t \le b$$

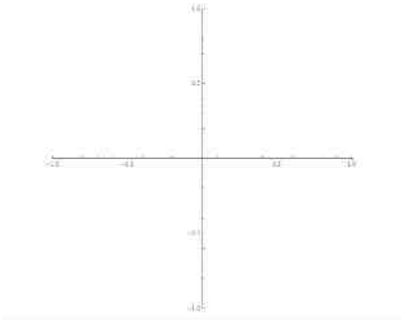
where t is a parameter.

Contours

Example: the parametric representation

$$z(t) = \cos(t) + i\sin(t) \text{ for } 0 \le t \le 2\pi$$

represents a circle in the complex plane (observe orientation).

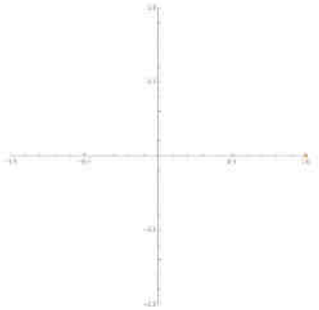


Contours

Example: the parametric representation

$$z(t) = \cos(t) - i\sin(t) \text{ for } 0 \le t \le 2\pi$$

represents a circle in the complex plane (observe orientation).

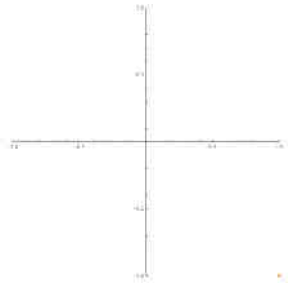


Contours

Example: the parametric representation

$$z(t) = t^2 + it \text{ for } -1 \le t \le 1$$

represents a parabola in the complex plane (observe orientation).



Contours

A curve in the plane is defined as

$$C: z(t) = x(t) + iy(t), \quad \text{for} \quad a \le t \le b$$

where t is a parameter.

There is orientation implicitly given in the definition.

Contours

A curve in the plane is defined as

$$C: z(t) = x(t) + iy(t), \quad \text{for} \quad a \le t \le b$$

where t is a parameter.

$z(a)$ is called the **initial**.

$z(b)$ is called **terminal point**.

Contours

Simple curve: A curve

$$C: z(t) = x(t) + iy(t), \quad \text{for} \quad a \le t \le b$$

is simple if it does not cross itself, that is,

$$z(t_1) \ne z(t_2), \text{ whenever } t_1 \ne t_2$$

except possibly when $t_1 = a$ and $t_2 = b$.

Contours

Closed curve: A curve

$$C: z(t) = x(t) + iy(t), \quad \text{for} \quad a \le t \le b$$

is closed if

$$z(a) = z(b)\,.$$

Contours

Simple closed curve or Jordan curve: A curve

$$C: z(t) = x(t) + iy(t), \quad \text{for} \quad a \le t \le b$$

is simple if it does not cross itself, that is,

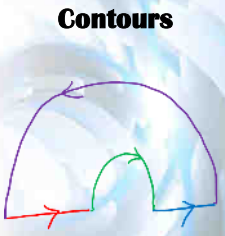
$z(t_1) \neq z(t_2)$, whenever $t_1 \neq t_2$

and $z(a) = z(b)$.

Contours

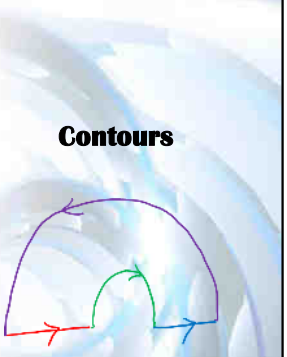
Examples

Contours



Contours

Topic No. 87



Contours

Contours

Contours

A curve in the plane is defined as

$$C: z(t) = x(t) + iy(t), \quad \text{for} \quad a \le t \le b$$

where t is a parameter.

Contours

Differentiability

The complex-valued function $f(t) = u(t) + iv(t)$ is said to be differentiable $[a, b]$ if both $x(t)$ and $y(t)$ both are differentiable on $[a, b]$.

Here we require one-sided differentiability at the end points $x = a$ and $x = b$.

Contours

Example: Check differentiability of the curve C, where

$$C: z(t) = |t| + i\,t^2 \text{ for } -1 \le t \le 1$$

Contours

Example: Check differentiability of the curve C, where

$$C: z(t) = \cos(t) + i\sin(t) \text{ for } 0 \le t \le 2\pi$$

Contours

Smooth curve: A curve

$$C: z(t) = x(t) + iy(t), \quad \text{for} \quad a \le t \le b$$

is said to be smooth curve if $z'(t)$ is continuous and nonzero on the interval.

Contours

Example: Check whether C is a smooth curve, where C is given by

$$C: z(t) = t^2 + it^3 \text{ for } -1 \le t \le 1.$$

Contours

Contour: A curve C is said to be a contour if it is constructed by joining finitely many smooth curves end to end.

Contours

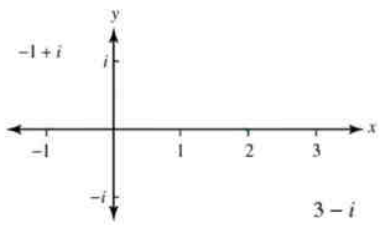
Contour: A curve C is said to be a contour if it is constructed by joining finitely many smooth curves end to end.

We express a contour obtained by joining sequence of curves $C_1, C_2, \dots, C_n$ in the following way

$$C = C_1 + C_2 + \cdots + C_n$$

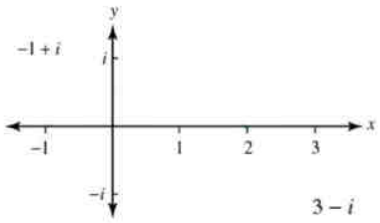
Contours

Example: Find parametrization of the a polygonal path C from $-1 + i$ to $3 - i$.



Contours

Example: Find parametrization of the a polygonal path C from $-1 + i$ to $3 - i$.



Let C_1 be line segment from $-1 + i$ to -1
Let C_2 be line segment from -1 to $1 + i$
Let C_3 be line segment from $1 + i$ to $3 - i$

Contours

Parametrization of line segment from z_0 to z_1 is given by

$$z_0 + t(z_1 - z_0), \qquad 0 \le t \le 1$$

Contours

Parametrization of line segment from z_0 to z_1 is given by

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Equation of C_1, line segment from $-1 + i$ to -1

Contours

Parametrization of line segment from z_0 to z_1 is given by

$$z_0 + t(z_1 - z_0), \qquad 0 \le t \le 1$$

Equation of C_1, line segment from $-1 + i$ to -1

$$C_1: z_1(t) = -1 + i(1 - t), \qquad for\ 0 \le t \le 1$$

Equation of C_2, line segment from -1 to $1 + i$

Contours

Parametrization of line segment from z_0 to z_1 is given by

$$z_0 + t(z_1 - z_0), \qquad 0 \le t \le 1$$

Equation of C_1, line segment from $-1 + i$ to -1

$$C_1: z_1(t) = -1 + i(1 - t), \qquad for\ 0 \le t \le 1$$

Equation of C_2, line segment from -1 to $1 + i$

$$C_2: z_2(t) = (-1 + 2t) + i\,t, \qquad for\ 0 \le t \le 1$$

Equation of C_3, line segment from $1 + i$ to $3 - i$

Contours

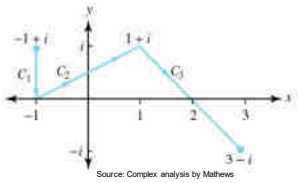
Example: Find parametrization of the a polygonal path C from $-1 + i$ to $3 - i$.

Solution: The polygonal path is the contour C, given by

$$C = C_1 + C_2 + C_3$$

where

$$C_1: z_1(t) = -1 + i(1 - t), \qquad for\ 0 \le t \le 1$$
$$C_2: z_2(t) = (-1 + 2t) + i\,t, \qquad for\ 0 \le t \le 1$$
$$C_3: z_3(t) = (1 + 2t) + i(1 - 2t), for\ 0 \le t \le 1$$



Source: Complex analysis by Mathews

Contours

Opposite curve: Consider a curve C

$$C: z_1(t) = x(t) + iy(t), \qquad \text{for} \quad a \le t \le b$$

The opposite curve $-C$ is given by

$$-C: z_2(t) = x(-t) + iy(-t), \qquad \text{for} \quad -b \le t \le -a$$

Geometrically $-C$ traces out the same set of points in the plane as C, but in the reverse order.

Contours

Opposite curve: Consider a curve C

$$C: z_1(t) = x(t) + iy(t), \qquad \text{for} \quad a \le t \le b$$

The opposite curve $-C$ is given by

$$-C: z_2(t) = x(-t) + iy(-t), \qquad \text{for} \quad -b \le t \le -a$$

Contours

Arc length

$$C: z(t) = x(t) + iy(t), \qquad \text{for} \quad a \le t \le b$$

We can find arc length of C from $z(a)$ to $z(b)$ in the following way:

$$L(C) = \int_a^b \sqrt{(x'(t))^2 + (y'(t))^2}\ dt$$

Contours

Since C is smooth so $z'(t)$, given by

$$z'(t) = x'(t) + iy'(t) \text{ for } \quad a \le t \le b$$

is continuous. Hence the following function is continuous

$$\sqrt{(x'(t))^2 + (y'(t))^2}$$

and hence it is integrable on $[a, b]$.

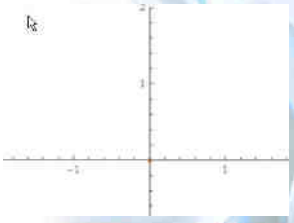
Contours

Example: Find arc length of circle

$$z(t) = \cos(t) + i\sin(t)\,, for\ 0 \le t \le 2\pi$$

Contours

Contours



Source: wolfram demonstration

Complex Integration

Topic No. 88

Complex Integration

$$\int$$

Complex Integration

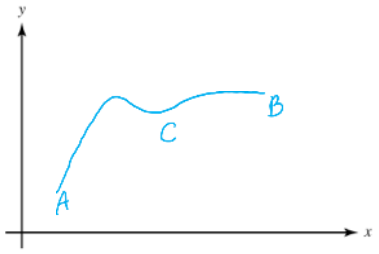
Contours

Complex Integration

Aim Define and Evaluate integrals of the form

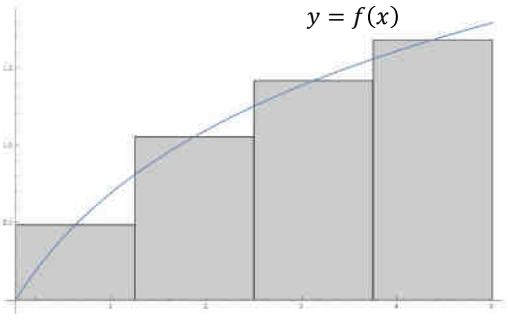
$$\int_C f(z)\,dz\ ,$$

where $f(z)$ is a complex valued function, and C is a contour in the plane from point A to point B.



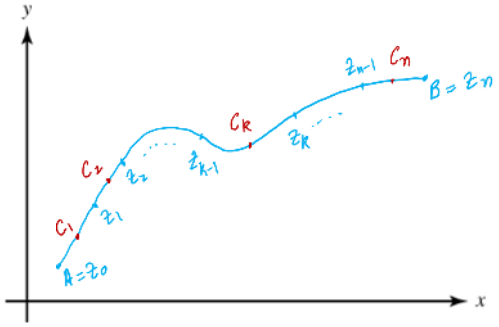
Complex Integration

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{k=1}^{n} f(x_k^*)\,\Delta x_k$$



Complex Integration

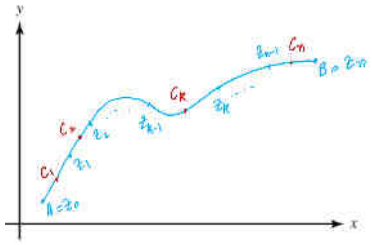
Divide C



Complex Integration

Hence we a partition

$$P_n = \{z_0 = A, z_1, z_2, \dots, z_n = B\}$$



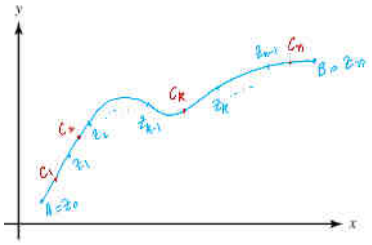
Complex Integration

Hence we a partition

$$P_n = \{z_0 = A, z_1, z_2, \dots, z_n = B\}$$

with differences

$$\Delta z_k = z_k - z_{k-1}\,, for\ k = 1,2, \dots, n$$



Complex Integration

Hence we a partition

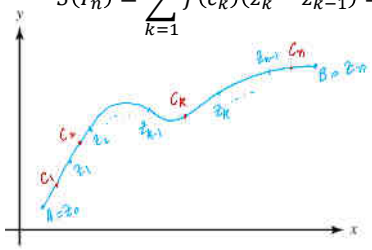
$$P_n = \{z_0 = A, z_1, z_2, \dots, z_n = B\}$$

with differences

$$\Delta z_k = z_k - z_{k-1}\,, for\ k = 1,2, \dots, n$$

and the Riemann sum corresponding to partition P_n is

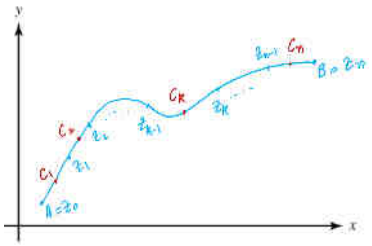
$$S(P_n) = \sum_{k=1}^{n} f(c_k)(z_k - z_{k-1}) = \sum_{k=1}^{n} f(c_k)\Delta z_k$$



Complex Integration

$$S(P_n) = \sum_{k=1}^{n} f(c_k)(z_k - z_{k-1}) = \sum_{k=1}^{n} f(c_k)\Delta z_k$$

$S(P_n)$ depends on n so it forms a sequence $\{S(P_n)\}_{n=1}^{\infty}$.



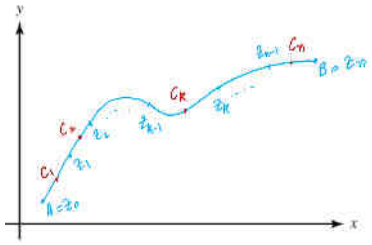
Complex Integration

$$S(P_n) = \sum_{k=1}^{n} f(c_k)(z_k - z_{k-1}) = \sum_{k=1}^{n} f(c_k)\Delta z_k$$

$S(P_n)$ depends on n so it forms a sequence $\{S(P_n)\}_{n=1}^{\infty}$.

Assume the sequence converges with limit L, that is,

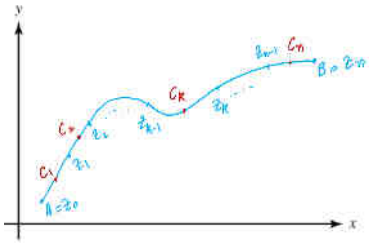
$$L = \lim_{n\to\infty} S(P_n)$$



Complex Integration

The value L depends on the partition chosen

$$L = \lim_{n\to\infty} S(P_n)$$



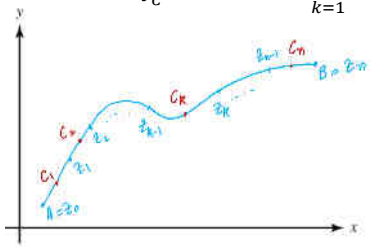
Complex Integration

The value L depends on the partition chosen

$$L = \lim_{n\to\infty} S(P_n)$$

If we get the same limit L for every partition of the contour then we say that

$$\int_C f(z)dt = \lim_{n\to\infty} \sum_{k=1}^{n} f(c_k)\Delta z_k$$

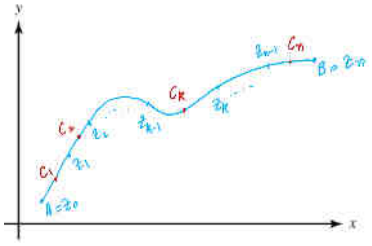


Complex Integration

Definition (Complex Integral)

Let $f(z)$ be a complex valued function and C be a contour then

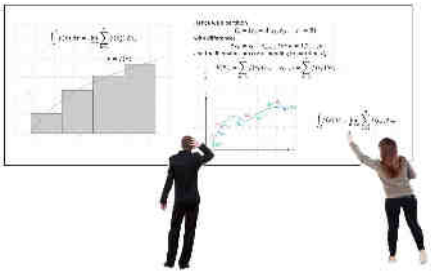
$$\int_C f(z)dt = \lim_{n\to\infty} \sum_{k=1}^{n} f(c_k)\Delta z_k$$



Complex Integration

Problem: The expression is too complicated to be calculated

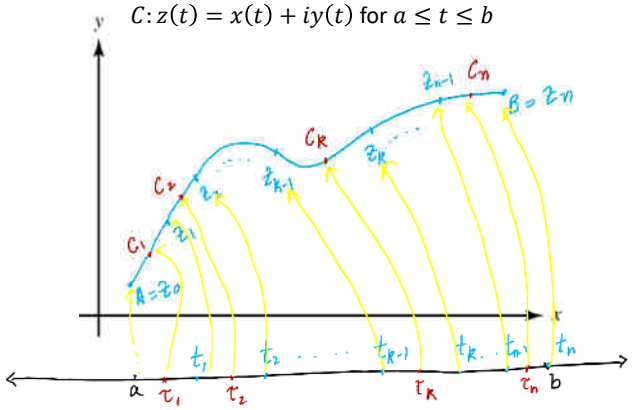
$$\int_C f(z)dt = \lim_{n\to\infty}\sum_{k=1}^{n} f(c_k)\Delta z_k$$



Complex Integration

The contour C is defined as

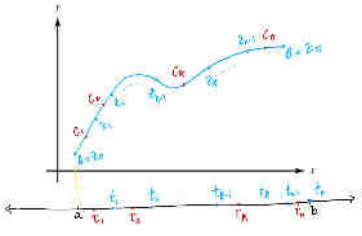
$$C: z(t) = x(t) + iy(t) \text{ for } a \le t \le b$$



Complex Integration

$$\lim_{n\to\infty}\sum_{k=1}^{n} f(c_k)\Delta z_k$$

$$= \lim_{n\to\infty}\sum_{k=1}^{n} f(c_k)(z_k - z_{k-1})$$

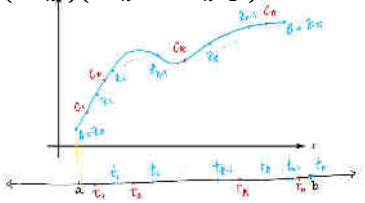


Complex Integration

$$\lim_{n\to\infty}\sum_{k=1}^{n} f(c_k)\Delta z_k$$

$$= \lim_{n\to\infty}\sum_{k=1}^{n} f(c_k)(z_k - z_{k-1})$$

$$= \lim_{n\to\infty}\sum_{k=1}^{n} f\big(z(\tau_k)\big)\big(z(t_k) - z(t_{k-1})\big)$$



Complex Integration

$$\lim_{n\to\infty}\sum_{k=1}^{n} f(c_k)\Delta z_k$$

$$= \lim_{n\to\infty}\sum_{k=1}^{n} f(c_k)(z_k - z_{k-1})$$

$$= \lim_{n\to\infty}\sum_{k=1}^{n} f\big(z(\tau_k)\big)\big(z(t_k) - z(t_{k-1})\big)$$

$$= \lim_{n\to\infty}\sum_{k=1}^{n} f\big(z(\tau_k)\big)\left(\frac{z(t_k) - z(t_{k-1})}{t_k - t_{k-1}}\right)(t_k - t_{k-1})$$

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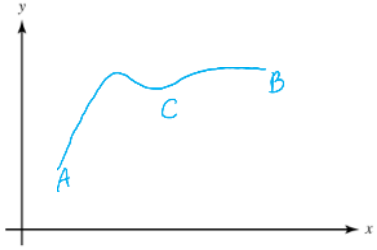
$$= \int_a^b f\big(z(t)\big)\, z'(t)\, dt$$

Complex Integration

Theorem: Suppose that $f(z)$ is a continuous complex-valued function defined on a set containing the contour C. Let $z(t)$ be any parametrization of C for $a \le t \le b$ then

$$\int_C f(z) = \int_a^b f\big(z(t)\big)\, z'(t)\, dt$$



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Note: we are allowed to choose any parametrization of the contour as we want.

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The above integral involves integrating complex functions over real intervals.

Complex Integration

Next Task

Understand integration of complex functions on real intervals

Complex Integration

Complex Integration

$$\int$$

Complex Integration

Topic No. 89

Complex Integrals over real intervals Part I

$$\int_a^b f(t)\, dt$$

Complex Integration

Recall

Let $f(z)$ is a continuous complex-valued function defined on a set containing the contour C, then

$$\int_C f(z)\, dz = \int_a^b f\big(z(t)\big)\, z'(t)\, dt$$

where

$C: z(t) = x(t) + iy(t)$ for $a \le t \le b$

Complex Integration

Next Task

Understand integration of complex functions on real intervals

Complex Integration

Aim

Find a way to evaluate the following

$$\int_a^b f(t)\, dt$$

where

$f(t) = u(t) + i\, v(t)$ for $a \le t \le b$

Complex Integration

Definition (Integral of $f(t)$)

Let $f(t) = u(t) + iv(t)$, where $u(t)$ and $v(t)$ are real-valued functions of the real variable t for $a \le t \le b$. Then

$$\int_a^b f(t)\,dt = \int_a^b u(t)\,dt + i\int_a^b v(t)\,dt$$

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Improper integrals of $f(t)$ over unbounded intervals can be defined in a similar way.

Complex Integration

Example: Evaluate the following

$$\int_1^2 (t + i\,t^2)\,dt$$

Complex Integration

Example: Evaluate the following

$$\int_1^2 (t + i\,t^2)\,dt$$

Complex Integration

Example: Evaluate the following

$$\int_1^2 (t + i\,t^2)\,dt$$

Complex Integration

Question

Under what conditions the following integral exists

$$\int_a^b f(t)\,dt$$

where $f(t) = u(t) + iv(t)$.

Complex Integration

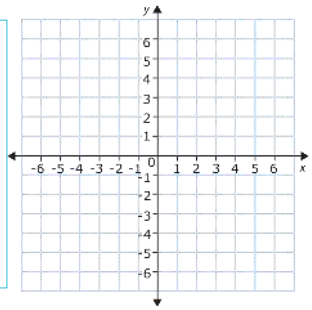
Answer

We know from real analysis that

"$\int_a^b h(x)\ dx$ exists if h(x) is piecewise continuous on $[a, b]$"

Recall

A real-valued function $h(x)$ is piecewise continuous if it has only finitely many discontinuities $p_1, p_2, \dots, p_n$, such that left hand limit and right-hand-limit exists at each of these points.



Complex Integration

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We know from real analysis that

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So

$$\int_a^b u(t)\,dt \ \text{ and } \int_a^b v(t)\,dt$$

exist if $u(t)$ and $v(t)$ are piecewise continuous on $[a, b]$ respectively.

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We know from real analysis that

"$\int_a^b h(x)\ dx$ exists if h(x) is piecewise continuous on $[a,b]$"

So

$$\int_a^b u(t)\,dt \quad \text{and} \int_a^b v(t)\,dt$$

exist if $u(t)$ and $v(t)$ are piecewise continuous on $[a,b]$ respectively.

If $u(t)$ and $v(t)$ are piecewise continuous then we say that $f(t)=u(t)+iv(t)$ has the same property.

Complex Integration

Question

Under what conditions the following integral exists

$$\int_a^b f(t)\,dt$$

where $f(t)=u(t)+iv(t)$.

Answer: The above integral exists if $f(t)$ is piecewise continuous on $[a,b]$.

Complex Integration

Example: Evaluate the following

$$\int_0^{\pi/2} \exp(t+it)\,dt$$

Complex Integration

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$$\int_0^{\pi/2} e^{t+it}dt$$
$$= \int_0^{\pi/2} e^t e^{it}dt$$

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$$= \int_0^{\pi/2} e^t(\cos t+\mathrm{i}\sin t)dt$$

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The above integral can be evaluated by evaluating two real integrals, that is,

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$$= \int_0^{\pi/2} e^t \cos t\,dt + i\int_0^{\pi/2} e^t \sin t\,dt$$

Complex Integration

Solution cont. ($\int_0^{\pi/2} \exp(t+it)\,dt$)

$$\int_0^{\pi/2} e^t \cos t\,dt$$

Complex Integration

Solution cont. ($\int_0^{\pi/2} \exp(t+it)\,dt$)

$$\int_0^{\pi/2} e^t \sin t\,dt$$

Complex Integration

Solution cont. ($\int_0^{\pi/2} \exp(t+it)\,dt$)

Complex Integration

Complex Integrals over real intervals Part I

$$\int_a^b f(t)\,dt$$

Complex Integration

Complex Integrals over real intervals Part II

Topic No. 90

$$\int_a^b f(t)\,dt$$

Complex Integration

Properties of

$$\int_a^b f(t)\,dt$$

where

$$f(t) = u(t) + iv(t)$$

Complex Integration

Recall

Definition (Integral of $f(t)$)

Let $f(t) = u(t) + iv(t)$, where $u(t)$ and $v(t)$ are real-valued functions of the real variable t for $a \le t \le b$. Then

$$\int_a^b f(t)\,dt = \int_a^b u(t)\,dt + i\int_a^b v(t)\,dt$$

Complex Integration

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Let $f(t) = u(t) + iv(t)$, where $u(t)$ and $v(t)$ are real-valued functions of the real variable t for $a \le t \le b$. Then

$$\int_a^b f(t)\,dt = \int_a^b u(t)\,dt + i\int_a^b v(t)\,dt$$

Hence

$$Re\left[\int_a^b f(t)\,dt\right] = \int_a^b Re[f(t)]\,dt$$

Complex Integration

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Let $f(t) = u(t) + iv(t)$, where $u(t)$ and $v(t)$ are real-valued functions of the real variable t for $a \le t \le b$. Then

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Hence

$$Re\left[\int_a^b f(t)\,dt\right] = \int_a^b Re[f(t)]\,dt$$

Similarly

$$Im\left[\int_a^b f(t)\,dt\right] = \int_a^b Im[f(t)]\,dt$$

Complex Integration

For
$$f(t) = u(t) + iv(t)$$
and
$$g(t) = p(t) + iq(t)$$
we have
$$\int_a^b \big(f(t) + g(t)\big)\, dt = \int_a^b f(t)\, dt + \int_a^b g(t)\, dt$$

Complex Integration

For
$$f(t) = u(t) + iv(t)$$
and a partition of the interval $[a, b]$ given as
$$a \le t \le d, d \le t \le b$$
we have
$$\int_a^b f(t)\, dt = \int_a^d f(t)\, dt + \int_d^b f(t)\, dt$$

Complex Integration

Fundamental theorem of calculus (FTC) for Complex functions

Let $f(t) = u(t) + iv(t)$
and
$$F'(t) = f(t)$$
such that $f(t)$ and $F(t)$ are continuous on $[a, b]$, then
$$\int_a^b f(t)dt = F(t)|_{t=a}^{t=b} = F(b) - F(a)$$

Complex Integration

A consequence of FTC for complex functions

For a differentiable function
$$f(t) = u(t) + iv(t)$$
and $a \le t \le b$, we have
$$\int_a^b f'(t)dt = f(b) - f(a)$$

Complex Integration

The following is a useful bridge for evaluating real integrals as well
$$\int_a^b f(t)\, dt = \int_a^b u(t)\, dt + i\int_a^b v(t)\, dt$$

Complex Integration

Example: Evaluate the following
$$\int_0^{\pi/2} \exp(t + it)\, dt$$

Complex Integration

Example: Evaluate the following
$$\int_0^{\pi/2} \exp(t + it)\, dt$$
In the last module we evaluated the above integral by evaluating two real integrals, that is,
$$\int_0^{\pi/2} e^{t+it} dt = \int_0^{\pi/2} e^t e^{it} dt = \int_0^{\pi/2} e^t(\cos t + i\sin t) dt = \int_0^{\pi/2} e^t \cos t\, dt + i\int_0^{\pi/2} e^t \sin t\, dt$$

Complex Integration

Example: Evaluate the following
$$\int_0^{\pi/2} \exp(t + it)\, dt$$
The above integral can also be calculated in much simple way using properties of complex integrals over real intervals.

Complex Integration

Example: Evaluate the following

$$\int_0^{\pi/2} \exp(t + it)\, dt$$

Complex Integration

Although there are many similarities between real and complex integrals over a real interval. But there are some important differences as well.

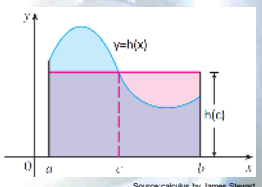
Source: https://www.100freedownloads.com

Complex Integration

Mean value theorem for real integrals

If $h(x)$ is continuous on $[a, b]$ then there exist $c \in [a, b]$ such that

$$\int_a^b h(x)\, dx = h(c)\,(b - a)$$



Source:calculus by James Stewart

Complex Integration

Example: Mean value theorem for real integrals does not hold in case of integrals of complex functions over real intervals.

Complex Integration

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$$f(t) = e^{it} \quad \text{for } 0 \le t \le 2\pi$$

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$$\int_0^{2\pi} e^{it}\, dt$$
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$$= e^{i2\pi} - e^0$$
$$= 0$$

For any $c \in [0,2\pi]$ we have

$$|f(c)(2\pi - 0)| = |e^{ic}||2\pi| = 2\pi$$

Complex Integration

Complex Integrals over real intervals Part II

$$\int_a^b f(t)\, dt$$

Contour Integrals

Contour Integrals Examples Part I

Topic No. 91

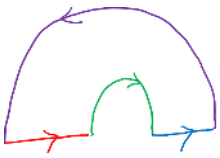
$$\int_C$$

Contour Integrals

Recall:

Contour: A curve C is said to be a contour if it is constructed by joining finitely many smooth curves end to end.
We express a contour obtained by joining sequence of curves $C_1, C_2, \dots, C_n$ in the following way

$$C = C_1 + C_2 + \cdots + C_n$$



Contour Integrals

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Theorem: Suppose that $f(z)$ is a continuous complex-valued function defined on a set containing the contour C. Let $z(t)$ be any parametrization of C for $a \le t \le b$ then

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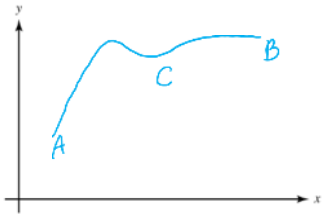

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Note: we are allowed to choose any parametrization of the contour as we want.

The above integral involves integrating complex functions over real intervals.

Contour Integrals

Recall:

Definition (Integral of $f(t)$)

Let $f(t) = u(t) + iv(t)$, where $u(t)$ and $v(t)$ are real-valued functions of the real variable t for $a \le t \le b$. Then

$$\int_a^b f(t)\, dt = \int_a^b u(t)\, dt + i \int_a^b v(t)\, dt$$

Contour Integrals

Example: Evaluate the integral

$$\int_C \exp(z)\, dz$$

where the contour C is a line segment from $0 + i0$ to $a + ib$.

Solution: The parametric representation of C is given by

Contour Integrals

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Solution: The parametric representation of C is given by

$$\mathrm{C}: \; z(t) = (a + ib)t \;\text{ for }\; 0 \le t \le 1$$

Which implies

$$z'(t) = (a + ib)$$

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Which implies

$$z'(t) = (a + ib)$$

Hence

$$\int_C \exp(z)\; dz = \int_0^1 \exp\big(z(t)\big)\;\, z'(t)\, dt$$

Contour Integrals

Example: Evaluate the integral

$$\int_C \exp(z)\, dz$$

where the contour C is a line segment from $0 + i0$ to $a + ib$.

Solution cont.:

$$\int_C \exp(z)\; dz = \int_0^1 \exp\big((a + ib)t\big)\;\, (a + ib)\, dt$$

Contour Integrals

Example: Evaluate the integral

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where the contour C is a line segment from $0 + i0$ to $a + ib$.

Solution cont.:

$$\int_C \exp(z)\; dz = \int_0^1 \exp\big((a + ib)t\big)\;\, (a + ib)\, dt$$

$$= (a + ib) \int_0^1 \exp\big((a + ib)t\big)\; dt$$

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$$= (a + ib) \int_0^1 \exp\big((a + ib)t\big)\; dt$$

$$= (a + ib)\; \frac{\exp\big((a + ib)t\big)}{(a + ib)}\Bigg]_0^1$$

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$$= \exp(a + ib) - 1$$

Contour Integrals

Contour Integrals
Examples
Part I

$$\int_C$$

Contour Integrals

Contour Integrals
Examples
Part II

Topic No. 92

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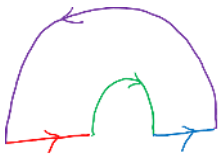

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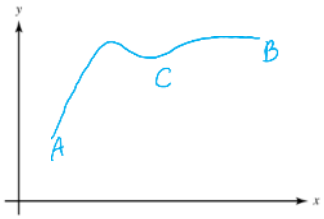

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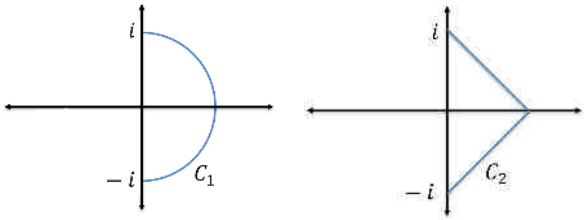
Contour Integrals

Example1: Evaluate the integrals

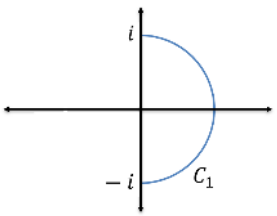
$$\int_{C_1} z\, dz \quad and \int_{C_2} z\, dz$$

where the contour C_1 is semicircle from $-i$ to i (counterclockwise), and C_2 is the polygonal path from $-i$ to i as shown in the figure.



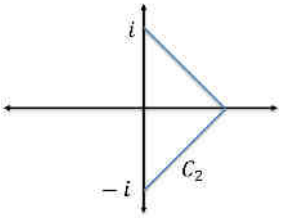
Contour Integrals

Solution ($\int_{C_1} z\, dz$):



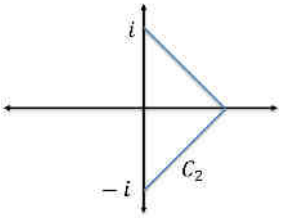
Contour Integrals

Solution cont. ($\int_{C_2} z\,dz$):



Contour Integrals

Solution cont. ($\int_{C_2} z\,dz$):

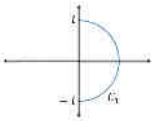


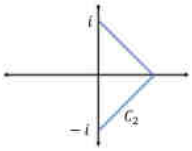
Contour Integrals

Example1: Evaluate the integrals

$$\int_{C_1} z\,dz \quad and \int_{C_2} z\,dz$$

where the contour C_1 is semicircle from $-i$ to i (counterclockwise), and C_2 is the polygonal path from $-i$ to i as shown in the figure.





Contour Integrals

When a contour integral is independent of path?

Contour Integrals

Contour Integrals Examples Part II

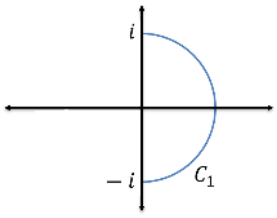
$$\int_C$$

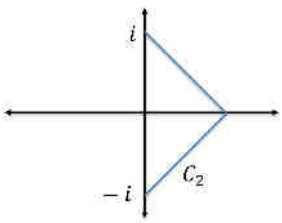
Contour Integrals

Example2: Evaluate the integrals

$$\int_{C_1} \bar{z}\,dz \quad and \int_{C_2} \bar{z}\,dz$$

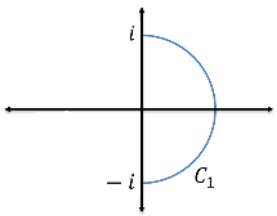
where the contour C_1 is semicircle from $-i$ to i (counterclockwise), and C_2 is the polygonal path from $-i$ to i as shown in the figure.





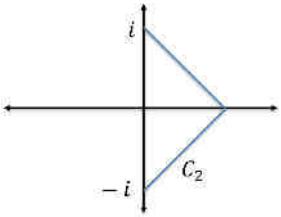
Contour Integrals

Solution ($\int_{C_1} \bar{z}\,dz$):



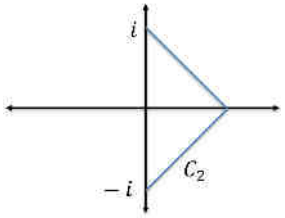
Contour Integrals

Solution cont. ($\int_{C_2} \bar{z}\,dz$):



Contour Integrals

Solution cont. ($\int_{C_2} \bar{z}\,dz$):



Contour Integrals

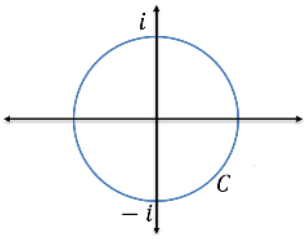
When a contour integral is independent of path?

Contour Integrals

Example3: Evaluate the integrals

$$\int_C z\,dz$$

where the contour C is circle from $-i$ to $-i$ (counterclockwise) as shown in the figure.



Contour Integrals

Contour Integrals
Examples
Part II

$$\int_C$$

Contour Integrals

Contour Integrals
Examples
Part III

Topic No. 93

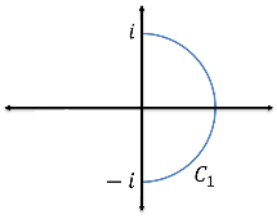
$$\int_C$$

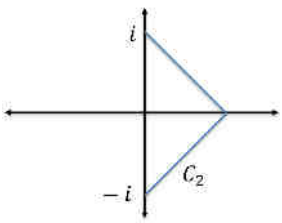
Contour Integrals

Recall: Example1: Evaluate the integrals

$$\int_{C_1} z\,dz \qquad and \qquad \int_{C_2} z\,dz$$

where the contour C_1 is semicircle from $-i$ to i (counterclockwise), and C_2 is the polygonal path from $-i$ to i as shown in the figure.





Contour Integrals

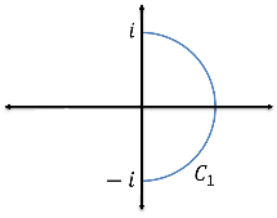
When a contour integral is independent of path?

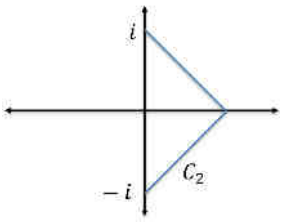
Contour Integrals

Example2: Evaluate the integrals

$$\int_{C_1} \bar{z}\,dz \quad and \int_{C_2} \bar{z}\,dz$$

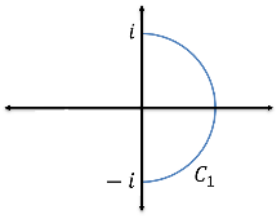
where the contour C_1 is semicircle from $-i$ to i (counterclockwise), and C_2 is the polygonal path from $-i$ to i as shown in the figure.





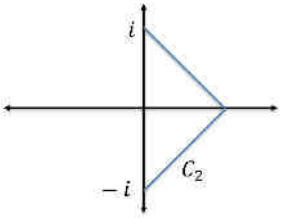
Contour Integrals

Solution ($\int_{C_1} \bar{z}\, dz$):



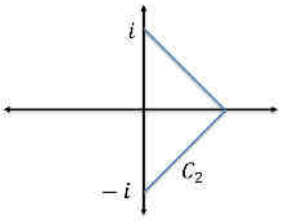
Contour Integrals

Solution cont. ($\int_{C_2} \bar{z}\, dz$):



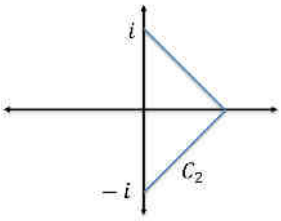
Contour Integrals

Solution cont. ($\int_{C_2} \bar{z}\, dz$):



Contour Integrals

Solution cont. ($\int_{C_2} \bar{z}\, dz$):



Contour Integrals

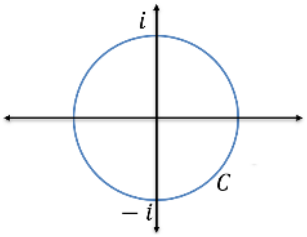
When a contour integral is independent of path?

Contour Integrals

Example3: Evaluate the integral

$$\int_C z\, dz$$

where the contour C is circle from $-i$ to $-i$ (counterclockwise) as shown in the figure.

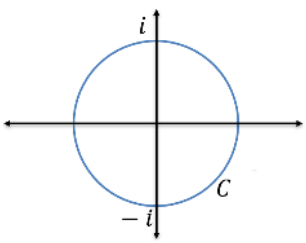


Contour Integrals

Example3: Evaluate the integral

$$\int_C z\, dz$$

where the contour C is circle from $-i$ to $-i$ (counterclockwise) as shown in the figure.



Contour Integrals

When a contour integral around a closed contour is zero?

Contour Integrals

Contour Integrals
Examples
Part III

$$\int_C$$

Contour Integrals

Contour Integrals
Examples
Part IV
(Contours with a branch cut passing through them)

Topic No. 94

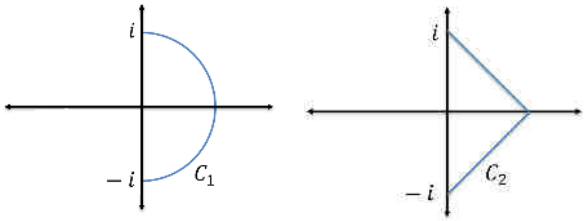
$$\int_C$$

Contour Integrals

Recall: Example1: Evaluate the integrals

$$\int_{C_1} z\,dz \quad and \int_{C_2} z\,dz$$

where the contour C_1 is semicircle from $-i$ to i (counterclockwise), and C_2 is the polygonal path from $-i$ to i as shown in the figure.

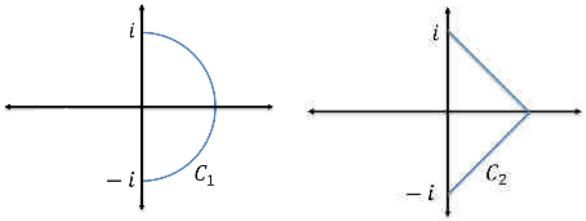


Contour Integrals

Recall: Example2: Evaluate the integrals

$$\int_{C_1} \bar{z}\,dz \quad and \quad \int_{C_2} \bar{z}\,dz$$

where the contour C_1 is semicircle from $-i$ to i (counterclockwise), and C_2 is the polygonal path from $-i$ to i as shown in the figure.

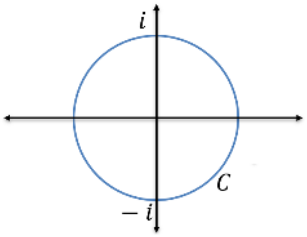


Contour Integrals

Recall: Example3: Evaluate the integrals

$$\int_C z\,dz$$

where the contour C is circle from $-i$ to $-i$ (counterclockwise) as shown in the figure.



Contour Integrals

When a contour integral is independent of path?
When a contour integral around a closed contour is zero?

Contour Integrals

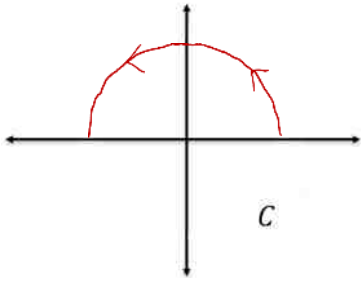
How to compute contour integrals if a branch cut passes through our contour?

Contour Integrals

Example4: Evaluate the integral

$$\int_C z^a\,dz$$

where the contour C is a semicircle from 2 to -2 (counterclockwise) as shown in the figure.



Contour Integrals

Solution: We know that

$$z^a = \exp(a\log(z))$$

Contour Integrals

Solution: We know that

$$z^a = \exp(a\log(z))$$

and $\log(z)$ is a multivalued function defined as

$$\log(z) = \ln|z| + i\arg(z),$$

Contour Integrals

Solution: We know that

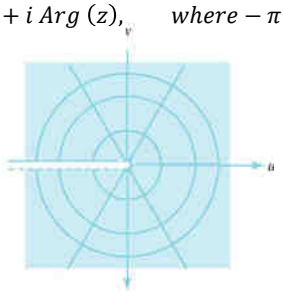
$$z^a = \exp(a\log(z))$$

and $\log(z)$ is a multivalued function defined as

$$\log(z) = \ln|z| + i\arg(z),$$

and the principal branch of $\log(z)$ is given by

$$\text{Log}\,(z) = \ln|z| + i\,Arg\,(z), \qquad where\ -\pi < Arg\,(z) \leq \pi$$



Contour Integrals

Solution: We know that

$$z^a = \exp(a\log(z))$$

and $\log(z)$ is a multivalued function defined as

$$\log(z) = \ln|z| + i\arg(z),$$

and the principal branch of $\log(z)$ is given by

$$\text{Log}\,(z) = \ln|z| + i\,Arg\,(z), \qquad where\ -\pi < Arg\,(z) \leq \pi$$

So the principal branch of z^a is given by

$$z^a = \exp(a\,\text{Log}(z))$$

where

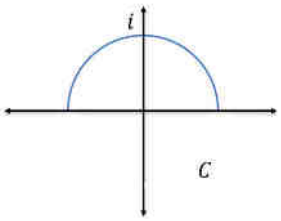
$$-\pi < Arg\,(z) \leq \pi$$

Contour Integrals

Solution (cont.): The contour is given by:

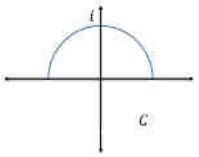
$C: z(t) = 2e^{i\theta}$, where $0 \leq \theta \leq \pi$

So the contour passes through a branch cut.



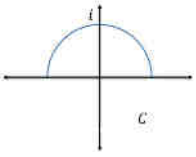
Contour Integrals

Solution (cont.) Method 1



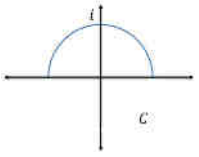
Contour Integrals

Solution (cont.) Method 1



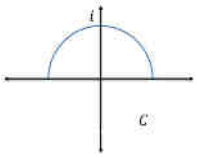
Contour Integrals

Solution (cont.) Method 1



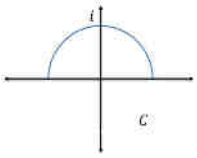
Contour Integrals

Solution (cont.) Method 2



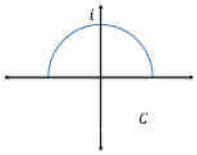
Contour Integrals

Solution (cont.) Method 2



Contour Integrals

Solution (cont.) Method 2



Contour Integrals

Similar procedure can be applied to compute the following contour integral

$$\int_C z^a \, dz$$

where

$$C: z(t) = 2e^{i\theta} \text{ for } -\pi \leq \theta \leq \pi \, .$$

Same But different

Contour Integrals

Contour Integrals Examples Part IV (Contours with a branch cut passing through them)

$$\int_C$$

Properties of Contour Integrals

Topic No. 95

Properties of contour integrals

$$\int_C$$

Properties of Contour Integrals

$$\int_{-C} f(z) \, dz = -\int_C f(z) \, dz$$

Properties of Contour Integrals

$$\int_{-C} f(z) \, dz = -\int_C f(z) \, dz$$

Properties of Contour Integrals

$$\int_{-C} f(z)\,dz = -\int_{C} f(z)\,dz$$

Consider a curve C

$$C: z_1(t) = x(t) + iy(t), \quad \text{for} \quad a \le t \le b$$

Properties of Contour Integrals

$$\int_{-C} f(z)\,dz = -\int_{C} f(z)\,dz$$

Consider a curve C

$$C: z_1(t) = x(t) + iy(t), \quad \text{for} \quad a \le t \le b$$

The opposite curve $-C$ is given by

$$-C: z_2(t) = x(-t) + iy(-t), \quad \text{for} \quad -b \le t \le -a$$

Properties of Contour Integrals

$$\int_{-C} f(z)\,dz = -\int_{C} f(z)\,dz$$

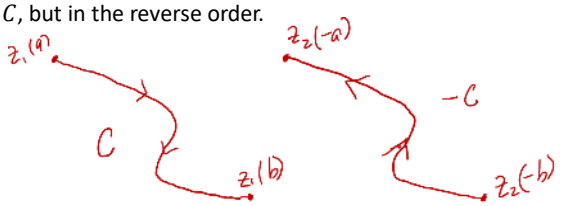
Consider a curve C

$$C: z_1(t) = x(t) + iy(t), \quad \text{for} \quad a \le t \le b$$

The opposite curve $-C$ is given by

$$-C: z_2(t) = x(-t) + iy(-t), \quad \text{for} \quad -b \le t \le -a$$

Geometrically $-C$ traces out the same set of points in the plane as C, but in the reverse order.



Properties of Contour Integrals

Proof($\int_C f(z)\,dz = -\int_{-C} f(z)\,dz$)

Here

$$C: z_1(t) = x(t) + iy(t), \quad \text{for} \quad a \le t \le b$$

$$-C: z_2(t) = x(-t) + iy(-t), \quad \text{for} \quad -b \le t \le -a$$

Hence

$$z_2(-t) = z_1(t)$$

Properties of Contour Integrals

Proof cont. ($\int_{-C} f(z)\,dz = -\int_C f(z)\,dz$)

$$\int_{-C} f(z)\,dz = \int_{-b}^{-a} f(z_2(t))\frac{d}{dt}(z_2(t))dt$$

Properties of Contour Integrals

Proof cont. ($\int_{-C} f(z)\,dz = -\int_C f(z)\,dz$)

$$\int_{-C} f(z)\,dz = \int_{-b}^{-a} f(z_2(t))\frac{d}{dt}(z_2(t))dt$$

Take $t = -s$

$$\int_{-C} f(z)\,dz = \int_{b}^{a} f(z_2(-s))\left[\frac{d}{d(-s)}(z_2(-s))\right](-ds)$$

Properties of Contour Integrals

Proof cont. ($\int_{-C} f(z)\,dz = -\int_C f(z)\,dz$)

$$\int_{-C} f(z)\,dz = \int_{-b}^{-a} f(z_2(t))\frac{d}{dt}(z_2(t))dt$$

Take $t = -s$

$$\int_{-C} f(z)\,dz = \int_{b}^{a} f(z_2(-s))\left[\frac{d}{d(-s)}(z_2(-s))\right](-ds)$$

$$= -\int_{b}^{a} f(z_1(s))\left[\frac{d}{d(-s)}(z_1(s))\right]ds$$

Properties of Contour Integrals

Proof cont. ($\int_{-C} f(z)\,dz = -\int_C f(z)\,dz$)

$$\int_{-C} f(z)\,dz = \int_{-b}^{-a} f(z_2(t))\frac{d}{dt}(z_2(t))dt$$

Take $t = -s$

$$\int_{-C} f(z)\,dz = \int_{b}^{a} f(z_2(-s))\left[\frac{d}{d(-s)}(z_2(-s))\right](-ds)$$

$$= -\int_{b}^{a} f(z_1(s))\left[\frac{d}{d(-s)}(z_1(s))\right]ds$$

$$= \int_{a}^{b} f(z_1(s))\left[\frac{d}{ds}(z_1(s))\,\frac{d\,s}{d(-s)}\right]ds$$

Properties of Contour Integrals

Proof cont. ($\int_{-C} f(z)\,dz = -\int_C f(z)\,dz$)

$$\int_{-C} f(z)\,dz = \int_{-b}^{-a} f(z_2(t))\frac{d}{dt}(z_2(t))dt$$

Take $t = -s$

$$\int_{-C} f(z)\,dz = \int_b^a f(z_2(-s))\left[\frac{d}{d(-s)}(z_2(-s))\right](-ds)$$
$$= -\int_b^a f(z_1(s))\left[\frac{d}{d(-s)}(z_1(s))\right]ds$$
$$= \int_a^b f(z_1(s))\left[\frac{d}{ds}(z_1(s))\ \frac{d\,s}{d(-s)}\right]ds$$
$$= \int_a^b f(z_1(s))\left[\frac{d}{ds}(z_1(s))\ 1/\left(\frac{d\,(-s)}{d(s)}\right)\right]ds$$

Properties of Contour Integrals

Proof cont. ($\int_{-C} f(z)\,dz = -\int_C f(z)\,dz$)

$$\int_{-C} f(z)\,dz = \int_{-b}^{-a} f(z_2(t))\frac{d}{dt}(z_2(t))dt$$

Take $t = -s$

$$\int_{-C} f(z)\,dz = \int_b^a f(z_2(-s))\left[\frac{d}{d(-s)}(z_2(-s))\right](-ds)$$
$$= -\int_b^a f(z_1(s))\left[\frac{d}{d(-s)}(z_1(s))\right]ds$$
$$= \int_a^b f(z_1(s))\left[\frac{d}{ds}(z_1(s))\ \frac{d\,s}{d(-s)}\right]ds$$
$$= \int_a^b f(z_1(s))\left[\frac{d}{ds}(z_1(s))\ 1/\left(\frac{d\,(-s)}{d(s)}\right)\right]ds$$
$$= -\int_a^b f(z_1(s))\,z_1'(s)\,ds = -\int_C f(z)\,dz$$

Properties of Contour Integrals

$$\int_{-C} f(z)\,dz = -\int_C f(z)\,dz$$

Example:

Properties of Contour Integrals

$$\int_C \big(f(z) + g(z)\big)dz = \int_C f(z)dz + \int_C g(z)dz$$

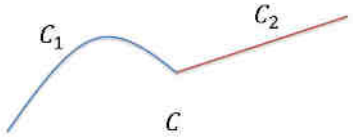
Example:

Properties of Contour Integrals

$$\int_C (a+ib)\,f(z)dz = (a+ib)\int_C f(z)dz$$

Properties of Contour Integrals

What happens if a contour consists of many contours (called legs)



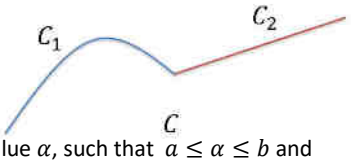
Properties of Contour Integrals

What happens if a contour consists of many contours (called legs)

Consider a contour

$$C: z(t), \qquad a \le t \le b$$

such that C has two legs, as shown in the figure

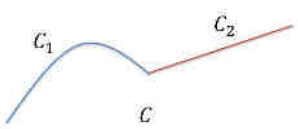


There exist a value α, such that $a \le \alpha \le b$ and

$$C_1: z(t), \qquad a \le t \le \alpha$$
$$C_2: z(t), \qquad \alpha \le t \le b$$

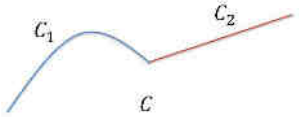
Properties of Contour Integrals

$$\int_{C_1+C_2} f(z)dz = \int_C f(z)dz$$



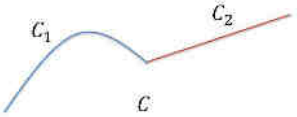
Properties of Contour Integrals

$$\int_{C_1+C_2} f(z)dz$$
$$= \int_C f(z)dz$$
$$= \int_a^b f\big(z(t)\big)\, z'(t)dz$$



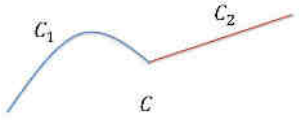
Properties of Contour Integrals

$$\int_{C_1+C_2} f(z)dz$$
$$= \int_C f(z)dz$$
$$= \int_a^b f\big(z(t)\big)\, z'(t)dz$$
$$= \int_a^\alpha f\big(z(t)\big)\, z'(t)dz + \int_\alpha^b f\big(z(t)\big)\, z'(t)dz$$



Properties of Contour Integrals

$$\int_{C_1+C_2} f(z)dz$$
$$= \int_C f(z)dz$$
$$= \int_a^b f\big(z(t)\big)\, z'(t)dz$$
$$= \int_a^\alpha f\big(z(t)\big)\, z'(t)dz + \int_\alpha^b f\big(z(t)\big)\, z'(t)dz$$
$$= \int_{C_1} f(z)dz + \int_{C_2} f(z)dz$$



Properties of Contour Integrals

Hence we have

$$\int_{C_1+C_2} f(z)dz = \int_{C_1} f(z)dz + \int_{C_2} f(z)dz$$

Properties of Contour Integrals

Effect of reparameterization of a contour on contour integral

Properties of Contour Integrals

Effect of reparameterization of a contour on contour integral

Let C be a contour given by

$$C: z_1(t) = x_1(t) + iy_1(t), \qquad a \le t \le b$$

Properties of Contour Integrals

Effect of reparameterization of a contour on contour integral

Let C be a contour given by

$$C: z_1(t) = x_1(t) + iy_1(t), \qquad a \le t \le b$$

A reparameterization of C, given by

$$C: z_2(s) = x_2(s) + iy_2(s), \qquad c \le s \le d$$

Properties of Contour Integrals

Effect of reparameterization of a contour on contour integral

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Properties of Contour Integrals

Effect of reparameterization of a contour on contour integral

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A reparameterization of C, given by

$$C: z_2(s) = x_2(s) + iy_2(s), \qquad c \le s \le d$$

There exist a differentiable function t $= h(s)$, such that

$$h(a) = c, \qquad h(b) = d \quad \text{and} \quad h'(s) > 0$$

and

$$z_1(t) = z_1\big(h(s)\big) = z_2(s)$$

Properties of Contour Integrals

Effect of reparameterization of a contour on contour integral

Let C be a contour given by

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and

$$z_1(t) = z_1\big(h(s)\big) = z_2(s)$$

Then

$$\int_a^b f(z_1(t))z_1'(t)dt = \int_c^d f(z_2(s))z_2'(s)\,ds$$

Properties of Contour Integrals

Properties of contour integrals

$$\int_C$$

Inequalities Involving Contour Integrals

Inequalities involving moduli of contour integrals

Topic No. 96

$$\left|\int_C f(z)\,dz\right|$$

Inequalities Involving Contour Integrals

Absolute value inequality

For a piecewise continuous complex-valued function of real variable t

$$f(t) = u(t) + i\,v(t)$$
$$a \le t \le b$$

We have

$$\left|\int_a^b f(t)dt\right| \le \int_a^b |f(t)|dt$$

Inequalities Involving Contour Integrals

Theorem (Absolute value inequality)

For a piecewise continuous complex-valued function of real variable t

$$f(t) = u(t) + i\,v(t), \qquad a \le t \le b$$

We have

$$\left|\int_a^b f(t)dt\right| \le \int_a^b |f(t)|dt$$

Proof:

Inequalities Involving Contour Integrals

Proof (Absolute value inequality)

Inequalities Involving Contour Integrals

Proof (Absolute value inequality)

Inequalities Involving Contour Integrals

Theorem (ML inequality)

If $f(z)$ is continuous on contour C, then

$$\left|\int_C f(z)\,dz\right| \le ML$$

Where L is length of the contour and M is upper bound for the modulus $|f(z)|$ on C, that is,

$$|f(z)| \le M, \qquad for\ all\ z \in C\ .$$

Proof:

Inequalities Involving Contour Integrals

Proof ($\left|\int_C f(z)\,dz\right| \le ML$)

Inequalities Involving Contour Integrals

Example ($\left|\int_C f(z)\,dz\right| \le ML$): Find upper bound of

$$\int_C \bar{z}\,dz$$

where $C: z(t) = e^{it},\ 0 \le t \le 2\pi.$

Inequalities Involving Contour Integrals

Example:

Inequalities Involving Contour Integrals

Inequalities involving moduli of Contour integrals

$$\left|\int_C f(z)\,dz\right|$$

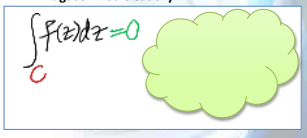
The Cauchy-Goursat Theorem

Topic No. 97

The Cauchy-Goursat Theorem Part I

Augustin-Louis Cauchy

Édouard Goursat



The Cauchy-Goursat Theorem

Let us recall some Topology

Source: wolfram demonstration

The Cauchy-Goursat Theorem

Recall:

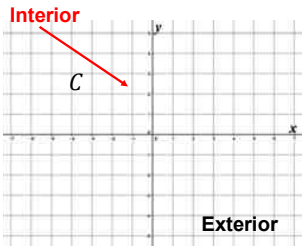
Domain is a connected open set.

The Cauchy-Goursat Theorem

Interior and exterior of a curve C

(Jordan Curve Theorem)

Each simple closed curve C divides the plane into two domains (connected open sets). One is bounded and is called interior of C. The other is unbounded and is called the exterior of C.



The Cauchy-Goursat Theorem

Interior and exterior of a curve C

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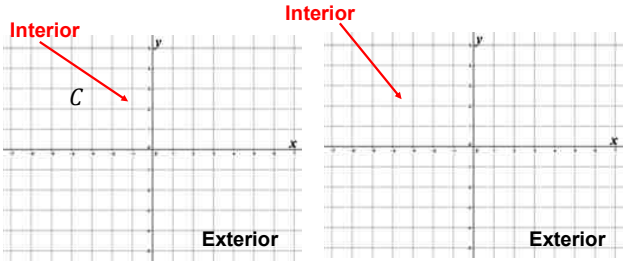
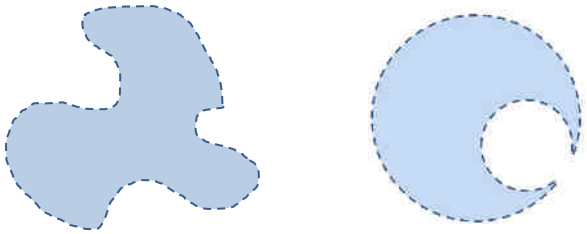
Each simple closed curve C divides the plane into two domains (connected open sets). One is bounded and is called interior of C. The other is unbounded and is called the exterior of C.



The Cauchy-Goursat Theorem

Simply connected domain

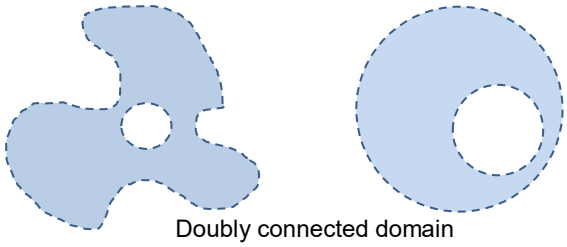
A domain D (connected open set) is called simply connected if interior of every simple closed contour C is contained in D.



The Cauchy-Goursat Theorem

Simply connected domain

A domain D (connected open set) that is not simply connected is called **multiply connected**.

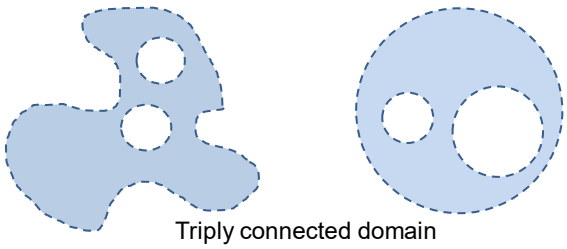


Doubly connected domain

The Cauchy-Goursat Theorem

Simply connected domain

A domain D (connected open set) that is not simply connected is called **multiply connected**.



Triply connected domain

The Cauchy-Goursat Theorem

Positive direction of a contour C is the direction corresponding to increasing values of the parameter t.

The Cauchy-Goursat Theorem

Positive direction of a contour C is the direction corresponding to increasing values of the parameter t.

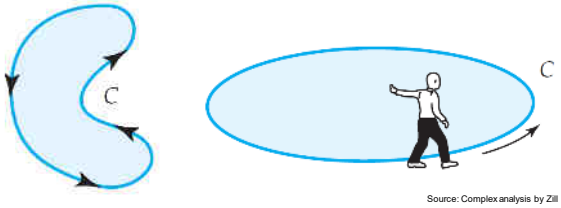
Negative direction of a contour C is the direction opposite to the positive direction.

The Cauchy-Goursat Theorem

Positive direction of a contour C is the direction corresponding to increasing values of the parameter t.

Negative direction of a contour C is the direction opposite to the positive direction.

In case of simple closed contour positive orientation, roughly speaking, corresponds to counter-clockwise direction. And negative direction corresponds to clockwise direction.



Source: Complex analysis by Zill

The Cauchy-Goursat Theorem

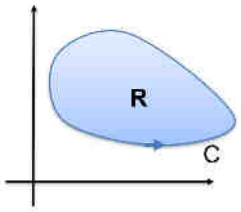
Positive direction of a contour C is the direction corresponding to increasing values of the parameter t.
Negative direction of a contour C is the direction opposite to the positive direction.
In case of simple closed contour positive orientation, roughly speaking, corresponds to counter-clockwise direction. And negative direction corresponds to clockwise direction.
If C is positively oriented then $-C$ is negatively oriented.

The Cauchy-Goursat Theorem

Green's Theorem

Let C be a simple closed contour with positive orientation and let R be the domain that forms the interior of C. If P and Q are continuous and have continuous partial derivatives P_x, P_y, Q_x, Q_y at all points on C and R, then

$$\int_C P(x,y)dx + Q(x,y)dy = \int_R \int \big(Q_x(x,y) - P_y(x,y)\big)dxdy$$

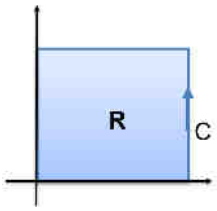


The Cauchy-Goursat Theorem

Example (Green's Theorem): Calculate the following line integral

$$\int_C y\,dx + x\,dy$$

where C is the boundary of a unit square as shown in the following diagram.

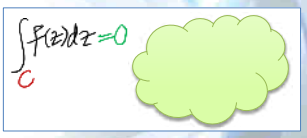


The Cauchy-Goursat Theorem

Cauchy's theorem

Let f be analytic in a simply connected domain D and f' is continuous in D. If C is a simple closed contour that lies in D, then

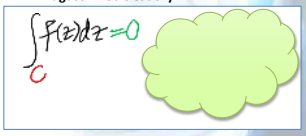
$$\int_C f(z)\ dz = 0$$



The Cauchy-Goursat Theorem

The Cauchy-Goursat Theorem Part I

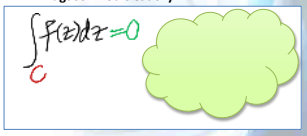
Augustin-Louis Cauchy — Édouard Goursat



The Cauchy-Goursat Theorem

The Cauchy-Goursat Theorem Part II

Augustin-Louis Cauchy — Édouard Goursat



The Cauchy-Goursat Theorem

Theorem (Cauchy's theorem): Let f be analytic in a simply connected domain D and f' is continuous in D. If C is a simple closed contour that lies in D, then

$$\int_C f(z)\ dz = 0$$

Proof:

The Cauchy-Goursat Theorem

Proof (Cauchy's theorem):

The Cauchy-Goursat Theorem

Proof cont.(Cauchy's theorem):

The Cauchy-Goursat Theorem

Proof cont.(Cauchy's theorem):

The Cauchy-Goursat Theorem

Example(Cauchy's theorem)

Show that

$$\int_C \exp(z^2+1)dz = 0$$

where C is a simple closed contour in complex plane.

The Cauchy-Goursat Theorem

Example(Cauchy's theorem)

Show that

$$\int_C (a_n z^n + a_{n-1}z^{n-1} + \cdots + a_1 z + a_0)\, dz = 0$$

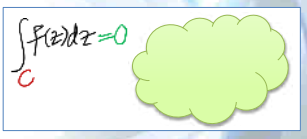
where C is a simple closed contour in complex plane.

The Cauchy-Goursat Theorem

Cauchy's theorem

Let f be analytic in a simply connected domain D and f' is continuous in D. If C is a simple closed contour that lies in D, then

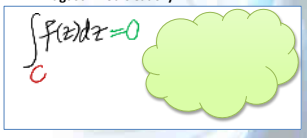
$$\int_C f(z)\ dz = 0$$



The Cauchy-Goursat Theorem

The Cauchy-Goursat Theorem Part II

Augustin-Louis Cauchy Édouard Goursat

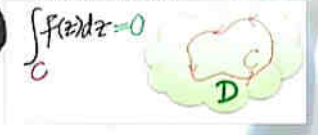


The Cauchy-Goursat Theorem

Topic No. 98

The Cauchy-Goursat Theorem Part II

Augustin-Louis Cauchy Édouard Goursat



The Cauchy-Goursat Theorem

Theorem (Cauchy's theorem): Let f be analytic in a simply connected domain D and f' is continuous in D. If C is a simple closed contour that lies in D, then

$$\int_C f(z)\ dz = 0$$

Proof: $f(z) = u(z) + iv(z)$

$C: z(t) = x(t) + iy(t), \quad a \le t \le b$ (Positively oriented)

$$\int_C f(z)dz = \int_a^b f(z(t))\, z'(t)\, dt$$

$$= \int_a^b \left[u(z(t)) + iv(z(t))\right]\left(x'(t) + iy'(t)\right) dt$$

$$= \int_a^b (ux' - vy')\, dt + i\int_a^b (uy' + vx')\, dt$$

The Cauchy-Goursat Theorem

Proof (Cauchy's theorem):

$$\int_C f(z)dz = \int_a^b (ux' - vy')dt + i\int_a^b (uy' + vx')dt$$

$$= \int_a^b u\frac{dx}{dt}dt - v\frac{dy}{dt}dt + i\int_a^b u\frac{dy}{dt}dt + v\frac{dx}{dt}dt$$

$x = x(t) \Rightarrow dx = x'dt$, $y = y(t) \Rightarrow dy = y'dt$

$$\int_C f(z)dz = \int_a^b u\,dx - v\,dy + i\int_a^b u\,dy + v\,dx$$

I II

C

VU

The Cauchy-Goursat Theorem

Proof cont.(Cauchy's theorem):

$$\int_C u\,dx - v\,dy = \iint_R (-v_x - u_y)\,dx\,dy \quad -(i)$$

$$\int_C u\,dy + v\,dx = \iint_R (u_x - v_y)\,dx\,dy \quad -(i*)$$

f exists. $\Rightarrow f$ satisfies C-R equations

$u_x = v_y$, $u_y = -v_x$

$\Rightarrow \int_C f(z)dz = \int_a^b u\,dx - v\,dy + i\int_a^b u\,dy + v\,dx = 0$

VU

The Cauchy-Goursat Theorem

Proof cont.(Cauchy's theorem):

C is positively oriented ✓

Case 2 If C is negatively oriented ✓

$\Rightarrow -C$ is positively oriented.

$$\int_{-C} f(z)dz = -\int_C f(z)dz = 0$$

VU

The Cauchy-Goursat Theorem

Example(Cauchy's theorem)

Show that

$$\int_C \exp(z^2+1)dz = 0$$

where C is a simple closed contour in complex plane. (simply connected)

$\exp(z)$ & $z+1$ are entire function.

$f(z) = \exp(z^2+1)$

$f'(z) = \exp(z^2+1)(2z)$

$\Rightarrow \int_C \exp(z^2+1)dz = 0$

VU

The Cauchy-Goursat Theorem

Example(Cauchy's theorem)

Show that

$$\int_C (a_n z^n + a_{n-1}z^{n-1} + \cdots + a_1 z + a_0)\,dz = 0$$

where C is a simple closed contour in complex plane.

Simply Connected

VU

The Cauchy-Goursat Theorem

Cauchy's theorem

Let f be analytic in a simply connected domain D and f' is continuous in D. If C is a simple closed contour that lies in D, then

$$\int_C f(z)\,dz = 0$$

VU

The Cauchy-Goursat Theorem

The Cauchy-Goursat Theorem Part III

Édouard Goursat

Augustin-Louis Cauchy

VU

The Cauchy-Goursat Theorem

Cauchy's theorem

Let f be analytic in a simply connected domain D and f' is continuous in D. If C is a simple closed contour that lies in D, then

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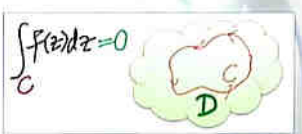


VU

The Cauchy-Goursat Theorem

Example:

$$\int_C \frac{1}{z}\,dz$$

where C is given by

$$C: z(t) = (2+\cos t) + i\,(3+\sin t), \qquad 0 \le t \le 2\pi$$

In fact C is a circle of radius 1 and center (2,3) with the following equation

$$(x-2)^2 + (y-3)^2 = 1$$

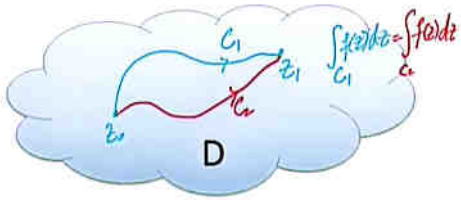
Since $\frac{1}{z}$ is analytic at every point except at $z = 0$ which is outside the circle C, hence

$$\int_C \frac{1}{z}\,dz = 0$$

The Cauchy-Goursat Theorem

Independence of path

Let z_0 and z_1 be points in a domain D. A contour integral $\int_C f(z)\,dz$ is said to be independent of the path if its valued is the same for all contours C in D with initial point z_0 and terminal point z_1.



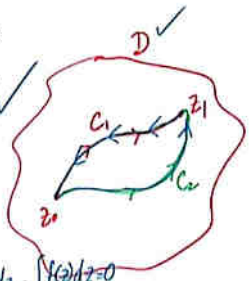
The Cauchy-Goursat Theorem

Fact: Let z_0 and z_1 be two points in a simply connected domain D. Let $f(z)$ be an analytic function. Then the contour integral

$$\int_C f(z)\,dz$$

is independent of path from z_0 to z_1.

Proof: To show: $\int_{C_1} f(z)dz = \int_{C_2} f(z)dz$



$C = C_2 - C_1$, anti-clockwise

$\int_C f(z)dz = 0 \Rightarrow \int_{C_2-C_1} f(z)dz = 0$

$\Rightarrow \int_{C_2} f(z)dz - \int_{C_1} f(z)dz = 0$

The Cauchy-Goursat Theorem

A generalization of Cauchy-Goursat theorem

Let f be analytic in a simply connected domain D ~~and f' is continuous in D~~. If C is a ~~simple~~ closed contour that lies in D, then

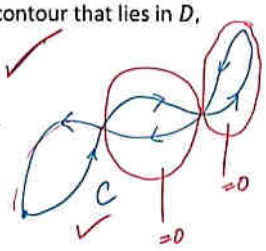
$$\int_C f(z)\,dz = 0$$

The Cauchy-Goursat Theorem

A generalization of Cauchy-Goursat theorem

Let f be analytic in a simply connected domain D ~~and f' is continuous in D~~. If C is a ~~simple~~ closed contour that lies in D, then

$$\int_C f(z)\,dz = 0$$



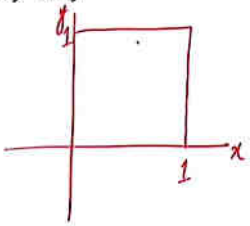
The Cauchy-Goursat Theorem

Example: Calculate

$$\int_C \frac{\exp(z^2)}{(z^2+4)^3}\,dz$$

Let C be a closed contour inside the unit square

$$A = \{(x,y): 0 \le x, y \le 1\}$$



The Cauchy-Goursat Theorem

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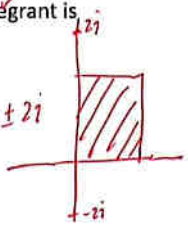
The integrand is analytic at every point except at points where the denominator is zero, that is, $z = \pm 2i$. So integrant is analytic inside A.

$z^2 + 4 = 0$

$(z-2i)(z+2i) = 0 \Rightarrow z = \pm 2i$



The Cauchy-Goursat Theorem

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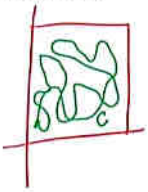


Generalizations of Cauchy-Goursat Theorem

Generalizations of Cauchy-Goursat Theorem

Topic No. 100

Édouard Goursat

Augustin-Louis Cauchy

$\int_C f(z)dz = 0$

Generalizations of Cauchy-Goursat Theorem

Cauchy-Goursat theorem

Let f be analytic in a simply connected domain D ~~and f' is continuous in D~~. If C is a simple closed contour that lies in D, then

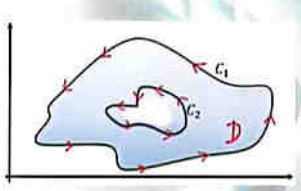
$$\int_C f(z)\,dz = 0$$

$\int_C f(z)dz$

Generalizations of Cauchy-Goursat Theorem

Theorem
(Deformation of contour)

$$\int_{C_1} f(z)\,dz = \int_{C_2} f(z)\,dz$$



Generalizations of Cauchy-Goursat Theorem

Theorem (Deformation of contour)

Let C_1 and C_2 be two simple closed positively oriented contours such that C_2 lies interior to C_1. If $f(z)$ is analytic in a domain D that contains both C_1 and C_2 and the region between them, as shown in the following figure, then

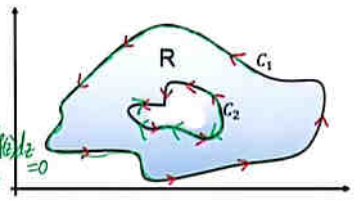
$$\int_{C_1} f(z)\,dz = \int_{C_2} f(z)\,dz$$

Cauchy-Goursat thm

$\int_C f(z)dz = 0$

$C = C_1 - C_2$

$\int_{C_1-C_2} f(z)dz = 0, \int_{C_1} f(z)dz + \int_{-C_2} f(z)dz = 0$



Generalizations of Cauchy-Goursat Theorem

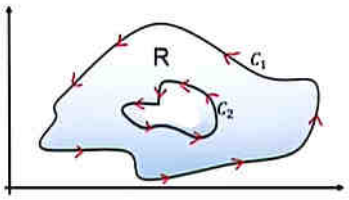
Theorem (Deformation of contour)

Let C_1 and C_2 be two simple closed positively oriented contours such that C_2 lies interior to C_1. If $f(z)$ is analytic in a domain D that contains both C_1 and C_2 and the region between them, as shown in the following figure, then

$$\int_{C_1} f(z)\,dz = \int_{C_2} f(z)\,dz$$

The contour $C_1 - C_2$ is positively oriented as boundary of the region R (between C_1 and C_2). Hence

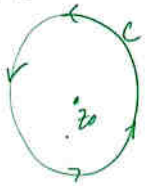
$$\int_{C_2} f(z)\,dz = 0$$



Generalizations of Cauchy-Goursat Theorem

Example: Let z_0 denote a fixed complex value. If C is a simple closed contour with positive orientation such that z_0 lies interior to C, then for integer n we have the following

$$\int_C \frac{dz}{(z-z_0)^n} = \begin{cases} 2\pi i & \text{if } n = 1 \\ 0 & \text{if } n \neq 1 \end{cases}$$

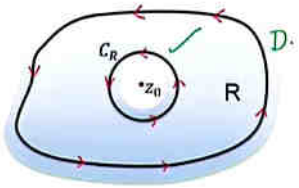


Generalizations of Cauchy-Goursat Theorem

Example: Let z_0 denote a fixed complex value. If C is a simple closed contour with positive orientation such that z_0 lies interior to C, then for integer n we have the following

$$\int_C \frac{dz}{(z-z_0)^n} = \begin{cases} 2\pi i & \text{if } n = 1 \\ 0 & \text{if } n \neq 1 \end{cases}$$

Solution:

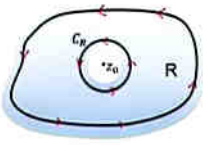


Generalizations of Cauchy-Goursat Theorem

Solution:

$C_R: z(t) = z_0 + Re^{it}$ where $0 \le t \le 2\pi$

$$\int_C f(z)dz = \int_{C_R} f(z)dz$$



Generalizations of Cauchy-Goursat Theorem

Solution:

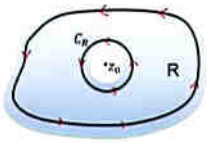
$C_R: z(t) = z_0 + Re^{it}$ where $0 \le t \le 2\pi$

$$\int_C \frac{dz}{(z-z_0)^n} = \int_{C_R} \frac{dz}{(z-z_0)^n}$$

$$= \int_0^{2\pi} \frac{1}{(z_0 + Re^{it} - z_0)^n} (iR\,e^{it})\,dt = i R^{1-n} \int_0^{2\pi} e^{i(1-n)t}\,dt$$

$\int_{C_1} f(z)\,dz = \int_0^{2\pi} f(z(t))\,z'(t)\,dt$

$f(z(t)) = R^n e^{int}$, $z'(t) = iR\,e^{it}$



Generalizations of Cauchy-Goursat Theorem

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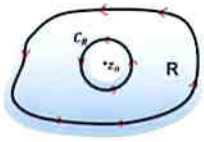
Case 1 ($n = 1$)

$$\int_C \frac{dz}{(z-z_0)^1}$$

$$= i R^0 \int_0^{2\pi} e^{it(0)}\,dt$$



The Fundamental Theorems of Integration

Topic No. 101

The fundamental theorems of integration

The Fundamental Theorems of Integration

Antiderivative of a complex-valued function can be constructed by using contour integration

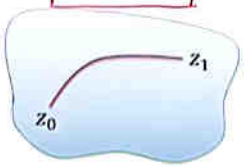
The Fundamental Theorems of Integration

Theorem (indefinite integrals): Let f be an analytic complex valued function in a simply connected domain D. If C is a contour from a fixed $z_0 \in D$ to an arbitrary point $z \in D$, then the function

$$F(z) = \int_C f(s)\,ds = \int_{z_0}^{z_1} f(s)\,ds$$

is well defined and an analytic function in D, such that

$$F'(z) = f(z)$$

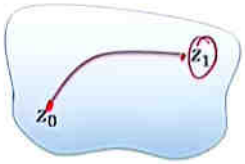


The Fundamental Theorems of Integration

Theorem (definite integrals): Let f be an analytic complex valued function in a simply connected domain D. Let $z_0, z_1 \in D$ and C be contour joining z_0 with z_1, then

$$\int_C f(s)\,ds = \int_{z_0}^{z_1} f(s)\,ds = F(z_1) - F(z_0)$$

where $F(z)$ is antiderivative of $f(z)$.



The Fundamental Theorems of Integration

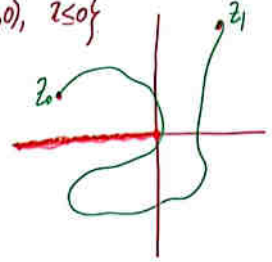
Example: Evaluate $\int_C \frac{dz}{z}$

C is a contour in $D = \mathbb{C} - \{(z,0),\ z \le 0\}$

Here D is simply connected.

$F(z) = \log(z)$, $F'(z) = \frac{1}{z}$

$$\int_{z_0}^{z_1} \frac{dz}{z} = \int_C \frac{dz}{z} = \log(z_1) - \log(z_0)$$

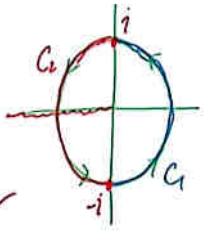


The Fundamental Theorems of Integration

Example: $\int_C \frac{dz}{z}$, $C: z(t) = e^{it}$, $0 \le t \le 2\pi$

$\int_{C_1} \frac{dz}{z}$, $\int_{C_2} \frac{dz}{z}$, $C = C_1 + C_2$

$$\Rightarrow \int_C \frac{dz}{z} = \int_{C_1} \frac{dz}{z} + \int_{C_2} \frac{dz}{z}$$



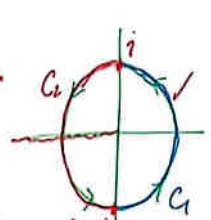
Case (C_1), $D = \mathbb{C} - \{(x,0),\ x \le 0\}$

$F(z) = \log(z) = \ln|z| + i\,\mathrm{Arg}(z)$, $-\pi < \mathrm{Arg}(z) < \pi$

$F'(z) = \frac{1}{z}$, $\int_{C_1} \frac{dz}{z} = \log(i) - \log(-i) = \pi i$

The Fundamental Theorems of Integration

Example: $\int_{C_2} \frac{dz}{z}$



$\log(z) = \ln|z| + i\theta$

$0 < \theta < 2\pi$

$f(z) = \log(z), \quad F(z) = \frac{1}{z}, \quad D = \mathbb{C} - \{(x,y), x \geq 0\}$

$\int_{-i}^{i} \frac{dz}{z} = -\int_{C_1} \frac{dz}{z} = \log(-i) - \log(i) = i\frac{3\pi}{2} - i\frac{\pi}{2} = \pi i$

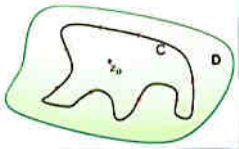
$\int_C \frac{dz}{z} = \int_{C_1} \frac{dz}{z} + \int_{C_2} \frac{dz}{z} = \pi i + \pi i = 2\pi i$

VU

Cauchy's Integral Formula

Topic No. 102

Cauchy's Integral Formula Part I



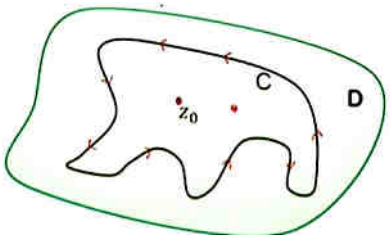
VU

Cauchy's Integral Formula

Theorem (Cauchy's integral formula):

Let f be analytic in the simply connected domain D and let C be a simple closed positively oriented contour that lies in D. If z_0 is a point that lies interior to C, then

$$f(z_0) = \frac{1}{2\pi i}\int_C \frac{f(z)}{z - z_0}\, dz$$



VU

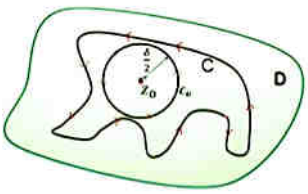
Cauchy's Integral Formula

Proof: Since $f(z)$ is analytic so it is continuous on D, and hence continuous at $z_0 \in D$. By definition of continuity we have

For $\varepsilon > 0$ there exist a $\delta > 0$ such that

$$|f(z) - f(z_0))| < \varepsilon \quad whenever\ |z - z_0| < \delta$$

We choose δ such that the circle $C_0 = \left\{z: |z - z_0| < \frac{\delta}{2}\right\}$ lies inside C.



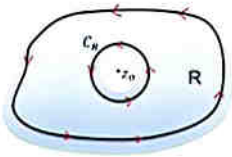
VU

Cauchy's Integral Formula

Recall

Example: Let z_0 denote a fixed complex value. If C is a simple closed contour with positive orientation such that z_0 lies interior to C, then for integer n we have the following

$$\int_C \frac{dz}{(z - z_0)^n} = \begin{cases} 2\pi i & if\ n = 1 \\ 0 & if\ n \neq 1 \end{cases}$$



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$$|f(z) - f(z_0))| < \varepsilon \quad whenever\ |z - z_0| < \delta$$

We choose δ such that the circle $C_0 = \left\{z: |z - z_0| < \frac{\delta}{2}\right\}$ lies inside C.

Since $f(z_0)$ is a fixed complex number so we can write

$$f(z_0) = \frac{f(z_0)}{2\pi i}\int_{C_0} \frac{dz}{z - z_0} = \frac{1}{2\pi i}\int_{C_0} \frac{f(z_0)dz}{z - z_0}$$

VU

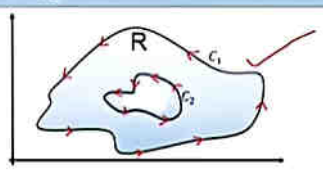
The Cauchy-Goursat Theorem

Recall

Theorem (Deformation of contour)

Let C_1 and C_2 be two simple closed positively oriented contours such that C_1 lies interior to C_2. If $f(z)$ is analytic in a domain D that contains both C_1 and C_2 and the region between them, as shown in the following figure, then

$$\int_{C_1} f(z)\, dz = \int_{C_2} f(z)\, dz$$



VU

Cauchy's Integral Formula

Proof: Since $f(z)$ is analytic so it is continuous on D, and hence continuous at $z_0 \in D$. By definition of continuity we have

For $\varepsilon > 0$ there exist a $\delta > 0$ such that

$$|f(z) - f(z_0))| < \varepsilon \quad whenever\ |z - z_0| < \delta$$

We choose δ such that the circle $C_0 = \left\{z: |z - z_0| < \frac{\delta}{2}\right\}$ lies inside C.

Since $f(z_0)$ is a fixed complex number so we can write

$$f(z_0) = \frac{f(z_0)}{2\pi i}\int_{C_0} \frac{dz}{z - z_0} = \frac{1}{2\pi i}\int_{C_0} \frac{f(z_0)dz}{z - z_0}$$

By theorem on deformation of contour, we get

$$\frac{1}{2\pi i}\int_C \frac{f(z)}{z - z_0}\, dz = \frac{1}{2\pi i}\int_{C_0} \frac{f(z)}{z - z_0}\, dz$$

VU

Cauchy's Integral Formula

$$|f(z) - f(z_0))| < \varepsilon \quad whenever \ |z - z_0| < \delta$$

$$f(z_0) = \frac{f(z_0)}{2\pi i}\int_{C_0}\frac{dz}{z-z_0} = \frac{1}{2\pi i}\int_{C_0}\frac{f(z_0)dz}{z-z_0}$$

$$\frac{1}{2\pi i}\int_C \frac{f(z)}{z-z_0}\,dz = \frac{1}{2\pi i}\int_{C_0}\frac{f(z)}{z-z_0}\,dz$$

Proof (cont.):

$$\left|\frac{1}{2\pi i}\int_C \frac{f(z)}{z-z_0}\,dz - f(z_0)\right|$$

$$= \left|\frac{1}{2\pi i}\int_C \frac{f(z)}{z-z_0}\,dz - \frac{1}{2\pi i}\int_{C_0}\frac{f(z_0)dz}{z-z_0}\right|$$

$$\leq \frac{1}{2\pi}\int_{C_0}\frac{|f(z)-f(z_0)|}{|z-z_0|}\,|dz|$$

$$\leq \frac{1}{2\pi}(\varepsilon)\left(\frac{1}{(1/2)\delta}\right)\left(2\pi\frac{\delta}{2}\right) = \varepsilon$$

Cauchy's Integral Formula

Proof (cont.):

$$\left|\frac{1}{2\pi i}\int_C \frac{f(z)}{z-z_0}\,dz - f(z_0)\right| = \varepsilon$$

Since ε is arbitrary so

$$\frac{1}{2\pi i}\int_C \frac{f(z)}{z-z_0}\,dz - f(z_0) = 0$$

Hence

$$f(z_0) = \frac{1}{2\pi i}\int_C \frac{f(z)}{z-z_0}\,dz$$

Cauchy's Integral Formula

Example (Cauchy's integral formula):Evaluate

$$\int_C \frac{dz}{z}$$

where $C: z(t) = e^{it}$, *for* $0 \leq t \leq 2\pi$.

Solution: Comparing the following two

$$\int_C \frac{dz}{z} = \int_C \frac{f(z)}{z-z_0}dz$$

we get $f(z) = 1, z_0 = 0$. Hence by Cauchy's integral formula.

$$\int_C \frac{dz}{z} = 2\pi i\ f(0)$$

$$= 2\pi i$$

Cauchy's Integral Formula

Example (Cauchy's integral formula):Evaluate

$$\int_C \frac{P_n(z)}{z-\alpha}dz$$

where C is a contour and α lies in the interior of C, and

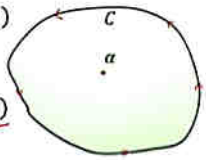
$$P_n(z) = a_n z^n + a_{n-1}z^{n-1} + \cdots + a_1 z + z_0 .$$

Solution: Here $P_n(z)$ is analytic in the simply connected domain $D = \mathbb{C}$. Also the contour C is simple and closed. Hence by Cauchy's integral formula

$$\frac{1}{2\pi i}\int_C \frac{P_n(z)}{z-\alpha}dz = P_n(\alpha)$$

Or

$$\int_C \frac{P_n(z)}{z-\alpha}dz = 2\pi i\ P_n(\alpha)$$



Cauchy's Integral Formula

Topic No. 109

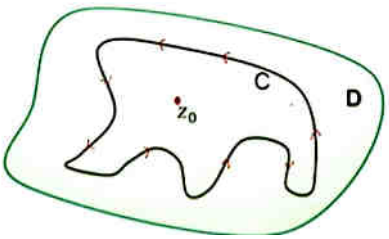
Cauchy's Integral Formula Part II

Cauchy's Integral Formula

Theorem (Cauchy's integral formula):

Let f be analytic in the simply connected domain D and let C be a simple closed positively oriented contour that lies in D. If z_0 is a point that lies interior to C, then

$$f(z_0) = \frac{1}{2\pi i}\int_C \frac{f(z)}{z-z_0}\,dz$$



Cauchy's Integral Formula

Recall

Example (Cauchy's integral formula):Evaluate

$$\int_C \frac{dz}{z}$$

where $C: z(t) = e^{it}$, *for* $0 \leq t \leq 2\pi$.

Cauchy's Integral Formula

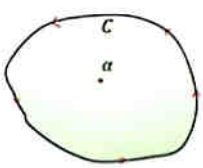
Recall

Example (Cauchy's integral formula):Evaluate

$$\int_C \frac{P_n(z)}{z-\alpha}dz$$

where C is a contour and α lies in the interior of C, and

$$P_n(z) = a_n z^n + a_{n-1}z^{n-1} + \cdots + a_1 z + z_0 .$$



Cauchy's Integral Formula

Example (Cauchy's integral formula): Evaluate

$$\int_C \frac{3z^2 + 10z - 14}{z^3 + 5z^2 - 14z}\,dz$$

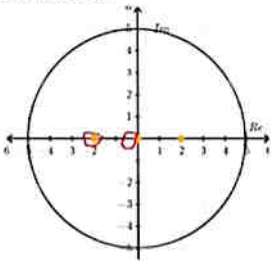
where $C: z(t) = 5e^{it}, 0 \le t \le 2\pi$.

Solution: The denominator can be written as

$$z^3 + 5z^2 - 14z = z(z+7)(z-2)$$

we can find partial fractions of the integrand as follows

$$\frac{3z^2 + 10z - 14}{z^3 + 5z^2 - 14z} = \frac{1}{z} + \frac{1}{z-2} + \frac{1}{z+7}$$



VU

Cauchy's Integral Formula

Solution (cont.):

$$\int_C \frac{3z^2 + 10z - 14}{z^3 + 5z^2 - 14z}\,dz = \int_C \frac{1}{z}\,dz + \int_C \frac{1}{z-2}\,dz + \int_C \frac{1}{z+7}\,dz$$

By Cauchy's integral formula

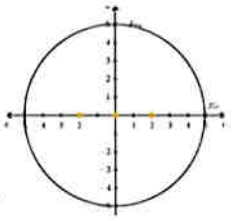
$$\int_C \frac{1}{z}\,dz = \int_C \frac{1}{z-2}\,dz = 2\pi i$$

By Cauchy-Goursat Theorem

$$\int_C \frac{1}{z+7}\,dz = 0$$

Hence

$$\int_C \frac{3z^2 + 10z - 14}{z^3 + 5z^2 - 14z}\,dz = 2\pi i + 2\pi i = 4\pi i$$



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Cauchy's Integral Formula

Example (Cauchy's integral formula): Evaluate

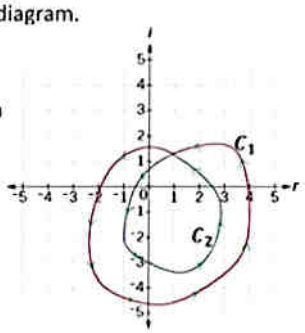
$$\int_C \frac{1}{z-1}\,dz$$

where C is given in the following diagram.

Solution: we can write C as union of two simple closed contours, namely, C_1, C_2 as shown in the diagram.

Hence

$$\int_C \frac{1}{z-1}\,dz = \int_{C_1} \frac{1}{z-1}\,dz + \int_{C_2} \frac{1}{z-1}\,dz$$



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Cauchy's Integral Formula

Solution (cont.) :

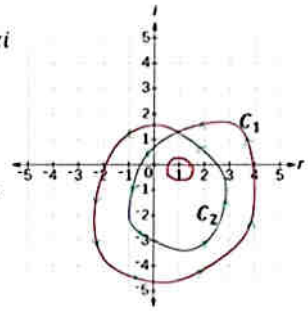
$$\int_C \frac{1}{z-1}\,dz = \int_{C_1} \frac{1}{z-1}\,dz + \int_{C_2} \frac{1}{z-1}\,dz$$

By Cauchy's integral formula

$$\int_{C_1} \frac{1}{z-1}\,dz = \int_{C_2} \frac{1}{z-1}\,dz = 2\pi i$$

Hence

$$\int_C \frac{1}{z-1}\,dz = \int_{C_1} \frac{1}{z-1}\,dz + \int_{C_2} \frac{1}{z-1}\,dz = 2\pi i + 2\pi i = 4\pi i$$

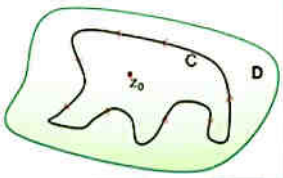


VU

Cauchy's Integral Formulas for Derivatives

Topic No. 104

Cauchy's integral formulas for derivatives



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Cauchy's Integral Formulas for Derivatives

Recall

Cauchy's integral formula

Let f be analytic in the simply connected domain D and let C be a simple closed positively oriented contour that lies in D. If z_0 is a point that lies interior to C, then

$$f(z_0) = \frac{1}{2\pi i}\int_C \frac{f(z)}{z - z_0}\,dz$$

VU

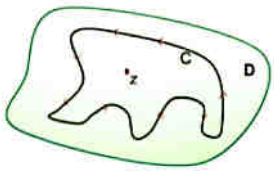
Cauchy's Integral Formulas for Derivatives

Theorem (Cauchy's integral formulas for derivatives)

Let f be analytic in the simply connected domain D and let C be a simple closed positively oriented contour that lies in D. If z is a point that lies interior to C, then for any integer $n \ge 0$,

$$f^{(n)}(z) = \frac{n!}{2\pi i}\int_C \frac{f(\xi)}{(\xi - z)^{n+1}}\,d\xi$$

Note: For $n = 0$ we have get Cauchy's integral formula.



VU

Cauchy's Integral Formulas for Derivatives

Example (Cauchy's integral formulas for derivatives)

Evaluate the following

$$\int_C \frac{z^3 + 3z^2 + 2z + 1}{(z-1)^3}\,dz$$

where $C: z(t) = 2e^{it}, 0 \le t \le 2\pi$.

Solution: The integrand is not analytic inside the contour since the denominator vanish at $z = 1$, which is a point inside the contour C.

The function $f(z) = z^3 + 3z^2 + 2z + 1$ is analytic on $D = \mathbb{C}$ (a simply connected domain). Also the contour is simple and closed with positive orientation.

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Cauchy's Integral Formulas for Derivatives

Solution (cont.): By Cauchy's integral formulas for derivatives

$$f^{(n)}(z) = \frac{n!}{2\pi i}\int_C \frac{f(\xi)}{(\xi - z)^{n+1}}\, d\xi$$

we get

$$\int_C \frac{z^3 + 3z^2 + 2z + 1}{(z-1)^3}\, dz = \frac{2\pi i}{2!} f''(1)$$

Since $f(z) = z^3 + 3z^2 + 2z + 1$ so

$$f'(z) = 3z^2 + 6z + 2, \qquad f''(z) = 6z + 6$$

Hence $f''(1) = 6(1) + 6 = 12$, so

$$\int_C \frac{z^3 + 3z^2 + 2z + 1}{(z-1)^3}\, dz = \frac{2\pi i}{2!}(12) = 12\pi i.$$

VU

Cauchy's Integral Formulas for Derivatives

Example (Cauchy's integral formulas for derivatives)

Let z_0 denote a fixed complex value. If C is a simple closed contour with positive orientation such that z_0 lies interior to C, then for integer n we have the following

$$\int_C \frac{dz}{(z - z_0)^n} = \begin{cases} 2\pi i & \text{if } n = 1 \\ 0 & \text{if } n \neq 1 \end{cases}$$

Solution:

Case 1 $n = 1$

Here $f(z) = 1$, by Cauchy's integral formula

$$\int_C \frac{dz}{z - z_0} = 2\pi i\ f(z_0) = 2\pi i$$

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Cauchy's Integral Formulas for Derivatives

Solution:

Case 2 $n \geq 0$, **and** $n \neq 1$

Here $f(z) = 1$, by Cauchy's integral formulas for derivatives

$$\int_C \frac{dz}{(z - z_0)^n} = \frac{2\pi i}{(n-1)!}\, f^{n-1}(z_0) = \frac{2\pi i}{(n-1)!}(0) = 0.$$

Case 2 $n < 0$

In this case the integrand is analytic function and hence by Cauchy-Goursat theorem

$$\int_C \frac{dz}{(z - z_0)^n} = \int_C (z - z_0)^{-n}\, dz = 0$$

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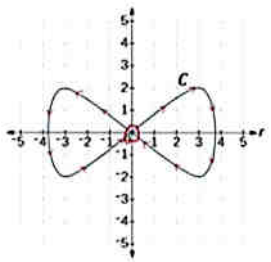
Cauchy's Integral Formulas for Derivatives

Example: Evaluate

$$\int_C \frac{z^3 - 6z^2 + 11z - 5}{(z^2 - 5z + 6)^2} dz$$

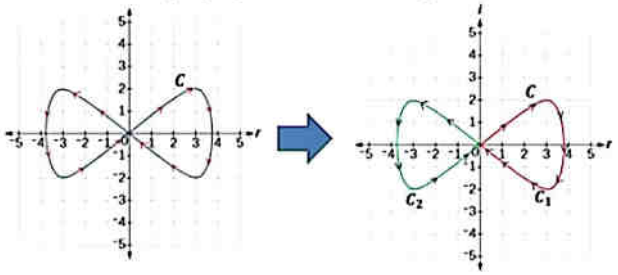
where C is given in the following diagram.



VU

Cauchy's Integral Formulas for Derivatives

Solution: The contour C can be written as union of two contours, namely, C_1, C_2, as shown in the figure.



Here $C = C_1 + C_2$, where C_1 is negatively oriented and hence $-C_1$ is positively oriented, and C_2 is positively oriented.

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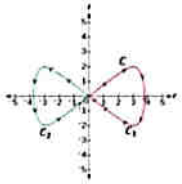
Cauchy's Integral Formulas for Derivatives

Solution (cont.): Hence we have

$$\int_C \frac{z^3 - 6z^2 + 11z - 5}{(z^2 - 5z + 6)^2} dz$$

$$= \int_{C_1} \frac{z^3 - 6z^2 + 11z - 5}{(z^2 - 5z + 6)^2} dz + \int_{C_2} \frac{z^3 - 6z^2 + 11z - 5}{(z^2 - 5z + 6)^2} dz$$

$$= -\int_{-C_1} \frac{z^3 - 6z^2 + 11z - 5}{(z^2 - 5z + 6)^2} dz + \int_{C_2} \frac{z^3 - 6z^2 + 11z - 5}{(z^2 - 5z + 6)^2} dz$$



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Cauchy's Integral Formulas for Derivatives

Solution (cont.): The integrand can be written in the following form

$$\frac{z^3 - 6z^2 + 11z - 5}{(z^2 - 5z + 6)^2} = \frac{1}{(z-3)^2} + \frac{1}{(z-2)} + \frac{1}{(z-2)^2}$$

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Cauchy's Integral Formulas for Derivatives

Solution (cont.): Hence

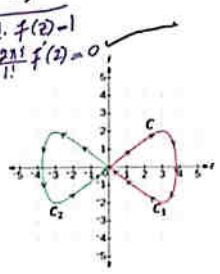
$$\int_{-C_1} \frac{z^3 - 6z^2 + 11z - 5}{(z^2 - 5z + 6)^2} dz$$

$$= \int_{-C_1} \frac{dz}{(z-3)^2} + \int_{-C_1} \frac{dz}{(z-2)} + \int_{-C_1} \frac{dz}{(z-2)^2}$$

$n=1,\ f(z)=1$ — Cauchy's Integral formula for derivatives $= \frac{2\pi i}{1!} f'(3) = 2\pi i\, f'(3) = 0$

$f(z)=1,\ z_0=2$ — Cauchy's Integral formula $= 2\pi i\, f(z_0) = 2\pi i$

$n=1,\ f(z)=1$ — Cauchy's Integral formula for derivatives $= \frac{2\pi i}{1!} f'(2) = 0$



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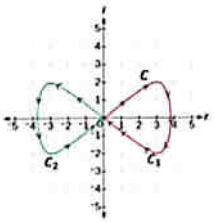
Cauchy's Integral Formulas for Derivatives

Solution (cont.):Hence

$$\int_{-C_1} \frac{z^3 - 6z^2 + 11z - 5}{(z^2 - 5z + 6)^2} dz$$

$$= \int_{-C_1} \frac{dz}{(z-3)^2} + \int_{-C_1} \frac{dz}{(z-2)} + \int_{-C_1} \frac{dz}{(z-2)^2}$$

$$= 0 + 2\pi i + 0 = 2\pi i$$



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Cauchy's Integral Formulas for Derivatives

Solution (cont.): Now let us consider the case of C_2

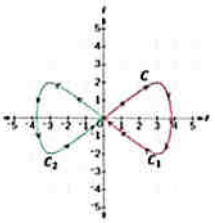
$$\int_{C_2} \frac{z^3 - 6z^2 + 11z - 5}{(z^2 - 5z + 6)^2} dz$$

$$= \int_{C_2} \frac{dz}{(z-3)^2} + \int_{C_2} \frac{dz}{(z-2)} + \int_{C_2} \frac{dz}{(z-2)^2}$$

Since all the integrands are analytic inside and on C_2 so by Cauchy-Goursat theorem

$$\int_{C_2} \frac{z^3 - 6z^2 + 11z - 5}{(z^2 - 5z + 6)^2} dz$$

$$= 0 + 0 + 0 = 0$$



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Cauchy's Integral Formulas for Derivatives

Corollary1: If f is analytic in the domain D, then for integers $n \geq 0$, all derivatives $f^{(n)}(z)$ exist for $z \in D$ (and hence analytic in D).

Edited By Mahar Afaq Safdar Muhammadi

MSc Math, WhatsApp No. 03494873115

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Cauchy's Integral Formulas for Derivatives

Corollary2: If u is a harmonic function at each point (x, y) in the domain D, then all partial derivatives $u_x, u_y, u_{xx}, u_{xy}, u_{yy}$ exist and are harmonic functions.

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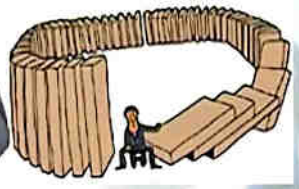
Some Consequences...

Some consequences of the Cauchy's integral formulas for derivatives

Topic No. 105



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Morera's Theorem

Morera's Theorem

Giacinto Morera
18 July 1856 – 8 February 1909

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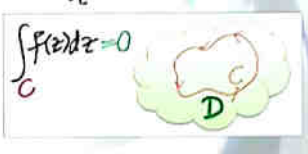
Morera's Theorem

Cauchy-Goursat theorem

Let f be analytic in a simply connected domain D ~~and f' is continuous in D~~. If C is a simple closed contour that lies in D, then

$$\int_C f(z)\, dz = 0$$



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Morera's Theorem

Converse of Cauchy-Goursat theorem is true?

If for simple closed contour C in a simply connected domain D

$$\int_C f(z)\, dz = 0$$

then f is analytic?

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Morera's Theorem

Theorem (Morera's theorem)

Let f be a continuous function in a simply connected domain D. If

$$\int_C f(z)dz = 0$$

for every closed contour C in D then f is analytic in D.

Morera's Theorem

Morera's theorem

No proof

The proof mainly uses the following fact:

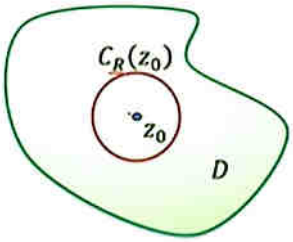
If f is analytic in the domain D, then for integers $n \geq 0$, all derivatives $f^{(n)}(z)$ exist for $z \in D$ (and hence analytic in D).

Gauss's Mean Value Theorem

Theorem (Gauss's mean value theorem)

If f is analytic in a simply connected domain D that contains the circle $C_R(z_0) = \{z: |z - z_0| = R\}$, then

$$f(z_0) = \frac{1}{2\pi}\int_0^{2\pi} f(z_0 + Re^{it})\,dt$$



Gauss's Mean Value Theorem

Proof: The circle $C_R(z_0)$ is given by

$$C_R(z_0): z(t) = z_0 + Re^{it}, 0 \leq t \leq 2\pi$$

Since $f(z)$ is analytic in D, so

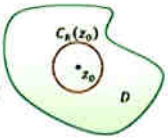
$$f(z_0) = \frac{1}{2\pi i}\int_{C_R} \frac{f(z)}{z - z_0}$$

$$= \frac{1}{2\pi i}\int_0^{2\pi} \frac{f(z_0 + Re^{it})\,(iR\,e^{it})}{(z_0 + Re^{it} - z_0)}\,dt$$

$$= \frac{i}{2\pi i}\int_0^{2\pi} f(z_0 + Re^{it})\,dt$$

Hence

$$f(z_0) = \frac{1}{2\pi}\int_0^{2\pi} f(z_0 + Re^{it})\,dt$$



Gauss's Mean Value Theorem

Example (Gauss's mean value theorem): Evaluate

$$\int_0^{2\pi} \exp(5e^{it} + i\pi)\,dt$$

Solution: The function $f(z) = \exp(z)$ is analytic hence, by Gauss's mean value theorem

$$\int_0^{2\pi} \exp(5e^{it} + \pi i)\,dt = (2\pi)\exp(i\pi) = -2\pi.$$

Some Consequences...

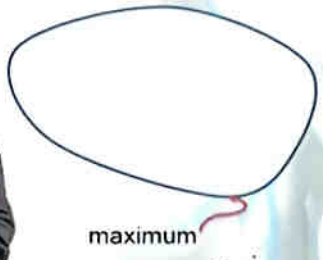
Some consequences of the Cauchy's integral formulas for derivatives

- **Morera's theorem**
- **Gauss's mean value theorem**

Maximum Modulus Principle

Topic No.106

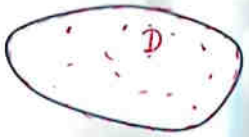
Maximum modulus principle



Maximum Modulus Principle

Maximum Modulus Principle

Let f be analytic and non-constant in the domain D. Then $|f(z)|$ does not attain a maximum valued at any point z_0 in D



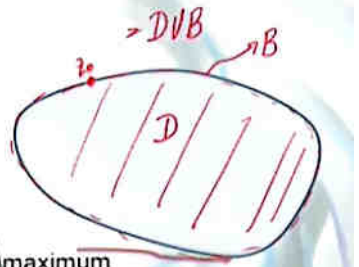
Maximum Modulus Principle

Maximum Modulus Principle

Let f be analytic and non-constant in the bounded domain D. If f is continuous on the closed region R that consists of D and all its boundary points B, then $|f(z)|$ assumes its maximum value, and does so only at point(s) z_0 on the boundary B.

Maximum Modulus Principle

Maximum Modulus Principle



Maximum Modulus Principle

Example (Maximum Modulus Principle): Consider

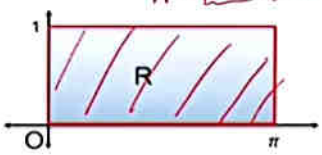
$$f(z) = \sin(z)$$

defined on the following region

$$R = \{(x, y): 0 \le x \le \pi, 0 \le y \le 1\}$$

Find points in R, where $|f(z)|$ has maximum value.

Solution:



Maximum Modulus Principle

Solution: Here

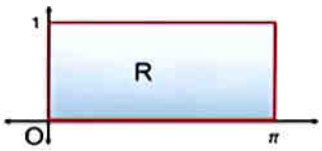
$$\begin{aligned} |\sin z|^2 &= |\sin(x+iy)|^2 \\ &= |\sin x \cosh y + i \cos x \sinh y|^2 \\ &= \sin^2 x \cosh^2 y + \cos^2 x \sinh^2 y \\ &= \sin^2 x(\cosh^2 y - \sinh^2 y) + \sinh^2 y(\cos^2 x + \sin^2 x) \end{aligned}$$

Using $\cosh^2 y - \sinh^2 y = 1$ and $\cos^2 x + \sin^2 x = 1$, we get

$$|\sin z|^2 = \sin^2 x + \sinh^2 y$$

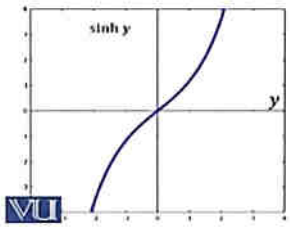
Hence

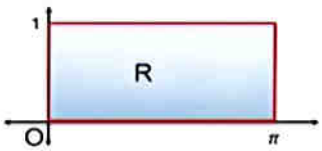
$$|f(z)| = \sqrt{\sin^2 x + \sinh^2 y}$$



Maximum Modulus Principle

Solution (cont.): Maximum value of $\sin x$ occurs at $x = \frac{\pi}{2}$. The function $\sinh y$ is an increasing function and hence, for $0 \le y \le 1$, has maximum value of at $y = 1$.





Maximum Modulus Principle

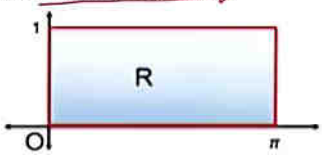
Solution (cont.): Maximum value of $\sin x$ occurs at $x = \frac{\pi}{2}$. The function $\sinh y$ is an increasing function and hence, for $0 \le y \le 1$, has maximum value of at $y = 1$. Hence the function

$$|f(z)| = \sqrt{\sin^2 x + \sinh^2 y}$$

has maximum value when $x = \frac{\pi}{2}, y = 1$, that is, at point

$$p = \left(\frac{\pi}{2}, 1\right).$$

The point p lies at the boundary. So the function $|f(z)|$ has maximum value at the boundary and at no other point.

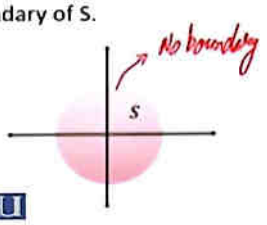


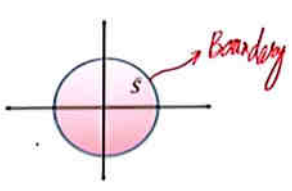
Maximum Modulus Principle

Example (Maximum modulus principle): Let $f(z) = az + b$. If we set our domain D to be $S = \{z: |z| < 1\}$. The function $f(z)$ is continuous on the region $\bar{S} = \{z: |z| \le 1\}$. Then

$$\max_{|z| \le 1} |f(z)| = |a| + |b|$$

and that this value is assumed by f at a point $z_0 = e^{it}$ on the boundary of S.





Maximum Modulus Principle

Example (Maximum modulus principle): Let $f(z) = az + b$. If we set our domain D to be $S = \{z: |z| < 1\}$. The function $f(z)$ is continuous on the region $\bar{S} = \{z: |z| \le 1\}$. Then

$$\max_{|z| \le 1} |f(z)| = |a| + |b|$$

and that this value is assumed by f at a point $z_0 = e^{it}$ on the boundary of S.

Solution: From the triangle inequality and the fact that

$$|z| \le 1$$

we get

$$|f(z)| = |az + b| \le |az| + |b| = |z||a| + |b| \le |a| + |b|$$

So maximum value of $|f(z)|$ is $|a| + |b|$. Now we need to show that this valued occurs at the boundary of S.

Maximum Modulus Principle

Solution (cont.) Consider a point z_0 on the boundary of S given in the following way,

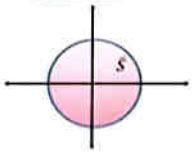
$$z_0 = e^{it_0}$$

where $t_0 = \arg b - \arg a$, then

$$\begin{aligned}\arg(az_0) &= \arg(a) + \arg(z_0)\\ &= \arg(a) + (\arg b - \arg a)\\ &= \arg b\end{aligned}$$

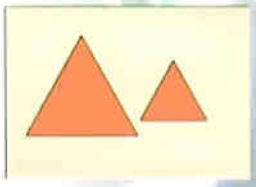
So the vectors az_0 and b lie on the same ray hence

$$|f(z_0)| = |az_0 + b| = |az_0| + |b| = |a| + |b|.$$



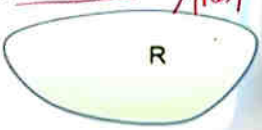
Maximum Modulus Principle

Using **maximum modulus principle** we see that **minimum values** of $|f(z)|$ have similar properties.



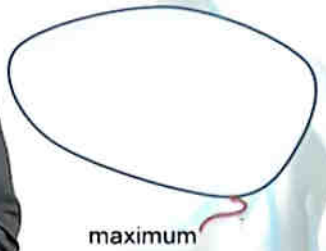
Maximum Minimum Modulus Principle

Let a function f be continuous on a closed bounded region R, and let it be analytic and not constant throughout the interior of R. If $f(z) \neq 0$ anywhere in R then $|f(z)|$ has a minimum value m in R which occurs on the boundary of R and never in the interior.



Maximum Modulus Principle

Maximum modulus principle



Cauchy's Inequality

Topic No. 107

Cauchy's Inequality

Cauchy's Inequality

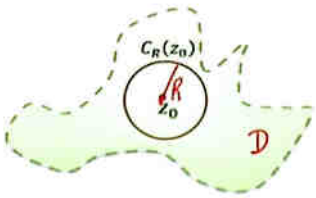
Theorem (Cauchy's Inequality): Let f be analytic in the simply connected domain D that contains the circle

$$C_R(z_0) = \{z: |z - z_0| = R\}.$$

If $|f(z)| \leq M$ for all points $z \in C_R(z_0)$ then

$$|f^{(n)}(z_0)| \leq \frac{n!\,M}{R^n}, \qquad for\ n = 1,2,\ldots$$

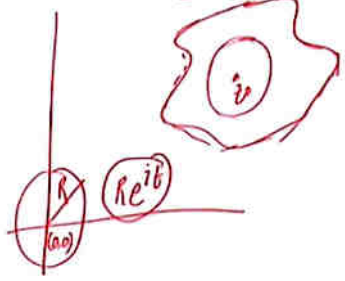
Proof:

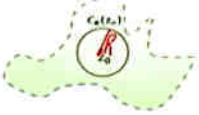


Cauchy's Inequality

Proof: The circle $C_R(z_0)$ has the following parametric representation

$$C_R(z_0): z(t) = z_0 + Re^{it}, 0 \leq t \leq 2\pi$$





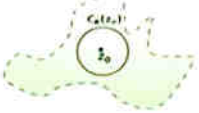
Cauchy's Inequality

Proof: The circle $C_R(z_0)$ has the following parametric representation

$$C_R(z_0): z(t) = z_0 + Re^{it}, 0 \leq t \leq 2\pi$$

Here $f(z)$ is analytic in the simply connected domain D, and $C_R(z_0)$ is a simple closed contour inside D. So for point z_0 lying inside $C_R(z_0)$, by Cauchy's integral formula for derivative, we have

$$\begin{aligned}f^{(n)}(z_0) &= \frac{n!}{2\pi i}\int_{C_R(z_0)} \frac{f(z)\,dz}{(z - z_0)^{n+1}}\\ &= \frac{n!}{2\pi i}\int_0^{2\pi} \frac{f(z_0 + Re^{it})\,(iRe^{it})\,dt}{(z_0 + Re^{it} - z_0)^{n+1}}\end{aligned}$$



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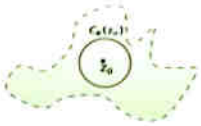
$$f^{(n)}(z_0) = \frac{n!}{2\pi i}\int_{C_R(z_0)} \frac{f(z)\,dz}{(z-z_0)^{n+1}}$$

$$= \frac{n!}{2\pi i}\int_0^{2\pi} \frac{f(z_0 + Re^{it})\ (iR\,e^{it})dt}{(z_0 + Re^{it} - z_0)^{n+1}}$$

$$= \frac{n!}{2\pi i}\int_0^{2\pi} \frac{f(z_0 + Re^{it})\ (iR\,e^{it})dt}{R^{n+1}\,e^{i(n+1)t}}$$



Recall

Theorem (Absolute value inequality) For a piecewise continuous complex-valued function of real variable t

$$f(t) = u(t) + i\,v(t), \qquad a \le t \le b$$

We have

$$\left|\int_a^b f(t)dt\right| \le \int_a^b |f(t)|dt$$

Cauchy's Inequality

Proof (cont.) :

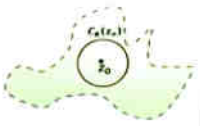
$$|f^{(n)}(z_0)| = \left|\frac{n!}{2\pi i}\int_0^{2\pi} \frac{f(z_0 + Re^{it})\ (iR\,e^{it})dt}{R^{n+1}\,e^{i(n+1)t}}\right|$$

$$\le \frac{n!}{2\pi}\int_0^{2\pi} |f(z_0 + Re^{it})|\left|\frac{(ie^{it})}{R^n\,e^{i(n+1)t}}\right| dt$$

$$\le \frac{n!}{2\pi i}\int_0^{2\pi} M\,\frac{1}{R^n}\,dt$$

$$\le \frac{n!}{2\pi}\frac{M}{R^n}\,(2\pi)$$

$$= \frac{n!\,M}{R^n}$$



Liouville's Theorem

Liouville's Theorem

Joseph Liouville

Liouville's Theorem

Liouville's Theorem

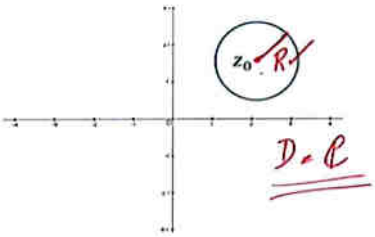
Let f be an entire function and is bounded for all values of z in the complex plane, then f is constant.

Liouville's Theorem

Theorem (Liouville's Theorem)

Let f be an entire function and is bounded for all values of z in the complex plane, then f is constant.

Proof:



Liouville's Theorem

Theorem (Liouville's Theorem)

Let f be an entire function and is bounded for all values of z in the complex plane, then f is constant.

Proof: Since f is bounded so there exist M, such that

$$|f(z)| \le M \ \ for\ all\ z \in \mathbb{C}$$

We let z_0 denote an arbitrary point in the complex plane. Using Cauchy's inequality by using the circle $C_R(z_0)$ and $n = 1$, we get

$$|f'(z_0)| \le \frac{M}{R}$$

Since R can be arbitrarily large so we have

$$|f'(z_0)| = 0 \Rightarrow f'(z_0) = 0$$

We choose z_0 an arbitrary point so we have

$$f'(z) = 0\ \forall\, z \in \mathbb{C}.$$

Liouville's Theorem

Proof (cont.): Since

$$f'(z) = 0\ \forall\, z \in \mathbb{C}.$$

So the function $f(z)$ must be constant.

Liouville's Theorem

Fundamental Theorem of Algebra

If P is a polynomial of degree $n \geq 1$, then P has at least one zero.

Liouville's Theorem

A consequence of Fundamental Theorem of Algebra

et P be a polynomial of degree $n \geq$ 1. Then P can be expressed as the product of linear factors. That is

$$z) = A(z - z_1)(z - z_2) \dots (z - z_n)$$

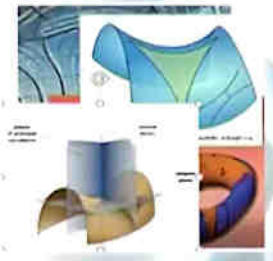
Cauchy's Inequality and Liouville's Thm

- **Cauchy's Inequality**
- **Liouville's Theorem**
- **Fundamental Theorem of algebra**

Introduction to Differential Geometry

Topic No. 108

Differential Geometry



Introduction to Differential Geometry

Geometry

- **What is geometry?**

Introduction to Differential Geometry

Euclid

Euclid of Alexandria
300 BC

Introduction to Differential Geometry

Archimedes

Archimedes of Syracuse
287 – c. 212 BC

Introduction to Differential Geometry

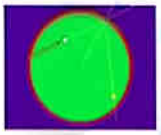
Muslim Era

Muslim Era

Introduction to Differential Geometry

Muslim Era

Hasan Ibn al-Haytham
965 – c. 1040



Introduction to Differential Geometry

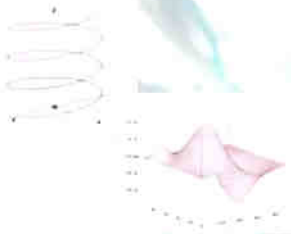
Geometry –Algebra bridge

Pierre de Fermat

René Descartes

Introduction to Differential Geometry

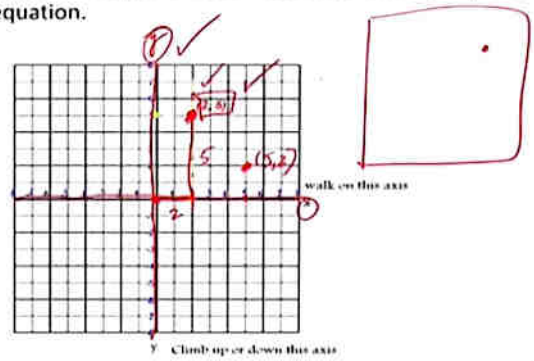
Geometry –Algebra bridge



Geometrical shapes can be represented with algebraic equations

Introduction to Differential Geometry

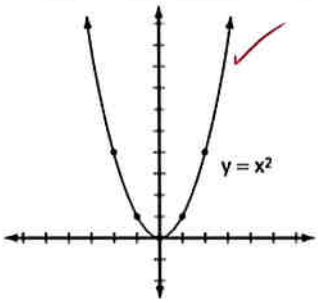
We can associate a point in plane with an ordered pair.
Using the coordinate plane we can associate a curve in plane with an equation.



Introduction to Differential Geometry

Graph of equation $y = x^2$

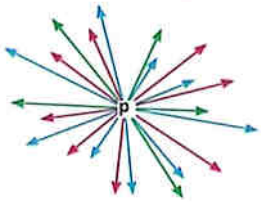
X	0	1	-1	2	-2
Y	0	1	-1	4	4



Tangent Space

Topic No. 109

Tangent space

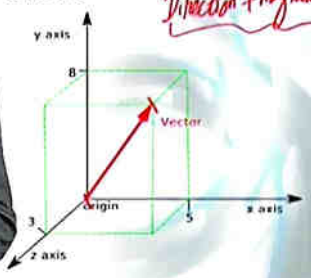


Vectors in $\mathbb{R}^3$

What are vector?

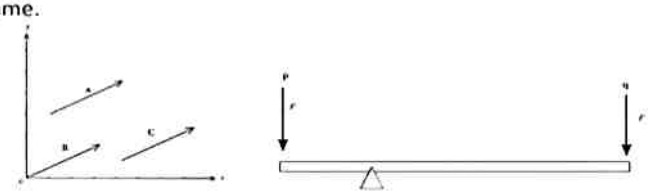
Intuitively a vector in $\mathbb{R}^3$ is an oriented line segment or "arrow".

Direction + magnitude



Vectors in $\mathbb{R}^3$

According to this intuitive description two translated vectors are same.



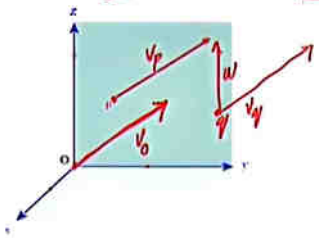
To define a vector v precisely we mention the starting point p of the vector as well.

Definition: A vector in $\mathbb{R}^3$ is $p + v$ where p is initial point of vector v.

Strictly speaking v is a point in $\mathbb{R}^3$.

Tangent Vectors

Definition: A tangent vector v_p to $\mathbb{R}^3$ consists of two points of $\mathbb{R}^3$; its vector part v and its point of application p.



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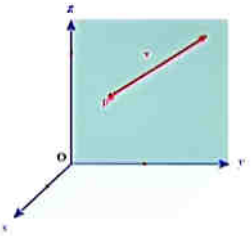


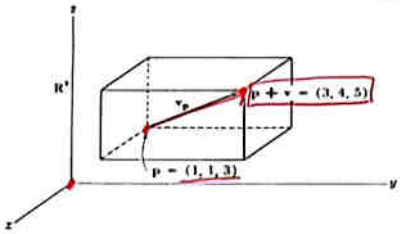
A tangent vector v_p is drawn as an arrow from the point p to the point $p + v$.

Tangent Vectors

Example: If $v = (2,3,2)$, and $p = (1,1,3)$ then the tangent vector

$$v_p = (2,3,2)_{(1,1,3)}$$

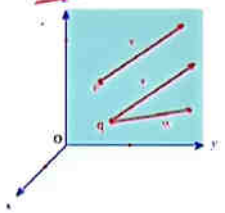
Starts in $(1,1,3)$ and ends in $(3,4,5)$.



Tangent Vectors

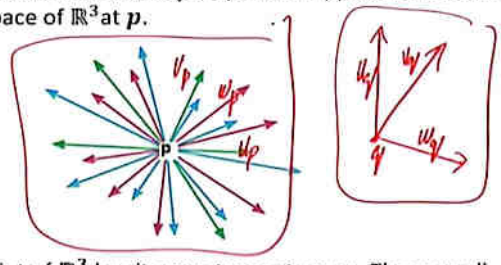
Equality of tangent vectors

- Two tangent vectors v_p and w_q are parallel if and only in $v = w$.
- Two tangent vectors v_p and w_q are equal if and only in $v = w$ and $p = q$.
- v_p and v_q are different tangent vectors if $p \neq q$.



Tangent Space

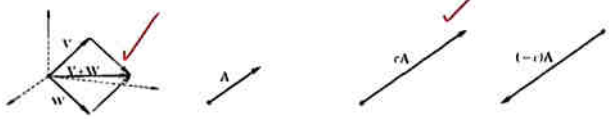
Definition: Let p be a point of $\mathbb{R}^3$. The set $T_p(\mathbb{R}^3)$ consisting of all tangent vectors that have p as point of application is called the tangent space of $\mathbb{R}^3$ at p.



Note: Each point of $\mathbb{R}^3$ has its own tangent space. They are all different from each other.

Tangent Space

- Given two vectors V and W in $\mathbb{R}^3$ we add them by parallelogram law.
- Let c be scalar, then for vector A, cA stretches or shrinks A by the factor c, and reverse the direction if $c < 0$.



For $v_p, w_p \in T_p(\mathbb{R}^3)$ then we can define:

$$v_p + w_p = (v + w)_p$$

And for any scalar $c \in \mathbb{R}^3$ we define

$$cv_p = (cv)_p$$

Tangent Space

Fact: $T_p(\mathbb{R}^3)$ is vector space under the above addition and scalar multiplication.

exercise.

Tangent Space

Recall

- A linear transformation $L: V \to W$ from a vector space V to vector space W is a function that satisfy the following conditions:
  for any vectors $v, w \in V$ and any scalar c

$$L(v + w) = L(v) + L(w)$$
$$L(cv) = cL(v)$$

- Two vector spaces V and W are isomorphic if there exist a bijective linear transformation between them $L: V \to W$. A bijective linear transformation L is also called an isomorphism.

Tangent Space

Lemma: $T_p(\mathbb{R}^3)$ is isomorphic to $\mathbb{R}^3$.

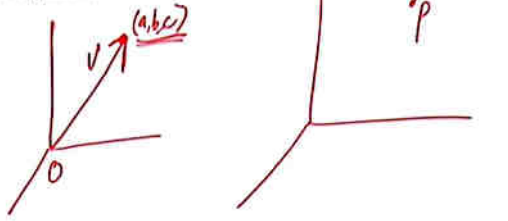
Proof: The following function

$$L: T_p(\mathbb{R}^3) \to \mathbb{R}^3$$

such that for $v_p \in T_p(\mathbb{R}^3)$

$$L(v_p) = v$$

is an isomorphism.



Vector Fields

Topic No. 110

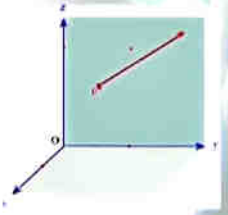
Vector Fields

Vector Fields

Recall

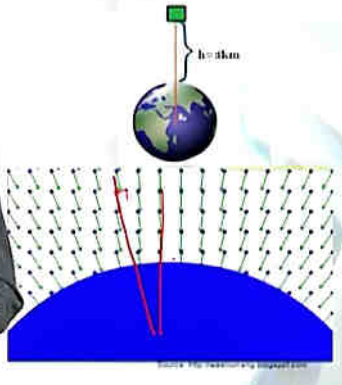
Tangent Vector

A tangent vector v_p to $\mathbb{R}^3$ consists of two points of $\mathbb{R}^3$; its vector part v and its point of application p.



Vector Fields

Gravitational Force



Vector Fields

Vector Field

A vector field V on $\mathbb{R}^3$ is a function that assigns to each point p of $\mathbb{R}^3$ a tangent vector $V(p)$ to $\mathbb{R}^3$ at p.

Vector Fields

Vector Field

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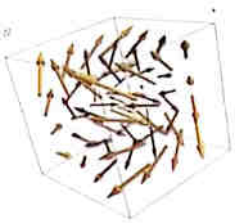


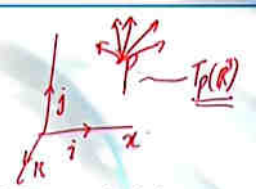
Vector Fields

Addition and Scalar (function) multiplication

Since a vector field is a function so we can add two vector fields just like we add functions.

- If V and W are two vector fields on $\mathbb{R}^3$ then $(V + W)(p) = V(p) + W(p)$
- For any real-valued function $f: \mathbb{R}^3 \to \mathbb{R}$ we define $(fV)(p) = f(p)V(p)$

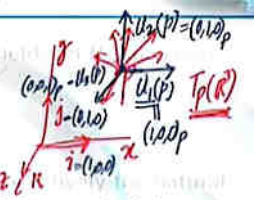
Vector Fields



Natural frame field on $\mathbb{R}^3$

To investigate vector fields on $\mathbb{R}^3$ we introduce a set of basic vector fields on $\mathbb{R}^3$.

Vector Fields



Natural frame field on $\mathbb{R}^3$

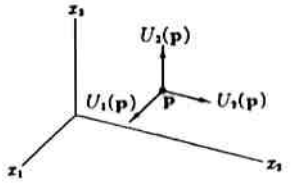
To investigate vector fields on $\mathbb{R}^3$ we introduce a set of basic vector fields on $\mathbb{R}^3$.

Natural Frame Field on $\mathbb{R}^3$

Definition: Let U_1, U_2, U_3 be the vector fields on $\boldsymbol{R^3}$ such that

$$U_1(p) = (1,0,0)_p$$
$$U_2(p) = (0,1,0)_p$$
$$U_3(p) = (0,0,1)_p$$

For each point p of $\mathbb{R}^3$. We call U_1, U_2, U_3 -collectively-the natural frame field on $\mathbb{R}^3$.

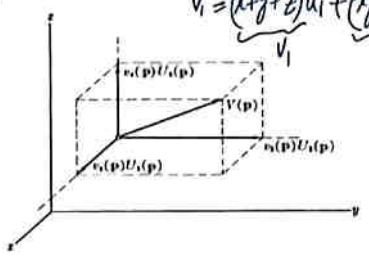


Vector Field

Lemma: If V is a vector field on $\mathbb{R}^3$, there are three uniquely determined real-valued functions, v_1, v_2, v_3 on $\mathbb{R}^3$ such that

$$V = v_1U_1 + v_2U_2 + v_3U_3$$

The functions v_1, v_2, v_3 are called the Euclidean coordinate functions of V.



Vector Field

Proof: Let p be a point of $\mathbb{R}^3$. By definition $V(p)$ is a tangent vector to $\mathbb{R}^3$ *at* p. Thus the vector part of $V(p)$ depends on p, we write

$$V(p) = \big(v_1(p), v_2(p), v_3(p)\big)_p$$

Now

$$V(p) = \big(v_1(p), v_2(p), v_3(p)\big)_p$$
$$V(p) = v_1(p)(1,0,0)_p + v_2(p)(0,1,0)_p + v_3(p)(0,0,1)_p$$

Vector Field

Proof: Let p be a point of $\mathbb{R}^3$. By definition $V(p)$ is a tangent vector to $\mathbb{R}^3$ *at* p. Thus the vector part of $V(p)$ depends on p, we write

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$$V(p) = v_1(p)(1,0,0)_p + v_2(p)(0,1,0)_p + v_3(p)(0,0,1)_p$$
$$V(p) = v_1(p)U_1(p) + v_2(p)U_2(p) + v_3(p)U_3(p)$$
$$V(p) = (v_1U_1)(p) + (v_2U_2)(p) + (v_3U_3)(p)$$

Vector Field

Proof: Let p be a point of $\mathbb{R}^3$. By definition $V(p)$ is a tangent vector to $\mathbb{R}^3$ *at* p. Thus the vector part of $V(p)$ depends on p, we write

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$$V(p) = (v_1U_1)(p) + (v_2U_2)(p) + (v_3U_3)(p)$$
$$V(p) = (v_1U_1 + v_2U_2 + v_3U_3)(p)$$

Hence

$$V = v_1U_1 + v_2U_2 + v_3U_3$$

Vector Field

Lemma: If V is a vector field on $\mathbb{R}^3$, there are three uniquely determined real-valued functions, v_1, v_2, v_3 on $\mathbb{R}^3$ such that

$$V = v_1U_1 + v_2U_2 + v_3U_3$$

The functions v_1, v_2, v_3 are called the Euclidean coordinate functions of V.

The sigma $\boldsymbol{\Sigma}$ notation

The following notation will help us in simplifying things.

$$\sum v_iU_i = v_1U_1 + v_2U_2 + v_3U_3$$

Vector Field

Example: Consider the following vector field

$$V(x,y,z) = xyU_1 + (x+y)U_2 + xyz\,U_3$$

It gives one vector for each point of $\mathbb{R}^3$. In particular for $(1,2,3)=p$ the following vector is assigned

$$V(1,2,3) = \left[(xy)(1,2,3)\right]U_1(1,2,3) + \left[(x+y)(1,2,3)\right]U_2(1,2,3) + \left[(xyz)(1,2,3)\right]U_3(1,2,3)$$

$$= 2(1,0,0)_p + 3(0,1,0)_p + 6(0,0,1)_p$$

$$= (2,0,0)_p + (0,3,0)_p + (0,0,6)_p = \boxed{(2,3,6)_{(1,2,3)}}$$

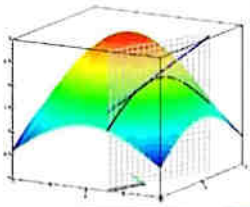
Vector Field

Definition: A vector field V is differentiable if each of its three Euclidean coordinate function v_1, v_2, v_3, is differentiable function (smooth, of class C^∞)

We always assume vector fields are differentiable.

Directional Derivatives

Directional Derivatives
Part I



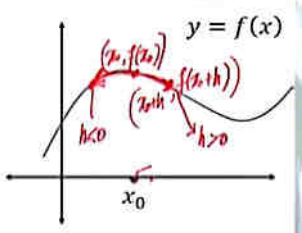
Topic No. 111

Directional Derivatives

Derivative (f: $\mathbb{R} \to \mathbb{R}$)

For $y = f(x)$ at point x_0 is

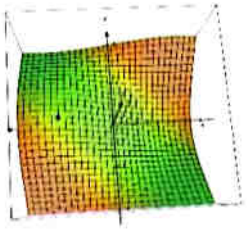
$$f'(x_0) = \lim_{h\to 0} \frac{f(x_0+h) - f(x_0)}{h}$$



Directional Derivatives

Derivatives for

$z = f(x,y): \mathbb{R}^2 \to \mathbb{R}$



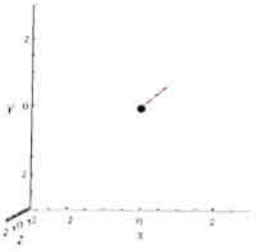
Directional Derivatives

Derivatives for

$f(x,y,z): \mathbb{R}^3 \to \mathbb{R}$

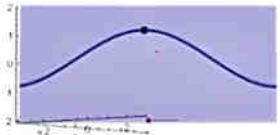
Directional Derivatives

What is directional derivative??



Directional Derivatives

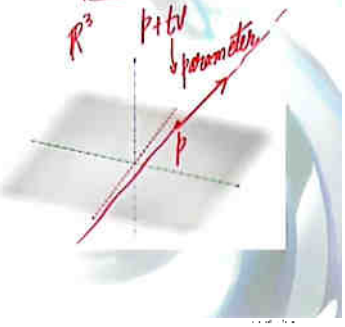
What is directional derivative??



Directional Derivatives

Associated with each tangent vector v_p to $\mathbb{R}^3$ is the straight line

$$t \to p + tv$$

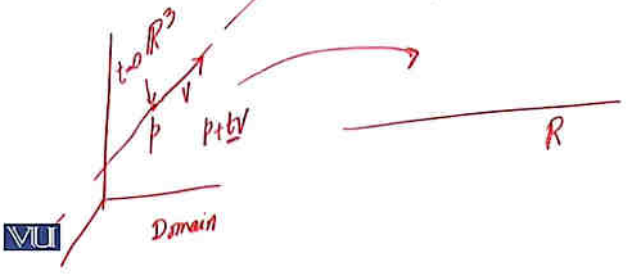


Directional Derivatives

Definition: Let f be a differentiable real-valued function on $\mathbb{R}^3$, and let v_p be a tangent vector to $\mathbb{R}^3$. Then the number

$$v_p[f] = \frac{d}{dt}\big(f(p+tv)\big)\Big|_{t=0}$$

is called the derivative of f with respect to v_p.



Directional Derivatives

Definition: Let f be a differentiable real-valued function on $\mathbb{R}^3$, and let v_p be a tangent vector to $\mathbb{R}^3$. Then the number

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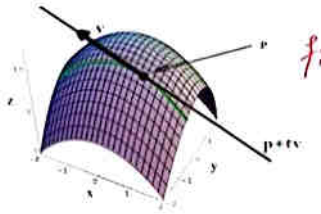


Directional Derivatives

- The number $v_p[f]$ is called the derivative of f with respect to v_p or directional derivative.
- The directional derivative $v_p[f]$ informs us about the change in the value of f as we move away from p in the direction v_p.

Directional Derivatives

Example: Compute $v_p[f]$ for the function $f = x^2yz$, with $p = (1,1,0)$ and $v = (1,0,-3)$.

Solution: First we compute the line defined by p and v

$$p + tv = (1,1,0) + t(1,0,-3) = (1+t, 1, -3t)$$

Now evaluating f along the line.

$$f(p+tv) = f(1+t, 1, -3t) = (1+t)^2(1)(-3t)$$
$$= -3t - 6t^2 - 3t^3$$

Now

$$\frac{d}{dt}\big(f(p+tv)\big) = \frac{d}{dt}(-3t - 6t^2 - 3t^3) = -3 - 12t - 9t^2$$

Now applying definition of $v_p[f]$ we get

$$v_p[f] = \frac{d}{dt}\big(f(p+tv)\big)\Big|_{t=0} = (-3 - 12t - 9t^2)\Big|_{t=0} = -3$$

Directional Derivatives

Lemma: If $v_p = (v_1, v_2, v_3)_p$ is a tangent vector to $\mathbb{R}^3$, then

$$v_p[f] = \sum v_i \frac{\partial f}{\partial x_i}(p)$$

$$= v_1 \frac{\partial f}{\partial x_1}(p) + v_2 \frac{\partial f}{\partial x_2}(p) + v_3 \frac{\partial f}{\partial x_3}(p)$$

Directional Derivatives

The same example as before:

Example: Compute $v_p[f]$ for the function $f = x^2yz$, with $p = (1,1,0)$ and $v = (1,0,-3)$.

Solution: Here $v_p = (v_1, v_2, v_3) = (1,0,-3)$.

$$\frac{\partial f}{\partial x} = 2xyz, \qquad \frac{\partial f}{\partial y} = x^2z, \qquad \frac{\partial f}{\partial z} = x^2y$$

Thus at point $p = (1,1,0)$

$$\frac{\partial f}{\partial x} = 0, \qquad \frac{\partial f}{\partial y} = 0, \qquad \frac{\partial f}{\partial z} = 1$$

Hence

$$v_p[f] = \sum v_i \frac{\partial f}{\partial x_i}(p) = 1(0) + 0(0) - 3(1) = -3$$

Directional Derivatives

Theorem: Let f and g be functions on $\mathbb{R}^3$, v_p and w_p tangent vectors, a and b numbers. Then

1)- $(av_p + bw_p)[f] = av_p[f] + bw_p[f]$

2)- $v_p[af + bg] = av_p[f] + bv_p[g]$.

3)- $v_p[fg] = v_p[f].g(p) + f(p).v_p[g]$

Proof:- Hint

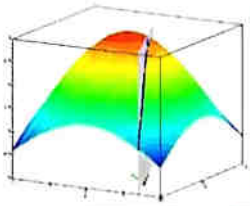
$v_p = (v_1, v_2, v_3)_p$, $w_p = (w_1, w_2, w_3)_p$.

$av_p + bw_p = (av_1 + bw_1, av_2 + bw_2, av_3 + bw_3)_p$.

$v_p[f] = \sum v_i \frac{\partial f}{\partial x_i}(p)$

Directional Derivatives

Directional Derivatives Part II



Topic No. 112

VU

Operation of a Vector Field on $f(x, y, z)$

Operation of a vector field on a function $f(x, y, z)$

VU

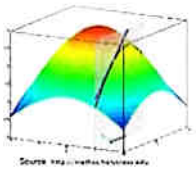
Operation of a Vector Field on $f(x, y, z)$

Recall

Definition: Let f be a differentiable real-valued function on $\mathbb{R}^3$, and let v_p be a tangent vector to $\mathbb{R}^3$. Then the number

$$v_p[f] = \frac{d}{dt}\big(f(p + tv)\big)\Big|_{t=0}$$

is called the derivative of f with respect to v_p.



VU

Operation of a Vector Field on $f(x, y, z)$

To calculate directional derivative at each point for $f(x, y, z)\colon \mathbb{R}^3 \to \mathbb{R}$ we need a vector at each point of $\mathbb{R}^3$. That is we need a vector field.

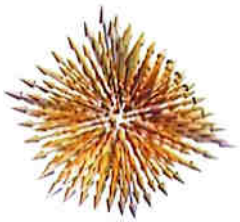
VU

Operation of a Vector Field on $f(x, y, z)$

Definition: Let V be a vector field, and let $f\colon \mathbb{R}^3 \to \mathbb{R}$ be a function. Then $V[f]$ be the function defined by

$$V[f]\colon \mathbb{R}^3 \to \mathbb{R},\ V[f](p) = V(p)[f]$$

So $V[f]$ is a function which to each point p in $\mathbb{R}^3$ gives us the derivative of f with respect to the tangent vector $V(p)$.



VU

Operation of a Vector Field on $f(x, y, z)$

Example: For the natural fame field U_1, U_2, U_3, we get

$$U_1[f](p) = U_1(p)[f] = (1,0,0)_p[f]$$

$U_1(p) = (1,0,0)_p$
$U_2(p) = (0,1,0)_p$
$U_3(p) = (0,0,1)_p$

$V[f](p) = V(p)[f]$

VU

Operation of a Vector Field on $f(x, y, z)$

Example: For the natural fame field U_1, U_2, U_3, we get

$$U_1[f](p) = U_1(p)[f] = (1,0,0)_p[f]$$

Using the following result

Lemma: If $v_p = (v_1, v_2, v_3)$ is a tangent vector to $\boldsymbol{R}^3$, then

$$v_p[f] = \sum_{i=1}^{3} v_i \frac{\partial f}{\partial x_i}(p)$$

we get

$$U_1[f](p) = U_1(p)[f] = (1,0,0)_p[f]$$
$$= 1\frac{\partial f}{\partial x}(p) + 0\frac{\partial f}{\partial y}(p) + 0\frac{\partial f}{\partial z}(p) = \frac{\partial f}{\partial x}(p)$$

VU

Operation of a Vector Field on $f(x, y, z)$

Corollary: Let V and W be vector fields on $\mathbb{R}^3$ and f, g and h are real-valued function, then

(1)- $(fV + gW)[h] = fV[h] + gW[h]$
(2)- $V[af + bg] = aV[f] + bV[g]$
(3)- $V[fg] = V[f].g + fV[g]$

$g, f : \mathbb{R}^3 \to \mathbb{R}$
$W, V : \mathbb{R}^3 \to$ tangent vector $\to$ vector field

VU

Operation of a Vector Field on $f(x,y,z)$

Corollary: Let V and W be vector fields on $\mathbb{R}^3$ and f, g and h are real-valued function, then

(1)- $(fV+gW)[h] = fV[h] + gW[h]$ ← equality of fuction

(2)- $V[af+bg] = aV[f] + bV[g]$

(3)- $V[fg] = V[f].g + fV[g]$

Proof (Hints): use the following properties.

1)- $(av_p + bw_p)[f] = av_p[f] + bw_p[f]$

2)- $v_p[af+bg] = av_p[f] + bv_p[g]$.

3)- $v_p[fg] = v_p[f].g(p) + f(p).v_p[g]$

$(fv+gw)[h](p) = (fv+gw)(p)[h]$

VU

Operation of a Vector Field on $f(x,y,z)$

Example: Let V be a vector field $V = x^2U_1 + yU_3$ and let f be the function $f = x^2y - y^2z$. Calculate $V[f]$.

Solution: We need to calculate the following:

$$V[f] = (x^2U_1 + yU_3)[x^2y - y^2z]$$

$V[f]: \mathbb{R}^3 \to \mathbb{R}$

Using the following two properties

1)- $(fV+gW)[h] = fV[h] + gW[h]$

2)- $V[af+bg] = aV[f] + bV[g]$

we get

$$\begin{aligned} V[f] &= (x^2U_1 + yU_3)[x^2y - y^2z] \\ &= (x^2U_1 + yU_3)[x^2y] - (x^2U_1 + yU_3)[y^2z] \\ &= x^2U_1[x^2y] + yU_3[x^2y] - x^2U_1[y^2z] - yU_3[y^2z] \\ &= x^2(2xy) + y.0 - x^2.0 - y(y^2) \\ &= 2x^3y - y^3 \end{aligned}$$

VU

Operation of a Vector Field on $f(x,y,z)$

Operation of a vector field on a function $f(x,y,z)$

VU

Curves in $\mathbb{R}^3$

Curves Part I

Topic No. 113

VU

Curves in $\mathbb{R}^3$

What is a curve?

A path of a moving object.

VU

Curves in $\mathbb{R}^3$

What is a curve?

Let for time t the bird is located at

$\alpha(t) = (\alpha_1(t), \alpha_2(t), \alpha_3(t))$, for $t \in (0,8)$.

VU

Curves in $\mathbb{R}^3$

What is a curve?

Hence path of this bird is a function

$\alpha: (0,8) \to \mathbb{R}^3$.

given by

$\alpha(t) = (\alpha_1(t), \alpha_2(t), \alpha_2(t))$

for $t \in (0,8)$.

VU

Curves in $\mathbb{R}^3$

A path in $\mathbb{R}^3$ is a function

$\alpha: I \to \mathbb{R}^3$

given by

$\alpha(t) = (\alpha_1(t), \alpha_2(t), \alpha_3(t))$, here $t \in I$ where $I \subset \mathbb{R}$ is an open interval.

The real-valued functions $\alpha_1(t)$, $\alpha_2(t)$, and $\alpha_3(t)$ are called Euclidean coordinate functions.

VU

Curves in $\mathbb{R}^3$

Differentiability of α

Function

$$\alpha: I \to \mathbb{R}^3$$

given by

$$\alpha(t) = (\alpha_1(t), \alpha_2(t), \alpha_3(t)),$$

is said to be **differentiable** if its coordinate functions $\alpha_1(t)$, $\alpha_2(t)$, and $\alpha_3(t)$ are differentiable.

Curves in $\mathbb{R}^3$

Curve

A curve in $\mathbb{R}^3$ is a differentiable function

$$\alpha: I \to \mathbb{R}^3$$

from an open interval $I \subset \mathbb{R}$ to $\mathbb{R}^3$.

Curves in $\mathbb{R}^3$

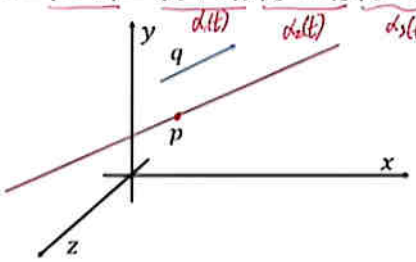
Example 1: Straight line

A line through point p in the direction of q is a curve

$$\alpha: \mathbb{R} \to \mathbb{R}^3$$

defined as

$$\alpha(t) = p + tq = (p_1 + tq_1, p_2 + tq_2, p_3 + tq_3).$$



Curves in $\mathbb{R}^3$

Example 3: Circle (clockwise)

The curve $\alpha(t) = (a\cos t, a\sin t, 0)$ travels around a circle of radius $a > 0$ in the xy-plane of $\mathbb{R}^3$.

$t \in [0, 2\pi]$

Circle of radius 2

Curves in $\mathbb{R}^3$

Example 3: Circle (clockwise)

The curve $\alpha(t) = (a\cos t, a\sin t, 0)$ travels around a circle of radius $a > 0$ in the xy-plane of $\mathbb{R}^3$.

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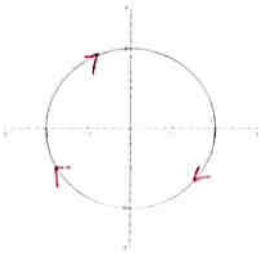


Circle of radius 2

Curves in $\mathbb{R}^3$

Example: Helix

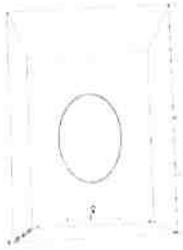
The function

$$\alpha: \mathbb{R} \to \mathbb{R}^3$$

defined as

$$\alpha(t) = (a\cos t, a\sin t, bt)$$

where $a > 0, b \neq 0$, geometrically represents a helix.



Curves in $\mathbb{R}^3$

Example: Helix

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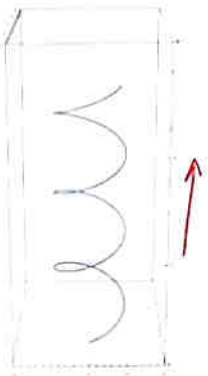
x, y



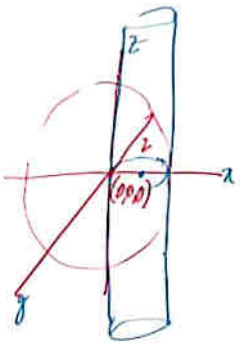
Curves in $\mathbb{R}^3$

Example: Viviani's curve

The curve $\alpha: \mathbb{R} \to \mathbb{R}^3$ defined by

$$\alpha(t) = \left(1 + \cos t, \sin t, 2\sin\frac{t}{2}\right)$$

is obtained by intersection of sphere of radius 2 and center (0,0,0) with cylinder over a circle in xy-plane of radius 1 and center (1,0,0).



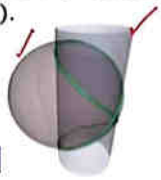
Curves in $\mathbb{R}^3$

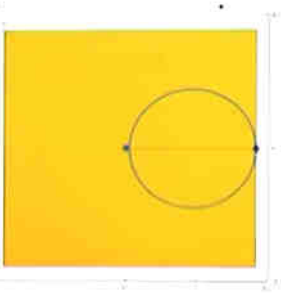
Example: Viviani's curve

The curve $\alpha: \mathbb{R} \to \mathbb{R}^3$ defined by

$$\alpha(t) = \left(1 + \cos t, \sin t, 2\sin\frac{t}{2}\right)$$

is obtained by intersection of sphere of radius 2 and center (0,0,0) with cylinder over a circle in xy-plane of radius 1 and center (1,0,0).



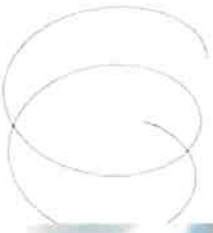


VU

Curves in $\mathbb{R}^3$

Curves
Part II

Topic No. 114



VU

Curves in $\mathbb{R}^3$

Curve

A curve in $\mathbb{R}^3$ is a differentiable function

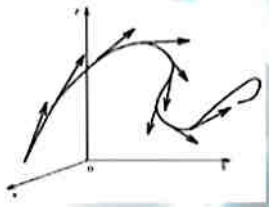
$$\alpha: I \to \mathbb{R}^3$$

from an open interval $I \subset \mathbb{R}$ to $\mathbb{R}^3$.

VU

Velocity Vector

Velocity vector



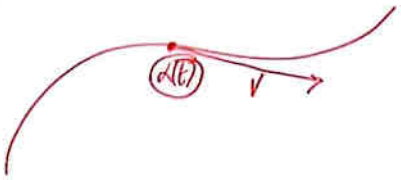
VU

Velocity Vector

Definition (Velocity Vector): Let $\alpha: I \to \mathbb{R}^3$ be a curve in $\mathbb{R}^3$ with $\alpha = (\alpha_1, \alpha_2, \alpha_3)$. For each number t in I, the velocity vector of α at t is the tangent vector

$$\alpha'(t) = \left(\frac{d\alpha_1}{dt}(t), \frac{d\alpha_2}{dt}(t), \frac{d\alpha_3}{dt}(t)\right)_{\alpha(t)}$$

at the point $\alpha(t)$ in $\mathbb{R}^3$.



VU

Velocity Vector

Example: Compute the velocity vector of the Viviani's curve for arbitrary t and for $t = 0$.

Solution: Viviani's curve $\alpha: \mathbb{R} \to \mathbb{R}^3$ is given as

$$\alpha(t) = \left(1 + \cos t, \sin t, 2\sin\frac{t}{2}\right).$$

Now by definition

$$\alpha'(t) = \left(\frac{d\alpha_1}{dt}(t), \frac{d\alpha_2}{dt}(t), \frac{d\alpha_3}{dt}(t)\right)_{\alpha(t)}$$

which in this case is

$$\alpha'(t) = \left(-\sin t, \cos t, \cos\frac{t}{2}\right)_{\alpha(t)}$$

At $t = 0$: $\alpha(0) = (2,0,0)$, and hence

$$\alpha'(0) = (0,1,1)_{(2,0,0)}$$

VU

Geometrical Interpretation

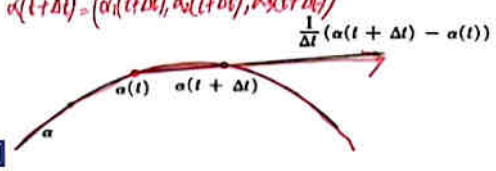
To see velocity geometrical, recall that derivative of a real-valued function on R is given by

$$\frac{df}{dt} = lim_{\Delta t \to 0} \frac{f(t + \Delta t) - f(t)}{\Delta t}$$

The same formula works for curves $\alpha: I \to \mathbb{R}^3$, that is, we have

$$\frac{d\alpha}{dt} = lim_{\Delta t \to 0} \frac{\alpha(t + \Delta t) - \alpha(t)}{\Delta t}$$

$\alpha(t+\Delta t) = (\alpha_1(t+\Delta t), \alpha_2(t+\Delta t), \alpha_3(t+\Delta t))$

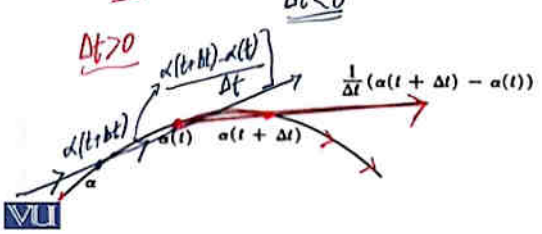


VU

Geometrical Interpretation

$$\frac{\alpha(t + \Delta t) - \alpha(t)}{\Delta t} = \left(\frac{\alpha_1(t + \Delta t) - \alpha_1(t)}{\Delta t}, \frac{\alpha_2(t + \Delta t) - \alpha_2(t)}{\Delta t}, \frac{\alpha_3(t + \Delta t) - \alpha_3(t)}{\Delta t}\right)$$

which is in fact a vector from $\alpha(t)$ to $\alpha(t + \Delta t)$, scalar multiple by $\frac{1}{\Delta t}$.

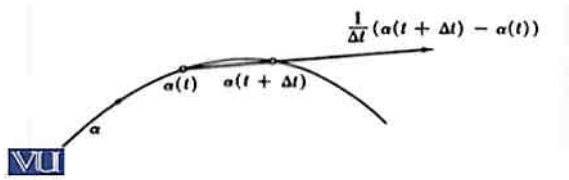


VU

Geometrical Interpretation

$$\frac{\alpha(t+\Delta t)-\alpha(t)}{\Delta t} = \left(\frac{\alpha_1(t+\Delta t)-\alpha_1(t)}{\Delta t}, \frac{\alpha_2(t+\Delta t)-\alpha_2(t)}{\Delta t}, \frac{\alpha_3(t+\Delta t)-\alpha_3(t)}{\Delta t}\right)$$

which is in fact a vector from $\alpha(t)$ to $\alpha(t+\Delta t)$, scalar multiple by $\frac{1}{\Delta t}$. It is important to note that the above vector is always in the direction of increasing parameter.



Geometrical Interpretation

$$\frac{\alpha(t+\Delta t)-\alpha(t)}{\Delta t} = \left(\frac{\alpha_1(t+\Delta t)-\alpha_1(t)}{\Delta t}, \frac{\alpha_2(t+\Delta t)-\alpha_2(t)}{\Delta t}, \frac{\alpha_3(t+\Delta t)-\alpha_3(t)}{\Delta t}\right)$$

which is in fact a vector from $\alpha(t)$ to $\alpha(t+\Delta t)$, scalar multiple by $\frac{1}{\Delta t}$. It is important to note that the above vector is always in the direction of increasing parameter.

Now as we take the limit $\Delta t \to 0$ we get the a tangent vector to the curve α at point $\alpha(t)$ given by

$$\alpha'(t) = \left(\frac{d\alpha_1}{dt}(t), \frac{d\alpha_2}{dt}(t), \frac{d\alpha_3}{dt}(t)\right)_{\alpha(t)}.$$

Velocity Vector

Velocity vector in a compact form

For any tangent vector v_p of R^3 at point p we can write

$$(v_1, v_2, v_3)_p = \sum v_i U_i(p)$$

$$= v_1(1,0,0)_p + v_2(0,1,0)_p + v_3(0,0,1)_p$$

$$= v_1 U_1(p) + v_2 U_2(p) + v_3 U_3(p)$$

$$= \sum_{i=1}^{3} v_i U_i(p)$$

Velocity Vector

Velocity vector in a compact form

For any tangent vector v_p of R^3 at point p we can write

$$(v_1, v_2, v_3)_p = \sum v_i U_i(p)$$

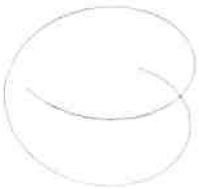
So the velocity vector to $\alpha(t) = (\alpha_1(t), \alpha_2(t), \alpha_3(t))$ can be written as

$$\alpha'(t) = \sum_{i=1}^{3} \frac{d\alpha_i}{dt} U_i(\alpha(t))$$

Curves in $\mathbb{R}^3$

Curves Part III

Topic No. 115



Reparametrization of Curves in $\mathbb{R}^3$

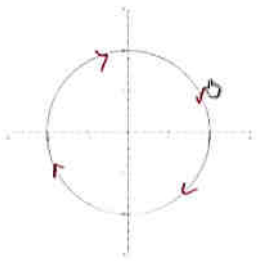
Reparametrization of curves in $\mathbb{R}^3$

Curves in $\mathbb{R}^3$

Example: Circle (clockwise)

The curve $\alpha(t) = (a\cos t, a\sin t, 0)$ travels around a circle of radius $a > 0$ in the xy-plane of $\mathbb{R}^3$.

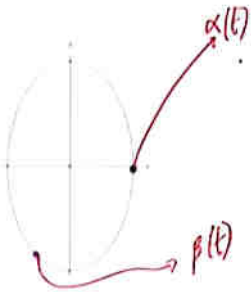
$t \in [0, 2\pi]$



Circle of radius 2

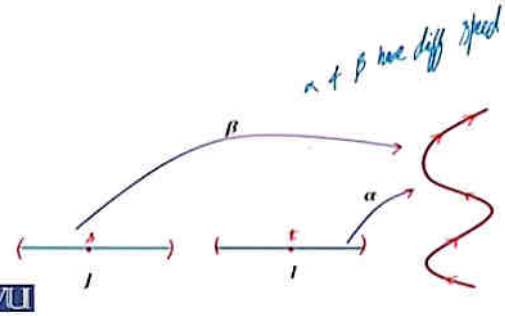
Reparametrization of Curves in $\mathbb{R}^3$

What is reparameterization



Reparametrization of Curves in $\mathbb{R}^3$

What is reparameterization

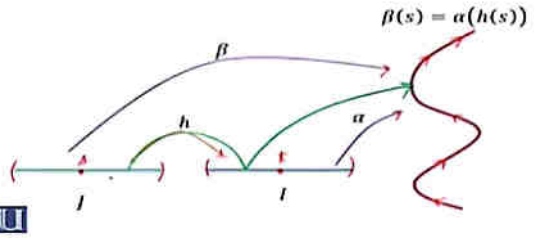


Reparametrization of Curves in $\mathbb{R}^3$

Definition: Let $\alpha: I \to \mathbb{R}^3$ be a curve. If $h: J \to I$ is a differentiable function on an open interval J, then the composite function

$$\beta = \alpha(h): J \to \mathbb{R}^3$$

is a curve called a reparameterization of α by h.



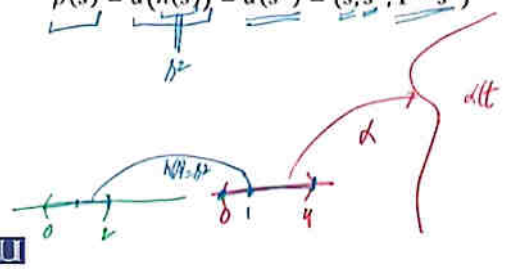
Reparametrization of Curves in $\mathbb{R}^3$

Example: Let the curve α is defined as

$$\alpha(t) = \left(\sqrt{t}, t\sqrt{t}, 1-t\right) \text{ on } I = (0,4)$$

If $h: J \to I$ be the function given as $h(s) = s^2$, where $J = (0,2)$, then the reparameterization is

$$\beta(s) = \alpha(h(s)) = \alpha(s^2) = (s, s^3, 1-s^2)$$



Reparametrization of Curves in $\mathbb{R}^3$

Lemma: If β is the reparameterization of α by h, then

$$\beta'(s) = \frac{dh}{ds}(s)\alpha'(h(s)).$$

Proof: If $\alpha = (\alpha_1, \alpha_2, \alpha_3)$, then

$$\beta(s) = \alpha(h(s)) = \left(\alpha_1(h(s)), \alpha_2(h(s)), \alpha_3(h(s))\right).$$

Now

$$\beta'(s) = \alpha'(h(s)) = \left(\alpha_1(h(s))', \alpha_2(h(s))', \alpha_3(h(s))'\right).$$

By Chain rule

$$\alpha_i(h(s))' = \alpha_i'(h(s))\, h'(s) \text{ for } i = 1,2,3.$$

Reparametrization of Curves in $\mathbb{R}^3$

Proof (cont.): Which yields

$$\begin{aligned}\beta'(s) &= \alpha'(h(s)) \\ &= \left(\alpha_1'(h(s))h'(s), \alpha_2'(h(s))h'(s), \alpha_3'(h(s))h'(s)\right) \\ &= h'(s)\left(\alpha_1'(h(s)), \alpha_2'(h(s)), \alpha_3'(h(s))\right) \\ &= h'(s)\alpha'(h(s))\end{aligned}$$

$$= h'(s)\frac{d}{ds}\alpha(h(s))$$

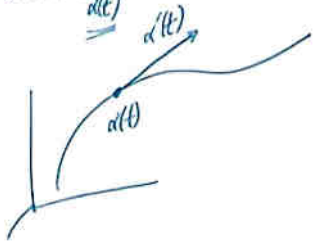
$$= h'(s)\,\alpha(h(s))'$$

Derivative with respect to Velocity

Lemma: Let α be a curve in $\mathbb{R}^3$ and let f be a differentiable function on $\mathbb{R}^3$. Then

$$\alpha'(t)[f] = \frac{d(f(\alpha))}{dt}(t).$$

$\alpha'(t) = (v_1, v_2, v_3)$ $d(t)$



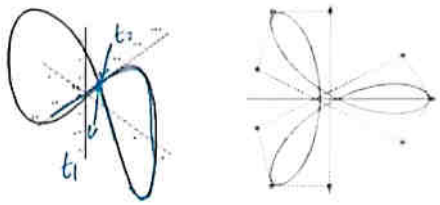
Properties of Curves

One-to-one curves

The function $\alpha: I \to \mathbb{R}^3$ is one-to one if

$$\alpha(t_1) \neq \alpha(t_2) \text{ for } t_1 \neq t_2$$

If α is one-to-one then the curve does not intersect itself, or repeat itself over after some time.



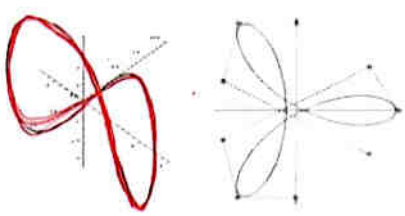
Properties of Curves

Periodic curves

A curve $\alpha: \mathbb{R} \to \mathbb{R}^3$ is called periodic if there is a positive number p such that

$$\alpha(t+p) = \alpha(t) \quad \text{for all} \quad t \in \mathbb{R}$$

The smallest such number p is the period of α.



Properties of Curves

Regular curves: A regular curve is a function $\alpha: I \to \mathbb{R}^3$ which if of class C^k for some $k \geq 1$ and

$$\alpha'(t) \neq (0,0,0)_{\alpha(t)} \text{ for all } t \in I. \quad (1)$$

In future we will consider regular curves to be of class C^3 unless otherwise stated.

Properties of Curves

Example: The curve $\alpha: \mathbb{R} \to \mathbb{R}^3$ given as

$$\alpha(t) = (t^2, t^3, 0)$$

is not regular.

(1) $\alpha(t) \in C^3$ ✓

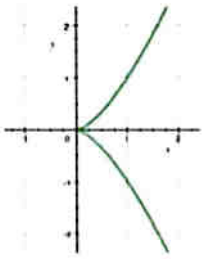
(2) $\alpha'(t) = (2t, 3t^2, 0)$

$(2t, 3t^2, 0) = (0,0,0)$

$2t = 0,\ 3t^2 = 0,\ 0 = 0$

$t = 0,\ t = 0$

$\alpha'(0) = 0$



A curve with a *cusp*

Curves in $\mathbb{R}^3$

Topic No. 116

Geometrical Properties of curves

Curves in $\mathbb{R}^3$

- **Speed of a curve α**
- **Arc-length of a curve α**

Speed of a Curve α in $\mathbb{R}^3$

Question

Consider an object moving in space with the following parametrization

$$\alpha(t) = \left(2t, t^2, \frac{t^3}{3}\right), t \in (-2,2)$$

1. What is speed of $\alpha(t)$ at any time instance?
2. What is the distance travelled by $\alpha(t)$ from $t = -1$ to $t = 1$

Speed of a Curve α in $\mathbb{R}^3$

Speed of a curve α

Speed of a Curve α in $\mathbb{R}^3$

Speed of a curve

Let $\alpha = (\alpha_1, \alpha_2, \alpha_3): I \to \mathbb{R}^3$ be a curve. The speed of α is defined as the length of the velocity of α

$\alpha'(t)$

$\alpha(t)$

Speed of a Curve α in $\mathbb{R}^3$

Speed of a curve

Let $\alpha = (\alpha_1, \alpha_2, \alpha_3): I \to \mathbb{R}^3$ be a curve. The speed of α is defined as the length of the velocity of α, given by

$$v = \|\alpha'\| = \sqrt{\left(\frac{d\alpha_1}{dt}\right)^2 + \left(\frac{d\alpha_2}{dt}\right)^2 + \left(\frac{d\alpha_3}{dt}\right)^2}$$

Speed of a Curve α in $\mathbb{R}^3$

Example: Consider an object moving in space with the following parametrization

$$\alpha(t) = \left(2t, t^2, \frac{t^3}{3}\right), t \in (-2,2)$$

What is speed of $\alpha(t)$ at any time instance?

$\|\alpha'(t)\|$

$\alpha'(t) = \left(2, 2t, \frac{3t^2}{3}\right) = (2, 2t, t^2)$

$\|\alpha'(t)\| = \sqrt{(2)^2 + (2t)^2 + (t^2)^2} = \sqrt{4 + 4t^2 + t^4} = \sqrt{(2+t^2)^2} = 2 + t^2$

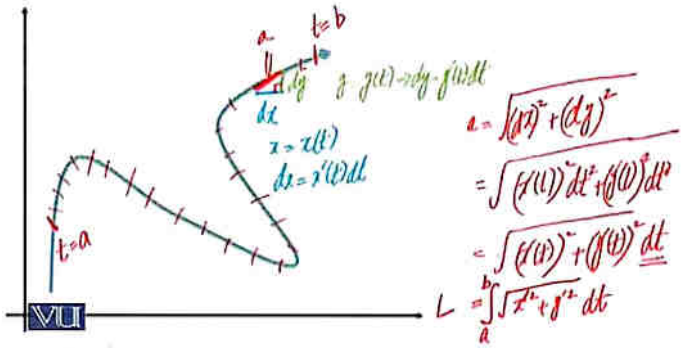
At $t = 0$, $\|\alpha'(0)\| = 2$

Arc-length of a Curve α

Arc-length of a curve α

Arc-length of a Curve α

Question: Find arc-length of curve $\alpha(t) = (x(t), y(t), 0)$ from $t = a$ to $t = b$.



$L = \sqrt{(dx)^2 + (dy)^2}$

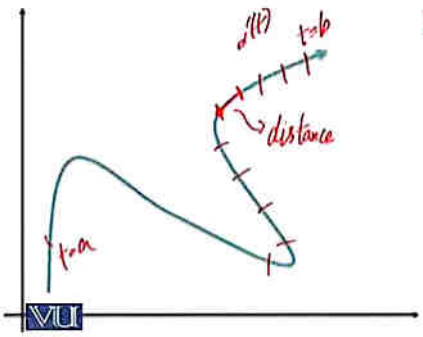
$= \sqrt{(x'(t))^2 dt^2 + (y'(t))^2 dt^2}$

$= \sqrt{(x'(t))^2 + (y'(t))^2}\, dt$

$L = \int_a^b \sqrt{x'^2 + y'^2}\, dt$

Arc-length of a Curve α

Question: Find arc-length of curve $\alpha(t) = (x(t), y(t), 0)$ from $t = a$ to $t = b$.



$\int_a^b \sqrt{x'(t)^2 + y'(t)^2}\, dt$

$\int_a^b \|\alpha'(t)\|\, dt$

distance travelled in time dt

Arc-length of a Curve α

Definition: The arc length of α from $t = a$ to $t = b$ is equal to

$$\int_a^b \|\alpha'\| dt = \int_a^b v(t) dt$$

Speed of a Curve α in $\mathbb{R}^3$

Example: Consider an object moving in space with the following parametrization

$$\alpha(t) = \left(2t, t^2, \frac{t^3}{3}\right), t \in (-2,2)$$

Find arc-length of α from $t = -1$ to $t = 1$.

Solution:

$\|\alpha'(t)\| = 2 + t^2$

$\int_{-1}^{1} \|\alpha'(t)\|\, dt = \int_{-1}^{1} (2+t^2) dt = \left[2t + \frac{t^3}{3}\right]_{-1}^{1}$

$= 2(1-(-1)) + \frac{1}{3}(1-(-1))$

$= 4 + \frac{2}{3} = \frac{14}{3}$ Ans

Speed Of A Curve

Definition: The restriction of a curve $\alpha: I \to \mathbb{R}^3$ to a closed interval $[a, b]$ is called a curve segment. If σ is a curve segment of α,

$$\sigma: [a, b] \to \mathbb{R}^3$$

then the length of σ is written as $L(\sigma)$.

$\sigma = \alpha|_{[a,b]}$

Curves in $\mathbb{R}^3$

Topic No. 117

Arc-length reparameterization

Arc-length Reparameterization

Unit speed curve

A curve β has unit speed if its speed is 1 at every point, that is,

$$\|\beta'\| = 1.$$

Arc-length Reparameterization

Definition: A curve β has unit speed if its speed is 1 at every point, that is,

$$\|\beta'\| = 1.$$

Example: Let $\beta: I \to \mathbb{R}^3$ be defined by

$$\beta(t) = (\cos t, \sin t, 10)$$

$\beta'(t) = (-\sin t, \cos t, 0)$

$\|\beta'(t)\| = \sqrt{\sin^2 t + \cos^2 t} = \sqrt{1} = 1$ ✓

Arc-length Reparameterization

Question

Given a curve α of any speed. Can we reparametrize α to β so that β is unit speed curve.

Answer: Yes, we have Arc-Length reparameterization

Arc-length Reparameterization

Theorem: Let $\alpha: I \to \mathbb{R}^3$ be a regular curve. Then there exist a reparameterization β of α such that

$$\|\beta'\| = 1.$$

$\alpha'(t) \neq (0,0,0), \forall t \in I$

$\alpha(t) \in C^3$

To find $\beta = \alpha(h(s))$

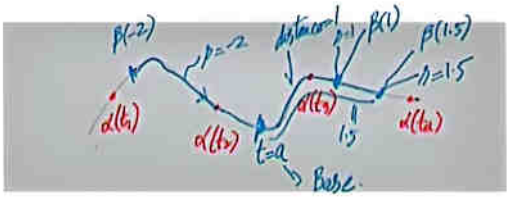
Arc-length Reparameterization

Theorem: Let $\alpha: I \to \boldsymbol{R}^3$ be a regular curve. Then there exist a reparameterization β of α such that

$$\|\beta'\| = 1.$$

Proof: Idea of the proof:

$\beta(h)$



Arc-length Reparameterization

Theorem: Let $\alpha: I \to \mathbb{R}^3$ be a regular curve. Then there exist a reparameterization β of α such that

$$\|\beta'\| = 1.$$

Proof: Consider the arc-length function

$$s(t) = \int_a^t \|\alpha'(u)\|\, du$$

By fundamental theorem of calculus $\frac{ds}{dt} = \|\alpha'\|$ (speed of α).

Since α is regular so $\alpha'(t) \neq 0, \forall\, t$, hence

$$\frac{ds}{dt} = \|\alpha'\| > 0$$

Which implies that $s(t)$ is strictly increasing function and hence a one-to-one function. So inverse of $s(t)$ exists, that is, $t = t(s)$.

Arc-length Reparameterization

Proof (cont.) : Consider a reparameterization of α, denoted by β, defined below

$$\beta(s) = \alpha(t(s))$$

Now

$$\|\beta'\| = \left\|\alpha(t(s))'\right\| = \left\|\alpha'(t(s))\frac{dt}{ds}(s)\right\|$$

$$= \|\alpha'(t(s))\| \left\|\frac{dt}{ds}(s)\right\| = \frac{ds}{dt}\frac{dt}{ds} = 1$$ ✓

Recall: Lemma: If β is the reparameterization of α by h, then $\beta'(s) = \frac{dh}{ds}(s)\alpha'(h(s))$.

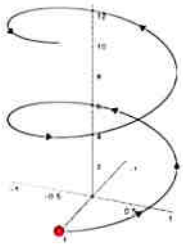
Arc-length Reparameterization

Example: Consider a helix $\alpha: \mathbb{R} \to \mathbb{R}^3$ defined by

$$\alpha(t) = (a\cos t, a\sin t, bt)$$

Find arc-length reparameterization of α.

$s(t) = \int_0^t \|\alpha'(u)\|\, du$



Arc-length Reparameterization

Example: Consider a helix $\alpha: \mathbb{R} \to \mathbb{R}^3$ defined by

$$\alpha(t) = (a\cos t\,, a\sin t, bt)$$

Find arc-length reparameterization of α.

Solution: We start from arc-length function

$$s(t) = \int_0^t \|\alpha'(u)\|\, du$$

Here

$$\alpha'(u) = (-a\sin u\,, a\cos u, b)$$

Hence

$$\|\alpha'(u)\| = \sqrt{a^2\sin^2 u + a^2\cos^2 u + b^2} = \sqrt{a^2+b^2}$$

Arc-length Reparameterization

Solution (cont.): We start from arc-length function

$$s(t) = \int_0^t \sqrt{a^2+b^2}\, du$$

$$s(t) = \sqrt{a^2+b^2}\int_0^t du$$

$$s(t) = \sqrt{a^2+b^2}\, t$$

Inverse of $s(t)$, that is, $t(s)$ is given by

$$t = \frac{s}{\sqrt{a^2+b^2}}$$

Hence arc-length reparameterization of α is given by

$$\beta(s) = \alpha(t(s))$$

$$= \left(a\cos\left(\frac{s}{\sqrt{a^2+b^2}}\right), a\sin\left(\frac{s}{\sqrt{a^2+b^2}}\right), b\left(\frac{s}{\sqrt{a^2+b^2}}\right)\right)$$

Arc-length Reparameterization

Orientation of a reparameterization

A reparameterization of a curve α given by

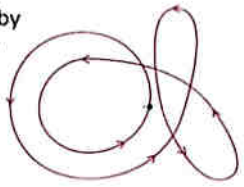
$$\beta(t) = \alpha(h(s))$$

Is orientation preserving if

$$h'(s) \geq 0$$

It is orientation reversing if

$$h'(s) \leq 0$$

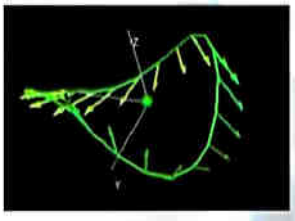


FACT: Unit speed reparameterization is always orientation preserving.

Vector Fields on Curves

Vector fields on curves

Topic No. 118

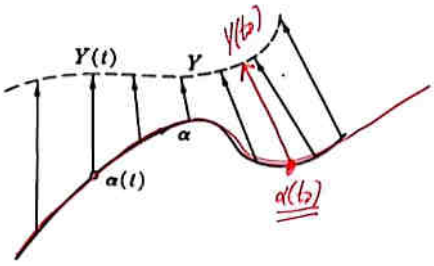


Vector Fields on Curves

Let $\alpha: I \to \mathbb{R}^3$ be a curve. We define

Definition: A vector field Y on curve α is a function that assigns to each number t in I a tangent vector $Y(t)$ to $\mathbb{R}^3$ at the point $\alpha(t)$.

$t_0 \in I$



Vector Fields on Curves

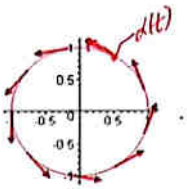
Example: Consider $\alpha: [0, 2\pi] \to \mathbb{R}^3$ defined by

$$\alpha(t) = (\cos t\,, \sin t\,, 0)\,.$$

The tangent vector field for α is given by

$$Y(t) = (-\sin t\,, \cos t\,, 0)_{\alpha(t)}$$

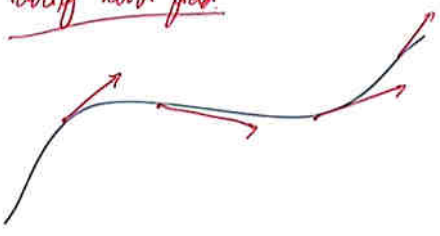
$$= (-\sin t\,, \cos t\,, 0)_{(\cos t, \sin t, 0)}$$



Vector Fields on Curves

Example: For curve α the velocity α' satisfies the definition of vector field on a curve.

Velocity vector field.



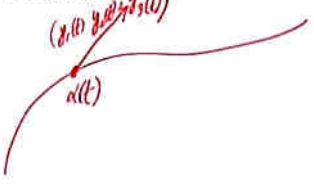
Vector Fields on Curves

Properties of vector fields on curves

Definition: If Y is a vector field on $\alpha: I \to \mathbb{R}^3$, then for each t in I we can write

$$Y(t) = (y_1(t), y_2(t), y_3(t))_{\alpha(t)} = \sum_{i=1}^{3} y_i(t)\, U_i(\alpha(t))$$

The real-valued function $y_1, y_2 y_3: \mathbb{R}^3 \to \mathbb{R}$ are called Euclidean coordinate functions.



Vector Fields on Curves

Addition of vector fields

Let

$$Y(t) = y_1U_1 + y_2U_2 + y_3U_3$$

and

$$Z(t) = z_1U_1 + z_2U_2 + z_3U_3$$

be two vector valued functions on $\alpha: I \to \mathbb{R}^3$. Then the addition of $Y(t)$ and $Z(t)$ is defined as

$$(Y + Z)(t) = (y_1 + z_1)U_1 + (y_2 + z_2)\,U_2 + (y_3 + z_3)\,U_3$$

Vector Fields on Curves

Addition of vector fields

Example: Let

$Y(t) = t^2U_1 - tU_3$, $\quad Z(t) = (1 - t^2)U_2 + tU_3$

Then $Y + Z$ is given by:

$$(Y+Z)(t) = (t^2+0)U_1 + (0+(1-t^2))U_2 + (-t+t)U_3$$

$$= t^2U_1 + (1-t^2)U_2 + 0U_3$$

Vector Fields on Curves

Scalar Multiplication

Let

$$Y(t) = y_1U_1 + y_2U_2 + y_3U_3$$

be a vector field on $\alpha: I \to \mathbb{R}^3$ and $f(t)$ be a real valued functions. Then we define fY as follows

$$(fY)(t) = fy_1U_1 + fy_2\,U_2 + fy_3\,U_3\,.$$

Vector Fields on Curves

Scalar Multiplication

Example: Let

$$Y(t) = t^2U_1 - tU_3\,, \qquad f(t) = \frac{t+1}{t}$$

Then fY is given by:

$$(fY)(t) = t^2\left(\frac{t+1}{t}\right)U_1 - t\left(\frac{t+1}{t}\right)U_3$$

$$(fY)(t) = t(t+1)U_1 - (t+1)U_3$$

Vector Fields on Curves

Dot product of two vector fields

Let

$$Y(t) = y_1U_1 + y_2U_2 + y_3U_3$$

and

$$Z(t) = z_1U_1 + z_2U_2 + z_3U_3$$

be two vector fields on $\alpha: I \to \mathbb{R}^3$. Then the dot product of of $Y(t)$ and $Z(t)$ is defined as:

$$Y.Z(t) = y_1z_1 + y_2z_2 + y_3z_3\,.$$

Vector Fields on Curves

Dot product of two vector fields

Example:

Let

$Y(t) = t^2U_1 - tU_3$, $\quad Z(t) = (1 - t^2)U_2 + tU_3$

Then $Y.Z$ is given by:

$$Y(t) = t^2U_1 + 0U_2 - tU_3$$

$$Z(t) = 0U_1 + (1-t^2)U_2 + tU_3$$

$$(Y.Z)(t) = (t^2)(0) + (0)(1-t^2) + (-t)(t)$$

$$= -t^2.$$

Vector Fields on Curves

Cross product of two vector fields

Let

$$Y(t) = y_1U_1 + y_2U_2 + y_3U_3$$

And

$$Z(t) = z_1U_1 + z_2U_2 + z_3U_3$$

be two vector fields on $\alpha: I \to \mathbb{R}^3$. Then the cross product of of $Y(t)$ and $Z(t)$ is defined as:

$$Y \times Z = \begin{vmatrix} U_1 & U_2 & U_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{vmatrix} = \begin{vmatrix} U_1(\alpha(t)) & U_2(\alpha(t)) & U_3(\alpha(t)) \\ y_1(t) & y_2(t) & y_3(t) \\ z_1(t) & z_2(t) & z_3(t) \end{vmatrix}$$

Vector Fields on Curves

Cross product of two vector fields

Example: Let

$Y(t) = t^2U_1 - tU_3$, $\quad Z(t) = (1 - t^2)U_2 + tU_3$

Then $Y \times Z$ is given below:

$$(Y \times Z)(t) = \begin{vmatrix} U_1(\alpha(t)) & U_2(\alpha(t)) & U_3(\alpha(t)) \\ t^2 & 0 & -t \\ 0 & (1-t^2) & t \end{vmatrix}$$

$$= (t)(1-t^2)\,U_1 - (t^3 - 0)\,U_2 + (t^2(1-t^2))\,U_3$$

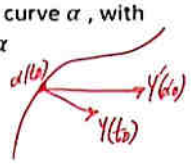
Differentiation of Vector Fields

Differentiation of a vector field on a curve

Definition: Let $Y(t)$ be a vector field on a curve α, with $Y(t) = \sum y_i U_i$. Then the new vector filed on α

$$Y'(t) = \sum \frac{dy_i}{dt} U_i\,(\alpha(t))$$

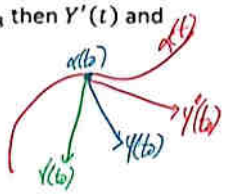
is the derivative of $Y(t)$.



Example: Let $Y(t) = 2tU_1 - (t^3 + 1)U_3$ then $Y'(t)$ and $Y''(t)$.

$Y'(t) = 2U_1 - (3t^2)U_3$

$\therefore Y''(t) = -6tU_3$



Differentiation of Vector Fields

Properties of differentiation

Let

$$Y(t) = y_1U_1 + y_2U_2 + y_3U_3$$

and

$$Z(t) = z_1U_1 + z_2U_2 + z_3U_3$$

be two vector valued functions on $\alpha: I \to \mathbb{R}^3$. Let $f: \mathbb{R} \to \mathbb{R}$ be a real-valued function. Let a and b be two real numbers then

Linearity:

$$(aY + bZ)' = aY' + bZ'$$

Leibnizian properties

$$(Y.Z)' = Y'.Z + Y.Z'$$
$$(fY)' = f'Y + fY'$$

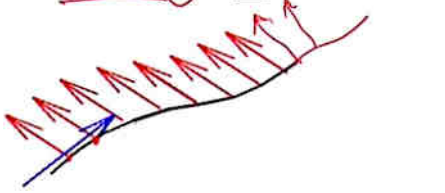
Differentiation of Vector Fields

Parallel vector fields

A vector field on a curve is parallel if all its values are parallel tangent vectors.

That is

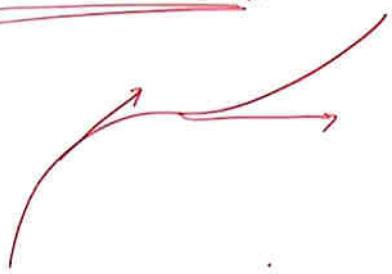
$$Y(t) = (c_1, c_2, c_3)_{\alpha(t)} = \sum c_i U_i(\alpha(t))$$



Differentiation of Vector Fields

Definition: Let $\alpha: I \to \boldsymbol{R}^3$ be a curve with coordinates $\alpha(t) = (\alpha_1(t), \alpha_2(t), \alpha_3(t))$ then the acceleration of α is defined as

$$\alpha'' = \left(\frac{d^2\alpha_1}{dt^2}, \frac{d^2\alpha_2}{dt^2}, \frac{d^2\alpha_3}{dt^2}\right)_{\alpha(t)}$$



Differentiation of Vector Fields

Vanishing derivatives

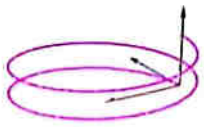
Lemma: Let $\alpha: I \to \mathbb{R}^3$ be a curve then

- α is constant if and only if $\alpha' = 0$.
- If α is not constant, then α is a straight line if and only if $\alpha'' = 0$.
- A vector field Y on a curve α is parallel if and only if $Y' = 0$.

Geometry of Curves in $\mathbb{R}^3$

Topic No. 119

Curvature



Geometry of Curves in $\mathbb{R}^3$

Aim

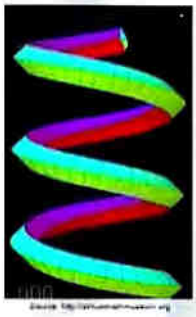
To study mathematical measurements of turning and twisting

Geometry of Curves in $\mathbb{R}^3$

Can we find the difference in the following curves?

Geometry of Curves in $\mathbb{R}^3$

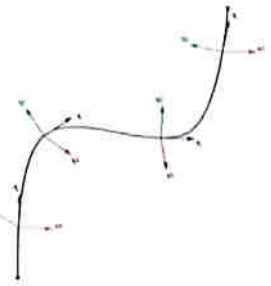
We want to quantify turning and twist in a curve.



Geometry of Curves in $\mathbb{R}^3$

Main tools

Three vector fields on a given curve



Geometry of Curves in $\mathbb{R}^3$

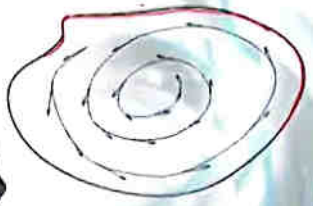
Speed limit

We will assume that the curves are unit speed curves, that is,

$$\|\beta'(t)\| = 1$$

Curvature

Curvature

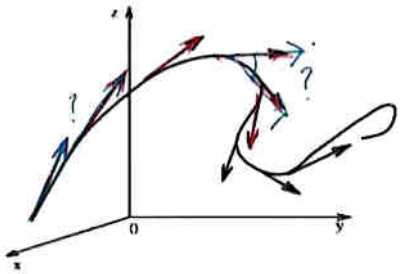


Curvature

How to calculate curvature?

$\|\beta'\| = 1$, $T = \beta'$

$\frac{d}{ds}\beta'$

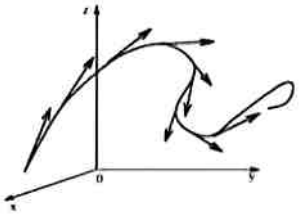


Curvature

Unit tangent: Let $\beta: I \to \mathbb{R}^3$ be a unit speed curve. The velocity

$$T = \beta'$$

is called the unit tangent vector field on β.



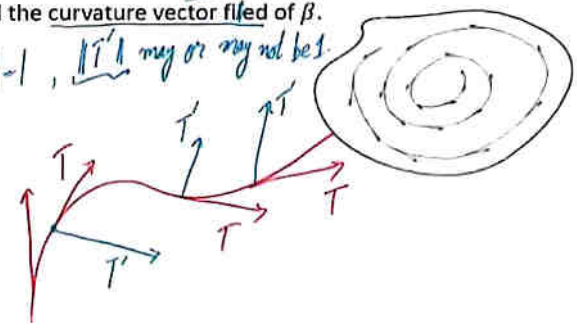
Curvature

Definition: The rate of change of tangent, that is,

$$T' = \beta''$$

is called the curvature vector filed of β.

$\|T\| = 1$, $\|T'\|$ may or may not be 1.



Curvature

Definition (Curvature): The function $\kappa: I \to \mathbb{R}$ defined by

$$\kappa(t) = \|T'(t)\|$$

is called the curvature function of β.

The curvature function satisfies $\kappa(t) \geq 0$ for all $t \in I$.

Curvature

- When κ is small then the curve is close to straight line.
- When κ is large then the turn in the curve is sharp.

curvature value is large.

curvature is small

Curvature

Example: Find curvature of the following unit speed curve at an arbitrary point

$$\alpha: (0,2\pi) \to \mathbb{R}^3$$

define by

$$\alpha(s) = (\cos s, \sin s, 0)$$

Solution:

$\kappa = \|T'\| = ?$

$T = \alpha'(s) = (-\sin s, \cos s, 0)$

$T' = \alpha''(s) = (-\cos s, -\sin s, 0)$

$\kappa = \|T'\| = \sqrt{\cos^2 s + \sin^2 s} = 1$

Curvature

Example: Find curvature of the following unit speed curve at an arbitrary point

$$\alpha: \mathbb{R} \to \mathbb{R}^3$$

define by

$$\alpha(s) = p + tv$$

Solution: $p = (p_1, p_2, p_3)$, $v = (v_1, v_2, v_3)$

$\|v\| = 1$

$\alpha(t) = (p_1 + tv_1, p_2 + tv_2, p_3 + tv_3)$

$T = \alpha'(t) = (v_1, v_2, v_3)$

$T' = \alpha''(t) = (0,0,0)$ $\quad \kappa = \|T'\| = 0$

Geometry of Curves in $\mathbb{R}^3$

Frenet-Serret frame
Or
TNB frame

Topic No. 120

Geometry of Curves in $\mathbb{R}^3$

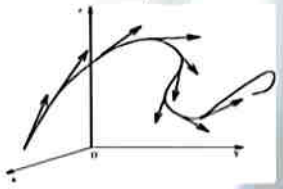
Recall

Unit tangent vector field

Let $\beta: I \to \mathbb{R}^3$ be a unit speed curve. The velocity

$$T = \beta'$$

is called the unit tangent vector field on β.



Geometry of Curves in $\mathbb{R}^3$

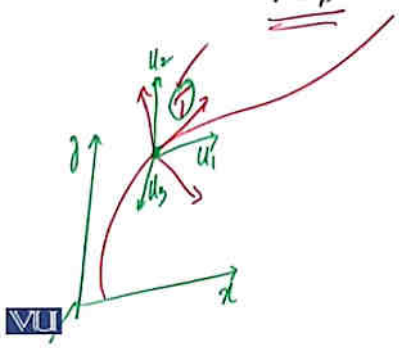
We need more vector fields on curves

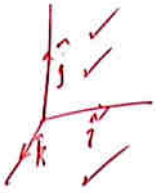
- So far we have defined unit tangent vector field on a unit speed curve, which gives curvature.
- To further explore geometry of curves in $\mathbb{R}^3$ we need more vector fields (on curves).

Three Vector Fields on Curves

Let $\beta: I \to \mathbb{R}^3$ be a unit speed curve. The unit tangent field is given by

$$T = \beta'$$





Three Vector Fields on Curves

Let $\beta: I \to \mathbb{R}^3$ be a unit speed curve. The unit tangent field is given by

$$T = \beta'$$

Since

$$\|T\|^2 = 1$$

So

$$T.T = 1 \quad (1)$$

$(V.W)' = V'.W + V.W' = 0$

Differentiating both sides of (1)

$T'.T + T.T' = 0$

$\Rightarrow T'.T = 0$

Three Vector Fields on Curves

Let $\beta: I \to \mathbb{R}^3$ be a unit speed curve. The unit tangent field is given by

$$T = \beta'$$

Since

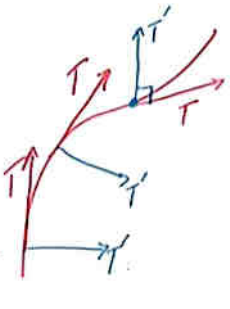
$$\|T\| = 1$$

So

$$T.T = 1$$

and this gives us

$$T'.T = 0$$



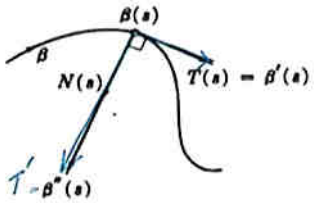
Three Vector Fields on Curves

A vector field V on β is called normal to β if $V.\beta' = 0$.

Since

$$T'.T = 0$$

so curvature vector field is normal to β.



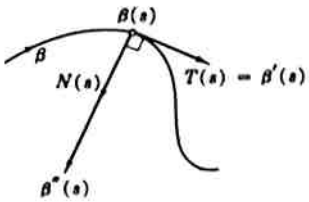
Three Vector Fields on Curves

Principal normal

Let β be a curve. Let $\kappa > 0$, that is, curve is never straight.
We define principal normal vector field N as

$$N = \frac{T'}{\kappa}$$

We can see that $\|N\| = 1$, hence a unit vector.



Three Vector Fields on Curves

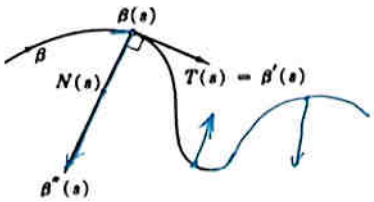
Principal normal

Let β be a curve. Let $\kappa > 0$, that is, curve is never straight.
We define principal normal vector field N as

$$N = \frac{T'}{\kappa}$$

We can see that $\|N\| = 1$, hence a unit vector.

The principal normal vector tells us the direction in which the curve turns.



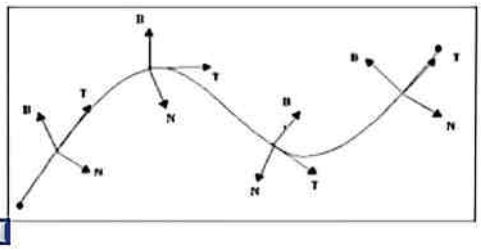
Three Vector Fields on Curves

Binormal vector field

The vector field on curve β

$$B = T \times N$$

is called the binormal vector field of β.



Three Vector Fields on Curves

Lemma: If β is a unit-speed curve in $\mathbb{R}^3$ with $\kappa > 0$. Then the three vector fields T, N, B on β are unit vectors that are mutually orthogonal at each point.

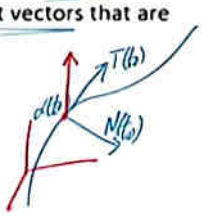
Proof:

$$\|T\| = \|\beta'\| = 1,$$

$$\|N\| = \left\|\frac{T'}{\kappa}\right\| = \frac{\|T'\|}{\kappa} = \frac{\kappa}{\kappa} = 1$$

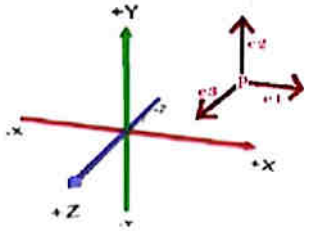
$$\|B\| = \|T \times N\| = \|T\|\,\|N\| \sin 90 = 1$$

$$T.N = 0; \quad T.B = T.T \times N = 0, \quad N.B = N.T \times N = 0$$



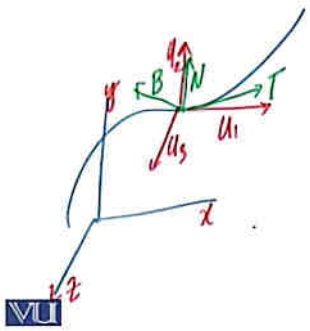
Frame

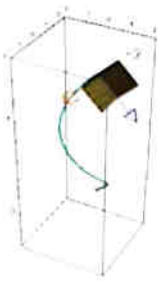
Definition: A set e_1, e_2, e_3 of three mutually orthogonal unit vectors tangent to $\mathbb{R}^3$ at point p is called a frame at the point p.



Frenet-Serret Frame Field

The triplet (T, B, N) provide a frame at each and every point of a unit speed curve.





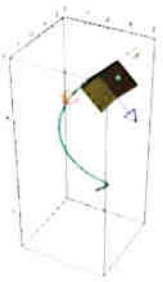
Frenet-Serret Frame Field

The triplet (T, B, N) provide a frame at each and every point of a unit speed curve.

TN-plane is called osculating plane.

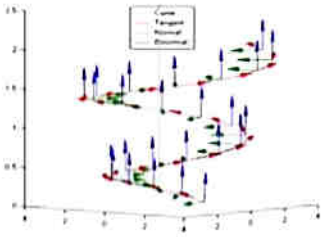
TB-plane is called rectifying plane.

NB −plane is called normal plane.



Frenet-Serret Frame Field

The triplet (T, B, N) is called the Frenet-Serret frame field on β or sometimes known as TNB frame field on β.



Frenet-Serret Formulas

Topic No. 121

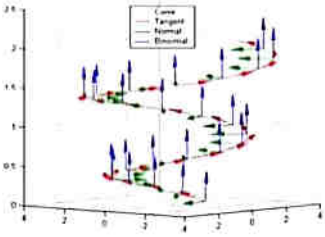
Frenet-Serret formulas

$$\begin{pmatrix} \vec{T}' \\ \vec{N}' \\ \vec{B}' \end{pmatrix} = \begin{pmatrix} 0 & k & 0 \\ -k & 0 & \tau \\ 0 & -\tau & 0 \end{pmatrix} \begin{pmatrix} \vec{T} \\ \vec{N} \\ \vec{B} \end{pmatrix}$$

Frenet-Serret Formulas

Recall

The triplet (T, B, N) is called the Frenet-Serret frame field on β or sometimes known as TNB frame field on β.



Frenet-Serret Formulas

The derivatives of a T, N and B

The derivative of T, that is, curvature vector field provide us geometric information about a unit speed curve in $\mathbb{R}^3$.

Now we want to express derivatives of T, N, B in terms of T, N, B.

Frenet-Serret Formulas

The Frenet-Serret Formulas

Theorem: If β is a unit speed curve with curvature κ and torsion τ then

$$\begin{aligned} T' &= \kappa N \\ N' &= -\kappa T + \tau B \\ B' &= -\tau N \end{aligned}$$

$$\begin{pmatrix} T' \\ N' \\ B' \end{pmatrix} = \begin{pmatrix} 0 & \kappa & 0 \\ -\kappa & 0 & \tau \\ 0 & -\tau & 0 \end{pmatrix} \begin{pmatrix} T \\ N \\ B \end{pmatrix}$$

$A^t = -A$

Frenet-Serret Formulas

T': The derivative of T

We know that

$$N = \frac{T'}{\kappa}$$

So

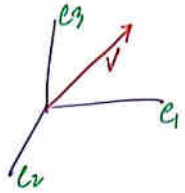
$$T' = \kappa N$$

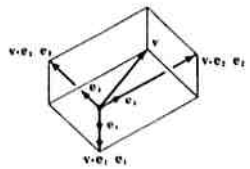
Frenet-Serret Formulas

Recall

Theorem: Let e_1, e_2, e_3 be a frame at a point p of $\boldsymbol{R^3}$. If v is any tangent vector to $\boldsymbol{R^3}$ at p, then

$$v = (v.e_1)e_1 + (v.e_2)e_2 + (v.e_3)e_3$$





Frenet-Serret Formulas

B' : The derivative of B

We can write B' in the following form

$$B' = (B'.T)T + (B'.N)N + (B'.B)B$$

$\|B\| = 1$

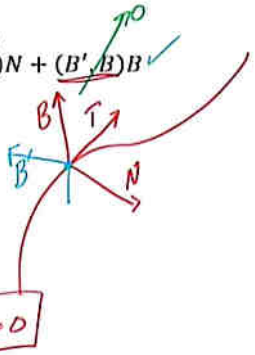
$\Rightarrow \|B\|^2 = 1$

$\Rightarrow B.B = 1$

Differentiate

$B'.B + B.B' = 0$

$\Rightarrow 2B'.B = 0 \Rightarrow \boxed{B'.B = 0}$



VU

Frenet-Serret Formulas

B' : The derivative of B

We can write B' in the following form

$$B' = (B'.T)T + (B'.N)N + (B'.B)B$$

$B.T = 0$

Differentiating, we get

$B'.T + B.T' = 0$ $\quad T' = KN$

$B'.T = -B.(KN)$

$= -K(B.N)$

$= -K(0) = 0$

VU

Frenet-Serret Formulas

B' : The derivative of B

We can write B' in the following form

$$B' = (B'.T)T + (B'.N)N + (B'.B)B$$

$\Rightarrow B' = (B'.N)N$

$\Rightarrow B'$ is parallel to N.

VU

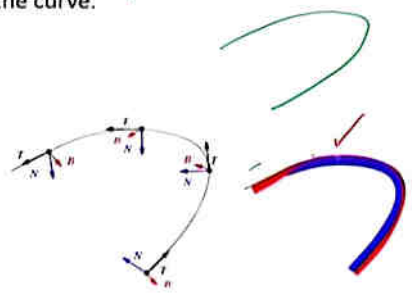
Frenet-Serret Formulas

The torsion of a curve

The function $\tau: I \to \mathbb{R}$ such that

$$B' = -\tau N$$

is called the torsion of the curve.



VU

Frenet-Serret Formulas

N' : The derivative of N

We can write N′ in the following form

$$N' = (N'.T)T + (N'.N)N + (N'.B)B$$

$\|N\| = 1$

$\|N\|^2 = 1$

$N.N = 1$

$\Rightarrow N'.N = 0$

VU

Frenet-Serret Formulas

N' : The derivative of N

We can write N′ in the following form

$$N' = (N'.T)T + (N'.N)N + (N'.B)B$$

$N.T = 0$ $\quad -K$

Differentiating

$N'.T + N.T' = 0$

$N'.T = -N.(KN)$ $\quad T' = KN$

$= -K\,N.N$

$N'.T = -K$

VU

Frenet-Serret Formulas

N' : The derivative of N

We can write N′ in the following form

$$N' = (N'.T)T + (N'.N)N + (N'.B)B$$

$N.B = 0$

$N'.B + N.B' = 0$

$N'.B = -N.(-\tau N)$ $\quad B' = -\tau N$

$= \tau\, N.N$

$\boxed{N'.B = \tau}$

VU

Frenet-Serret Formulas

Example: Computer Frenet Frame, cuvature and torsion for the following unit speed curve

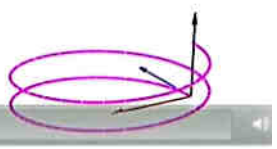
$$\beta(s) = \left(a\cos\frac{s}{c}, a\sin\frac{s}{c}, \frac{bs}{c}\right),$$

where $c = \sqrt{a^2 + b^2}$ and $a > 0$.

Solution:

$T(s) = \beta'(s) = \left(\frac{-a}{c}\sin\frac{s}{c}, \frac{a}{c}\cos\frac{s}{c}, \frac{b}{c}\right)$

$T'(s) = \left(\frac{-a}{c^2}\cos\frac{s}{c}, \frac{-a}{c^2}\sin\frac{s}{c}, 0\right)$



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Frenet-Serret Formulas

Solution (contd.)

$$\kappa = \|T'(s)\| = \left\| \left(\frac{-a}{c^2}\cos\frac{s}{c}, \frac{-a}{c^2}\sin\frac{s}{c}, 0 \right) \right\|$$

$$= \frac{a}{c^2} = \frac{a}{a^2+b^2} > 0$$

$$T' = \kappa N \Rightarrow N = \frac{1}{\kappa} T'$$

$$N = \left(-\cos\frac{s}{c}, -\sin\frac{s}{c}, 0 \right).$$

VU

Frenet-Serret Formulas

Solution (contd.)

$$B = T \times N = \begin{vmatrix} i_1 & i_2 & i_3 \\ -\frac{a}{c}\sin\frac{s}{c} & \frac{a}{c^2}\sin\frac{s}{c} & \frac{b}{c} \\ -\cos\frac{s}{c} & -\sin\frac{s}{c} & 0 \end{vmatrix}$$

$$\Rightarrow B = \left(\frac{b}{c}\sin\frac{s}{c}, -\frac{b}{c}\cos\frac{s}{c}, \frac{a}{c} \right). \quad B' = -\tau N$$

$$B' = \left(\frac{b}{c^2}\cos\frac{s}{c}, \frac{b}{c^2}\sin\frac{s}{c}, 0 \right)$$

$$N = \left(-\cos\frac{s}{c}, -\sin\frac{s}{c}, 0 \right) \Rightarrow \tau = \frac{b}{c^2}$$

VU

Complex Analysis & Differential Geometry

MTH632

End

دعاؤں میں یاد رکھیے گا۔

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